MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3830 REAGENTS – FLOTATION RGENT EQUIP. DESIGN OF FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
RGENT.	:	Reagents
EQUIP.	:	Equipment

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1. INTENT

This document is to prepare design of foundation for **Flotation Reagents Building (3830)**.The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in STAAD PRO. For design of raft force values are directly taken from STAAD & designed in excel sheet. Column is designed in STAAD.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Doc. No. MD-216-3830-EG-ST-GAD-2350 attached in Attachment 1. Loads are also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

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5.3 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation	50
Exposed to weather or contact with ground	Column	50

5.4 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundation design has been considered 350 Kn/m^2 at 1.2 m below EGL. (Refer Table 5 - BH-10 of Geotechnical Report : MD-216-3000-EG-CV-RPT-1008).

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6. DESIGN CALCULATION FOR FOUNDATION

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DESIGN CALCULATION OF FOOTING F1

Size of Columns at node	Along-X	Along-Z
2	0.50m	0.85m

Size of fdn. raft:	2.75m	4.75m
Thickness of raft:	0.60	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.80	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

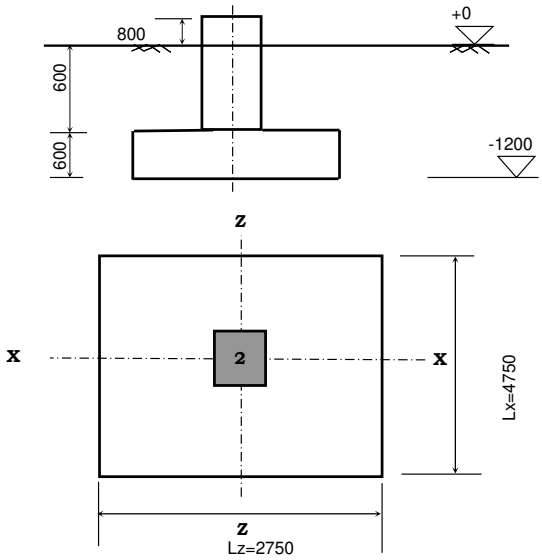
Weight of raft = (2.75*4.75*0.6*24) = 188.10KN

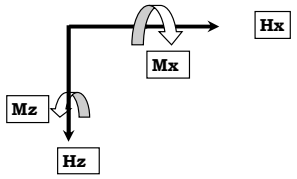
Weight of soil = ((2.75*4.75)-(0.5*0.85))*0.6*20 = 151.65KN

Total = 339.75KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F1

Load at bottom of footing	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	43.25	0.00	0.00	3.00	29.00
202 DL+L+TL1	41.25	0.00	0.00	3.00	29.00
203 DL+L+TL2	45.25	0.00	0.00	3.00	29.00
204 DL+LR	104.25	0.00	0.00	5.00	43.00
205 DL+0.75L+0.75TL1+0.75LR	87.50	0.00	0.00	4.50	39.50
206 DL+0.75L+0.75TL2+0.75LR	90.50	0.00	0.00	4.50	39.50
207 DL+0.6WL1	-26.95	0.00	0.00	17.40	173.60
208 DL+WL2	-30.75	0.00	0.00	30.00	290.00
209 DL+WL3	-73.75	0.00	0.00	27.00	270.00
210 DL+WL4	-30.75	0.00	0.00	30.00	290.00
211 DL+WL5	-61.75	0.00	0.00	-35.00	-317.00
212 DL+WL6	-18.75	0.00	0.00	-32.00	-297.00
213 DL+WL7	-61.75	0.00	0.00	-35.00	-317.00
214 DL+WL8	-18.75	0.00	0.00	-32.00	-297.00
215 DL+WL9	-75.75	0.00	0.00	-1.00	-1.00
216 DL+WL10	-32.75	0.00	0.00	1.00	17.00
217 DL+WL11	-75.75	0.00	0.00	-8.00	-56.00
218 DL+WL12	-31.75	0.00	0.00	-5.00	-35.00
219 DL+0.75WL1+0.75L+0.75LR	1.25	0.00	0.00	22.50	220.25
220 DL+0.75WL2+0.75L+0.75LR	33.50	0.00	0.00	24.75	235.25
221 DL+0.75WL3+0.75L+0.75LR	1.25	0.00	0.00	22.50	220.25
222 DL+0.75WL4+0.75L+0.75LR	33.50	0.00	0.00	24.75	235.25
223 DL+0.75WL5+0.75L+0.75LR	10.25	0.00	0.00	-24.00	-220.00
224 DL+0.75WL67+0.75L+0.75LR	42.50	0.00	0.00	-21.75	-205.00
225 DL+0.75WL7+0.75L+0.75LR	10.25	0.00	0.00	-24.00	-220.00
226 DL+0.75WL8+0.75L+0.75LR	42.50	0.00	0.00	-21.75	-205.00
227 DL+0.75WL9+0.75L+0.75LR	-0.25	0.00	0.00	1.50	17.00
228 DL+0.75WL10+0.75L+0.75LR	32.00	0.00	0.00	3.00	30.50
229 DL+0.75WL11+0.75L+0.75LR	-0.25	0.00	0.00	-3.75	-24.25
230 DL+0.75WL12+0.75L+0.75LR	32.75	0.00	0.00	-1.50	-8.50
231 0.6DL+WL1	-91.05	0.00	0.00	25.80	258.40
232 0.6DL+WL2	-48.05	0.00	0.00	28.80	278.40
233 0.6DL+WL3	-91.05	0.00	0.00	25.80	258.40
234 0.6DL+WL4	-48.05	0.00	0.00	28.80	278.40
235 0.6DL+WL5	-79.05	0.00	0.00	-36.20	-328.60
236 0.6DL+WL6	-36.05	0.00	0.00	-33.20	-308.60
237 0.6DL+WL7	-79.05	0.00	0.00	-36.20	-328.60
238 0.6DL+WL8	-36.05	0.00	0.00	-33.20	-308.60
239 0.6DL+WL9	-93.05	0.00	0.00	-2.20	-12.60
240 0.6DL+WL10	-50.05	0.00	0.00	-0.20	5.40
241 0.6DL+WL11	-93.05	0.00	0.00	-9.20	-67.60
242 0.6DL+WL12	-49.05	0.00	0.00	-6.20	-46.60
243 0.6DL+0.7SL1	25.95	0.00	0.00	2.50	23.70
244 0.6DL+0.7SL2	25.95	0.00	0.00	1.10	11.10
245 DL+0.75WL1+0.75L+0.75LR+CR1	36.25	0.00	0.00	31.50	278.25
246 DL+0.75WL2+0.75L+0.75LR+CR1	68.50	0.00	0.00	33.75	293.25
247 DL+0.75WL3+0.75L+0.75LR+CR1	36.25	0.00	0.00	31.50	278.25
248 DL+0.75WL4+0.75L+0.75LR+CR1	68.50	0.00	0.00	33.75	293.25

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258 DL+0.75WL2+0.75L+0.75LR+CR2	59.50	0.00	0.00	31.75	282.25
259 DL+0.75WL3+0.75L+0.75LR+CR2	27.25	0.00	0.00	29.50	267.25
260 DL+0.75WL4+0.75L+0.75LR+CR2	59.50	0.00	0.00	31.75	282.25
261 DL+0.75WL5+0.75L+0.75LR+CR2	36.25	0.00	0.00	-17.00	-173.00
262 DL+0.75WL67+0.75L+0.75LR+CR2	68.50	0.00	0.00	-14.75	-158.00
263 DL+0.75WL7+0.75L+0.75LR+CR2	36.25	0.00	0.00	-17.00	-173.00
264 DL+0.75WL8+0.75L+0.75LR+CR2	68.50	0.00	0.00	-14.75	-158.00
265 DL+0.75WL9+0.75L+0.75LR+CR2	25.75	0.00	0.00	8.50	64.00
266 DL+0.75WL10+0.75L+0.75LR+CR2	58.00	0.00	0.00	10.00	77.50
267 DL+0.75WL11+0.75L+0.75LR+CR2	25.75	0.00	0.00	3.25	22.75
268 DL+0.75WL12+0.75L+0.75LR+CR2	58.75	0.00	0.00	5.50	38.50
269 DL+0.75WL1+0.75L+0.75LR+CR3	1.25	0.00	0.00	22.50	220.25
270 DL+0.75WL2+0.75L+0.75LR+CR3	33.50	0.00	0.00	24.75	235.25
271 DL+0.75WL3+0.75L+0.75LR+CR3	1.25	0.00	0.00	22.50	220.25
272 DL+0.75WL4+0.75L+0.75LR+CR3	33.50	0.00	0.00	24.75	235.25
273 DL+0.75WL5+0.75L+0.75LR+CR3	10.25	0.00	0.00	-24.00	-220.00
274 DL+0.75WL67+0.75L+0.75LR+CR3	42.50	0.00	0.00	-21.75	-205.00
275 DL+0.75WL7+0.75L+0.75LR+CR3	10.25	0.00	0.00	-24.00	-220.00
276 DL+0.75WL8+0.75L+0.75LR+CR3	42.50	0.00	0.00	-21.75	-205.00
277 DL+0.75WL9+0.75L+0.75LR+CR3	-0.25	0.00	0.00	1.50	17.00
278 DL+0.75WL10+0.75L+0.75LR+CR3	32.00	0.00	0.00	3.00	30.50
279 DL+0.75WL11+0.75L+0.75LR+CR3	-0.25	0.00	0.00	-3.75	-24.25
280 DL+0.75WL12+0.75L+0.75LR+CR3	32.75	0.00	0.00	-1.50	-8.50
281 DL+0.75WL1+0.75L+0.75LR+CR4	1.25	0.00	0.00	22.50	220.25
282 DL+0.75WL2+0.75L+0.75LR+CR4	33.50	0.00	0.00	24.75	235.25
283 DL+0.75WL3+0.75L+0.75LR+CR4	1.25	0.00	0.00	22.50	220.25
284 DL+0.75WL4+0.75L+0.75LR+CR4	33.50	0.00	0.00	24.75	235.25
285 DL+0.75WL5+0.75L+0.75LR+CR4	10.25	0.00	0.00	-24.00	-220.00
286 DL+0.75WL67+0.75L+0.75LR+CR4	42.50	0.00	0.00	-21.75	-205.00
287 DL+0.75WL7+0.75L+0.75LR+CR4	10.25	0.00	0.00	-24.00	-220.00
288 DL+0.75WL8+0.75L+0.75LR+CR4	42.50	0.00	0.00	-21.75	-205.00
289 DL+0.75WL9+0.75L+0.75LR+CR4	-0.25	0.00	0.00	1.50	17.00
290 DL+0.75WL10+0.75L+0.75LR+CR4	32.00	0.00	0.00	3.00	30.50
291 DL+0.75WL11+0.75L+0.75LR+CR4	-0.25	0.00	0.00	-3.75	-24.25
292 DL+0.75WL12+0.75L+0.75LR+CR4	32.75	0.00	0.00	-1.50	-8.50

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	383.0	3.00	29.0	0.0	0.0	NA	31.4	NA	51.1
202 DL+L+TL1	381.0	3.00	29.0	0.0	0.0	NA	31.2	NA	50.8
203 DL+L+TL2	385.0	3.00	29.0	0.0	0.0	NA	31.5	NA	51.3
204 DL+LR	444.0	5.00	43.0	0.0	0.0	NA	24.5	NA	35.5
205 DL+0.75L+0.75TL1+0.75LR	427.3	4.50	39.5	0.0	0.0	NA	25.7	NA	38.0
206 DL+0.75L+0.75TL2+0.75LR	430.3	4.50	39.5	0.0	0.0	NA	25.9	NA	38.2
207 DL+0.6WL1	312.8	17.40	173.6	0.0	0.0	NA	4.3	NA	7.2
208 DL+WL2	309.0	30.00	290.0	0.0	0.0	NA	2.5	NA	4.1
209 DL+WL3	266.0	27.00	270.0	0.0	0.0	NA	2.3	NA	3.9
210 DL+WL4	309.0	30.00	290.0	0.0	0.0	NA	2.5	NA	4.1
211 DL+WL5	278.0	-35.00	-317.0	0.0	0.0	NA	2.1	NA	3.2
212 DL+WL6	321.0	-32.00	-297.0	0.0	0.0	NA	2.6	NA	4.0
213 DL+WL7	278.0	-35.00	-317.0	0.0	0.0	NA	2.1	NA	3.2
214 DL+WL8	321.0	-32.00	-297.0	0.0	0.0	NA	2.6	NA	4.0
215 DL+WL9	264.0	-1.00	-1.0	0.0	0.0	NA	627.0	NA	105.6
216 DL+WL10	307.0	1.00	17.0	0.0	0.0	NA	42.9	NA	122.8
217 DL+WL11	264.0	-8.00	-56.0	0.0	0.0	NA	11.2	NA	13.2
218 DL+WL12	308.0	-5.00	-35.0	0.0	0.0	NA	20.9	NA	24.6
219 DL+0.75WL1+0.75L+0.75LR	341.0	22.50	220.3	0.0	0.0	NA	3.7	NA	6.1
220 DL+0.75WL2+0.75L+0.75LR	373.3	24.75	235.3	0.0	0.0	NA	3.8	NA	6.0
221 DL+0.75WL3+0.75L+0.75LR	341.0	22.50	220.3	0.0	0.0	NA	3.7	NA	6.1
222 DL+0.75WL4+0.75L+0.75LR	373.3	24.75	235.3	0.0	0.0	NA	3.8	NA	6.0
223 DL+0.75WL5+0.75L+0.75LR	350.0	-24.00	-220.0	0.0	0.0	NA	3.8	NA	5.8
224 DL+0.75WL67+0.75L+0.75LR	382.3	-21.75	-205.0	0.0	0.0	NA	4.4	NA	7.0
225 DL+0.75WL7+0.75L+0.75LR	350.0	-24.00	-220.0	0.0	0.0	NA	3.8	NA	5.8
226 DL+0.75WL8+0.75L+0.75LR	382.3	-21.75	-205.0	0.0	0.0	NA	4.4	NA	7.0
227 DL+0.75WL9+0.75L+0.75LR	339.5	1.50	17.0	0.0	0.0	NA	47.4	NA	90.5
228 DL+0.75WL10+0.75L+0.75LR	371.8	3.00	30.5	0.0	0.0	NA	28.9	NA	49.6
229 DL+0.75WL11+0.75L+0.75LR	339.5	-3.75	-24.3	0.0	0.0	NA	33.3	NA	36.2
230 DL+0.75WL12+0.75L+0.75LR	372.5	-1.50	-8.5	0.0	0.0	NA	104.1	NA	99.3
231 0.6DL+WL1	248.7	25.80	258.4	0.0	0.0	NA	2.3	NA	3.9
232 0.6DL+WL2	291.7	28.80	278.4	0.0	0.0	NA	2.5	NA	4.1
233 0.6DL+WL3	248.7	25.80	258.4	0.0	0.0	NA	2.3	NA	3.9
234 0.6DL+WL4	291.7	28.80	278.4	0.0	0.0	NA	2.5	NA	4.1
235 0.6DL+WL5	260.7	-36.20	-328.6	0.0	0.0	NA	1.9	NA	2.9
236 0.6DL+WL6	303.7	-33.20	-308.6	0.0	0.0	NA	2.3	NA	3.7
237 0.6DL+WL7	260.7	-36.20	-328.6	0.0	0.0	NA	1.9	NA	2.9
238 0.6DL+WL8	303.7	-33.20	-308.6	0.0	0.0	NA	2.3	NA	3.7
239 0.6DL+WL9	246.7	-2.20	-12.6	0.0	0.0	NA	46.5	NA	44.9
240 0.6DL+WL10	289.7	-0.20	5.4	0.0	0.0	NA	127.4	NA	579.4
241 0.6DL+WL11	246.7	-9.20	-67.6	0.0	0.0	NA	8.7	NA	10.7
242 0.6DL+WL12	290.7	-6.20	-46.6	0.0	0.0	NA	14.8	NA	18.8

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243 0.6DL+0.7SL1	365.7	2.50	23.7	0.0	0.0	NA	36.6	NA	58.5
244 0.6DL+0.7SL2	365.7	1.10	11.1	0.0	0.0	NA	78.2	NA	133.0
245 DL+0.75WL1+0.75L+0.75LR+CR1	376.0	31.50	278.3	0.0	0.0	NA	3.2	NA	4.8
246 DL+0.75WL2+0.75L+0.75LR+CR1	408.3	33.75	293.3	0.0	0.0	NA	3.3	NA	4.8
247 DL+0.75WL3+0.75L+0.75LR+CR1	376.0	31.50	278.3	0.0	0.0	NA	3.2	NA	4.8
248 DL+0.75WL4+0.75L+0.75LR+CR1	408.3	33.75	293.3	0.0	0.0	NA	3.3	NA	4.8
258 DL+0.75WL2+0.75L+0.75LR+CR2	399.3	31.75	282.3	0.0	0.0	NA	3.4	NA	5.0
259 DL+0.75WL3+0.75L+0.75LR+CR2	367.0	29.50	267.3	0.0	0.0	NA	3.3	NA	5.0
260 DL+0.75WL4+0.75L+0.75LR+CR2	399.3	31.75	282.3	0.0	0.0	NA	3.4	NA	5.0
261 DL+0.75WL5+0.75L+0.75LR+CR2	376.0	-17.00	-173.0	0.0	0.0	NA	5.2	NA	8.8
262 DL+0.75WL67+0.75L+0.75LR+CR2	408.3	-14.75	-158.0	0.0	0.0	NA	6.1	NA	11.1
263 DL+0.75WL7+0.75L+0.75LR+CR2	376.0	-17.00	-173.0	0.0	0.0	NA	5.2	NA	8.8
264 DL+0.75WL8+0.75L+0.75LR+CR2	408.3	-14.75	-158.0	0.0	0.0	NA	6.1	NA	11.1
265 DL+0.75WL9+0.75L+0.75LR+CR2	365.5	8.50	64.0	0.0	0.0	NA	13.6	NA	17.2
266 DL+0.75WL10+0.75L+0.75LR+CR2	397.8	10.00	77.5	0.0	0.0	NA	12.2	NA	15.9
267 DL+0.75WL11+0.75L+0.75LR+CR2	365.5	3.25	22.8	0.0	0.0	NA	38.2	NA	45.0
268 DL+0.75WL12+0.75L+0.75LR+CR2	398.5	5.50	38.5	0.0	0.0	NA	24.6	NA	29.0
269 DL+0.75WL1+0.75L+0.75LR+CR3	341.0	22.50	220.3	0.0	0.0	NA	3.7	NA	6.1
270 DL+0.75WL2+0.75L+0.75LR+CR3	373.3	24.75	235.3	0.0	0.0	NA	3.8	NA	6.0
271 DL+0.75WL3+0.75L+0.75LR+CR3	341.0	22.50	220.3	0.0	0.0	NA	3.7	NA	6.1
272 DL+0.75WL4+0.75L+0.75LR+CR3	373.3	24.75	235.3	0.0	0.0	NA	3.8	NA	6.0
273 DL+0.75WL5+0.75L+0.75LR+CR3	350.0	-24.00	-220.0	0.0	0.0	NA	3.8	NA	5.8
274 DL+0.75WL67+0.75L+0.75LR+CR3	382.3	-21.75	-205.0	0.0	0.0	NA	4.4	NA	7.0
275 DL+0.75WL7+0.75L+0.75LR+CR3	350.0	-24.00	-220.0	0.0	0.0	NA	3.8	NA	5.8
276 DL+0.75WL8+0.75L+0.75LR+CR3	382.3	-21.75	-205.0	0.0	0.0	NA	4.4	NA	7.0
277 DL+0.75WL9+0.75L+0.75LR+CR3	339.5	1.50	17.0	0.0	0.0	NA	47.4	NA	90.5
278 DL+0.75WL10+0.75L+0.75LR+CR3	371.8	3.00	30.5	0.0	0.0	NA	28.9	NA	49.6
279 DL+0.75WL11+0.75L+0.75LR+CR3	339.5	-3.75	-24.3	0.0	0.0	NA	33.3	NA	36.2
280 DL+0.75WL12+0.75L+0.75LR+CR3	372.5	-1.50	-8.5	0.0	0.0	NA	104.1	NA	99.3
281 DL+0.75WL1+0.75L+0.75LR+CR4	341.0	22.50	220.3	0.0	0.0	NA	3.7	NA	6.1
282 DL+0.75WL2+0.75L+0.75LR+CR4	373.3	24.75	235.3	0.0	0.0	NA	3.8	NA	6.0
283 DL+0.75WL3+0.75L+0.75LR+CR4	341.0	22.50	220.3	0.0	0.0	NA	3.7	NA	6.1
284 DL+0.75WL4+0.75L+0.75LR+CR4	373.3	24.75	235.3	0.0	0.0	NA	3.8	NA	6.0
285 DL+0.75WL5+0.75L+0.75LR+CR4	350.0	-24.00	-220.0	0.0	0.0	NA	3.8	NA	5.8
286 DL+0.75WL67+0.75L+0.75LR+CR4	382.3	-21.75	-205.0	0.0	0.0	NA	4.4	NA	7.0
287 DL+0.75WL7+0.75L+0.75LR+CR4	350.0	-24.00	-220.0	0.0	0.0	NA	3.8	NA	5.8
288 DL+0.75WL8+0.75L+0.75LR+CR4	382.3	-21.75	-205.0	0.0	0.0	NA	4.4	NA	7.0
289 DL+0.75WL9+0.75L+0.75LR+CR4	339.5	1.50	17.0	0.0	0.0	NA	47.4	NA	90.5
290 DL+0.75WL10+0.75L+0.75LR+CR4	371.8	3.00	30.5	0.0	0.0	NA	28.9	NA	49.6
291 DL+0.75WL11+0.75L+0.75LR+CR4	339.5	-3.75	-24.3	0.0	0.0	NA	33.3	NA	36.2
292 DL+0.75WL12+0.75L+0.75LR+CR4	372.5	-1.50	-8.5	0.0	0.0	NA	104.1	NA	99.3

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Base pressure check of footing F1

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	55.35	0
Min Px	2	201	0	55.35	0
Max Py	529	246	0	81.56	0
Min Py	458	235	0	0.00	0
Max Pz	2	201	0	55.35	0
Min Pz	2	201	0	55.35	0

Gross Maximum Base pressure, p_{max} = 81.6 KN/m^2

Net Allowable Safe Bearing Capacity = 350 KN/m^2

Depth of soil above footing = 0.60 m

Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 362 KN/m^2

> 81.6 KN/m^2

Hence OK

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Design of foundation raft F1

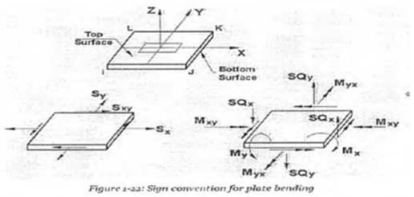
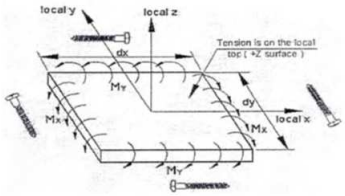


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F1_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	414	0.422	-0.293	0	0	0	49	112.258	-27.616	76.838	139.874	123:0.9DL+1.6WL5
Min Qx	417	-0.422	-0.293	0	0	0	49.222	112.258	27.616	76.838	139.874	123:0.9DL+1.6WL5
Max Qy	346	-0.084	0.428	0	0	0	33.017	134.42	-7.262	40.279	141.682	133:1.2DL+1.6WL1+1.0L+0.5LR+1.0CR1
Min Qy	416	-0.09	-0.489	0	0	0	41.306	149.549	7.052	48.358	156.601	123:0.9DL+1.6WL5
Max Sx	603	0	0	0	0	0	-0.005	-0.005	0.008	-0.013	-0.013	101:1.4DL
Min Sx	603	0	0	0	0	0	-0.005	-0.005	0.008	-0.013	-0.013	101:1.4DL
Max Sy	603	0	0	0	0	0	-0.005	-0.005	0.008	-0.013	-0.013	101:1.4DL
Min Sy	603	0	0	0	0	0	-0.005	-0.005	0.008	-0.013	-0.013	101:1.4DL
Max Sxy	603	0	0	0	0	0	-0.005	-0.005	0.008	-0.013	-0.013	101:1.4DL
Min Sxy	603	0	0	0	0	0	-0.005	-0.005	0.008	-0.013	-0.013	101:1.4DL
Max Mx	417	-0.422	-0.293	0	0	0	49.222	112.258	27.616	76.838	139.874	123:0.9DL+1.6WL5
Min Mx	417	0.382	0.241	0	0	0	-39.879	-96.774	-30.82	-70.699	-127.594	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Max My	416	-0.09	-0.489	0	0	0	41.306	149.549	7.052	48.358	156.601	123:0.9DL+1.6WL5
Min My	416	0.085	0.418	0	0	0	-27.225	-133.234	-8.319	-35.544	-141.553	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Max Mxy	333	0.15	0.247	0	0	0	-14.023	2.998	36.061	-50.084	39.059	123:0.9DL+1.6WL5
Min Mxy	327	-0.15	0.247	0	0	0	-14.023	2.998	-36.061	-50.084	39.059	123:0.9DL+1.6WL5

PLATE CENTER STRESS : FDN_F1_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	422	0.104	0.034	0	0	0	3.418	50.821	-6.47	9.888	57.291	123:0.9DL+1.6WL5
Min Qx	419	-0.104	0.034	0	0	0	3.418	50.821	6.47	9.888	57.291	123:0.9DL+1.6WL5
Max Qy	366	-0.025	0.161	0	0	0	27.456	45.329	1.496	28.952	46.825	133:1.2DL+1.6WL1+1.0L+0.5LR+1.0CR1
Min Qy	396	-0.028	-0.185	0	0	0	32.325	49.393	-1.909	34.234	51.302	123:0.9DL+1.6WL5
Max Sx	603	0	0	0	0	0	-0.005	-0.005	0.008	-0.013	-0.013	101:1.4DL
Min Sx	603	0	0	0	0	0	-0.005	-0.005	0.008	-0.013	-0.013	101:1.4DL
Max Sy	603	0	0	0	0	0	-0.005	-0.005	0.008	-0.013	-0.013	101:1.4DL
Min Sy	603	0	0	0	0	0	-0.005	-0.005	0.008	-0.013	-0.013	101:1.4DL
Max Sxy	603	0	0	0	0	0	-0.005	-0.005	0.008	-0.013	-0.013	101:1.4DL
Min Sxy	603	0	0	0	0	0	-0.005	-0.005	0.008	-0.013	-0.013	101:1.4DL
Max Mx	396	-0.028	-0.185	0	0	0	32.325	49.393	-1.909	34.234	51.302	123:0.9DL+1.6WL5
Min Mx	396	0.025	0.154	0	0	0	-25.498	-44.168	1.197	-26.695	-45.365	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Max My	422	0.104	0.034	0	0	0	3.418	50.821	-6.47	9.888	57.291	123:0.9DL+1.6WL5
Min My	396	0.025	0.154	0	0	0	-25.498	-44.168	1.197	-26.695	-45.365	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Max Mxy	337	0.019	0.128	0	0	0	0.609	0.237	17.441	18.05	17.678	123:0.9DL+1.6WL5
Min Mxy	342	-0.019	0.128	0	0	0	0.609	0.237	-17.441	18.05	17.678	123:0.9DL+1.6WL5

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		600	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

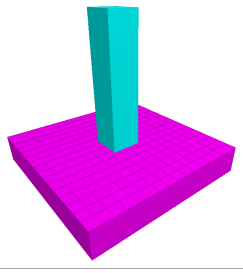
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c * (d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
123:0.9D	76.84	top face	50	544	76.838	0.35	0.0008	451.0	652.8	12	173.32	150	754.3	10.64	13.29	0.12	tension control
134:1.2D	-70.7	bottom face	50	542	70.699	0.32	0.0008	416.4		16	308.12	150	1341	18.91	23.64	0.066	tension control
123:0.9D	-50.08	bottom face	50	542	50.084	0.23	0.0005	294.5		16	308.12	150	1341	18.91	23.64	0.066	tension control
123:0.9D	-50.08	bottom face	50	542	50.084	0.23	0.0005	294.5		16	308.12	150	1341	18.91	23.64	0.066	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qx	0.104	1	62.4	9.888	0.001	0.544	558.957	527.759	933.321	527.76	1989	> 62.4	ok	395.8	φVc>Vu	not required	
	Min Qx	0.104	1	62.4	9.888			558.957			527.76	1989	> 62.4	ok	395.8	φVc>Vu	not required	

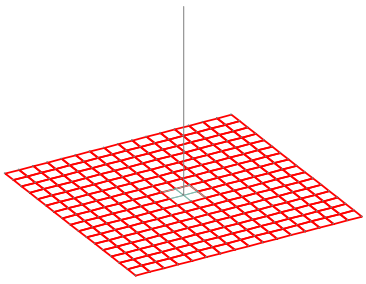
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c * (d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
123:0.9D	156.60	top face	50	544	156.6	0.71	0.0000	18.3	652.8	12	173.32	150	754.3	10.64	13.29	0.12	tension control
134:1.2D	-141.55	bottom face	50	542	141.55	0.64	0.0015	838.3		16	239.95	150	1341	18.91	23.64	0.066	tension control
123:0.9D	39.06	top face	50	544	39.059	0.18	0.0004	228.6		12	173.32	150	754.3	10.64	13.29	0.12	tension control
123:0.9D	39.06	top face	50	544	39.059	0.18	0.0004	228.6		12	173.32	150	754.3	10.64	13.29	0.12	tension control

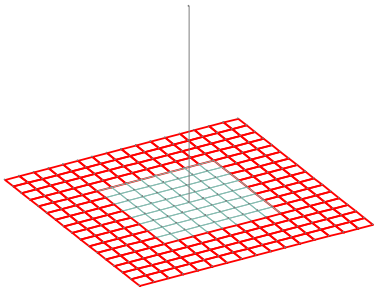
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qy	0.161	1	96.6	46.825	0.001	0.544	529.327	527.759	fc'933.3	527.76	555	> 96.6	ok	395.8	φVc>Vu	not required	
	Min Qy	0.185	1	111	51.302			530.029			527.76	555	> 111	ok	395.8	φVc>Vu	not required	



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
At Bottom **16ϕT** **150** c/c
At Top **12ϕT** **150** c/c

Along Z Direction :
At Bottom **16ϕT** **150** c/c
At Top **12ϕT** **150** c/c

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DESIGN CALCULATION OF FOOTING F1A

Size of Columns at node	Along-X	Along-Z
2	0.50m	0.85m

Size of fdn. raft:	2.10m	5.00m
Thickness of raft:	0.60	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.80	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

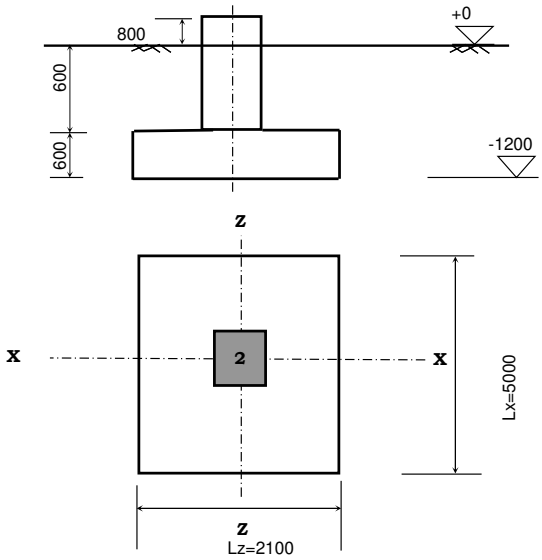
Weight of raft = (2.1*5*0.6*24) = 151.20KN

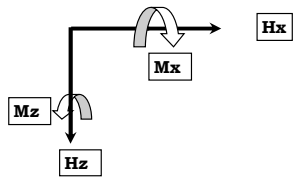
Weight of soil = ((2.1*5)-(0.5*0.85))*0.6*20 = 220.90KN

Total = 372.10KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F1A

Load at bottom of footing	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	43.25	0.00	0.00	3.00	29.00
202 DL+L+TL1	41.25	0.00	0.00	3.00	29.00
203 DL+L+TL2	45.25	0.00	0.00	3.00	29.00
204 DL+LR	104.25	0.00	0.00	5.00	43.00
205 DL+0.75L+0.75TL1+0.75LR	87.50	0.00	0.00	4.50	39.50
206 DL+0.75L+0.75TL2+0.75LR	90.50	0.00	0.00	4.50	39.50
207 DL+0.6WL1	-26.95	0.00	0.00	17.40	173.60
208 DL+WL2	-30.75	0.00	0.00	30.00	290.00
209 DL+WL3	-73.75	0.00	0.00	27.00	270.00
210 DL+WL4	-30.75	0.00	0.00	30.00	290.00
211 DL+WL5	-61.75	0.00	0.00	-35.00	-317.00
212 DL+WL6	-18.75	0.00	0.00	-32.00	-297.00
213 DL+WL7	-61.75	0.00	0.00	-35.00	-317.00
214 DL+WL8	-18.75	0.00	0.00	-32.00	-297.00
215 DL+WL9	-75.75	0.00	0.00	-1.00	-1.00
216 DL+WL10	-32.75	0.00	0.00	1.00	17.00
217 DL+WL11	-75.75	0.00	0.00	-8.00	-56.00
218 DL+WL12	-31.75	0.00	0.00	-5.00	-35.00
219 DL+0.75WL1+0.75L+0.75LR	1.25	0.00	0.00	22.50	220.25
220 DL+0.75WL2+0.75L+0.75LR	33.50	0.00	0.00	24.75	235.25
221 DL+0.75WL3+0.75L+0.75LR	1.25	0.00	0.00	22.50	220.25
222 DL+0.75WL4+0.75L+0.75LR	33.50	0.00	0.00	24.75	235.25
223 DL+0.75WL5+0.75L+0.75LR	10.25	0.00	0.00	-24.00	-220.00
224 DL+0.75WL67+0.75L+0.75LR	42.50	0.00	0.00	-21.75	-205.00
225 DL+0.75WL7+0.75L+0.75LR	10.25	0.00	0.00	-24.00	-220.00
226 DL+0.75WL8+0.75L+0.75LR	42.50	0.00	0.00	-21.75	-205.00
227 DL+0.75WL9+0.75L+0.75LR	-0.25	0.00	0.00	1.50	17.00
228 DL+0.75WL10+0.75L+0.75LR	32.00	0.00	0.00	3.00	30.50
229 DL+0.75WL11+0.75L+0.75LR	-0.25	0.00	0.00	-3.75	-24.25
230 DL+0.75WL12+0.75L+0.75LR	32.75	0.00	0.00	-1.50	-8.50
231 0.6DL+WL1	-91.05	0.00	0.00	25.80	258.40
232 0.6DL+WL2	-48.05	0.00	0.00	28.80	278.40
233 0.6DL+WL3	-91.05	0.00	0.00	25.80	258.40
234 0.6DL+WL4	-48.05	0.00	0.00	28.80	278.40
235 0.6DL+WL5	-79.05	0.00	0.00	-36.20	-328.60
236 0.6DL+WL6	-36.05	0.00	0.00	-33.20	-308.60
237 0.6DL+WL7	-79.05	0.00	0.00	-36.20	-328.60
238 0.6DL+WL8	-36.05	0.00	0.00	-33.20	-308.60
239 0.6DL+WL9	-93.05	0.00	0.00	-2.20	-12.60
240 0.6DL+WL10	-50.05	0.00	0.00	-0.20	5.40
241 0.6DL+WL11	-93.05	0.00	0.00	-9.20	-67.60
242 0.6DL+WL12	-49.05	0.00	0.00	-6.20	-46.60
243 0.6DL+0.7SL1	25.95	0.00	0.00	2.50	23.70
244 0.6DL+0.7SL2	25.95	0.00	0.00	1.10	11.10
245 DL+0.75WL1+0.75L+0.75LR+CR1	36.25	0.00	0.00	31.50	278.25
246 DL+0.75WL2+0.75L+0.75LR+CR1	68.50	0.00	0.00	33.75	293.25
247 DL+0.75WL3+0.75L+0.75LR+CR1	36.25	0.00	0.00	31.50	278.25
248 DL+0.75WL4+0.75L+0.75LR+CR1	68.50	0.00	0.00	33.75	293.25

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258 DL+0.75WL2+0.75L+0.75LR+CR2	59.50	0.00	0.00	31.75	282.25
259 DL+0.75WL3+0.75L+0.75LR+CR2	27.25	0.00	0.00	29.50	267.25
260 DL+0.75WL4+0.75L+0.75LR+CR2	59.50	0.00	0.00	31.75	282.25
261 DL+0.75WL5+0.75L+0.75LR+CR2	36.25	0.00	0.00	-17.00	-173.00
262 DL+0.75WL67+0.75L+0.75LR+CR2	68.50	0.00	0.00	-14.75	-158.00
263 DL+0.75WL7+0.75L+0.75LR+CR2	36.25	0.00	0.00	-17.00	-173.00
264 DL+0.75WL8+0.75L+0.75LR+CR2	68.50	0.00	0.00	-14.75	-158.00
265 DL+0.75WL9+0.75L+0.75LR+CR2	25.75	0.00	0.00	8.50	64.00
266 DL+0.75WL10+0.75L+0.75LR+CR2	58.00	0.00	0.00	10.00	77.50
267 DL+0.75WL11+0.75L+0.75LR+CR2	25.75	0.00	0.00	3.25	22.75
268 DL+0.75WL12+0.75L+0.75LR+CR2	58.75	0.00	0.00	5.50	38.50
269 DL+0.75WL1+0.75L+0.75LR+CR3	1.25	0.00	0.00	22.50	220.25
270 DL+0.75WL2+0.75L+0.75LR+CR3	33.50	0.00	0.00	24.75	235.25
271 DL+0.75WL3+0.75L+0.75LR+CR3	1.25	0.00	0.00	22.50	220.25
272 DL+0.75WL4+0.75L+0.75LR+CR3	33.50	0.00	0.00	24.75	235.25
273 DL+0.75WL5+0.75L+0.75LR+CR3	10.25	0.00	0.00	-24.00	-220.00
274 DL+0.75WL67+0.75L+0.75LR+CR3	42.50	0.00	0.00	-21.75	-205.00
275 DL+0.75WL7+0.75L+0.75LR+CR3	10.25	0.00	0.00	-24.00	-220.00
276 DL+0.75WL8+0.75L+0.75LR+CR3	42.50	0.00	0.00	-21.75	-205.00
277 DL+0.75WL9+0.75L+0.75LR+CR3	-0.25	0.00	0.00	1.50	17.00
278 DL+0.75WL10+0.75L+0.75LR+CR3	32.00	0.00	0.00	3.00	30.50
279 DL+0.75WL11+0.75L+0.75LR+CR3	-0.25	0.00	0.00	-3.75	-24.25
280 DL+0.75WL12+0.75L+0.75LR+CR3	32.75	0.00	0.00	-1.50	-8.50
281 DL+0.75WL1+0.75L+0.75LR+CR4	1.25	0.00	0.00	22.50	220.25
282 DL+0.75WL2+0.75L+0.75LR+CR4	33.50	0.00	0.00	24.75	235.25
283 DL+0.75WL3+0.75L+0.75LR+CR4	1.25	0.00	0.00	22.50	220.25
284 DL+0.75WL4+0.75L+0.75LR+CR4	33.50	0.00	0.00	24.75	235.25
285 DL+0.75WL5+0.75L+0.75LR+CR4	10.25	0.00	0.00	-24.00	-220.00
286 DL+0.75WL67+0.75L+0.75LR+CR4	42.50	0.00	0.00	-21.75	-205.00
287 DL+0.75WL7+0.75L+0.75LR+CR4	10.25	0.00	0.00	-24.00	-220.00
288 DL+0.75WL8+0.75L+0.75LR+CR4	42.50	0.00	0.00	-21.75	-205.00
289 DL+0.75WL9+0.75L+0.75LR+CR4	-0.25	0.00	0.00	1.50	17.00
290 DL+0.75WL10+0.75L+0.75LR+CR4	32.00	0.00	0.00	3.00	30.50
291 DL+0.75WL11+0.75L+0.75LR+CR4	-0.25	0.00	0.00	-3.75	-24.25
292 DL+0.75WL12+0.75L+0.75LR+CR4	32.75	0.00	0.00	-1.50	-8.50

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	415.4	3.00	29.0	0.0	0.0	NA	35.8	NA	55.4
202 DL+L+TL1	413.4	3.00	29.0	0.0	0.0	NA	35.6	NA	55.1
203 DL+L+TL2	417.4	3.00	29.0	0.0	0.0	NA	36.0	NA	55.6
204 DL+LR	476.4	5.00	43.0	0.0	0.0	NA	27.7	NA	38.1
205 DL+0.75L+0.75TL1+0.75LR	459.6	4.50	39.5	0.0	0.0	NA	29.1	NA	40.9
206 DL+0.75L+0.75TL2+0.75LR	462.6	4.50	39.5	0.0	0.0	NA	29.3	NA	41.1
207 DL+0.6WL1	345.2	17.40	173.6	0.0	0.0	NA	5.0	NA	7.9
208 DL+WL2	341.4	30.00	290.0	0.0	0.0	NA	2.9	NA	4.6
209 DL+WL3	298.4	27.00	270.0	0.0	0.0	NA	2.8	NA	4.4
210 DL+WL4	341.4	30.00	290.0	0.0	0.0	NA	2.9	NA	4.6
211 DL+WL5	310.4	-35.00	-317.0	0.0	0.0	NA	2.4	NA	3.5
212 DL+WL6	353.4	-32.00	-297.0	0.0	0.0	NA	3.0	NA	4.4
213 DL+WL7	310.4	-35.00	-317.0	0.0	0.0	NA	2.4	NA	3.5
214 DL+WL8	353.4	-32.00	-297.0	0.0	0.0	NA	3.0	NA	4.4
215 DL+WL9	296.4	-1.00	-1.0	0.0	0.0	NA	740.9	NA	118.5
216 DL+WL10	339.4	1.00	17.0	0.0	0.0	NA	49.9	NA	135.7
217 DL+WL11	296.4	-8.00	-56.0	0.0	0.0	NA	13.2	NA	14.8
218 DL+WL12	340.4	-5.00	-35.0	0.0	0.0	NA	24.3	NA	27.2
219 DL+0.75WL1+0.75L+0.75LR	373.4	22.50	220.3	0.0	0.0	NA	4.2	NA	6.6
220 DL+0.75WL2+0.75L+0.75LR	405.6	24.75	235.3	0.0	0.0	NA	4.3	NA	6.6
221 DL+0.75WL3+0.75L+0.75LR	373.4	22.50	220.3	0.0	0.0	NA	4.2	NA	6.6
222 DL+0.75WL4+0.75L+0.75LR	405.6	24.75	235.3	0.0	0.0	NA	4.3	NA	6.6
223 DL+0.75WL5+0.75L+0.75LR	382.4	-24.00	-220.0	0.0	0.0	NA	4.3	NA	6.4
224 DL+0.75WL67+0.75L+0.75LR	414.6	-21.75	-205.0	0.0	0.0	NA	5.1	NA	7.6
225 DL+0.75WL7+0.75L+0.75LR	382.4	-24.00	-220.0	0.0	0.0	NA	4.3	NA	6.4
226 DL+0.75WL8+0.75L+0.75LR	414.6	-21.75	-205.0	0.0	0.0	NA	5.1	NA	7.6
227 DL+0.75WL9+0.75L+0.75LR	371.9	1.50	17.0	0.0	0.0	NA	54.7	NA	99.2
228 DL+0.75WL10+0.75L+0.75LR	404.1	3.00	30.5	0.0	0.0	NA	33.1	NA	53.9
229 DL+0.75WL11+0.75L+0.75LR	371.9	-3.75	-24.3	0.0	0.0	NA	38.3	NA	39.7
230 DL+0.75WL12+0.75L+0.75LR	404.9	-1.50	-8.5	0.0	0.0	NA	119.1	NA	108.0
231 0.6DL+WL1	281.1	25.80	258.4	0.0	0.0	NA	2.7	NA	4.4
232 0.6DL+WL2	324.1	28.80	278.4	0.0	0.0	NA	2.9	NA	4.5
233 0.6DL+WL3	281.1	25.80	258.4	0.0	0.0	NA	2.7	NA	4.4
234 0.6DL+WL4	324.1	28.80	278.4	0.0	0.0	NA	2.9	NA	4.5
235 0.6DL+WL5	293.1	-36.20	-328.6	0.0	0.0	NA	2.2	NA	3.2
236 0.6DL+WL6	336.1	-33.20	-308.6	0.0	0.0	NA	2.7	NA	4.0
237 0.6DL+WL7	293.1	-36.20	-328.6	0.0	0.0	NA	2.2	NA	3.2
238 0.6DL+WL8	336.1	-33.20	-308.6	0.0	0.0	NA	2.7	NA	4.0
239 0.6DL+WL9	279.1	-2.20	-12.6	0.0	0.0	NA	55.4	NA	50.7
240 0.6DL+WL10	322.1	-0.20	5.4	0.0	0.0	NA	149.1	NA	644.1
241 0.6DL+WL11	279.1	-9.20	-67.6	0.0	0.0	NA	10.3	NA	12.1
242 0.6DL+WL12	323.1	-6.20	-46.6	0.0	0.0	NA	17.3	NA	20.8

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243 0.6DL+0.7SL1	398.1	2.50	23.7	0.0	0.0	NA	42.0	NA	63.7
244 0.6DL+0.7SL2	398.1	1.10	11.1	0.0	0.0	NA	89.7	NA	144.7
245 DL+0.75WL1+0.75L+0.75LR+CR1	408.4	31.50	278.3	0.0	0.0	NA	3.7	NA	5.2
246 DL+0.75WL2+0.75L+0.75LR+CR1	440.6	33.75	293.3	0.0	0.0	NA	3.8	NA	5.2
247 DL+0.75WL3+0.75L+0.75LR+CR1	408.4	31.50	278.3	0.0	0.0	NA	3.7	NA	5.2
248 DL+0.75WL4+0.75L+0.75LR+CR1	440.6	33.75	293.3	0.0	0.0	NA	3.8	NA	5.2
258 DL+0.75WL2+0.75L+0.75LR+CR2	431.6	31.75	282.3	0.0	0.0	NA	3.8	NA	5.4
259 DL+0.75WL3+0.75L+0.75LR+CR2	399.4	29.50	267.3	0.0	0.0	NA	3.7	NA	5.4
260 DL+0.75WL4+0.75L+0.75LR+CR2	431.6	31.75	282.3	0.0	0.0	NA	3.8	NA	5.4
261 DL+0.75WL5+0.75L+0.75LR+CR2	408.4	-17.00	-173.0	0.0	0.0	NA	5.9	NA	9.6
262 DL+0.75WL67+0.75L+0.75LR+CR2	440.6	-14.75	-158.0	0.0	0.0	NA	7.0	NA	11.9
263 DL+0.75WL7+0.75L+0.75LR+CR2	408.4	-17.00	-173.0	0.0	0.0	NA	5.9	NA	9.6
264 DL+0.75WL8+0.75L+0.75LR+CR2	440.6	-14.75	-158.0	0.0	0.0	NA	7.0	NA	11.9
265 DL+0.75WL9+0.75L+0.75LR+CR2	397.9	8.50	64.0	0.0	0.0	NA	15.5	NA	18.7
266 DL+0.75WL10+0.75L+0.75LR+CR2	430.1	10.00	77.5	0.0	0.0	NA	13.9	NA	17.2
267 DL+0.75WL11+0.75L+0.75LR+CR2	397.9	3.25	22.8	0.0	0.0	NA	43.7	NA	49.0
268 DL+0.75WL12+0.75L+0.75LR+CR2	430.9	5.50	38.5	0.0	0.0	NA	28.0	NA	31.3
269 DL+0.75WL1+0.75L+0.75LR+CR3	373.4	22.50	220.3	0.0	0.0	NA	4.2	NA	6.6
270 DL+0.75WL2+0.75L+0.75LR+CR3	405.6	24.75	235.3	0.0	0.0	NA	4.3	NA	6.6
271 DL+0.75WL3+0.75L+0.75LR+CR3	373.4	22.50	220.3	0.0	0.0	NA	4.2	NA	6.6
272 DL+0.75WL4+0.75L+0.75LR+CR3	405.6	24.75	235.3	0.0	0.0	NA	4.3	NA	6.6
273 DL+0.75WL5+0.75L+0.75LR+CR3	382.4	-24.00	-220.0	0.0	0.0	NA	4.3	NA	6.4
274 DL+0.75WL67+0.75L+0.75LR+CR3	414.6	-21.75	-205.0	0.0	0.0	NA	5.1	NA	7.6
275 DL+0.75WL7+0.75L+0.75LR+CR3	382.4	-24.00	-220.0	0.0	0.0	NA	4.3	NA	6.4
276 DL+0.75WL8+0.75L+0.75LR+CR3	414.6	-21.75	-205.0	0.0	0.0	NA	5.1	NA	7.6
277 DL+0.75WL9+0.75L+0.75LR+CR3	371.9	1.50	17.0	0.0	0.0	NA	54.7	NA	99.2
278 DL+0.75WL10+0.75L+0.75LR+CR3	404.1	3.00	30.5	0.0	0.0	NA	33.1	NA	53.9
279 DL+0.75WL11+0.75L+0.75LR+CR3	371.9	-3.75	-24.3	0.0	0.0	NA	38.3	NA	39.7
280 DL+0.75WL12+0.75L+0.75LR+CR3	404.9	-1.50	-8.5	0.0	0.0	NA	119.1	NA	108.0
281 DL+0.75WL1+0.75L+0.75LR+CR4	373.4	22.50	220.3	0.0	0.0	NA	4.2	NA	6.6
282 DL+0.75WL2+0.75L+0.75LR+CR4	405.6	24.75	235.3	0.0	0.0	NA	4.3	NA	6.6
283 DL+0.75WL3+0.75L+0.75LR+CR4	373.4	22.50	220.3	0.0	0.0	NA	4.2	NA	6.6
284 DL+0.75WL4+0.75L+0.75LR+CR4	405.6	24.75	235.3	0.0	0.0	NA	4.3	NA	6.6
285 DL+0.75WL5+0.75L+0.75LR+CR4	382.4	-24.00	-220.0	0.0	0.0	NA	4.3	NA	6.4
286 DL+0.75WL67+0.75L+0.75LR+CR4	414.6	-21.75	-205.0	0.0	0.0	NA	5.1	NA	7.6
287 DL+0.75WL7+0.75L+0.75LR+CR4	382.4	-24.00	-220.0	0.0	0.0	NA	4.3	NA	6.4
288 DL+0.75WL8+0.75L+0.75LR+CR4	414.6	-21.75	-205.0	0.0	0.0	NA	5.1	NA	7.6
289 DL+0.75WL9+0.75L+0.75LR+CR4	371.9	1.50	17.0	0.0	0.0	NA	54.7	NA	99.2
290 DL+0.75WL10+0.75L+0.75LR+CR4	404.1	3.00	30.5	0.0	0.0	NA	33.1	NA	53.9
291 DL+0.75WL11+0.75L+0.75LR+CR4	371.9	-3.75	-24.3	0.0	0.0	NA	38.3	NA	39.7
292 DL+0.75WL12+0.75L+0.75LR+CR4	404.9	-1.50	-8.5	0.0	0.0	NA	119.1	NA	108.0

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Base pressure check of footing F1A

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	62.52	0
Min Px	2	201	0	62.52	0
Max Py	533	246	0	94.62	0
Min Py	553	231	0	0.00	0
Max Pz	2	201	0	62.52	0
Min Pz	2	201	0	62.52	0

Gross Maximum Base pressure, p_{max} = 94.6 KN/m^2

Net Allowable Safe Bearing Capacity = 350 KN/m^2

Depth of soil above footing = 0.60 m

Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 362 KN/m^2

> 94.6 KN/m^2

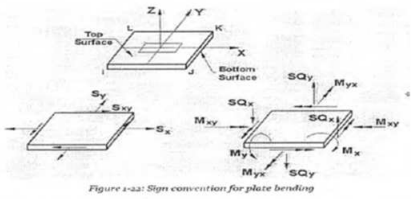
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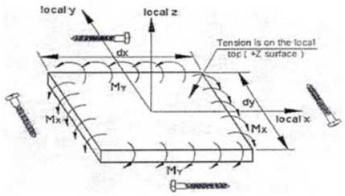
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Design of foundation raft F1A





Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F1A_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	414	0.37	-0.2	0	0	0	35	106.682	-26.089	61.032	132.771	123:0.9DL+1.6WL5
Min Qx	417	-0.397	-0.273	0	0	0	41.629	123.82	18.568	60.197	142.388	123:0.9DL+1.6WL5
Max Qy	346	-0.093	0.435	0	0	0	29.01	147.088	-3.484	32.494	150.572	133:1.2DL+1.6WL1+1.0L+0.5LR+1.0CR1
Min Qy	416	-0.1	-0.498	0	0	0	35.531	161.533	2.199	37.73	163.732	123:0.9DL+1.6WL5
Max Sx	556	0	0	0	0	0	-0.01	-0.018	0.045	-0.055	-0.063	101:1.4DL
Min Sx	556	0	0	0	0	0	-0.01	-0.018	0.045	-0.055	-0.063	101:1.4DL
Max Sy	556	0	0	0	0	0	-0.01	-0.018	0.045	-0.055	-0.063	101:1.4DL
Min Sy	556	0	0	0	0	0	-0.01	-0.018	0.045	-0.055	-0.063	101:1.4DL
Max Sxy	556	0	0	0	0	0	-0.01	-0.018	0.045	-0.055	-0.063	101:1.4DL
Min Sxy	556	0	0	0	0	0	-0.01	-0.018	0.045	-0.055	-0.063	101:1.4DL
Max Mx	417	-0.397	-0.273	0	0	0	41.629	123.82	18.568	60.197	142.388	123:0.9DL+1.6WL5
Min Mx	417	0.361	0.218	0	0	0	-35.314	-106.532	-22.035	-57.349	-128.567	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Max My	416	-0.1	-0.498	0	0	0	35.531	161.533	2.199	37.73	163.732	123:0.9DL+1.6WL5
Min My	416	0.096	0.424	0	0	0	-24.702	-143.808	-4.03	-28.732	-147.838	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Max Mxy	414	-0.3	0.121	0	0	0	-25.167	-73.221	28.829	-53.996	-102.05	133:1.2DL+1.6WL1+1.0L+0.5LR+1.0CR1
Min Mxy	344	-0.316	-0.118	0	0	0	-25.389	-75.424	-33.752	-59.141	-109.176	123:0.9DL+1.6WL5

PLATE CENTER STRESS : FDN_F1A_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	413	0.123	0.053	0	0	0	4.69	63.998	-6.586	11.276	70.584	123:0.9DL+1.6WL5
Min Qx	413	-0.106	-0.076	0	0	0	-3.413	-48.345	8.166	-11.579	-56.511	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Max Qy	337	0.012	0.215	0	0	0	-0.082	-4.149	2.406	-2.488	-6.555	123:0.9DL+1.6WL5
Min Qy	337	-0.019	-0.203	0	0	0	0.22	24.742	-2.176	2.396	26.918	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Max Sx	556	0	0	0	0	0	-0.01	-0.018	0.045	-0.055	-0.063	101:1.4DL
Min Sx	556	0	0	0	0	0	-0.01	-0.018	0.045	-0.055	-0.063	101:1.4DL
Max Sy	556	0	0	0	0	0	-0.01	-0.018	0.045	-0.055	-0.063	101:1.4DL
Min Sy	556	0	0	0	0	0	-0.01	-0.018	0.045	-0.055	-0.063	101:1.4DL
Max Sxy	556	0	0	0	0	0	-0.01	-0.018	0.045	-0.055	-0.063	101:1.4DL
Min Sxy	556	0	0	0	0	0	-0.01	-0.018	0.045	-0.055	-0.063	101:1.4DL
Max Mx	395	0.032	-0.186	0	0	0	24.041	59.979	-0.974	25.015	60.953	123:0.9DL+1.6WL5
Min Mx	395	-0.03	0.155	0	0	0	-19.838	-54.426	1.235	-21.073	-55.661	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Max My	419	-0.046	0.087	0	0	0	0.807	74.922	-1.755	2.562	76.677	123:0.9DL+1.6WL5
Min My	409	0.031	-0.008	0	0	0	-0.72	-63.054	2.757	-3.477	-65.811	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Max Mxy	423	-0.061	-0.157	0	0	0	-1.743	-27.188	18.393	-20.136	-45.581	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Min Mxy	326	-0.039	0.166	0	0	0	-0.818	-8.039	-20.661	-21.479	-28.7	123:0.9DL+1.6WL5

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		600	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

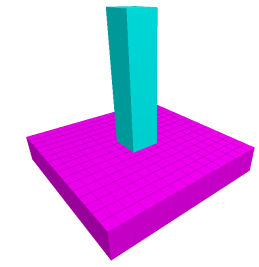
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
123:0.9D	60.20	top face	50	544	60.197	0.27	0.0006	352.9	652.8	12	173.32	150	754.3	10.64	13.29	0.12	tension control
134:1.2D	-57.3	bottom face	50	542	57.349	0.26	0.0006	337.4		16	308.12	150	1341	18.91	23.64	0.066	tension control
133:1.2D	-54.00	bottom face	50	542	53.996	0.25	0.0006	317.6		16	308.12	150	1341	18.91	23.64	0.066	tension control
123:0.9D	-59.14	bottom face	50	542	59.141	0.27	0.0006	348.0		16	308.12	150	1341	18.91	23.64	0.066	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.123	1	73.8	11.276	0.001	0.544	560.591	527.759	933.321	527.76	1989	> 73.8	ok	395.8	φVc>Vu	not required
	Min Qx	0.106	1	63.6	11.579			553.251			527.76	1989	> 63.6	ok	395.8	φVc>Vu	not required

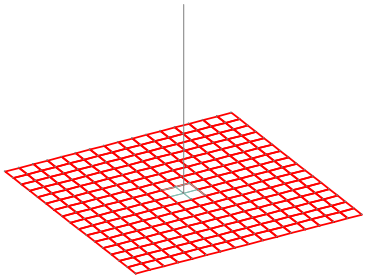
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
123:0.9D	163.73	top face	50	544	163.73	0.74	0.0001	61.0	652.8	12	173.32	150	754.3	10.64	13.29	0.12	tension control
134:1.2D	-147.84	bottom face	50	542	147.84	0.67	0.0016	875.9		16	229.64	150	1341	18.91	23.64	0.066	tension control
133:1.2D	-102.05	bottom face	50	542	102.05	0.46	0.0011	602.5		16	308.12	150	1341	18.91	23.64	0.066	tension control
123:0.9D	-109.18	bottom face	50	542	109.18	0.50	0.0012	644.9		16	308.12	150	1341	18.91	23.64	0.066	tension control

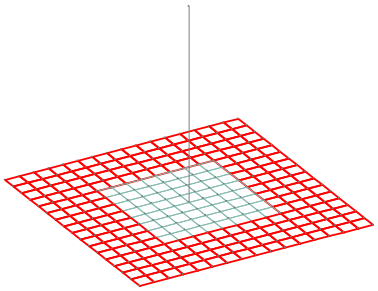
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.215	1	129	6.555	0.001	0.544	652.216	527.759	fc'933.3	527.76	555	> 129	ok	395.8	φVc>Vu	not required
	Min Qy	0.203	1	121.8	26.918			546.5			527.76	555	> 121.8	ok	395.8	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
At Bottom **16ϕT** **150** c/c
At Top **12ϕT** **150** c/c
Along Z Direction :
At Bottom **16ϕT** **150** c/c
At Top **12ϕT** **150** c/c

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DESIGN CALCULATION OF FOOTING F1B

Size of Columns at node	Along-X	Along-Z
2	0.50m	0.85m

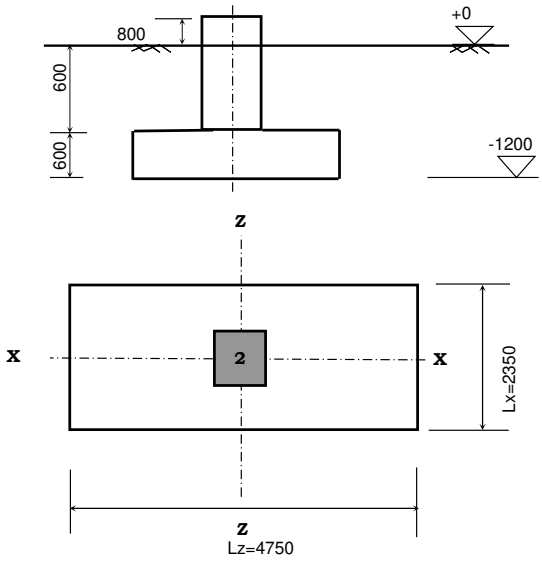
Size of fdn. raft:	4.75m	2.35m
Thickness of raft:	0.60	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.80	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

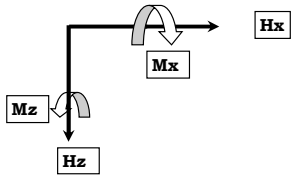
Calculation of Weight of Foundation :

Weight of raft = (4.75*2.35*0.6*24) =	160.74KN
Weight of pedestal = 0.5*0.85*1.4*24 =	14.30KN
Wt of soil + surcharge = 100 + ((4.75*2.35)-(0.5*0.85))*0.6*20 =	328.85KN
Total =	503.89KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F1B

Load at bottom of footing	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	43.25	0.00	0.00	3.00	29.00
202 DL+L+TL1	41.25	0.00	0.00	3.00	29.00
203 DL+L+TL2	45.25	0.00	0.00	3.00	29.00
204 DL+LR	104.25	0.00	0.00	5.00	43.00
205 DL+0.75L+0.75TL1+0.75LR	87.50	0.00	0.00	4.50	39.50
206 DL+0.75L+0.75TL2+0.75LR	90.50	0.00	0.00	4.50	39.50
207 DL+0.6WL1	113.45	0.00	0.00	17.40	173.60
208 DL+WL2	-30.75	0.00	0.00	30.00	290.00
209 DL+WL3	-73.75	0.00	0.00	27.00	270.00
210 DL+WL4	-30.75	0.00	0.00	30.00	290.00
211 DL+WL5	-61.75	0.00	0.00	-35.00	-317.00
212 DL+WL6	-18.75	0.00	0.00	-32.00	-297.00
213 DL+WL7	-61.75	0.00	0.00	-35.00	-317.00
214 DL+WL8	-18.75	0.00	0.00	-32.00	-297.00
215 DL+WL9	-75.75	0.00	0.00	-1.00	-1.00
216 DL+WL10	-32.75	0.00	0.00	1.00	17.00
217 DL+WL11	-75.75	0.00	0.00	-8.00	-56.00
218 DL+WL12	-31.75	0.00	0.00	-5.00	-35.00
219 DL+0.75WL1+0.75L+0.75LR	176.75	0.00	0.00	22.50	220.25
220 DL+0.75WL2+0.75L+0.75LR	33.50	0.00	0.00	24.75	235.25
221 DL+0.75WL3+0.75L+0.75LR	1.25	0.00	0.00	22.50	220.25
222 DL+0.75WL4+0.75L+0.75LR	33.50	0.00	0.00	24.75	235.25
223 DL+0.75WL5+0.75L+0.75LR	10.25	0.00	0.00	-24.00	-220.00
224 DL+0.75WL67+0.75L+0.75LR	42.50	0.00	0.00	-21.75	-205.00
225 DL+0.75WL7+0.75L+0.75LR	10.25	0.00	0.00	-24.00	-220.00
226 DL+0.75WL8+0.75L+0.75LR	42.50	0.00	0.00	-21.75	-205.00
227 DL+0.75WL9+0.75L+0.75LR	-0.25	0.00	0.00	1.50	17.00
228 DL+0.75WL10+0.75L+0.75LR	32.00	0.00	0.00	3.00	30.50
229 DL+0.75WL11+0.75L+0.75LR	-0.25	0.00	0.00	-3.75	-24.25
230 DL+0.75WL12+0.75L+0.75LR	32.75	0.00	0.00	-1.50	-8.50
231 0.6DL+WL1	142.95	0.00	0.00	25.80	258.40
232 0.6DL+WL2	-48.05	0.00	0.00	28.80	278.40
233 0.6DL+WL3	-91.05	0.00	0.00	25.80	258.40
234 0.6DL+WL4	-48.05	0.00	0.00	28.80	278.40
235 0.6DL+WL5	-79.05	0.00	0.00	-36.20	-328.60
236 0.6DL+WL6	-36.05	0.00	0.00	-33.20	-308.60
237 0.6DL+WL7	-79.05	0.00	0.00	-36.20	-328.60
238 0.6DL+WL8	-36.05	0.00	0.00	-33.20	-308.60
239 0.6DL+WL9	-93.05	0.00	0.00	-2.20	-12.60
240 0.6DL+WL10	-50.05	0.00	0.00	-0.20	5.40
241 0.6DL+WL11	-93.05	0.00	0.00	-9.20	-67.60
242 0.6DL+WL12	-49.05	0.00	0.00	-6.20	-46.60
243 0.6DL+0.7SL1	25.95	0.00	0.00	2.50	23.70
244 0.6DL+0.7SL2	25.95	0.00	0.00	1.10	11.10
245 DL+0.75WL1+0.75L+0.75LR+CR1	211.75	0.00	0.00	31.50	278.25
246 DL+0.75WL2+0.75L+0.75LR+CR1	68.50	0.00	0.00	33.75	293.25
247 DL+0.75WL3+0.75L+0.75LR+CR1	36.25	0.00	0.00	31.50	278.25
248 DL+0.75WL4+0.75L+0.75LR+CR1	68.50	0.00	0.00	33.75	293.25

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258 DL+0.75WL2+0.75L+0.75LR+CR2	59.50	0.00	0.00	31.75	282.25
259 DL+0.75WL3+0.75L+0.75LR+CR2	27.25	0.00	0.00	29.50	267.25
260 DL+0.75WL4+0.75L+0.75LR+CR2	59.50	0.00	0.00	31.75	282.25
261 DL+0.75WL5+0.75L+0.75LR+CR2	36.25	0.00	0.00	-17.00	-173.00
262 DL+0.75WL67+0.75L+0.75LR+CR2	68.50	0.00	0.00	-14.75	-158.00
263 DL+0.75WL7+0.75L+0.75LR+CR2	36.25	0.00	0.00	-17.00	-173.00
264 DL+0.75WL8+0.75L+0.75LR+CR2	68.50	0.00	0.00	-14.75	-158.00
265 DL+0.75WL9+0.75L+0.75LR+CR2	25.75	0.00	0.00	8.50	64.00
266 DL+0.75WL10+0.75L+0.75LR+CR2	58.00	0.00	0.00	10.00	77.50
267 DL+0.75WL11+0.75L+0.75LR+CR2	25.75	0.00	0.00	3.25	22.75
268 DL+0.75WL12+0.75L+0.75LR+CR2	58.75	0.00	0.00	5.50	38.50
269 DL+0.75WL1+0.75L+0.75LR+CR3	176.75	0.00	0.00	22.50	220.25
270 DL+0.75WL2+0.75L+0.75LR+CR3	33.50	0.00	0.00	24.75	235.25
271 DL+0.75WL3+0.75L+0.75LR+CR3	1.25	0.00	0.00	22.50	220.25
272 DL+0.75WL4+0.75L+0.75LR+CR3	33.50	0.00	0.00	24.75	235.25
273 DL+0.75WL5+0.75L+0.75LR+CR3	10.25	0.00	0.00	-24.00	-220.00
274 DL+0.75WL67+0.75L+0.75LR+CR3	42.50	0.00	0.00	-21.75	-205.00
275 DL+0.75WL7+0.75L+0.75LR+CR3	10.25	0.00	0.00	-24.00	-220.00
276 DL+0.75WL8+0.75L+0.75LR+CR3	42.50	0.00	0.00	-21.75	-205.00
277 DL+0.75WL9+0.75L+0.75LR+CR3	-0.25	0.00	0.00	1.50	17.00
278 DL+0.75WL10+0.75L+0.75LR+CR3	32.00	0.00	0.00	3.00	30.50
279 DL+0.75WL11+0.75L+0.75LR+CR3	-0.25	0.00	0.00	-3.75	-24.25
280 DL+0.75WL12+0.75L+0.75LR+CR3	32.75	0.00	0.00	-1.50	-8.50
281 DL+0.75WL1+0.75L+0.75LR+CR4	176.75	0.00	0.00	22.50	220.25
282 DL+0.75WL2+0.75L+0.75LR+CR4	33.50	0.00	0.00	24.75	235.25
283 DL+0.75WL3+0.75L+0.75LR+CR4	1.25	0.00	0.00	22.50	220.25
284 DL+0.75WL4+0.75L+0.75LR+CR4	33.50	0.00	0.00	24.75	235.25
285 DL+0.75WL5+0.75L+0.75LR+CR4	10.25	0.00	0.00	-24.00	-220.00
286 DL+0.75WL67+0.75L+0.75LR+CR4	42.50	0.00	0.00	-21.75	-205.00
287 DL+0.75WL7+0.75L+0.75LR+CR4	10.25	0.00	0.00	-24.00	-220.00
288 DL+0.75WL8+0.75L+0.75LR+CR4	42.50	0.00	0.00	-21.75	-205.00
289 DL+0.75WL9+0.75L+0.75LR+CR4	-0.25	0.00	0.00	1.50	17.00
290 DL+0.75WL10+0.75L+0.75LR+CR4	32.00	0.00	0.00	3.00	30.50
291 DL+0.75WL11+0.75L+0.75LR+CR4	-0.25	0.00	0.00	-3.75	-24.25
292 DL+0.75WL12+0.75L+0.75LR+CR4	32.75	0.00	0.00	-1.50	-8.50

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	547.1	3.00	29.0	0.0	0.0	NA	22.2	NA	73.0
202 DL+L+TL1	545.1	3.00	29.0	0.0	0.0	NA	22.1	NA	72.7
203 DL+L+TL2	549.1	3.00	29.0	0.0	0.0	NA	22.2	NA	73.2
204 DL+LR	608.1	5.00	43.0	0.0	0.0	NA	16.6	NA	48.7
205 DL+0.75L+0.75TL1+0.75LR	591.4	4.50	39.5	0.0	0.0	NA	17.6	NA	52.6
206 DL+0.75L+0.75TL2+0.75LR	594.4	4.50	39.5	0.0	0.0	NA	17.7	NA	52.8
207 DL+0.6WL1	617.3	17.40	173.6	0.0	0.0	NA	4.2	NA	14.2
208 DL+WL2	473.1	30.00	290.0	0.0	0.0	NA	1.9	NA	6.3
209 DL+WL3	430.1	27.00	270.0	0.0	0.0	NA	1.9	NA	6.4
210 DL+WL4	473.1	30.00	290.0	0.0	0.0	NA	1.9	NA	6.3
211 DL+WL5	442.1	-35.00	-317.0	0.0	0.0	NA	1.6	NA	5.1
212 DL+WL6	485.1	-32.00	-297.0	0.0	0.0	NA	1.9	NA	6.1
213 DL+WL7	442.1	-35.00	-317.0	0.0	0.0	NA	1.6	NA	5.1
214 DL+WL8	485.1	-32.00	-297.0	0.0	0.0	NA	1.9	NA	6.1
215 DL+WL9	428.1	-1.00	-1.0	0.0	0.0	NA	503.1	NA	171.3
216 DL+WL10	471.1	1.00	17.0	0.0	0.0	NA	32.6	NA	188.5
217 DL+WL11	428.1	-8.00	-56.0	0.0	0.0	NA	9.0	NA	21.4
218 DL+WL12	472.1	-5.00	-35.0	0.0	0.0	NA	15.9	NA	37.8
219 DL+0.75WL1+0.75L+0.75LR	680.6	22.50	220.3	0.0	0.0	NA	3.6	NA	12.1
220 DL+0.75WL2+0.75L+0.75LR	537.4	24.75	235.3	0.0	0.0	NA	2.7	NA	8.7
221 DL+0.75WL3+0.75L+0.75LR	505.1	22.50	220.3	0.0	0.0	NA	2.7	NA	9.0
222 DL+0.75WL4+0.75L+0.75LR	537.4	24.75	235.3	0.0	0.0	NA	2.7	NA	8.7
223 DL+0.75WL5+0.75L+0.75LR	514.1	-24.00	-220.0	0.0	0.0	NA	2.7	NA	8.6
224 DL+0.75WL67+0.75L+0.75LR	546.4	-21.75	-205.0	0.0	0.0	NA	3.1	NA	10.0
225 DL+0.75WL7+0.75L+0.75LR	514.1	-24.00	-220.0	0.0	0.0	NA	2.7	NA	8.6
226 DL+0.75WL8+0.75L+0.75LR	546.4	-21.75	-205.0	0.0	0.0	NA	3.1	NA	10.0
227 DL+0.75WL9+0.75L+0.75LR	503.6	1.50	17.0	0.0	0.0	NA	34.8	NA	134.3
228 DL+0.75WL10+0.75L+0.75LR	535.9	3.00	30.5	0.0	0.0	NA	20.6	NA	71.5
229 DL+0.75WL11+0.75L+0.75LR	503.6	-3.75	-24.3	0.0	0.0	NA	24.4	NA	53.7
230 DL+0.75WL12+0.75L+0.75LR	536.6	-1.50	-8.5	0.0	0.0	NA	74.2	NA	143.1
231 0.6DL+WL1	646.8	25.80	258.4	0.0	0.0	NA	2.9	NA	10.0
232 0.6DL+WL2	455.8	28.80	278.4	0.0	0.0	NA	1.9	NA	6.3
233 0.6DL+WL3	412.8	25.80	258.4	0.0	0.0	NA	1.9	NA	6.4
234 0.6DL+WL4	455.8	28.80	278.4	0.0	0.0	NA	1.9	NA	6.3
235 0.6DL+WL5	424.8	-36.20	-328.6	0.0	0.0	NA	1.5	NA	4.7
236 0.6DL+WL6	467.8	-33.20	-308.6	0.0	0.0	NA	1.8	NA	5.6
237 0.6DL+WL7	424.8	-36.20	-328.6	0.0	0.0	NA	1.5	NA	4.7
238 0.6DL+WL8	467.8	-33.20	-308.6	0.0	0.0	NA	1.8	NA	5.6
239 0.6DL+WL9	410.8	-2.20	-12.6	0.0	0.0	NA	38.3	NA	74.7
240 0.6DL+WL10	453.8	-0.20	5.4	0.0	0.0	NA	98.8	NA	907.7
241 0.6DL+WL11	410.8	-9.20	-67.6	0.0	0.0	NA	7.1	NA	17.9
242 0.6DL+WL12	454.8	-6.20	-46.6	0.0	0.0	NA	11.5	NA	29.3

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243 0.6DL+0.7SL1	529.8	2.50	23.7	0.0	0.0	NA	26.3	NA	84.8
244 0.6DL+0.7SL2	529.8	1.10	11.1	0.0	0.0	NA	56.1	NA	192.7
245 DL+0.75WL1+0.75L+0.75LR+CR1	715.6	31.50	278.3	0.0	0.0	NA	3.0	NA	9.1
246 DL+0.75WL2+0.75L+0.75LR+CR1	572.4	33.75	293.3	0.0	0.0	NA	2.3	NA	6.8
247 DL+0.75WL3+0.75L+0.75LR+CR1	540.1	31.50	278.3	0.0	0.0	NA	2.3	NA	6.9
248 DL+0.75WL4+0.75L+0.75LR+CR1	572.4	33.75	293.3	0.0	0.0	NA	2.3	NA	6.8
258 DL+0.75WL2+0.75L+0.75LR+CR2	563.4	31.75	282.3	0.0	0.0	NA	2.3	NA	7.1
259 DL+0.75WL3+0.75L+0.75LR+CR2	531.1	29.50	267.3	0.0	0.0	NA	2.3	NA	7.2
260 DL+0.75WL4+0.75L+0.75LR+CR2	563.4	31.75	282.3	0.0	0.0	NA	2.3	NA	7.1
261 DL+0.75WL5+0.75L+0.75LR+CR2	540.1	-17.00	-173.0	0.0	0.0	NA	3.7	NA	12.7
262 DL+0.75WL67+0.75L+0.75LR+CR2	572.4	-14.75	-158.0	0.0	0.0	NA	4.3	NA	15.5
263 DL+0.75WL7+0.75L+0.75LR+CR2	540.1	-17.00	-173.0	0.0	0.0	NA	3.7	NA	12.7
264 DL+0.75WL8+0.75L+0.75LR+CR2	572.4	-14.75	-158.0	0.0	0.0	NA	4.3	NA	15.5
265 DL+0.75WL9+0.75L+0.75LR+CR2	529.6	8.50	64.0	0.0	0.0	NA	9.7	NA	24.9
266 DL+0.75WL10+0.75L+0.75LR+CR2	561.9	10.00	77.5	0.0	0.0	NA	8.5	NA	22.5
267 DL+0.75WL11+0.75L+0.75LR+CR2	529.6	3.25	22.8	0.0	0.0	NA	27.4	NA	65.2
268 DL+0.75WL12+0.75L+0.75LR+CR2	562.6	5.50	38.5	0.0	0.0	NA	17.2	NA	40.9
269 DL+0.75WL1+0.75L+0.75LR+CR3	680.6	22.50	220.3	0.0	0.0	NA	3.6	NA	12.1
270 DL+0.75WL2+0.75L+0.75LR+CR3	537.4	24.75	235.3	0.0	0.0	NA	2.7	NA	8.7
271 DL+0.75WL3+0.75L+0.75LR+CR3	505.1	22.50	220.3	0.0	0.0	NA	2.7	NA	9.0
272 DL+0.75WL4+0.75L+0.75LR+CR3	537.4	24.75	235.3	0.0	0.0	NA	2.7	NA	8.7
273 DL+0.75WL5+0.75L+0.75LR+CR3	514.1	-24.00	-220.0	0.0	0.0	NA	2.7	NA	8.6
274 DL+0.75WL67+0.75L+0.75LR+CR3	546.4	-21.75	-205.0	0.0	0.0	NA	3.1	NA	10.0
275 DL+0.75WL7+0.75L+0.75LR+CR3	514.1	-24.00	-220.0	0.0	0.0	NA	2.7	NA	8.6
276 DL+0.75WL8+0.75L+0.75LR+CR3	546.4	-21.75	-205.0	0.0	0.0	NA	3.1	NA	10.0
277 DL+0.75WL9+0.75L+0.75LR+CR3	503.6	1.50	17.0	0.0	0.0	NA	34.8	NA	134.3
278 DL+0.75WL10+0.75L+0.75LR+CR3	535.9	3.00	30.5	0.0	0.0	NA	20.6	NA	71.5
279 DL+0.75WL11+0.75L+0.75LR+CR3	503.6	-3.75	-24.3	0.0	0.0	NA	24.4	NA	53.7
280 DL+0.75WL12+0.75L+0.75LR+CR3	536.6	-1.50	-8.5	0.0	0.0	NA	74.2	NA	143.1
281 DL+0.75WL1+0.75L+0.75LR+CR4	680.6	22.50	220.3	0.0	0.0	NA	3.6	NA	12.1
282 DL+0.75WL2+0.75L+0.75LR+CR4	537.4	24.75	235.3	0.0	0.0	NA	2.7	NA	8.7
283 DL+0.75WL3+0.75L+0.75LR+CR4	505.1	22.50	220.3	0.0	0.0	NA	2.7	NA	9.0
284 DL+0.75WL4+0.75L+0.75LR+CR4	537.4	24.75	235.3	0.0	0.0	NA	2.7	NA	8.7
285 DL+0.75WL5+0.75L+0.75LR+CR4	514.1	-24.00	-220.0	0.0	0.0	NA	2.7	NA	8.6
286 DL+0.75WL67+0.75L+0.75LR+CR4	546.4	-21.75	-205.0	0.0	0.0	NA	3.1	NA	10.0
287 DL+0.75WL7+0.75L+0.75LR+CR4	514.1	-24.00	-220.0	0.0	0.0	NA	2.7	NA	8.6
288 DL+0.75WL8+0.75L+0.75LR+CR4	546.4	-21.75	-205.0	0.0	0.0	NA	3.1	NA	10.0
289 DL+0.75WL9+0.75L+0.75LR+CR4	503.6	1.50	17.0	0.0	0.0	NA	34.8	NA	134.3
290 DL+0.75WL10+0.75L+0.75LR+CR4	535.9	3.00	30.5	0.0	0.0	NA	20.6	NA	71.5
291 DL+0.75WL11+0.75L+0.75LR+CR4	503.6	-3.75	-24.3	0.0	0.0	NA	24.4	NA	53.7
292 DL+0.75WL12+0.75L+0.75LR+CR4	536.6	-1.50	-8.5	0.0	0.0	NA	74.2	NA	143.1

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Base pressure check of footing F1B

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	61.15	0
Min Px	2	201	0	61.15	0
Max Py	346	235	0	245.71	0
Min Py	332	208	0	0.00	0
Max Pz	2	201	0	61.15	0
Min Pz	2	201	0	61.15	0

Gross Maximum Base pressure, p_{max} = 245.7 KN/m^2

Net Allowable Safe Bearing Capacity = 350 KN/m^2
Depth of soil above footing = 0.60 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 362 KN/m^2
> 245.7 KN/m^2
Hence OK

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Design of foundation raft F1B

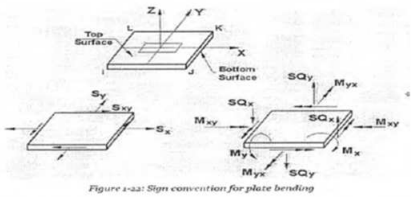
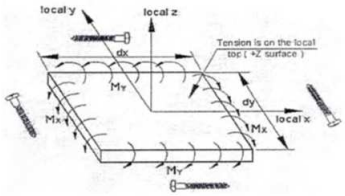


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F1B_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	417	0.531	0.331	0	0	0	-92	-109.918	-37.869	-130.308	-147.787	133:1.2DL+1.6WL1+1.0L+0.5LR+1.0CR1
Min Qx	414	-0.531	0.331	0	0	0	-92.439	-109.918	37.869	-130.308	-147.787	133:1.2DL+1.6WL1+1.0L+0.5LR+1.0CR1
Max Qy	416	0.269	0.757	0	0	0	-92.923	-190.069	-27.159	-120.082	-217.228	133:1.2DL+1.6WL1+1.0L+0.5LR+1.0CR1
Min Qy	416	-0.268	-0.67	0	0	0	58.361	174.062	32.939	91.3	207.001	123:0.9DL+1.6WL5
Max Sx	654	-0.001	0.001	0	0	0	-0.085	-0.02	0.049	-0.134	-0.069	101:1.4DL
Min Sx	654	-0.001	0.001	0	0	0	-0.085	-0.02	0.049	-0.134	-0.069	101:1.4DL
Max Sy	654	-0.001	0.001	0	0	0	-0.085	-0.02	0.049	-0.134	-0.069	101:1.4DL
Min Sy	654	-0.001	0.001	0	0	0	-0.085	-0.02	0.049	-0.134	-0.069	101:1.4DL
Max Sxy	654	-0.001	0.001	0	0	0	-0.085	-0.02	0.049	-0.134	-0.069	101:1.4DL
Min Sxy	654	-0.001	0.001	0	0	0	-0.085	-0.02	0.049	-0.134	-0.069	101:1.4DL
Max Mx	356	-0.092	0.239	0	0	0	82.193	33.325	-5.75	87.943	39.075	121:0.9DL+1.6WL3
Min Mx	406	0.105	0.418	0	0	0	-93.42	-85.656	-2.13	-95.55	-87.786	133:1.2DL+1.6WL1+1.0L+0.5LR+1.0CR1
Max My	416	-0.255	-0.663	0	0	0	46.917	177.419	32.612	79.529	210.031	109:1.2DL+1.6WL5+1.0L+0.5LR
Min My	416	0.246	0.638	0	0	0	-26.401	-191.332	-34.403	-60.804	-225.735	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Max Mxy	347	0.403	-0.217	0	0	0	-35.17	-78.225	60.653	-95.823	-138.878	123:0.9DL+1.6WL5
Min Mxy	344	-0.403	-0.217	0	0	0	-35.17	-78.225	-60.653	-95.823	-138.878	123:0.9DL+1.6WL5

PLATE CENTER STRESS : FDN_F1B_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	363	0.296	0.019	0	0	0	25.098	1.82	11.59	36.688	13.41	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Min Qx	368	-0.296	0.019	0	0	0	25.098	1.82	-11.59	36.688	13.41	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Max Qy	396	0.054	0.243	0	0	0	-83.714	-36.235	0.506	-84.22	-36.741	133:1.2DL+1.6WL1+1.0L+0.5LR+1.0CR1
Min Qy	396	-0.047	-0.196	0	0	0	53.047	32.616	1.19	54.237	33.806	109:1.2DL+1.6WL5+1.0L+0.5LR
Max Sx	654	-0.001	0.001	0	0	0	-0.085	-0.02	0.049	-0.134	-0.069	101:1.4DL
Min Sx	654	-0.001	0.001	0	0	0	-0.085	-0.02	0.049	-0.134	-0.069	101:1.4DL
Max Sy	654	-0.001	0.001	0	0	0	-0.085	-0.02	0.049	-0.134	-0.069	101:1.4DL
Min Sy	654	-0.001	0.001	0	0	0	-0.085	-0.02	0.049	-0.134	-0.069	101:1.4DL
Max Sxy	654	-0.001	0.001	0	0	0	-0.085	-0.02	0.049	-0.134	-0.069	101:1.4DL
Min Sxy	654	-0.001	0.001	0	0	0	-0.085	-0.02	0.049	-0.134	-0.069	101:1.4DL
Max Mx	366	-0.084	0.077	0	0	0	79.411	4.592	-2.314	81.725	6.906	121:0.9DL+1.6WL3
Min Mx	396	0.054	0.243	0	0	0	-83.714	-36.235	0.506	-84.22	-36.741	133:1.2DL+1.6WL1+1.0L+0.5LR+1.0CR1
Max My	396	-0.047	-0.196	0	0	0	53.047	32.616	1.19	54.237	33.806	109:1.2DL+1.6WL5+1.0L+0.5LR
Min My	396	0.028	0.189	0	0	0	-4.506	-46.838	-2.486	-6.992	-49.324	121:0.9DL+1.6WL3
Max Mxy	337	-0.002	0.075	0	0	0	8.452	-10.809	35.887	44.339	-46.696	123:0.9DL+1.6WL5
Min Mxy	342	0.002	0.075	0	0	0	8.452	-10.809	-35.887	44.339	-46.696	123:0.9DL+1.6WL5

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		600	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

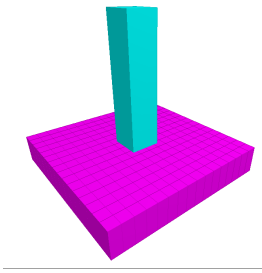
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
121:0.9D	87.94	top face	50	542	87.943	0.40	0.0010	518.6	975.6	16	206.17	150	1341	18.91	23.64	0.066	tension control
133:1.2D	-95.6	bottom face	50	542	95.55	0.43	0.0010	563.8		16	206.17	150	1341	18.91	23.64	0.066	tension control
123:0.9D	-95.82	top face	50	542	95.823	0.43	0.0010	565.4		16	206.17	150	1341	18.91	23.64	0.066	tension control
123:0.9D	-95.82	top face	50	542	95.823	0.43	0.0010	565.4		16	206.17	150	1341	18.91	23.64	0.066	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$	Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN		KN			
Load condition	Max Qx	0.296	1	177.6	36.688	0.002	0.542	572.855	535.839	929.889	535.84	1989 > 177.6	ok	401.9	φVc>Vu	not required
	Min Qx	0.296	1	177.6	36.688			572.855			535.84	1989 > 177.6	ok	401.9	φVc>Vu	not required

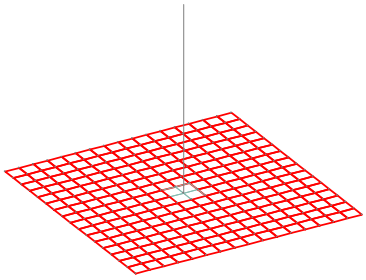
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
109:1.2D	210.03	top face	50	542	210.03	0.95	0.0023	1250.6	975.6	16	160.84	150	1341	18.91	23.64	0.066	tension control
134:1.2D	-225.74	bottom face	50	542	225.74	1.02	0.0016	894.1		16	206.17	150	1341	18.91	23.64	0.066	tension control
123:0.9D	-138.88	top face	50	542	138.88	0.63	0.0015	822.2		16	206.17	150	1341	18.91	23.64	0.066	tension control
123:0.9D	-138.88	top face	50	542	138.88	0.63	0.0015	822.2		16	206.17	150	1341	18.91	23.64	0.066	tension control

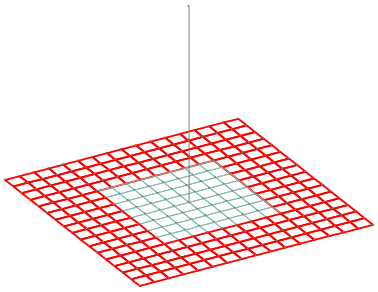
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$	Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN		KN			
Load condition	Max Qy	0.243	1	145.8	36.741	0.002	0.542	562.075	535.839	fc'929.9	535.84	561 > 145.8	ok	401.9	φVc>Vu	not required
	Min Qy	0.196	1	117.6	33.806			556.025			535.84	561 > 117.6	ok	401.9	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	16ϕT	150	c/c
At Top	16ϕT	150	c/c
Along Z Direction :			
At Bottom	16ϕT	150	c/c
At Top	16ϕT	150	c/c

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DESIGN CALCULATION OF FOOTING F2

Size of Columns at node	Along-X	Along-Z
2	0.50m	0.50m

Size of fdn. raft:	2.50m	3.20m
Thickness of raft:	0.50	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.80	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

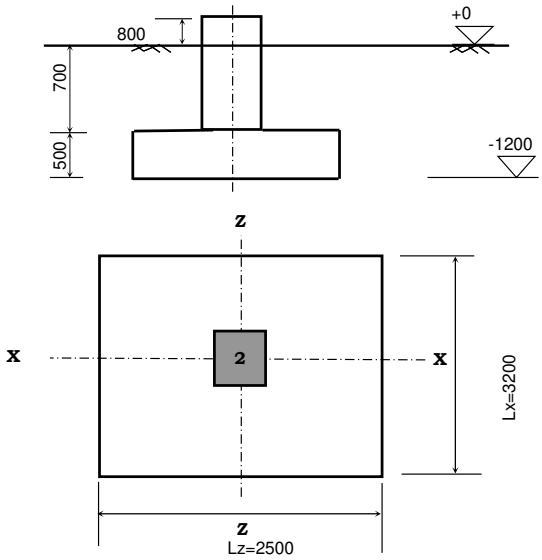
Weight of raft = (2.5*3.2*0.5*25) = 100.00KN

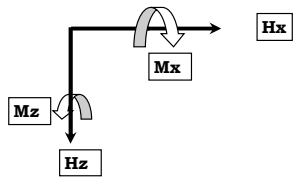
Wt of soil + surcharge = 100 + ((2.5*3.2)-(0.5*0.5))*0.7*20 = 208.50KN

Total = 308.50KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F2

Load at bottom of footing	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	72.50	0.00	0.00	2.00	4.00
202 DL+L+TL1	71.50	0.00	0.00	2.00	4.00
203 DL+L+TL2	73.50	0.00	0.00	2.00	4.00
204 DL+LR	101.50	0.00	0.00	7.00	14.00
205 DL+0.75L+0.75TL1+0.75LR	93.50	0.00	0.00	5.75	11.50
206 DL+0.75L+0.75TL2+0.75LR	95.00	0.00	0.00	5.75	11.50
207 DL+0.6WL1	22.70	0.00	0.00	24.20	48.40
208 DL+WL2	25.50	0.00	0.00	58.00	116.00
209 DL+WL3	-10.50	0.00	0.00	39.00	78.00
210 DL+WL4	25.50	0.00	0.00	58.00	116.00
211 DL+WL5	39.50	0.00	0.00	-33.00	-66.00
212 DL+WL6	74.50	0.00	0.00	-14.00	-28.00
213 DL+WL7	39.50	0.00	0.00	-33.00	-66.00
214 DL+WL8	74.50	0.00	0.00	-14.00	-28.00
215 DL+WL9	-66.00	-30.00	60.00	-27.00	-54.00
216 DL+WL10	-30.00	-30.00	60.00	-9.00	-18.00
217 DL+WL11	-85.00	-30.00	60.00	-26.00	-52.00
218 DL+WL12	-49.00	-30.00	60.00	-7.00	-14.00
219 DL+0.75WL1+0.75L+0.75LR	32.00	0.00	0.00	33.50	67.00
220 DL+0.75WL2+0.75L+0.75LR	59.00	0.00	0.00	47.75	95.50
221 DL+0.75WL3+0.75L+0.75LR	32.00	0.00	0.00	33.50	67.00
222 DL+0.75WL4+0.75L+0.75LR	59.00	0.00	0.00	47.75	95.50
223 DL+0.75WL5+0.75L+0.75LR	69.50	0.00	0.00	-20.50	-41.00
224 DL+0.75WL67+0.75L+0.75LR	95.75	0.00	0.00	-6.25	-12.50
225 DL+0.75WL7+0.75L+0.75LR	69.50	0.00	0.00	-20.50	-41.00
226 DL+0.75WL8+0.75L+0.75LR	95.75	0.00	0.00	-6.25	-12.50
227 DL+0.75WL9+0.75L+0.75LR	-9.62	-22.50	45.00	-16.00	-32.00
228 DL+0.75WL10+0.75L+0.75LR	17.38	-22.50	45.00	-2.50	-5.00
229 DL+0.75WL11+0.75L+0.75LR	-23.87	-22.50	45.00	-15.25	-30.50
230 DL+0.75WL12+0.75L+0.75LR	3.13	-22.50	45.00	-1.00	-2.00
231 0.6DL+WL1	-39.50	0.00	0.00	38.20	76.40
232 0.6DL+WL2	-3.50	0.00	0.00	57.20	114.40
233 0.6DL+WL3	-39.50	0.00	0.00	38.20	76.40
234 0.6DL+WL4	-3.50	0.00	0.00	57.20	114.40
235 0.6DL+WL5	10.50	0.00	0.00	-33.80	-67.60
236 0.6DL+WL6	45.50	0.00	0.00	-14.80	-29.60
237 0.6DL+WL7	10.50	0.00	0.00	-33.80	-67.60
238 0.6DL+WL8	45.50	0.00	0.00	-14.80	-29.60
239 0.6DL+WL9	-95.00	-30.00	60.00	-27.80	-55.60
240 0.6DL+WL10	-59.00	-30.00	60.00	-9.80	-19.60
241 0.6DL+WL11	-114.00	-30.00	60.00	-26.80	-53.60
242 0.6DL+WL12	-78.00	-30.00	60.00	-7.80	-15.60
243 0.6DL+0.7SL1	43.50	0.00	0.00	1.20	2.40
244 0.6DL+0.7SL2	43.50	0.00	0.00	1.20	2.40
245 DL+0.75WL1+0.75L+0.75LR+CR1	63.00	0.00	0.00	41.50	83.00
246 DL+0.75WL2+0.75L+0.75LR+CR1	90.00	0.00	0.00	55.75	111.50
247 DL+0.75WL3+0.75L+0.75LR+CR1	63.00	0.00	0.00	41.50	83.00
248 DL+0.75WL4+0.75L+0.75LR+CR1	90.00	0.00	0.00	55.75	111.50

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258 DL+0.75WL2+0.75L+0.75LR+CR2	58.00	0.00	0.00	47.75	95.50
259 DL+0.75WL3+0.75L+0.75LR+CR2	31.00	0.00	0.00	33.50	67.00
260 DL+0.75WL4+0.75L+0.75LR+CR2	58.00	0.00	0.00	47.75	95.50
261 DL+0.75WL5+0.75L+0.75LR+CR2	68.50	0.00	0.00	-20.50	-41.00
262 DL+0.75WL67+0.75L+0.75LR+CR2	94.75	0.00	0.00	-6.25	-12.50
263 DL+0.75WL7+0.75L+0.75LR+CR2	68.50	0.00	0.00	-20.50	-41.00
264 DL+0.75WL8+0.75L+0.75LR+CR2	94.75	0.00	0.00	-6.25	-12.50
265 DL+0.75WL9+0.75L+0.75LR+CR2	-10.62	-22.50	45.00	-16.00	-32.00
266 DL+0.75WL10+0.75L+0.75LR+CR2	16.38	-22.50	45.00	-2.50	-5.00
267 DL+0.75WL11+0.75L+0.75LR+CR2	-24.87	-22.50	45.00	-15.25	-30.50
268 DL+0.75WL12+0.75L+0.75LR+CR2	2.13	-22.50	45.00	-1.00	-2.00
269 DL+0.75WL1+0.75L+0.75LR+CR3	32.00	0.00	0.00	33.50	67.00
270 DL+0.75WL2+0.75L+0.75LR+CR3	59.00	0.00	0.00	47.75	95.50
271 DL+0.75WL3+0.75L+0.75LR+CR3	32.00	0.00	0.00	33.50	67.00
272 DL+0.75WL4+0.75L+0.75LR+CR3	59.00	0.00	0.00	47.75	95.50
273 DL+0.75WL5+0.75L+0.75LR+CR3	69.50	0.00	0.00	-20.50	-41.00
274 DL+0.75WL67+0.75L+0.75LR+CR3	95.75	0.00	0.00	-6.25	-12.50
275 DL+0.75WL7+0.75L+0.75LR+CR3	69.50	0.00	0.00	-20.50	-41.00
276 DL+0.75WL8+0.75L+0.75LR+CR3	95.75	0.00	0.00	-6.25	-12.50
277 DL+0.75WL9+0.75L+0.75LR+CR3	-9.62	-22.50	45.00	-16.00	-32.00
278 DL+0.75WL10+0.75L+0.75LR+CR3	17.38	-22.50	45.00	-2.50	-5.00
279 DL+0.75WL11+0.75L+0.75LR+CR3	-23.87	-22.50	45.00	-15.25	-30.50
280 DL+0.75WL12+0.75L+0.75LR+CR3	3.13	-22.50	45.00	-1.00	-2.00
281 DL+0.75WL1+0.75L+0.75LR+CR4	32.00	0.00	0.00	33.50	67.00
282 DL+0.75WL2+0.75L+0.75LR+CR4	59.00	0.00	0.00	47.75	95.50
283 DL+0.75WL3+0.75L+0.75LR+CR4	32.00	0.00	0.00	33.50	67.00
284 DL+0.75WL4+0.75L+0.75LR+CR4	59.00	0.00	0.00	47.75	95.50
285 DL+0.75WL5+0.75L+0.75LR+CR4	69.50	0.00	0.00	-20.50	-41.00
286 DL+0.75WL67+0.75L+0.75LR+CR4	95.75	0.00	0.00	-6.25	-12.50
287 DL+0.75WL7+0.75L+0.75LR+CR4	69.50	0.00	0.00	-20.50	-41.00
288 DL+0.75WL8+0.75L+0.75LR+CR4	95.75	0.00	0.00	-6.25	-12.50
289 DL+0.75WL9+0.75L+0.75LR+CR4	-9.62	-22.50	45.00	-16.00	-32.00
290 DL+0.75WL10+0.75L+0.75LR+CR4	17.38	-22.50	45.00	-2.50	-5.00
291 DL+0.75WL11+0.75L+0.75LR+CR4	-23.87	-22.50	45.00	-15.25	-30.50
292 DL+0.75WL12+0.75L+0.75LR+CR4	3.13	-22.50	45.00	-1.00	-2.00

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	381.0	2.00	4.0	0.0	0.0	NA	152.4	NA	76.2
202 DL+L+TL1	380.0	2.00	4.0	0.0	0.0	NA	152.0	NA	76.0
203 DL+L+TL2	382.0	2.00	4.0	0.0	0.0	NA	152.8	NA	76.4
204 DL+LR	410.0	7.00	14.0	0.0	0.0	NA	46.9	NA	23.4
205 DL+0.75L+0.75TL1+0.75LR	402.0	5.75	11.5	0.0	0.0	NA	55.9	NA	28.0
206 DL+0.75L+0.75TL2+0.75LR	403.5	5.75	11.5	0.0	0.0	NA	56.1	NA	28.1
207 DL+0.6WL1	331.2	24.20	48.4	0.0	0.0	NA	10.9	NA	5.5
208 DL+WL2	334.0	58.00	116.0	0.0	0.0	NA	4.6	NA	2.3
209 DL+WL3	298.0	39.00	78.0	0.0	0.0	NA	6.1	NA	3.1
210 DL+WL4	334.0	58.00	116.0	0.0	0.0	NA	4.6	NA	2.3
211 DL+WL5	348.0	-33.00	-66.0	0.0	0.0	NA	8.4	NA	4.2
212 DL+WL6	383.0	-14.00	-28.0	0.0	0.0	NA	21.9	NA	10.9
213 DL+WL7	348.0	-33.00	-66.0	0.0	0.0	NA	8.4	NA	4.2
214 DL+WL8	383.0	-14.00	-28.0	0.0	0.0	NA	21.9	NA	10.9
215 DL+WL9	242.5	-27.00	-54.0	-30.0	60.0	5.1	7.2	3.2	3.6
216 DL+WL10	278.5	-9.00	-18.0	-30.0	60.0	5.8	24.8	3.7	12.4
217 DL+WL11	223.5	-26.00	-52.0	-30.0	60.0	4.7	6.9	3.0	3.4
218 DL+WL12	259.5	-7.00	-14.0	-30.0	60.0	5.4	29.7	3.5	14.8
219 DL+0.75WL1+0.75L+0.75LR	340.5	33.50	67.0	0.0	0.0	NA	8.1	NA	4.1
220 DL+0.75WL2+0.75L+0.75LR	367.5	47.75	95.5	0.0	0.0	NA	6.2	NA	3.1
221 DL+0.75WL3+0.75L+0.75LR	340.5	33.50	67.0	0.0	0.0	NA	8.1	NA	4.1
222 DL+0.75WL4+0.75L+0.75LR	367.5	47.75	95.5	0.0	0.0	NA	6.2	NA	3.1
223 DL+0.75WL5+0.75L+0.75LR	378.0	-20.50	-41.0	0.0	0.0	NA	14.8	NA	7.4
224 DL+0.75WL67+0.75L+0.75LR	404.3	-6.25	-12.5	0.0	0.0	NA	51.7	NA	25.9
225 DL+0.75WL7+0.75L+0.75LR	378.0	-20.50	-41.0	0.0	0.0	NA	14.8	NA	7.4
226 DL+0.75WL8+0.75L+0.75LR	404.3	-6.25	-12.5	0.0	0.0	NA	51.7	NA	25.9
227 DL+0.75WL9+0.75L+0.75LR	298.9	-16.00	-32.0	-22.5	45.0	8.3	14.9	5.3	7.5
228 DL+0.75WL10+0.75L+0.75LR	325.9	-2.50	-5.0	-22.5	45.0	9.1	104.3	5.8	52.1
229 DL+0.75WL11+0.75L+0.75LR	284.6	-15.25	-30.5	-22.5	45.0	7.9	14.9	5.1	7.5
230 DL+0.75WL12+0.75L+0.75LR	311.6	-1.00	-2.0	-22.5	45.0	8.7	249.3	5.5	124.7
231 0.6DL+WL1	269.0	38.20	76.4	0.0	0.0	NA	5.6	NA	2.8
232 0.6DL+WL2	305.0	57.20	114.4	0.0	0.0	NA	4.3	NA	2.1
233 0.6DL+WL3	269.0	38.20	76.4	0.0	0.0	NA	5.6	NA	2.8
234 0.6DL+WL4	305.0	57.20	114.4	0.0	0.0	NA	4.3	NA	2.1
235 0.6DL+WL5	319.0	-33.80	-67.6	0.0	0.0	NA	7.6	NA	3.8
236 0.6DL+WL6	354.0	-14.80	-29.6	0.0	0.0	NA	19.1	NA	9.6
237 0.6DL+WL7	319.0	-33.80	-67.6	0.0	0.0	NA	7.6	NA	3.8
238 0.6DL+WL8	354.0	-14.80	-29.6	0.0	0.0	NA	19.1	NA	9.6
239 0.6DL+WL9	213.5	-27.80	-55.6	-30.0	60.0	4.4	6.1	2.8	3.1
240 0.6DL+WL10	249.5	-9.80	-19.6	-30.0	60.0	5.2	20.4	3.3	10.2
241 0.6DL+WL11	194.5	-26.80	-53.6	-30.0	60.0	4.1	5.8	2.6	2.9
242 0.6DL+WL12	230.5	-7.80	-15.6	-30.0	60.0	4.8	23.6	3.1	11.8

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243 0.6DL+0.7SL1	352.0	1.20	2.4	0.0	0.0	NA	234.7	NA	117.3
244 0.6DL+0.7SL2	352.0	1.20	2.4	0.0	0.0	NA	234.7	NA	117.3
245 DL+0.75WL1+0.75L+0.75LR+CR1	371.5	41.50	83.0	0.0	0.0	NA	7.2	NA	3.6
246 DL+0.75WL2+0.75L+0.75LR+CR1	398.5	55.75	111.5	0.0	0.0	NA	5.7	NA	2.9
247 DL+0.75WL3+0.75L+0.75LR+CR1	371.5	41.50	83.0	0.0	0.0	NA	7.2	NA	3.6
248 DL+0.75WL4+0.75L+0.75LR+CR1	398.5	55.75	111.5	0.0	0.0	NA	5.7	NA	2.9
258 DL+0.75WL2+0.75L+0.75LR+CR2	366.5	47.75	95.5	0.0	0.0	NA	6.1	NA	3.1
259 DL+0.75WL3+0.75L+0.75LR+CR2	339.5	33.50	67.0	0.0	0.0	NA	8.1	NA	4.1
260 DL+0.75WL4+0.75L+0.75LR+CR2	366.5	47.75	95.5	0.0	0.0	NA	6.1	NA	3.1
261 DL+0.75WL5+0.75L+0.75LR+CR2	377.0	-20.50	-41.0	0.0	0.0	NA	14.7	NA	7.4
262 DL+0.75WL67+0.75L+0.75LR+CR2	403.3	-6.25	-12.5	0.0	0.0	NA	51.6	NA	25.8
263 DL+0.75WL7+0.75L+0.75LR+CR2	377.0	-20.50	-41.0	0.0	0.0	NA	14.7	NA	7.4
264 DL+0.75WL8+0.75L+0.75LR+CR2	403.3	-6.25	-12.5	0.0	0.0	NA	51.6	NA	25.8
265 DL+0.75WL9+0.75L+0.75LR+CR2	297.9	-16.00	-32.0	-22.5	45.0	8.3	14.9	5.3	7.4
266 DL+0.75WL10+0.75L+0.75LR+CR2	324.9	-2.50	-5.0	-22.5	45.0	9.0	104.0	5.8	52.0
267 DL+0.75WL11+0.75L+0.75LR+CR2	283.6	-15.25	-30.5	-22.5	45.0	7.9	14.9	5.0	7.4
268 DL+0.75WL12+0.75L+0.75LR+CR2	310.6	-1.00	-2.0	-22.5	45.0	8.6	248.5	5.5	124.3
269 DL+0.75WL1+0.75L+0.75LR+CR3	340.5	33.50	67.0	0.0	0.0	NA	8.1	NA	4.1
270 DL+0.75WL2+0.75L+0.75LR+CR3	367.5	47.75	95.5	0.0	0.0	NA	6.2	NA	3.1
271 DL+0.75WL3+0.75L+0.75LR+CR3	340.5	33.50	67.0	0.0	0.0	NA	8.1	NA	4.1
272 DL+0.75WL4+0.75L+0.75LR+CR3	367.5	47.75	95.5	0.0	0.0	NA	6.2	NA	3.1
273 DL+0.75WL5+0.75L+0.75LR+CR3	378.0	-20.50	-41.0	0.0	0.0	NA	14.8	NA	7.4
274 DL+0.75WL67+0.75L+0.75LR+CR3	404.3	-6.25	-12.5	0.0	0.0	NA	51.7	NA	25.9
275 DL+0.75WL7+0.75L+0.75LR+CR3	378.0	-20.50	-41.0	0.0	0.0	NA	14.8	NA	7.4
276 DL+0.75WL8+0.75L+0.75LR+CR3	404.3	-6.25	-12.5	0.0	0.0	NA	51.7	NA	25.9
277 DL+0.75WL9+0.75L+0.75LR+CR3	298.9	-16.00	-32.0	-22.5	45.0	8.3	14.9	5.3	7.5
278 DL+0.75WL10+0.75L+0.75LR+CR3	325.9	-2.50	-5.0	-22.5	45.0	9.1	104.3	5.8	52.1
279 DL+0.75WL11+0.75L+0.75LR+CR3	284.6	-15.25	-30.5	-22.5	45.0	7.9	14.9	5.1	7.5
280 DL+0.75WL12+0.75L+0.75LR+CR3	311.6	-1.00	-2.0	-22.5	45.0	8.7	249.3	5.5	124.7
281 DL+0.75WL1+0.75L+0.75LR+CR4	340.5	33.50	67.0	0.0	0.0	NA	8.1	NA	4.1
282 DL+0.75WL2+0.75L+0.75LR+CR4	367.5	47.75	95.5	0.0	0.0	NA	6.2	NA	3.1
283 DL+0.75WL3+0.75L+0.75LR+CR4	340.5	33.50	67.0	0.0	0.0	NA	8.1	NA	4.1
284 DL+0.75WL4+0.75L+0.75LR+CR4	367.5	47.75	95.5	0.0	0.0	NA	6.2	NA	3.1
285 DL+0.75WL5+0.75L+0.75LR+CR4	378.0	-20.50	-41.0	0.0	0.0	NA	14.8	NA	7.4
286 DL+0.75WL67+0.75L+0.75LR+CR4	404.3	-6.25	-12.5	0.0	0.0	NA	51.7	NA	25.9
287 DL+0.75WL7+0.75L+0.75LR+CR4	378.0	-20.50	-41.0	0.0	0.0	NA	14.8	NA	7.4
288 DL+0.75WL8+0.75L+0.75LR+CR4	404.3	-6.25	-12.5	0.0	0.0	NA	51.7	NA	25.9
289 DL+0.75WL9+0.75L+0.75LR+CR4	298.9	-16.00	-32.0	-22.5	45.0	8.3	14.9	5.3	7.5
290 DL+0.75WL10+0.75L+0.75LR+CR4	325.9	-2.50	-5.0	-22.5	45.0	9.1	104.3	5.8	52.1
291 DL+0.75WL11+0.75L+0.75LR+CR4	284.6	-15.25	-30.5	-22.5	45.0	7.9	14.9	5.1	7.5
292 DL+0.75WL12+0.75L+0.75LR+CR4	311.6	-1.00	-2.0	-22.5	45.0	8.7	249.3	5.5	124.7

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Base pressure check of footing F2

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	59.09	0
Min Px	2	201	0	59.09	0
Max Py	427	246	0	84.14	0
Min Py	365	239	0	0.00	0
Max Pz	2	201	0	59.09	0
Min Pz	2	201	0	59.09	0

Gross Maximum Base pressure, p_{max} = 84.1 KN/m^2

Net Allowable Safe Bearing Capacity = 350 KN/m^2
Depth of soil above footing = 0.70 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 364 KN/m^2
> 84.1 KN/m^2
Hence OK

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Design of foundation raft F2

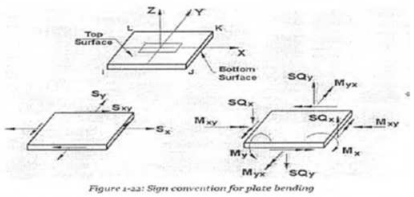
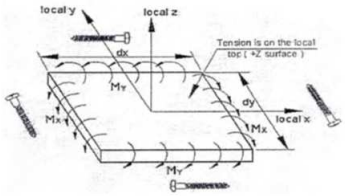


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F2_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	424	0.311	-0.005	0	0	0	42	34.212	-3.286	45.175	37.498	129:0.9DL+1.6WL11
Min Qx	414	-0.26	0.13	0	0	0	-19.402	-41.333	12.226	-31.628	-53.559	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Max Qy	416	0.137	0.35	0	0	0	-21.8	-69.77	-8.362	-30.162	-78.132	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Min Qy	415	-0.009	-0.306	0	0	0	24.019	47.918	-3.832	27.851	51.75	129:0.9DL+1.6WL11
Max Sx	524	-0.002	-0.002	0	0	0	-0.273	-0.503	0.223	-0.496	-0.726	101:1.4DL
Min Sx	524	-0.002	-0.002	0	0	0	-0.273	-0.503	0.223	-0.496	-0.726	101:1.4DL
Max Sy	524	-0.002	-0.002	0	0	0	-0.273	-0.503	0.223	-0.496	-0.726	101:1.4DL
Min Sy	524	-0.002	-0.002	0	0	0	-0.273	-0.503	0.223	-0.496	-0.726	101:1.4DL
Max Sxy	524	-0.002	-0.002	0	0	0	-0.273	-0.503	0.223	-0.496	-0.726	101:1.4DL
Min Sxy	524	-0.002	-0.002	0	0	0	-0.273	-0.503	0.223	-0.496	-0.726	101:1.4DL
Max Mx	424	0.311	-0.005	0	0	0	41.889	34.212	-3.286	45.175	37.498	129:0.9DL+1.6WL11
Min Mx	333	0.132	-0.049	0	0	0	-26.759	-2.672	5.538	-32.297	-8.21	142:1.2DL+1.6WL10+1.0L+0.5LR+1.0CR1
Max My	346	-0.125	0.27	0	0	0	8.542	58.661	-10.263	18.805	68.924	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Min My	416	0.137	0.35	0	0	0	-21.8	-69.77	-8.362	-30.162	-78.132	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Max Mxy	347	0.084	-0.08	0	0	0	-10.763	-4.912	20.805	-31.568	-25.717	129:0.9DL+1.6WL11
Min Mxy	347	-0.215	0.08	0	0	0	12.156	30.024	-15.696	27.852	45.72	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1

PLATE CENTER STRESS : FDN_F2_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	432	0.075	0.008	0	0	0	4.478	26.617	0.343	4.821	26.96	129:0.9DL+1.6WL11
Min Qx	394	-0.048	0.086	0	0	0	-11.327	-14.44	-2.135	-13.462	-16.575	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Max Qy	396	0.021	0.113	0	0	0	-15.599	-15.57	0.898	-16.497	-16.468	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Min Qy	395	-0.006	-0.096	0	0	0	14.999	10.003	-1.392	16.391	11.395	129:0.9DL+1.6WL11
Max Sx	524	-0.002	-0.002	0	0	0	-0.273	-0.503	0.223	-0.496	-0.726	101:1.4DL
Min Sx	524	-0.002	-0.002	0	0	0	-0.273	-0.503	0.223	-0.496	-0.726	101:1.4DL
Max Sy	524	-0.002	-0.002	0	0	0	-0.273	-0.503	0.223	-0.496	-0.726	101:1.4DL
Min Sy	524	-0.002	-0.002	0	0	0	-0.273	-0.503	0.223	-0.496	-0.726	101:1.4DL
Max Sxy	524	-0.002	-0.002	0	0	0	-0.273	-0.503	0.223	-0.496	-0.726	101:1.4DL
Min Sxy	524	-0.002	-0.002	0	0	0	-0.273	-0.503	0.223	-0.496	-0.726	101:1.4DL
Max Mx	395	-0.006	-0.096	0	0	0	14.999	10.003	-1.392	16.391	11.395	129:0.9DL+1.6WL11
Min Mx	396	0.021	0.113	0	0	0	-15.599	-15.57	0.898	-16.497	-16.468	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Max My	432	0.075	0.008	0	0	0	4.478	26.617	0.343	4.821	26.96	129:0.9DL+1.6WL11
Min My	422	-0.043	-0.013	0	0	0	-0.137	-15.776	4.543	-4.68	-20.319	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Max Mxy	342	0.004	-0.05	0	0	0	-0.561	-1.695	8.055	-8.616	-9.75	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
Min Mxy	337	-0.004	-0.05	0	0	0	-0.561	-1.695	-8.055	-8.616	-9.75	134:1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

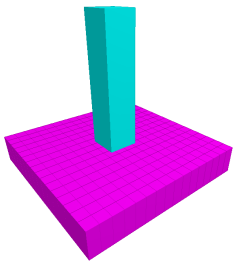
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
129:0.9D	45.18	top face	50	444	45.175	0.31	0.0007	324.7	532.8	12	212.36	200	565.8	7.978	9.972	0.131	tension control
142:1.2D	-32.3	bottom face	50	442	32.297	0.22	0.0005	232.8		16	377.52	200	1005.8	14.18	17.73	0.072	tension control
129:0.9D	-31.57	bottom face	50	442	31.568	0.22	0.0005	227.6		16	377.52	200	1005.8	14.18	17.73	0.072	tension control
134:1.2D	27.85	top face	50	444	27.852	0.19	0.0004	199.8		12	212.36	200	565.8	7.978	9.972	0.131	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.075	1	37.5	4.821	0.001	0.444	453.497	429.897	761.754	429.9	1623	> 37.5	ok	322.4	φVc>Vu	not required
	Min Qx	0.048	1	24	13.462			427.892			427.89	1621	> 24	ok	320.9	φVc>Vu	not required

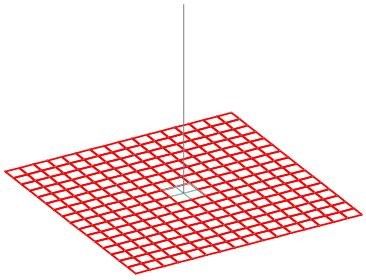
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
134:1.2D	68.92	top face	50	444	68.924	0.47	0.0011	496.7	532.8	12	212.36	200	565.8	7.978	9.972	0.131	tension control
134:1.2D	-78.13	bottom face	50	442	78.132	0.53	0.0013	566.3		16	355.19	200	1005.8	14.18	17.73	0.072	tension control
129:0.9D	-25.72	bottom face	50	442	25.717	0.18	0.0004	185.3		16	377.52	200	1005.8	14.18	17.73	0.072	tension control
134:1.2D	45.72	top face	50	444	45.72	0.31	0.0007	328.6		12	212.36	200	565.8	7.978	9.972	0.131	tension control

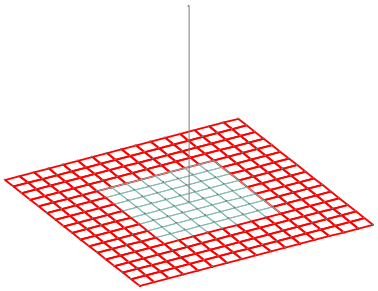
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.113	1	56.5	16.468	0.001	0.444	434.93	429.897	fc'761.8	429.9	452	> 56.5	ok	322.4	φVc>Vu	not required
	Min Qy	0.096	1	48	11.395			438.268			429.9	452	> 48	ok	322.4	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
At Bottom **16ϕT** **200** c/c
At Top **12ϕT** **200** c/c

Along Z Direction :
At Bottom **16ϕT** **200** c/c
At Top **12ϕT** **200** c/c

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DESIGN CALCULATION OF FOOTING F3

Size of Columns at node	Along-X	Along-Z
2	0.40m	0.40m

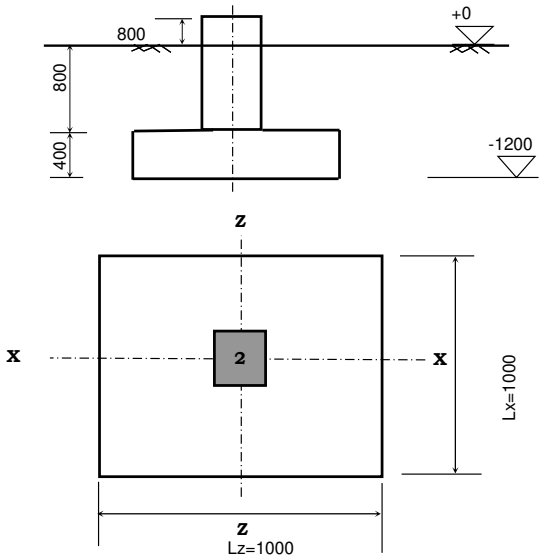
Size of fdn. raft:	1.00m	1.00m
Thickness of raft:	0.40	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.80	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

Weight of raft = (1*1*0.4*24) = 9.60KN

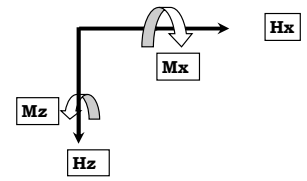
Weight of soil = ((1*1)-(0.4*0.4))*0.8*20 = 13.44KN

Total = 23.04KN



Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm



H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F3

Load at bottom of footing	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	16.00	0.00	0.00	2.00	4.00
202 DL+L	48.00	0.00	0.00	7.00	14.00

Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	39.0	2.00	4.0	0.0	0.0	NA	4.9	NA	7.8
202 DL+L	71.0	7.00	14.0	0.0	0.0	NA	2.5	NA	4.1
0	23.0	0.00	0.0	0.0	0.0	NA	NA	NA	NA

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Base pressure check of footing F3

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	68.12	0
Min Px	2	201	0	68.12	0
Max Py	401	202	0	174.46	0
Min Py	322	202	0	25.32	0
Max Pz	2	201	0	68.12	0
Min Pz	2	201	0	68.12	0

Gross Maximum Base pressure, p_{max} = 174.5 KN/m^2

Net Allowable Safe Bearing Capacity = 350 KN/m^2

Depth of soil above footing = 0.80 m

Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 366 KN/m^2

> 174.5 KN/m^2

Hence OK

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Design of foundation raft F3

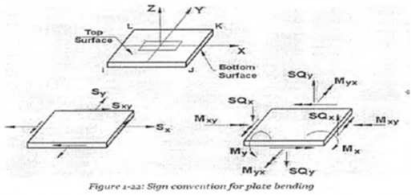
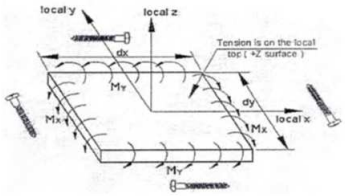


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F3_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	417	0.041	0.04	0	0	0	-2	-2.462	-0.088	-2.501	-2.55	102:1.4DL+1.7L
Min Qx	414	-0.041	0.04	0	0	0	-2.413	-2.462	0.088	-2.501	-2.55	102:1.4DL+1.7L
Max Qy	416	0.024	0.083	0	0	0	-6.38	-3.69	-0.047	-6.427	-3.737	102:1.4DL+1.7L
Min Qy	333	0.026	-0.005	0	0	0	-0.453	-5.826	-1.125	-1.578	-6.951	102:1.4DL+1.7L
Max Sx	427	0.011	0.002	0	0	0	-0.402	-1.171	0.048	-0.45	-1.219	101:1.4DL
Min Sx	427	0.011	0.002	0	0	0	-0.402	-1.171	0.048	-0.45	-1.219	101:1.4DL
Max Sy	427	0.011	0.002	0	0	0	-0.402	-1.171	0.048	-0.45	-1.219	101:1.4DL
Min Sy	427	0.011	0.002	0	0	0	-0.402	-1.171	0.048	-0.45	-1.219	101:1.4DL
Max Sxy	427	0.011	0.002	0	0	0	-0.402	-1.171	0.048	-0.45	-1.219	101:1.4DL
Min Sxy	427	0.011	0.002	0	0	0	-0.402	-1.171	0.048	-0.45	-1.219	101:1.4DL
Max Mx	333	0.008	-0.002	0	0	0	-0.154	-1.655	-0.305	-0.459	-1.96	101:1.4DL
Min Mx	416	0.024	0.083	0	0	0	-6.38	-3.69	-0.047	-6.427	-3.737	102:1.4DL+1.7L
Max My	346	-0.011	0.019	0	0	0	-3.259	1.794	-1.272	-4.531	3.066	102:1.4DL+1.7L
Min My	333	0.026	-0.005	0	0	0	-0.453	-5.826	-1.125	-1.578	-6.951	102:1.4DL+1.7L
Max Mxy	344	0.01	0.008	0	0	0	-0.465	-0.429	1.639	-2.104	-2.068	102:1.4DL+1.7L
Min Mxy	347	-0.01	0.008	0	0	0	-0.465	-0.429	-1.639	-2.104	-2.068	102:1.4DL+1.7L

PLATE CENTER STRESS : FDN_F3_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	417	0.041	0.04	0	0	0	-2.413	-2.462	-0.088	-2.501	-2.55	102:1.4DL+1.7L
Min Qx	414	-0.041	0.04	0	0	0	-2.413	-2.462	0.088	-2.501	-2.55	102:1.4DL+1.7L
Max Qy	416	0.024	0.083	0	0	0	-6.38	-3.69	-0.047	-6.427	-3.737	102:1.4DL+1.7L
Min Qy	333	0.026	-0.005	0	0	0	-0.453	-5.826	-1.125	-1.578	-6.951	102:1.4DL+1.7L
Max Sx	427	0.011	0.002	0	0	0	-0.402	-1.171	0.048	-0.45	-1.219	101:1.4DL
Min Sx	427	0.011	0.002	0	0	0	-0.402	-1.171	0.048	-0.45	-1.219	101:1.4DL
Max Sy	427	0.011	0.002	0	0	0	-0.402	-1.171	0.048	-0.45	-1.219	101:1.4DL
Min Sy	427	0.011	0.002	0	0	0	-0.402	-1.171	0.048	-0.45	-1.219	101:1.4DL
Max Sxy	427	0.011	0.002	0	0	0	-0.402	-1.171	0.048	-0.45	-1.219	101:1.4DL
Min Sxy	427	0.011	0.002	0	0	0	-0.402	-1.171	0.048	-0.45	-1.219	101:1.4DL
Max Mx	333	0.008	-0.002	0	0	0	-0.154	-1.655	-0.305	-0.459	-1.96	101:1.4DL
Min Mx	416	0.024	0.083	0	0	0	-6.38	-3.69	-0.047	-6.427	-3.737	102:1.4DL+1.7L
Max My	346	-0.011	0.019	0	0	0	-3.259	1.794	-1.272	-4.531	3.066	102:1.4DL+1.7L
Min My	333	0.026	-0.005	0	0	0	-0.453	-5.826	-1.125	-1.578	-6.951	102:1.4DL+1.7L
Max Mxy	344	0.01	0.008	0	0	0	-0.465	-0.429	1.639	-2.104	-2.068	102:1.4DL+1.7L
Min Mxy	347	-0.01	0.008	0	0	0	-0.465	-0.429	-1.639	-2.104	-2.068	102:1.4DL+1.7L

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

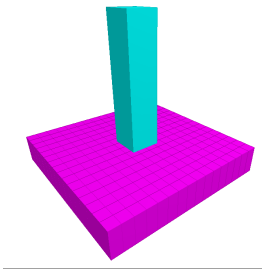
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
101:1.4D	-0.46	bottom face	50	344	0.459	0.01	0.0000	4.2	412.8	12	274.09	200	565.8	7.978	9.972	0.1	tension control
102:1.4D	-6.4	bottom face	50	344	6.427	0.07	0.0002	59.4		12	274.09	200	565.8	7.978	9.972	0.1	tension control
102:1.4D	-2.10	bottom face	50	344	2.104	0.02	0.0001	19.4		12	274.09	200	565.8	7.978	9.972	0.1	tension control
102:1.4D	-2.10	bottom face	50	344	2.104	0.02	0.0001	19.4		12	274.09	200	565.8	7.978	9.972	0.1	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN			
Load condition	Max Qx	0.041	1	16.4	2.501	0.002	0.344	347.318	335.24	590.188	335.24	1259	> 16.4	ok	251.4	φVc>Vu	not required
	Min Qx	0.041	1	16.4	2.501			347.318			335.24	1259	> 16.4	ok	251.4	φVc>Vu	not required

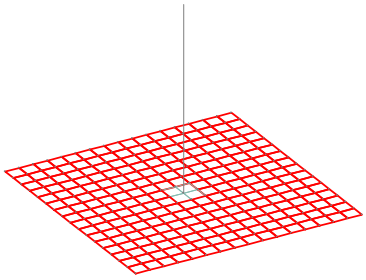
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
102:1.4D	3.07	top face	50	345	3.066	0.03	0.0001	28.2	414	10	190.34	200	392.9	5.54	6.925	0.146	tension control
102:1.4D	-6.95	bottom face	50	344	6.951	0.08	0.0002	64.2		12	274.09	200	565.8	7.978	9.972	0.1	tension control
102:1.4D	-2.07	bottom face	50	344	2.068	0.02	0.0001	19.1		12	274.09	200	565.8	7.978	9.972	0.1	tension control
102:1.4D	-2.07	bottom face	50	344	2.068	0.02	0.0001	19.1		12	274.09	200	565.8	7.978	9.972	0.1	tension control

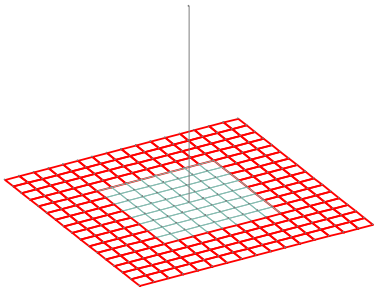
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN			
Load condition	Max Qy	0.083	1	33.2	3.737	0.001	0.345	355.017	335.24	fc'590.2	335.24	352	> 33.2	ok	251.4	φVc>Vu	not required
	Min Qy	0.005	1	2	6.951			326.573			326.57	346	> 2	ok	244.9	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
At Bottom **12ϕT** **200** c/c
At Top **10ϕT** **200** c/c
Along Z Direction :
At Bottom **12ϕT** **200** c/c
At Top **10ϕT** **200** c/c

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DESIGN CALCULATION OF FOOTING F4

Size of Columns at node	Along-X	Along-Z
2	0.70m	0.40m

Size of fdn. raft:	2.25m	2.00m
Thickness of raft:	0.40	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.80	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

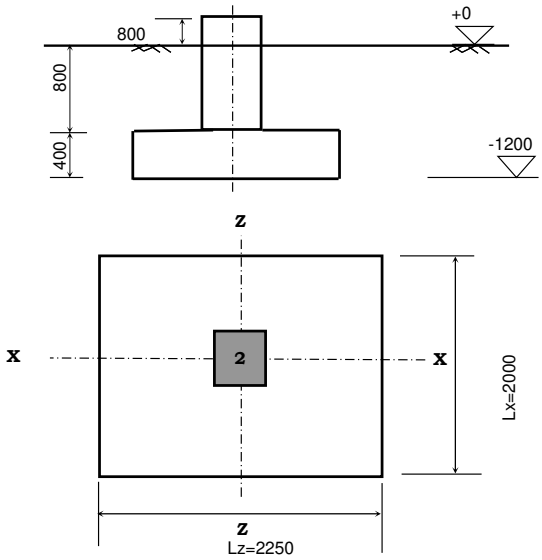
Weight of raft = (2.25*2*0.4*24) = 43.20KN

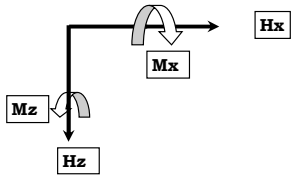
Wt of soil+surcharge = 100+ ((2.25*2)-(0.7*0.4))*0.8*20 = 167.52KN

Total = 210.72KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F4

Load at bottom of footing	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	19.00	0.00	0.00	0.00	0.00
202 DL+WL	19.00	-40.00	80.00	0.00	0.00
203 0.6DL+WL	11.40	-40.00	80.00	0.00	0.00

Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	229.7	0.00	0.0	0.0	0.0	NA	NA	NA	NA
202 DL+WL	229.7	0.00	0.0	-40.0	80.0	3.2	NA	2.3	NA
203 0.6DL+WL	222.1	0.00	0.0	-40.0	80.0	3.1	NA	2.2	NA

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Base pressure check of footing F4

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	56.65	0
Min Px	2	201	0	56.65	0
Max Py	418	202	0	101.81	0
Min Py	425	203	0	0.00	0
Max Pz	2	201	0	56.65	0
Min Pz	2	201	0	56.65	0

Gross Maximum Base pressure, p_{max} = 101.8 KN/m^2

Net Allowable Safe Bearing Capacity = 350 KN/m^2
Depth of soil above footing = 0.80 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 366 KN/m^2
> 101.8 KN/m^2
Hence OK

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Design of foundation raft F4

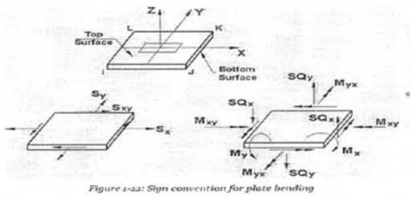
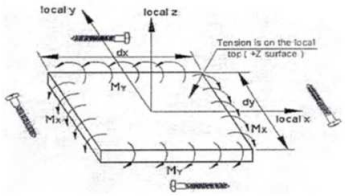


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F4_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	333	0.297	-0.112	0	0	0	-36	-15.14	2.869	-38.432	-18.009	102:1.2DL+1.6WL
Min Qx	415	-0.208	-0.059	0	0	0	-4.143	1.224	-6.404	-10.547	7.628	102:1.2DL+1.6WL
Max Qy	417	0.108	0.219	0	0	0	-21.465	-8.993	-5.46	-26.925	-14.453	102:1.2DL+1.6WL
Min Qy	347	0.108	-0.219	0	0	0	-21.533	-9.041	5.458	-26.991	-14.499	102:1.2DL+1.6WL
Max Sx	475	-0.006	0.004	0	0	0	-0.819	-0.494	-0.458	-1.277	-0.952	101:1.4DL
Min Sx	475	-0.006	0.004	0	0	0	-0.819	-0.494	-0.458	-1.277	-0.952	101:1.4DL
Max Sy	475	-0.006	0.004	0	0	0	-0.819	-0.494	-0.458	-1.277	-0.952	101:1.4DL
Min Sy	475	-0.006	0.004	0	0	0	-0.819	-0.494	-0.458	-1.277	-0.952	101:1.4DL
Max Sxy	475	-0.006	0.004	0	0	0	-0.819	-0.494	-0.458	-1.277	-0.952	101:1.4DL
Min Sxy	475	-0.006	0.004	0	0	0	-0.819	-0.494	-0.458	-1.277	-0.952	101:1.4DL
Max Mx	327	0.252	0.104	0	0	0	32.11	9.794	3.631	35.741	13.425	102:1.2DL+1.6WL
Min Mx	333	0.295	-0.11	0	0	0	-36.494	-14.197	3.015	-39.509	-17.212	103:0.9DL+1.6WL
Max My	327	0.258	0.106	0	0	0	30.125	10.613	3.5	33.625	14.113	103:0.9DL+1.6WL
Min My	333	0.297	-0.112	0	0	0	-35.563	-15.14	2.869	-38.432	-18.009	102:1.2DL+1.6WL
Max Mxy	445	-0.056	0.024	0	0	0	3.712	-0.288	7.457	11.169	-7.745	102:1.2DL+1.6WL
Min Mxy	465	-0.056	-0.024	0	0	0	3.711	-0.292	-7.462	11.173	-7.754	102:1.2DL+1.6WL

PLATE CENTER STRESS : FDN_F4_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	428	0.151	-0.035	0	0	0	-13.812	-11.927	-0.531	-14.343	-12.458	103:0.9DL+1.6WL
Min Qx	445	-0.056	0.024	0	0	0	3.712	-0.288	7.457	11.169	-7.745	102:1.2DL+1.6WL
Max Qy	432	0.092	0.074	0	0	0	-10.834	-6.901	0.807	-11.641	-7.708	102:1.2DL+1.6WL
Min Qy	430	0.092	-0.074	0	0	0	-10.851	-6.93	-0.818	-11.669	-7.748	102:1.2DL+1.6WL
Max Sx	475	-0.006	0.004	0	0	0	-0.819	-0.494	-0.458	-1.277	-0.952	101:1.4DL
Min Sx	475	-0.006	0.004	0	0	0	-0.819	-0.494	-0.458	-1.277	-0.952	101:1.4DL
Max Sy	475	-0.006	0.004	0	0	0	-0.819	-0.494	-0.458	-1.277	-0.952	101:1.4DL
Min Sy	475	-0.006	0.004	0	0	0	-0.819	-0.494	-0.458	-1.277	-0.952	101:1.4DL
Max Sxy	475	-0.006	0.004	0	0	0	-0.819	-0.494	-0.458	-1.277	-0.952	101:1.4DL
Min Sxy	475	-0.006	0.004	0	0	0	-0.819	-0.494	-0.458	-1.277	-0.952	101:1.4DL
Max Mx	437	0.125	0.031	0	0	0	11.926	9.13	-0.206	12.132	9.336	102:1.2DL+1.6WL
Min Mx	428	0.151	-0.035	0	0	0	-13.812	-11.927	-0.531	-14.343	-12.458	103:0.9DL+1.6WL
Max My	437	0.12	0.032	0	0	0	10.122	9.726	-0.275	10.397	10.001	103:0.9DL+1.6WL
Min My	428	0.149	-0.036	0	0	0	-13.15	-12.631	-0.612	-13.762	-13.243	102:1.2DL+1.6WL
Max Mxy	445	-0.056	0.024	0	0	0	3.712	-0.288	7.457	11.169	-7.745	102:1.2DL+1.6WL
Min Mxy	465	-0.056	-0.024	0	0	0	3.711	-0.292	-7.462	11.173	-7.754	102:1.2DL+1.6WL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

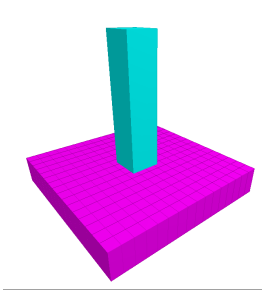
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
102:1.2D	35.74	top face	50	344	35.741	0.40	0.0010	332.1	619.2	12	182.72	150	754.3	10.64	13.29	0.075	tension control
103:0.9D	-39.5	bottom face	50	342	39.509	0.45	0.0011	369.6		16	324.84	150	1341	18.91	23.64	0.04	tension control
102:1.2D	11.17	top face	50	344	11.169	0.13	0.0003	103.3		12	182.72	150	754.3	10.64	13.29	0.075	tension control
102:1.2D	11.17	top face	50	344	11.173	0.13	0.0003	103.3		12	182.72	150	754.3	10.64	13.29	0.075	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN			
Load condition	Max Qx	0.151	1	60.4	14.343	0.002	0.344	344.197	338.444	590.188	338.44	1261	> 60.4	ok	253.8	φVc>Vu	not required
	Min Qx	0.056	1	22.4	11.169			334.468			334.47	1258	> 22.4	ok	250.9	φVc>Vu	not required

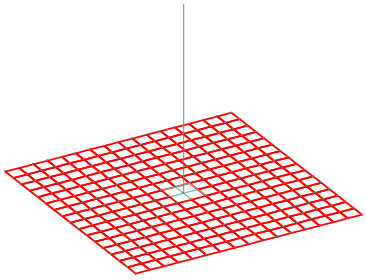
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
103:0.9D	14.11	top face	50	344	14.113	0.16	0.0004	130.6	619.2	12	182.72	150	754.3	10.64	13.29	0.075	tension control
102:1.2D	-18.01	bottom face	50	342	18.009	0.21	0.0005	167.7		16	324.84	150	1341	18.91	23.64	0.04	tension control
102:1.2D	-7.75	bottom face	50	342	7.745	0.09	0.0002	72.0		16	324.84	150	1341	18.91	23.64	0.04	tension control
102:1.2D	-7.75	bottom face	50	342	7.754	0.09	0.0002	72.1		16	324.84	150	1341	18.91	23.64	0.04	tension control

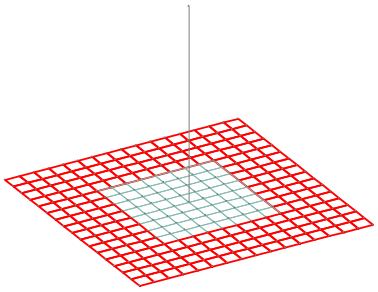
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN			
Load condition	Max Qy	0.074	1	29.6	7.708	0.002	0.344	342.561	338.444	fc'590.2	338.44	355	> 29.6	ok	253.8	φVc>Vu	not required
	Min Qy	0.074	1	29.6	7.748			342.473			338.44	355	> 29.6	ok	253.8	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
At Bottom **16ϕT** **150** c/c
At Top **12ϕT** **150** c/c
Along Z Direction :
At Bottom **16ϕT** **150** c/c
At Top **12ϕT** **150** c/c

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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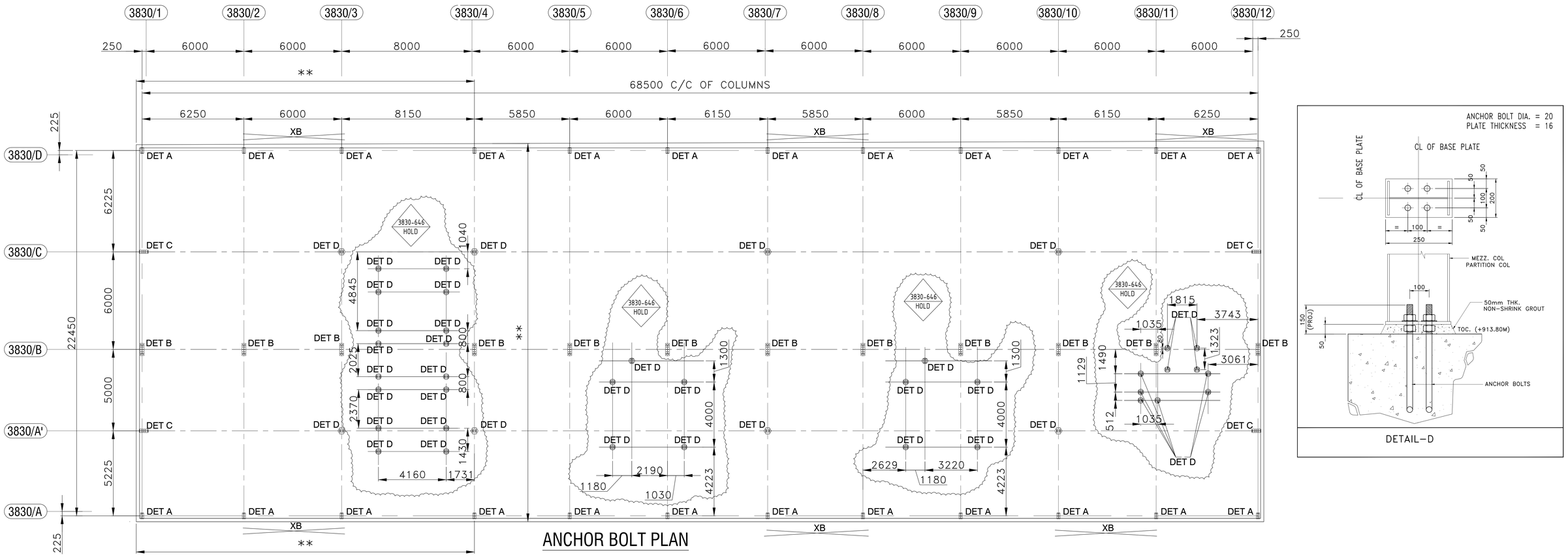
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ATTACHMENT 1 INPUT DRAWINGS

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DRAWING NUMBER MD-216-3830-EG-ST-GAD-2349 REV D01



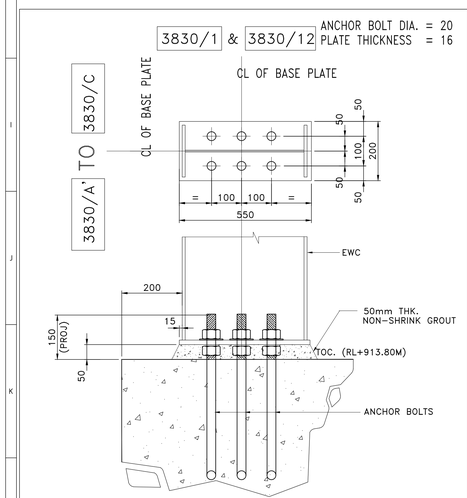
ANCHOR BOLT PLAN

XB DENOTES FULL HEIGHT DIAGONAL BRACING

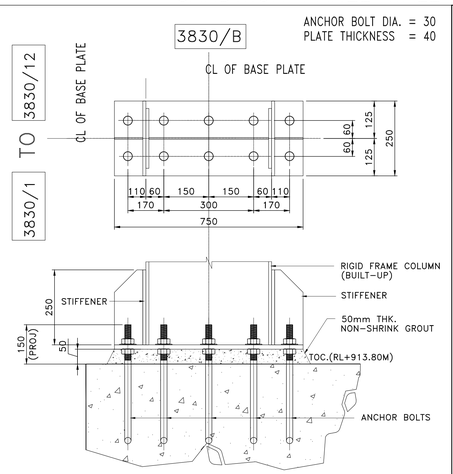
DENOTES MEZZANINE COLUMN LOCATION UP TO MEZZANINE LEVEL

DENOTES PARTITION COLUMN LOCATION UP TO ROOF LEVEL

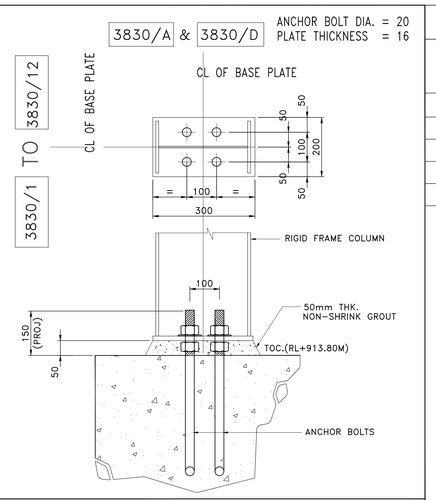
** SANDWICH PANEL WALL WITH 75 mm INSULATION ROCKWALL



DETAIL–C



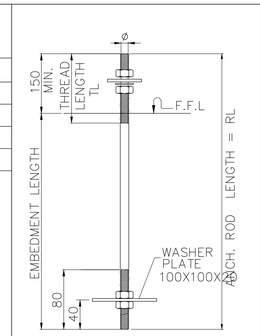
DETAIL–B



DETAIL–A

NOTE: ANCHOR BOLT SPECIFICATION (AS PER ASTM A36)

QTY.	DIA. OF BOLT=ø	DIA. OF HOLE	THREAD LENGTH	HOOK LENGTH	HOOK WIDTH	RADIUS	VERTICAL LENGTH (VL)	ANC.ROD LENGTH
--	ø	D	TL	HL	HW	R	VL	AL
312	M20	26	150	110	100	30	575	710
--	M24	30	150	140	128	40	675	851
120	M30	36	200	170	160	50	975	1189
--	M36	42	200	210	192	60	1075	1338



ANCHOR BOLT DETAIL

LEGEND
TOC–TOP OF CONCRETE
CL–CENTER LINE
F.F.L.–FLOOR FINISH LEVEL
DET–DETAIL
PROJ–PROJECTION

NOTES:–
1. ALL DIMENSIONS ARE IN MILLIMETER UNLESS NOTED OTHERWISE.
2. ALL LEVELS ARE IN METER.

MATERIAL SPECIFICATIONS

NO.	MATERIALS		GRADE & SPECIFICATION	STRENGTH
01	ANCHOR BOLTS	J-type	ASTM F1554 Grade A36M, Hot dip galvanized to ASTM A153M, Heavy Hex Nuts A563M-10S Washers-ASTMF436	Fu = 400 N/mm
		STRAIGHT -type	BS EN ISO 898-1 GRDE 8.8, Hot dip galvanized to BS EN ISO1461 BS EN ISO 898-2 10 STYLE2 WASHER-BS4320	Fu = 800 N/mm
02	BUILT-UP SECTIONS		ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm
03	HOT ROLLED SECTIONS	Tubes	E10025 S355HR or equivalent	Fy = 345 N/mm
		I-Sections	ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm
		Channels	E10025 S355JR or equivalent	Fy = 355 N/mm
		Angles	ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm
		Pipes	ASTM A53 Grade B or equivalent	Fy = 240 N/mm
04	Primary Connection bolts		ASTM A325M TYPE-1; Fully threaded hot dip galvanized	Ft = 827 N/mm
05	DIAGONAL 'X' BRACING	HR Angles	ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm
06	Cold formed secondary members		ASTM A653M SS Grade 340 Class 1: Coating Z275 (Grade 90)	Fy = 344 N/mm
07	Secondary Connection bolts (for purlins,girts and flange braces)		Grade 8.8 Fully threaded hot dip galvanized	Ft = 400 N/mm
08	Sag Rods		ASTM Grade A36M Electro galvanized, Fully threaded with 4 nuts Coating Z275 (Grade 90)	Fy = 245 N/mm
09	Mezzanine Deck Panels		ASTM A653M SS Grade 340 Class 1: Coating Z275 (Grade 90)	Fy = 344 N/mm

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3830-EG-ST-GAD-1001-A01	3D view	D01	29/11/2023	AS BUILT	--	SYC	--	PRI	
MD-216-3830-EG-ST-GAD-1003-A01	STEEL STRUCTURE GUIDE DRAWING	C01	13/08/2020	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS DATED 30/03/2020 & AS MKD.	--	SYC	--	PRI	
MD-216-3830-EG-ST-GAD-1002-A01	STEEL STRUCTURE GUIDE DRAWING								
MD-216-3830-EG-ST-GAD-1002_ELE inputs	STEEL STRUCTURE GUIDE DRAWING	A01	06/07/2020	ISSUED FOR REFERENCE	--	SYC	--	PRI	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

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DRAWN	--	--	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SYC	04/07/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	--	--	FLOTATION REAGENTS BUILDING
APPROVED	PRI	06/07/2020	ANCHOR BOLT PLAN AND DETAILS
EMPLOYER			
SCALE	NTS	A0	

DRAWING NUMBER	REVISION
MD-P20131-B05-A00	00

	DATE		DATE		DATE
DRAWN	02/06/2020 7029 7021	CHECKED	03/06/2020 7013	APPROVED	04/06/2020 7001

LARSEN & TOUBRO DRAWING NUMBER	REVISION
019045_450_S_G_26071_001	2

OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707_GAU01_AA02_MTL01_38311	OU100676480	2

DRAWING NUMBER	REV
MD-216-3830-EG-ST-GAD-2349	D01

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: BECHTEL-WTRAN-003987,30 July 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 30 July 2020	Comment by	EPC Response DATE: 13 August 2020	Status	Remarks
			General Comments (cc)				
MD-216-3830-EG-ST-GAD-2349-A01	3830	1	As the structural design criteria for this project, the live load for the platform is wrong	Mr. Doron Golan	Inside the building shed, Platforms are kept in hold.	Closed	

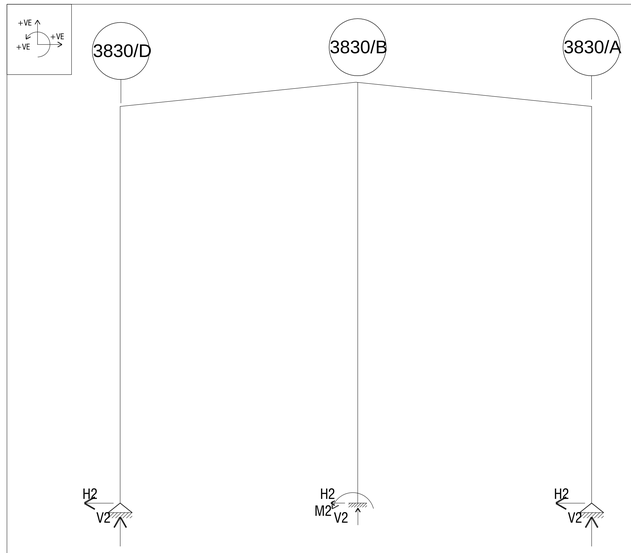
DRAWING NUMBER MD-216-3830-EG-ST-GAD-2350 REV D01

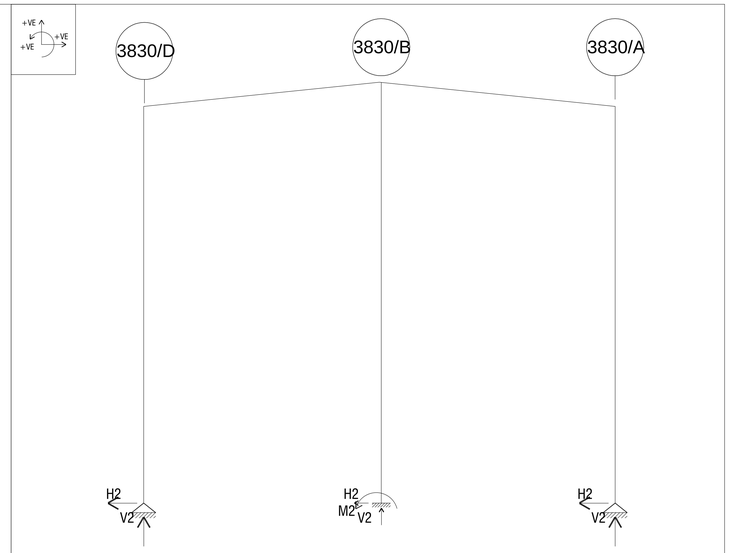
NOTES FOR REACTIONS
Building reactions are based on the following building data:

ROOF LOADS			
DEAD LOAD (PURLIN+PANEL) (kN/m2)			0.24
ROOF LIVE LOAD (kN/m2)		=	1.00
			REDUCIBLE AS PER SBC
COLLATERAL LOAD (kN/m2)		=	0.00
BASIC LOADS			
WIND LOAD	Wind Speed (3 sec)		=155 (Km/Hr)
	Wind Code		= SBC-301(2018)
	EXPOSURE	=	C
	CLASSIFICATION	= ENCLOSED	
SEIS LOAD IBC 2006	Ss	=	0.1g
	S1	=	0.02g
	SITE CLASS	=	C
TEMPERATURE LOAD	TEMPERATURE LOAD	=	± 25 C
CRANE LOADS			
MONORAILS OF CAPACITIES 2.0 & 5.0 TON			
PLATFORM LOADS			
DEAD LOAD (kN/m2)		=	0.50
LIVE LOAD (kN/m2)		=	4.00

LOADS LEGEND	
Wind_Left1	Denotes wind load from left to right with internal pressure
Wind_Right1	Denotes wind load from right to left with internal pressure
Wind_Left2	Denotes wind load from left to right with internal suction
Wind_Right2	Denotes wind load from right to left with internal suction
Wind_Left2	Denotes wind load from left to right with internal pressure
Wind_Right2	Denotes wind load from right to left with internal pressure
Wind_Left3	Denotes wind load from left to right with internal suction
Wind_Right4	Denotes wind load from right to left with internal suction
Wind_Long1	Denotes wind load parallel to ridge from end wall with internal pressure
Wind_Long2	Denotes wind load parallel to ridge from end wall with internal suction
Seismic_L	Denotes seismic load from left
Seismic_R	Denotes seismic load from right

IMPORTANT NOTES :
(1) BEARING CAPACITY OF CONCRETE UNDER BASE PLATE SHOULD NOT BE LESS THAN 1.05 KN/CM2.
(2) GIVEN REACTIONS ARE FOR UNFACTORED LOADS.
(3) BRACING REACTIONS TO BE ADDED WITH MAIN FRAME COLUMN REACTIONS FOR THE DESIGN OF FOUNDATION.
(4) LOADS IN THE SAME CATEGORY , NOT TO BE COMBINED IN THE SAME LOAD COMBINATION .
(5) WIND LOAD IS CALCULATED BASED ON SBC-301-2018.





FRAME CROSS SECTION ALONG GL-3830/(7,10,11,12)

CATEGORY	LOAD CASE	LEFT COL.@ GL 3830/D		INT COL.@ GL 3830/B			RIGHT COL.@ GL 3830/A	
		H1 (KN)	V1 (KN)	H2 (KN)	V2 (KN)	M2 (KN.M)	H3 (KN)	V3 (KN)
Dead	Dead	1	17	-1	22	3	-1	11
Collateral	Collateral	1	1	2	-1	-17	-3	25
Roof Live	Live	4	25	-2	51	9	-3	19
Wind	Wind_Left1	-32	-70	-21	-99	163	-32	-19
	Wind_Left2	-48	-40	-23	-62	176	-14	7
	Wind_Left3	-32	-70	-21	-99	163	-32	-19
	Wind_Left4	-48	-40	-23	-62	176	-14	7
	Wind_Right1	30	-28	32	-89	-229	29	-64
	Wind_Right2	14	2	30	-52	-217	47	-38
	Wind_Right3	30	-28	32	-89	-229	29	-64
	Wind_Right4	14	2	30	-52	-217	47	-38
	Wind_Long1	25	-69	4	-101	-19	-29	-73
	Wind_Long2	9	-39	1	-64	-7	-11	-47
	Wind_Long3	23	-86	9	-101	-53	-29	-60
	Wind_Long4	8	-55	7	-64	-41	-11	-34
Seismic	Seismic_L	0	0	-1	0	7	0	1
	Seismic_R	0	0	1	0	-7	0	-1
Temperature	25Degree	1	1	0	-2	0	-1	1
	-25Degree	-1	-1	0	2	0	1	-1

FRAME CROSS SECTION ALONG GL-3830/(1,2,3,4,5,6,8,9)

CATEGORY	LOAD CASE	LEFT COL.@ GL 3830/D		INT COL.@ GL 3830/B			RIGHT COL.@ GL 3830/A	
		H1 (KN)	V1 (KN)	H2 (KN)	V2 (KN)	M2 (KN.M)	H1 (KN)	V1 (KN)
Dead	Dead	1	18	-1	23	3	-1	14
Collateral	Collateral	1	2	4	-1	-26	-5	37
Roof Live	Live	5	29	-2	61	10	-3	23
Crane	Crane1	8	31	-9	35	40	1	-3
	Crane2	0	-1	7	26	-33	-7	38
Wind	Wind_Left1	-37	-83	-24	-117	193	-38	-22
	Wind_Left2	-56	-47	-27	-74	207	-17	9
	Wind_Left3	-37	-83	-24	-117	193	-38	-22
	Wind_Left4	-56	-47	-27	-74	207	-17	9
	Wind_Right1	35	-33	38	-105	-270	34	-75
	Wind_Right2	16	2	35	-62	-256	56	-44
	Wind_Right3	35	-33	38	-105	-270	34	-75
	Wind_Right4	16	2	35	-62	-256	56	-44
	Wind_Long1	29	-82	4	-119	-22	-34	-86
	Wind_Long2	11	-46	2	-76	-8	-12	-55
	Wind_Long3	28	-101	11	-119	-63	-35	-70
	Wind_Long4	9	-65	8	-75	-48	-13	-40
Seismic	Seismic_L	0	0	-1	0	7	0	1
	Seismic_R	0	0	1	0	-7	0	-1
Temperature	25Degree	0	1	0	-2	0	0	1
	-25Degree	0	-1	0	2	0	0	-1

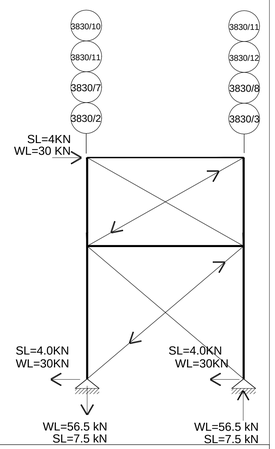
PLATFORM COLUMN REACTION



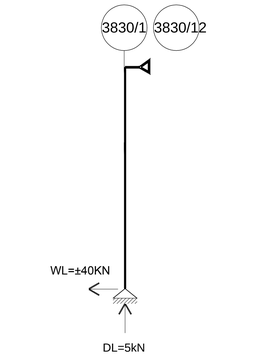
PARTITION COLUMN REACTION



SIDEWALL (XB) BRACING REACTION AT GL 3830/1& 3830/4



ENDWALL COLUMN REACTION AT GL 3240/A



MD-216-3830-EG-ST-GAD-1001-A01	3D view	D01	29/11/2023	AS BUILT	--	SYC	--	PRI											
MD-216-3830-EG-ST-GAD-1003-A01	STEEL STRUCTURE GUIDE DRAWING	C01	13/08/2020	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S	--	SYC	--	PRI											
MD-216-3830-EG-ST-GAD-1002-A01	STEEL STRUCTURE GUIDE DRAWING			COMMENTS DATED 30/07/2020															
MD-216-3830-EG-ST-GAD-1002_ELE inputs	STEEL STRUCTURE GUIDE DRAWING	A01	06/07/2020	ISSUED FOR REFERENCE	--	SYC	--	PRI											
DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP	DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

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DRAWING NUMBER		REVISION	LARSEN & TOUBRO DRAWING NUMBER		REVISION
MD-P20131-B05-A00		00	019045_450_S_G_26071_002		2
DATE 02/06/2020	DATE 03/06/2020	DATE 04/06/2020	OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
DRAWN 7023 7021	CHECKED 7013	APPROVED 7001	900707_GAU01_AA02_MTL01_38312	OU100676481	2

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DRAWN	--	--	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SYC	04/07/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	--	--	FLOTATION REAGENTS BUILDING
APPROVED	PRI	06/07/2020	COLUMN REACTIONS
EMPLOYER			
SCALE	NTS	A0	DRAWING NUMBER MD-216-3830-EG-ST-GAD-2350 REV D01

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

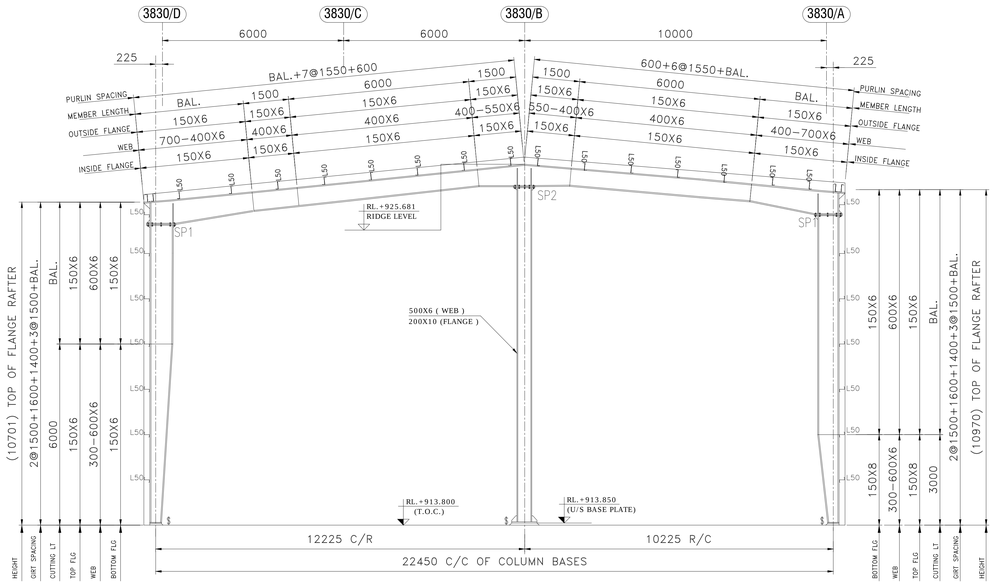
PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: BECHTEL-WTRAN-003987,30 July 2020

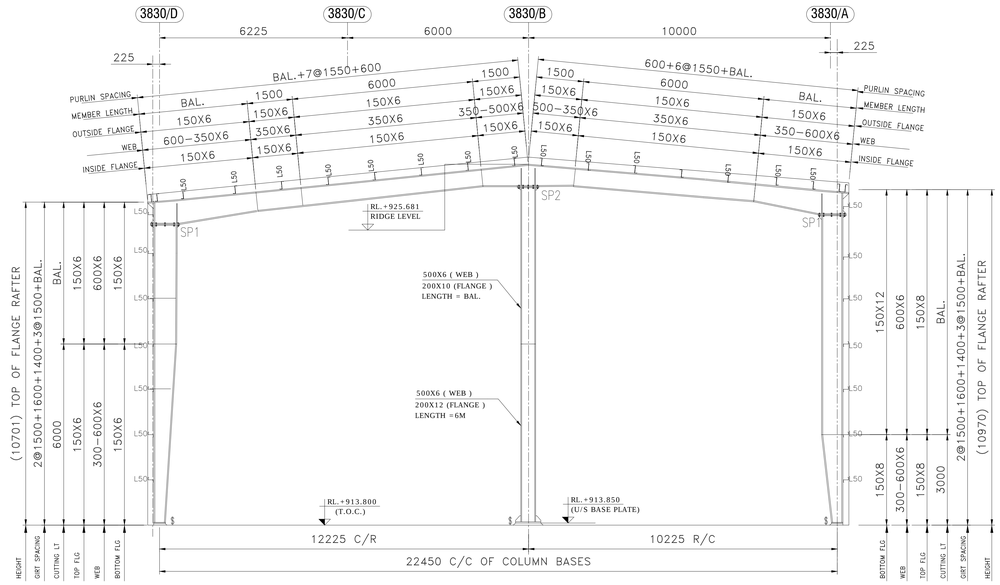
STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 30 July 2020	Comment by	EPC Response DATE: 13 August 2020	Status	Remarks
			General Comments (cc)				
MD-216-3830-EG-ST-GAD-2350-A01	3830	1	As the structural design criteria for this project, the live load for the platform is wrong	Mr. Doron Golan	Inside the building shed, Platforms are kept in hold.	Closed	



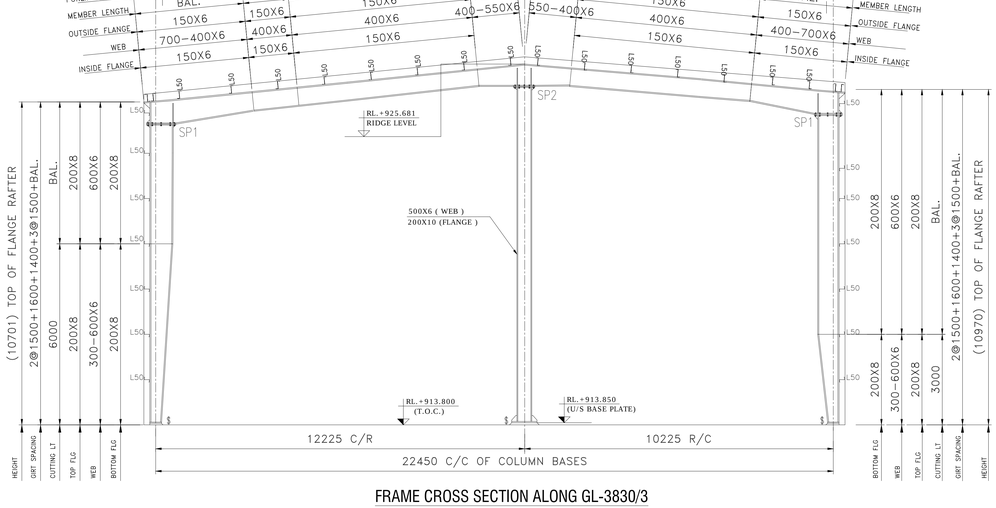
FRAME CROSS SECTION ALONG GL-3830/(1,2,4,5,6,8,9)

$ DENOTES 50MM THICK GROUT
L50: DENOTES L50X50X3 FLANGE BRACE ON ONE SIDE.



FRAME CROSS SECTION ALONG GL-3830/(7,10,11,12)

$ DENOTES 50MM THICK GROUT
L50: DENOTES L50X50X3 FLANGE BRACE ON ONE SIDE.



FRAME CROSS SECTION ALONG GL-3830/3

$ DENOTES 50MM THICK GROUT
L50: DENOTES L50X50X3 FLANGE BRACE ON ONE SIDE.

NOTE:

1. ALL DIMENSIONS ARE IN MILLIMETERS AND LEVELS & CO-ORDINATE ARE IN METERS (U.N.O.).

LEGEND

FFL - FINISHED FLOOR LEVEL
GL - GRID LINE
O/O - OUT TO OUT
C/C - CENTER TO CENTER
O/C - OUT TO CENTER
C/O - CENTER TO OUT
℄ - CENTER LINE
TYP - TYPICAL
TOC - TOP OF CONCRETE.
TOS - TOP OF STEEL.
LVL - LEVEL

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3830-EG-ST-GAD-1001-A01	3D view								
MD-216-3830-EG-ST-GAD-1003-A01	STEEL STRUCTURE GUIDE DRAWING	D01	29/11/2023	AS BUILT	--	SYC	--	PRI	
MD-216-3830-EG-ST-GAD-1002-A01	STEEL STRUCTURE GUIDE DRAWING	C01	13/08/2020	ISSUED FOR CONSTRUCTION	--	SYC	--	PRI	
MD-216-3830-EG-ST-GAD-1002_ELE inputs	STEEL STRUCTURE GUIDE DRAWING	A01	09/07/2020	ISSUED FOR REFERENCE	--	SYC	--	PRI	

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DRAWING NUMBER						REVISION
MD-P20131-B05-A00						0
DATE	01/06/2020	DATE	02/06/2020	DATE	02/06/2020	
DRAWN	3020 3021	CHECKED	7013	APPROVED	7001	

LARSEN & TOUBRO DRAWING NUMBER		REVISION
019045_450_S_G_26071_003		2
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
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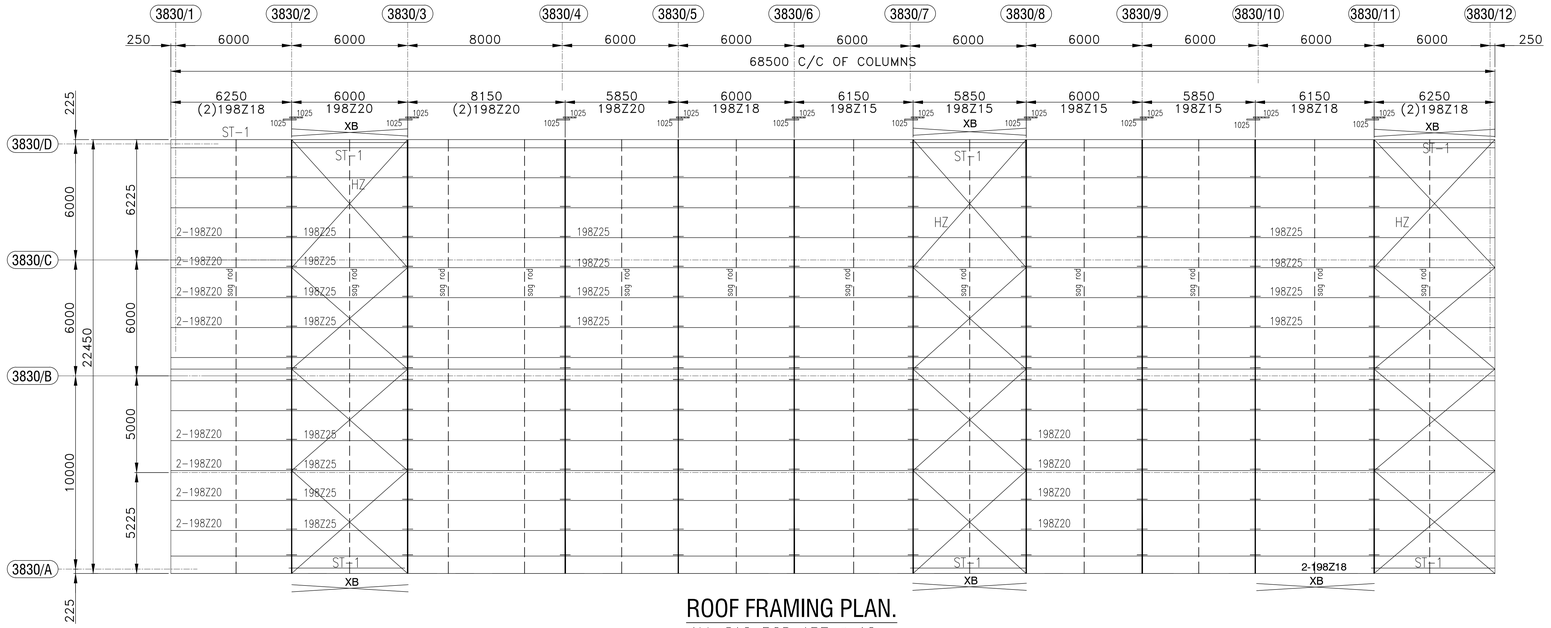
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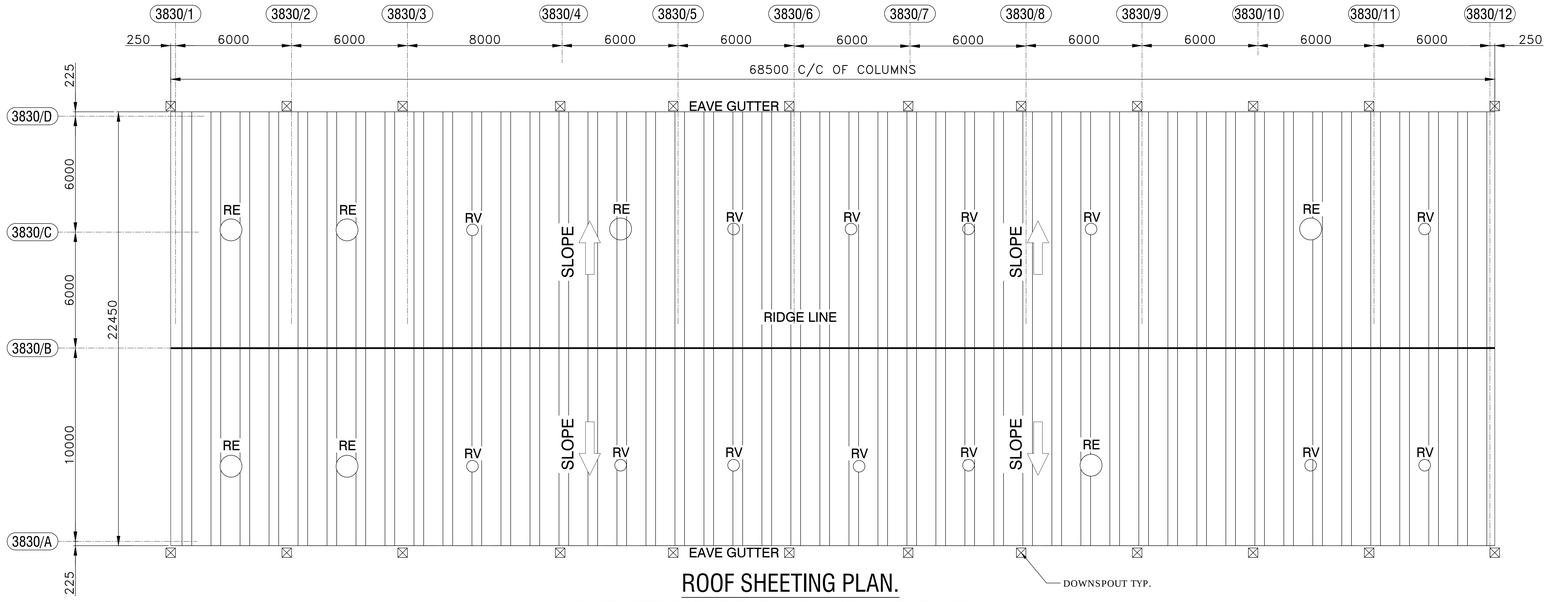
DRAWN	--	--	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SYC	09/07/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	--	--	FLOTATION REAGENTS BUILDING
APPROVED	PRI	09/07/2020	FRAME CROSS SECTION @GL(1) TO GL(12)
EMPLOYER			
SCALE	NTS	A0	DRAWING NUMBER: MD-216-3830-EG-ST-GAD-2351 REV: D01

DRAWING NUMBER MD-216-3830-EG-ST-GAD-2352 | REV D01



ROOF FRAMING PLAN.

ALL SAG ROD ARE ø 12.
HZ BRACING DOUBLE ANGLE 75X75X6
ST-1 DENOTES STRUT PURLIN 198Z15



ROOF SHEETING PLAN.

RV : DENOTES ROOF VENTILATORS EACH 20 kg, WITH DIA=600mm
RE : DENOTES ROOF EXTRACTOR EACH 200 kg, WITH DIA=1130mm

ROOF SHEETING SPECIFICATION		
DOUBLE SKIN	PROFILE : CORRUGATED	TOTAL COLORED THICKNESS (mm) : 0.50
	PROFILE 0.5TCT - PRE COLOUR COATED	FINISH : POLYESTER PAINTED
	100mm THK. OF ROOKWOOL IN BETWEEN WITH DENSITY 48KG/M3	

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETERS AND LEVELS & CO-ORDINATE ARE IN METERS (U.N.O.).

LEGEND
FFL - FINISHED FLOOR LEVEL
GL - GRID LINE
O/O - OUT TO OUT
C/C - CENTER TO CENTER
O/C - OUT TO CENTER
C/O - CENTER TO OUT
℄ - CENTER LINE
TYP - TYPICAL
TOC - TOP OF CONCRETE.
TOS - TOP OF STEEL.
LVL - LEVEL

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3830-EG-ST-GAD-1001-A01	3D view								
MD-216-3830-EG-ST-GAD-1003-A01	STEEL STRUCTURE GUIDE DRAWING	D01	29/11/2023	AS BUILT	--	SYC	--	PRI	
MD-216-3830-EG-ST-GAD-1002-A01	STEEL STRUCTURE GUIDE DRAWING	C01	13/08/2020	ISSUED FOR CONSTRUCTION	--	SYC	--	PRI	
MD-216-3830-EG-ST-GAD-1002_ELE inputs	STEEL STRUCTURE GUIDE DRAWING	A01	09/07/2020	ISSUED FOR REFERENCE	--	SYC	--	PRI	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

Arabian International Company
P.O.Box : 41017 Jeddah 21521,K.S.A
Tel. +966 2 691 0999
Fax +966 2 691 0099
E-mail aic@aicsteel.com
Site www.aicsteel.com

DRAWING NUMBER MD-P20131-B05-A00 | REVISION 0
DATE 02/06/2020 | DATE 02/06/2020 | DATE 02/06/2020
DRAWN 7029 7021 | CHECKED 7013 | APPROVED 7001

LARSEN & TOUBRO DRAWING NUMBER 019045_450_S_G_26071_004 | REVISION 2
OUTOTEC DRAWING NUMBER 900707_GAU01_AA02_MTL01_38310 | DOCUMENT ID OU100673724 | REVISION 2

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DRAWN	--	--	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SYC	09/07/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	--	--	FLOTATION REAGENTS BUILDING
APPROVED	PRI	09/07/2020	ROOF FRAMING & SHEETING PLAN
EMPLOYER			
SCALE NTS	A0		DRAWING NUMBER MD-216-3830-EG-ST-GAD-2352 REV D01

ATTACHMENT 2 STAAD INPUT – FOUNDATION F1

PROPRIETARY & CONFIDENTIAL

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 04/09/2020
JOB NAME FDN. MKD. F1
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
SET NL 200
SET ITERLIM 15
SET PRINT 1
UNIT METER KN
JOINT COORDINATES
2 0 911.8 3.175; 6 0.75 911.8 3.175; 10 -0.75 911.8 3.175; 34 -0.5 911.8 3.175;
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185 2 128;
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578 503 580 573 496; 579 504 581 580 503; 580 502 579 582 505;
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593 589 510 517 596; 594 597 518 519 598; 595 594 515 518 597;
596 599 520 513 592; 597 600 521 520 599; 598 598 519 522 601;
599 602 523 521 600; 600 601 522 524 603; 601 604 525 523 602;
602 603 524 546 605; 603 606 568 525 604;
START GROUP DEFINITION
ELEMENT
_RAFT 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 -
393 TO 399 402 TO 409 412 TO 419 422 TO 429 432 TO 536 545 547 549 TO 556 -
565 567 TO 603
_RAFT_MOM 326 327 333 334 337 342 TO 349 352 TO 359 362 TO 369 372 -
393 TO 399 402 TO 409 412 TO 419 422 TO 424 427 TO 429 432 TO 536 545 547 -
549 TO 556 565 567 TO 603
_RAFT_SHEAR 337 342 349 352 359 362 TO 369 372 393 TO 399 402 409 412 419 -
422 429 432 TO 536 545 547 549 TO 556 565 567 TO 603
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 TO 536 545 547 549 TO 556 565 -
567 TO 603 THICKNESS 0.6
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.5 ZD 0.85
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 TO 536 545 547 549 TO 556 565 -
567 TO 603 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 TO 536 545 547 549 TO 556 565 -
567 TO 603 PR GY -16
* LOAD DUE TO SURCHARGE
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 TO 536 545 547 549 TO 556 565 -
567 TO 603 PR GY -20
JOINT LOAD
128 FY -22 FZ -3 MX -23
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE ROOF LIVE LOAD ( LR )
JOINT LOAD
128 FY -61 FZ -2 MX -10
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FY -0.001
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE CRANE LOAD  ( CR1 )
JOINT LOAD
128 FY -35 FZ -9 MX -40
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE CRANE LOAD  ( CR2 )
JOINT LOAD
```



```
JOINT LOAD
128 FY -26 FZ -7 MX -33
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE CRANE LOAD  ( CR3 )
JOINT LOAD
128 FY -0.0001
*********************************************************************************************
LOAD 7 LOADTYPE None  TITLE CRANE LOAD  ( CR4 )
JOINT LOAD
128 FY -0.0001
*********************************************************************************************
LOAD 8 LOADTYPE None  TITLE WIND LOAD  ( WL1 )
JOINT LOAD
128 FY 117 FZ -24 MX -193
*********************************************************************************************
LOAD 9 LOADTYPE None  TITLE WIND LOAD  ( WL2 )
JOINT LOAD
128 FY 74 FZ -27 MX -207
*********************************************************************************************
LOAD 10 LOADTYPE None  TITLE WIND LOAD  ( WL3 )
JOINT LOAD
128 FY 117 FZ -24 MX -193
*********************************************************************************************
LOAD 11 LOADTYPE None  TITLE WIND LOAD  ( WL4 )
JOINT LOAD
128 FY 74 FZ -27 MX -207
*********************************************************************************************
LOAD 12 LOADTYPE None  TITLE WIND LOAD  ( WL5 )
JOINT LOAD
128 FY 105 FZ 38 MX 270
*********************************************************************************************
LOAD 13 LOADTYPE None  TITLE WIND LOAD  ( WL6 )
JOINT LOAD
128 FY 62 FZ 35 MX 256
*********************************************************************************************
LOAD 14 LOADTYPE None  TITLE WIND LOAD  ( WL7 )
JOINT LOAD
128 FY 105 FZ 38 MX 270
*********************************************************************************************
LOAD 15 LOADTYPE None  TITLE WIND LOAD  ( WL8 )
JOINT LOAD
128 FY 62 FZ 35 MX 256
*********************************************************************************************
LOAD 16 LOADTYPE None  TITLE WIND LOAD  ( WL9 )
JOINT LOAD
128 FY 119 FZ 4 MX 22
*********************************************************************************************
LOAD 17 LOADTYPE None  TITLE WIND LOAD  ( WL10 )
JOINT LOAD
128 FY 76 FZ 2 MX 8
*********************************************************************************************
LOAD 18 LOADTYPE None  TITLE WIND LOAD  ( WL11 )
JOINT LOAD
128 FY 119 FZ 11 MX 63
*********************************************************************************************
LOAD 19 LOADTYPE None  TITLE WIND LOAD  ( WL12 )
JOINT LOAD
128 FY 75 FZ 8 MX 48
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LOAD 20 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FZ -1 MX -7
*********************************************************************************************
LOAD 21 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FZ 1 MX 7
*********************************************************************************************
LOAD 22 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL1 )
JOINT LOAD
128 FY 2
*********************************************************************************************
LOAD 23 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL2 )
JOINT LOAD
128 FY -2
PERFORM ANALYSIS
CHANGE
SUPPORTS
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 TO 536 545 547 549 TO 556 565 -
```

```
402 TO 409 412 TO 419 422 TO 429 432 TO 536 545 547 549 TO 556 565 -
567 TO 603 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
********************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD 201 DL
REPEAT LOAD
1 1.0
PERFORM ANALYSIS
CHANGE
LOAD 202 DL+L+TL1
REPEAT LOAD
1 1.0 3 1.0 22 1.0
PERFORM ANALYSIS
CHANGE
LOAD 203 DL+L+TL2
REPEAT LOAD
1 1.0 3 1.0 23 1.0
PERFORM ANALYSIS
CHANGE
LOAD 204 DL+LR
REPEAT LOAD
1 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 205 DL+0.75L+0.75TL1+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 22 0.75
PERFORM ANALYSIS
CHANGE
LOAD 206 DL+0.75L+0.75TL2+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 23 0.75
PERFORM ANALYSIS
CHANGE
LOAD 207 DL+0.6WL1
REPEAT LOAD
1 1.0 8 0.6
PERFORM ANALYSIS
CHANGE
LOAD 208 DL+WL2
REPEAT LOAD
1 1.0 9 1.0
PERFORM ANALYSIS
CHANGE
LOAD 209 DL+WL3
REPEAT LOAD
1 1.0 10 1.0
PERFORM ANALYSIS
CHANGE
LOAD 210 DL+WL4
REPEAT LOAD
1 1.0 11 1.0
PERFORM ANALYSIS
CHANGE
LOAD 211 DL+WL5
REPEAT LOAD
1 1.0 12 1.0
PERFORM ANALYSIS
CHANGE
LOAD 212 DL+WL6
REPEAT LOAD
1 1.0 13 1.0
PERFORM ANALYSIS
CHANGE
LOAD 213 DL+WL7
REPEAT LOAD
1 1.0 14 1.0
PERFORM ANALYSIS
CHANGE
LOAD 214 DL+WL8
REPEAT LOAD
1 1.0 15 1.0
PERFORM ANALYSIS
CHANGE
LOAD 215 DL+WL9
REPEAT LOAD
1 1.0 16 1.0
PERFORM ANALYSIS
CHANGE
```



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CHANGE
LOAD 216 DL+WL10
REPEAT LOAD
1 1.0 17 1.0
PERFORM ANALYSIS
CHANGE
LOAD 217 DL+WL11
REPEAT LOAD
1 1.0 18 1.0
PERFORM ANALYSIS
CHANGE
LOAD 218 DL+WL12
REPEAT LOAD
1 1.0 19 1.0
PERFORM ANALYSIS
CHANGE
LOAD 219 DL+0.75WL1+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75
PERFORM ANALYSIS
CHANGE
LOAD 220 DL+0.75WL2+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75
PERFORM ANALYSIS
CHANGE
LOAD 221 DL+0.75WL3+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75
PERFORM ANALYSIS
CHANGE
LOAD 222 DL+0.75WL4+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75
PERFORM ANALYSIS
CHANGE
LOAD 223 DL+0.75WL5+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75
PERFORM ANALYSIS
CHANGE
LOAD 224 DL+0.75WL67+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75
PERFORM ANALYSIS
CHANGE
LOAD 225 DL+0.75WL7+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75
PERFORM ANALYSIS
CHANGE
LOAD 226 DL+0.75WL8+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75
PERFORM ANALYSIS
CHANGE
LOAD 227 DL+0.75WL9+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75
PERFORM ANALYSIS
CHANGE
LOAD 228 DL+0.75WL10+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75
PERFORM ANALYSIS
CHANGE
LOAD 229 DL+0.75WL11+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75
PERFORM ANALYSIS
CHANGE
LOAD 230 DL+0.75WL12+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75
PERFORM ANALYSIS
CHANGE
LOAD 231 0.6DL+WL1
REPEAT LOAD
1 0.6 8 1.0
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1 0.6 8 1.0
PERFORM ANALYSIS
CHANGE
LOAD 232 0.6DL+WL2
REPEAT LOAD
1 0.6 9 1.0
PERFORM ANALYSIS
CHANGE
LOAD 233 0.6DL+WL3
REPEAT LOAD
1 0.6 10 1.0
PERFORM ANALYSIS
CHANGE
LOAD 234 0.6DL+WL4
REPEAT LOAD
1 0.6 11 1.0
PERFORM ANALYSIS
CHANGE
LOAD 235 0.6DL+WL5
REPEAT LOAD
1 0.6 12 1.0
PERFORM ANALYSIS
CHANGE
LOAD 236 0.6DL+WL6
REPEAT LOAD
1 0.6 13 1.0
PERFORM ANALYSIS
CHANGE
LOAD 237 0.6DL+WL7
REPEAT LOAD
1 0.6 14 1.0
PERFORM ANALYSIS
CHANGE
LOAD 238 0.6DL+WL8
REPEAT LOAD
1 0.6 15 1.0
PERFORM ANALYSIS
CHANGE
LOAD 239 0.6DL+WL9
REPEAT LOAD
1 0.6 16 1.0
PERFORM ANALYSIS
CHANGE
LOAD 240 0.6DL+WL10
REPEAT LOAD
1 0.6 17 1.0
PERFORM ANALYSIS
CHANGE
LOAD 241 0.6DL+WL11
REPEAT LOAD
1 0.6 18 1.0
PERFORM ANALYSIS
CHANGE
LOAD 242 0.6DL+WL12
REPEAT LOAD
1 0.6 19 1.0
PERFORM ANALYSIS
CHANGE
LOAD 243 0.6DL+0.7SL1
REPEAT LOAD
1 0.6 20 0.7
PERFORM ANALYSIS
CHANGE
LOAD 244 0.6DL+0.7SL2
REPEAT LOAD
1 0.6 21 0.7
PERFORM ANALYSIS
CHANGE
LOAD 245 DL+0.75WL1+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 246 DL+0.75WL2+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 247 DL+0.75WL3+0.75L+0.75LR+CR1
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LOAD 247 DL+0.75WL3+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 248 DL+0.75WL4+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 249 DL+0.75WL5+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 250 DL+0.75WL67+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 251 DL+0.75WL7+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 252 DL+0.75WL8+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 253 DL+0.75WL9+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 254 DL+0.75WL10+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 255 DL+0.75WL11+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 256 DL+0.75WL12+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 257 DL+0.75WL1+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 258 DL+0.75WL2+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 259 DL+0.75WL3+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 260 DL+0.75WL4+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 261 DL+0.75WL5+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 262 DL+0.75WL67+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
PERFORM ANALYSIS
```

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PERFORM ANALYSIS
CHANGE
LOAD 263 DL+0.75WL7+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 264 DL+0.75WL8+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 265 DL+0.75WL9+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 266 DL+0.75WL10+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 267 DL+0.75WL11+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 268 DL+0.75WL12+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 269 DL+0.75WL1+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 270 DL+0.75WL2+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 271 DL+0.75WL3+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 272 DL+0.75WL4+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 273 DL+0.75WL5+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 274 DL+0.75WL67+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 275 DL+0.75WL7+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  276 DL+0.75WL8+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 277 DL+0.75WL9+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  278 DL+0.75WL10+0.75L+0.75LR+CR3
REPEAT LOAD
```



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REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 279 DL+0.75WL11+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 280 DL+0.75WL12+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 281 DL+0.75WL1+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  282 DL+0.75WL2+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 283 DL+0.75WL3+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 284 DL+0.75WL4+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  285 DL+0.75WL5+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 286 DL+0.75WL67+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 287 DL+0.75WL7+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 288 DL+0.75WL8+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 289 DL+0.75WL9+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 290 DL+0.75WL10+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 291 DL+0.75WL11+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 292 DL+0.75WL12+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD 101 1.4DL
REPEAT LOAD
1 1.4
```

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1 1.4
PERFORM ANALYSIS
CHANGE
LOAD 102 1.4DL+1.4TL1+1.7L+0.5LR
REPEAT LOAD
1 1.4 22 1.4 3 1.7 2 0.5
PERFORM ANALYSIS
CHANGE
LOAD  103 1.4DL+1.4TL2+1.7L+0.5LR
REPEAT LOAD
1 1.4 23 1.4 3 1.7 2 0.5
PERFORM ANALYSIS
CHANGE
LOAD  104 1.2DL+1.6LR+1.0L
REPEAT LOAD
1 1.2 2 1.6 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD  105 1.2DL+1.6WL1+1.0L+0.5LR
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 106 1.2DL+1.6WL2+1.0L+0.5LR
REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 107 1.2DL+1.6WL3+1.0L+0.5LR
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 108 1.2DL+1.6WL4+1.0L+0.5LR
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 109 1.2DL+1.6WL5+1.0L+0.5LR
REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 110 1.2DL+1.6WL6+1.0L+0.5LR
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 111 1.2DL+1.6WL7+1.0L+0.5LR
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 112 1.2DL+1.6WL8+1.0L+0.5LR
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD  113 1.2DL+1.6WL9+1.0L+0.5LR
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD  114 1.2DL+1.6WL10+1.0L+0.5LR
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD  115 1.2DL+1.6WL11+1.0L+0.5LR
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD  116 1.2DL+1.6WL12+1.0L+0.5LR
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD  117 1.2DL+1.0SL1+1.0L
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LOAD  117 1.2DL+1.0SL1+1.0L
REPEAT LOAD
1 1.2 20 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD  118 1.2DL+1.0SL2+1.0L
REPEAT LOAD
1 1.2 21 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 119 0.9DL+1.6WL1
REPEAT LOAD
1 0.9 8 1.6
PERFORM ANALYSIS
CHANGE
LOAD  120 0.9DL+1.6WL2
REPEAT LOAD
1 0.9 9 1.6
PERFORM ANALYSIS
CHANGE
LOAD 121 0.9DL+1.6WL3
REPEAT LOAD
1 0.9 10 1.6
PERFORM ANALYSIS
CHANGE
LOAD  122 0.9DL+1.6WL4
REPEAT LOAD
1 0.9 11 1.6
PERFORM ANALYSIS
CHANGE
LOAD  123 0.9DL+1.6WL5
REPEAT LOAD
1 0.9 12 1.6
PERFORM ANALYSIS
CHANGE
LOAD  124 0.9DL+1.6WL6
REPEAT LOAD
1 0.9 13 1.6
PERFORM ANALYSIS
CHANGE
LOAD  125 0.9DL+1.6WL7
REPEAT LOAD
1 0.9 14 1.6
PERFORM ANALYSIS
CHANGE
LOAD  126 0.9DL+1.6WL8
REPEAT LOAD
1 0.9 15 1.6
PERFORM ANALYSIS
CHANGE
LOAD  127 0.9DL+1.6WL9
REPEAT LOAD
1 0.9 16 1.6
PERFORM ANALYSIS
CHANGE
LOAD  128 0.9DL+1.6WL10
REPEAT LOAD
1 0.9 17 1.6
PERFORM ANALYSIS
CHANGE
LOAD 129 0.9DL+1.6WL11
REPEAT LOAD
1 0.9 18 1.6
PERFORM ANALYSIS
CHANGE
LOAD  130 0.9DL+1.6WL12
REPEAT LOAD
1 0.9 19 1.6
PERFORM ANALYSIS
CHANGE
LOAD  131 0.9DL+1.6SL1
REPEAT LOAD
1 0.9 20 1.6
PERFORM ANALYSIS
CHANGE
LOAD  132 0.9DL+1.6SL2
REPEAT LOAD
1 0.9 21 1.6
PERFORM ANALYSIS
```

```
PERFORM ANALYSIS
CHANGE
LOAD  133 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  134 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  135 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 136 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  137 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  138 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  139 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  140 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  141 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  142 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  143 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  144 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 145 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  146 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  147 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  148 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR2
REPEAT LOAD
```



```
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  149 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  150 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  151 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 152 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  153 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  154 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  155 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  156 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  157 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  158 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  159 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  160 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  161 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  162 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  163 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
```

```
CHANGE
LOAD  164 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  165 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  166 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  167 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 168 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  169 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  170 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  171 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  172 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  173 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  174 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  175 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 176 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  177 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 178 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  179 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0 7 1.0
```



```
1 1.2 18 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  180 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
*************************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 180
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185
CLB 0.05 MEMB 185
CLS 0.05 ALL
CLT 0.05 MEMB 185 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 -
372 393 TO 399 402 TO 409 412 TO 419 422 TO 429 432 497 TO 504 513 TO 520 -
568 TO 575 586 TO 593
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 3 STAAD INPUT – FOUNDATION F1A

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 04/09/2020
JOB NAME FDN. MKD. F1A
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
SET NL 200
SET ITERLIM 15
SET PRINT 1
UNIT METER KN
JOINT COORDINATES
2 0 911.8 3.175; 6 0.75 911.8 3.175; 10 -0.9 911.8 3.175; 34 -0.6 911.8 3.175;
35 -0.3 911.8 3.175; 41 0.25 911.8 3.175; 42 0.5 911.8 3.175;
128 0 913.8 3.175; 302 -0.6 911.8 3.49938; 303 -0.9 911.8 3.49938;
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307 0.5 911.8 3.49938; 308 0.75 911.8 3.49938; 309 0.95 911.8 3.175;
313 0.95 911.8 3.49938; 316 -1.15 911.8 3.175; 320 -1.15 911.8 3.49938;
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327 0.75 911.8 3.79938; 328 0.95 911.8 3.79938; 331 -1.15 911.8 3.79938;
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338 0.75 911.8 4.04938; 339 0.95 911.8 4.04938; 342 -1.15 911.8 4.04938;
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465 -0.9 911.8 5.04938; 466 -0.6 911.8 5.04938; 467 -0.3 911.8 5.04938;
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554 0.75 911.8 5.67438; 555 0.95 911.8 5.67438; 556 -1.15 911.8 5.67438;
MEMBER INCIDENCES
185 2 128;
ELEMENT INCIDENCES SHELL
326 303 302 34 10; 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2;
333 306 307 42 41; 334 307 308 6 42; 337 308 313 309 6; 342 320 303 10 316;
343 321 322 302 303; 344 322 323 304 302; 345 323 324 305 304;
346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307;
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346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307;
349 327 328 313 308; 352 331 321 303 320; 353 332 333 322 321;
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357 336 337 326 325; 358 337 338 327 326; 359 338 339 328 327;
362 342 332 321 331; 363 343 344 333 332; 364 344 345 334 333;
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368 348 349 338 337; 369 349 350 339 338; 372 353 343 332 342;
393 387 388 366 365; 394 388 389 369 366; 395 389 390 371 369;
396 390 391 373 371; 397 391 392 375 373; 398 392 393 377 375;
399 393 394 379 377; 402 397 387 365 384; 403 398 399 388 387;
404 399 400 389 388; 405 400 401 390 389; 406 401 402 391 390;
407 402 403 392 391; 408 403 404 393 392; 409 404 405 394 393;
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415 411 412 401 400; 416 412 413 402 401; 417 413 414 403 402;
418 414 415 404 403; 419 415 416 405 404; 422 419 409 398 408;
423 10 34 410 409; 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412;
427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415; 432 316 10 409 419;
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439 432 434 435 433; 440 436 420 423 437; 441 365 366 421 420;
442 366 369 424 421; 443 369 371 426 424; 444 371 373 428 426;
445 373 375 430 428; 446 375 377 432 430; 447 377 379 434 432;
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463 453 454 445 444; 464 455 447 438 446; 465 423 422 456 457;
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472 437 423 457 464; 473 465 466 448 447; 474 466 467 449 448;
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487 462 463 481 480; 488 464 457 475 482; 489 483 484 466 465;
490 484 485 467 466; 491 485 486 468 467; 492 486 487 469 468;
493 487 488 470 469; 494 488 489 471 470; 495 489 490 472 471;
496 491 483 465 473; 529 475 474 526 527; 530 474 476 528 526;
531 476 477 529 528; 532 477 478 530 529; 533 478 479 531 530;
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549 548 549 484 483; 550 549 550 485 484; 551 550 551 486 485;
552 551 552 487 486; 553 552 553 488 487; 554 553 554 489 488;
555 554 555 490 489; 556 556 548 483 491;
START GROUP DEFINITION
ELEMENT
_RAFT 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 -
393 TO 399 402 TO 409 412 TO 419 422 TO 429 432 TO 496 529 TO 536 -
549 TO 556
_RAFT_MOM 326 327 333 334 337 342 TO 349 352 TO 359 362 TO 369 372 -
393 TO 399 402 TO 409 412 TO 419 422 TO 424 427 TO 429 432 TO 496 -
529 TO 536 549 TO 556
_RAFT_SHEAR 326 337 342 343 349 352 353 359 362 TO 369 372 393 TO 399 402 -
403 409 412 413 419 422 423 429 432 TO 496 529 TO 536 549 TO 556
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 TO 496 529 TO 536 549 TO 555 -
556 THICKNESS 0.6
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.5 ZD 0.85
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 TO 496 529 TO 536 549 TO 555 -
556 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
**********************************************************************************************
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********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 TO 496 529 TO 536 549 TO 555 -
556 PR GY -22
* LOAD DUE TO SURCHARGE
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 TO 496 529 TO 536 549 TO 555 -
556 PR GY -20
JOINT LOAD
128 FY -22 FZ -3 MX -23
********************************************************************************************
LOAD 2 LOADTYPE None  TITLE ROOF LIVE LOAD ( LR )
JOINT LOAD
128 FY -61 FZ -2 MX -10
********************************************************************************************
LOAD 3 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FY -0.001
********************************************************************************************
LOAD 4 LOADTYPE None  TITLE CRANE LOAD  ( CR1 )
JOINT LOAD
128 FY -35 FZ -9 MX -40
********************************************************************************************
LOAD 5 LOADTYPE None  TITLE CRANE LOAD  ( CR2 )
JOINT LOAD
128 FY -26 FZ -7 MX -33
********************************************************************************************
LOAD 6 LOADTYPE None  TITLE CRANE LOAD  ( CR3 )
JOINT LOAD
128 FY -0.0001
********************************************************************************************
LOAD 7 LOADTYPE None  TITLE CRANE LOAD  ( CR4 )
JOINT LOAD
128 FY -0.0001
********************************************************************************************
LOAD 8 LOADTYPE None  TITLE WIND LOAD  ( WL1 )
JOINT LOAD
128 FY 117 FZ -24 MX -193
********************************************************************************************
LOAD 9 LOADTYPE None  TITLE WIND LOAD  ( WL2 )
JOINT LOAD
128 FY 74 FZ -27 MX -207
********************************************************************************************
LOAD 10 LOADTYPE None  TITLE WIND LOAD  ( WL3 )
JOINT LOAD
128 FY 117 FZ -24 MX -193
********************************************************************************************
LOAD 11 LOADTYPE None  TITLE WIND LOAD  ( WL4 )
JOINT LOAD
128 FY 74 FZ -27 MX -207
********************************************************************************************
LOAD 12 LOADTYPE None  TITLE WIND LOAD  ( WL5 )
JOINT LOAD
128 FY 105 FZ 38 MX 270
********************************************************************************************
LOAD 13 LOADTYPE None  TITLE WIND LOAD  ( WL6 )
JOINT LOAD
128 FY 62 FZ 35 MX 256
********************************************************************************************
LOAD 14 LOADTYPE None  TITLE WIND LOAD  ( WL7 )
JOINT LOAD
128 FY 105 FZ 38 MX 270
********************************************************************************************
LOAD 15 LOADTYPE None  TITLE WIND LOAD  ( WL8 )
JOINT LOAD
128 FY 62 FZ 35 MX 256
********************************************************************************************
LOAD 16 LOADTYPE None  TITLE WIND LOAD  ( WL9 )
JOINT LOAD
128 FY 119 FZ 4 MX 22
********************************************************************************************
LOAD 17 LOADTYPE None  TITLE WIND LOAD  ( WL10 )
JOINT LOAD
128 FY 76 FZ 2 MX 8
********************************************************************************************
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********************************************************************************************
LOAD 18 LOADTYPE None  TITLE WIND LOAD  ( WL11 )
JOINT LOAD
128 FY 119 FZ 11 MX 63
********************************************************************************************
LOAD 19 LOADTYPE None  TITLE WIND LOAD  ( WL12 )
JOINT LOAD
128 FY 75 FZ 8 MX 48
********************************************************************************************
LOAD 20 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FZ -1 MX -7
********************************************************************************************
LOAD 21 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FZ 1 MX 7
********************************************************************************************
LOAD 22 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL1 )
JOINT LOAD
128 FY 2
********************************************************************************************
LOAD 23 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL2 )
JOINT LOAD
128 FY -2
PERFORM ANALYSIS
CHANGE
SUPPORTS
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 TO 496 529 TO 536 549 TO 555 -
556 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD 201 DL
REPEAT LOAD
1 1.0
PERFORM ANALYSIS
CHANGE
LOAD 202 DL+L+TL1
REPEAT LOAD
1 1.0 3 1.0 22 1.0
PERFORM ANALYSIS
CHANGE
LOAD 203 DL+L+TL2
REPEAT LOAD
1 1.0 3 1.0 23 1.0
PERFORM ANALYSIS
CHANGE
LOAD 204 DL+LR
REPEAT LOAD
1 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 205 DL+0.75L+0.75TL1+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 22 0.75
PERFORM ANALYSIS
CHANGE
LOAD 206 DL+0.75L+0.75TL2+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 23 0.75
PERFORM ANALYSIS
CHANGE
LOAD 207 DL+0.6WL1
REPEAT LOAD
1 1.0 8 0.6
PERFORM ANALYSIS
CHANGE
LOAD 208 DL+WL2
REPEAT LOAD
1 1.0 9 1.0
PERFORM ANALYSIS
CHANGE
LOAD 209 DL+WL3
REPEAT LOAD
1 1.0 10 1.0
PERFORM ANALYSIS
CHANGE
LOAD 210 DL+WL4
REPEAT LOAD
```

```
REPEAT LOAD
1 1.0 11 1.0
PERFORM ANALYSIS
CHANGE
LOAD 211 DL+WL5
REPEAT LOAD
1 1.0 12 1.0
PERFORM ANALYSIS
CHANGE
LOAD 212 DL+WL6
REPEAT LOAD
1 1.0 13 1.0
PERFORM ANALYSIS
CHANGE
LOAD 213 DL+WL7
REPEAT LOAD
1 1.0 14 1.0
PERFORM ANALYSIS
CHANGE
LOAD 214 DL+WL8
REPEAT LOAD
1 1.0 15 1.0
PERFORM ANALYSIS
CHANGE
LOAD 215 DL+WL9
REPEAT LOAD
1 1.0 16 1.0
PERFORM ANALYSIS
CHANGE
LOAD 216 DL+WL10
REPEAT LOAD
1 1.0 17 1.0
PERFORM ANALYSIS
CHANGE
LOAD 217 DL+WL11
REPEAT LOAD
1 1.0 18 1.0
PERFORM ANALYSIS
CHANGE
LOAD 218 DL+WL12
REPEAT LOAD
1 1.0 19 1.0
PERFORM ANALYSIS
CHANGE
LOAD 219 DL+0.75WL1+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75
PERFORM ANALYSIS
CHANGE
LOAD 220 DL+0.75WL2+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75
PERFORM ANALYSIS
CHANGE
LOAD 221 DL+0.75WL3+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75
PERFORM ANALYSIS
CHANGE
LOAD 222 DL+0.75WL4+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75
PERFORM ANALYSIS
CHANGE
LOAD 223 DL+0.75WL5+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75
PERFORM ANALYSIS
CHANGE
LOAD 224 DL+0.75WL67+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75
PERFORM ANALYSIS
CHANGE
LOAD 225 DL+0.75WL7+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75
PERFORM ANALYSIS
CHANGE
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CHANGE
LOAD 226 DL+0.75WL8+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75
PERFORM ANALYSIS
CHANGE
LOAD 227 DL+0.75WL9+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75
PERFORM ANALYSIS
CHANGE
LOAD 228 DL+0.75WL10+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75
PERFORM ANALYSIS
CHANGE
LOAD 229 DL+0.75WL11+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75
PERFORM ANALYSIS
CHANGE
LOAD 230 DL+0.75WL12+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75
PERFORM ANALYSIS
CHANGE
LOAD 231 0.6DL+WL1
REPEAT LOAD
1 0.6 8 1.0
PERFORM ANALYSIS
CHANGE
LOAD 232 0.6DL+WL2
REPEAT LOAD
1 0.6 9 1.0
PERFORM ANALYSIS
CHANGE
LOAD 233 0.6DL+WL3
REPEAT LOAD
1 0.6 10 1.0
PERFORM ANALYSIS
CHANGE
LOAD 234 0.6DL+WL4
REPEAT LOAD
1 0.6 11 1.0
PERFORM ANALYSIS
CHANGE
LOAD 235 0.6DL+WL5
REPEAT LOAD
1 0.6 12 1.0
PERFORM ANALYSIS
CHANGE
LOAD 236 0.6DL+WL6
REPEAT LOAD
1 0.6 13 1.0
PERFORM ANALYSIS
CHANGE
LOAD 237 0.6DL+WL7
REPEAT LOAD
1 0.6 14 1.0
PERFORM ANALYSIS
CHANGE
LOAD 238 0.6DL+WL8
REPEAT LOAD
1 0.6 15 1.0
PERFORM ANALYSIS
CHANGE
LOAD 239 0.6DL+WL9
REPEAT LOAD
1 0.6 16 1.0
PERFORM ANALYSIS
CHANGE
LOAD 240 0.6DL+WL10
REPEAT LOAD
1 0.6 17 1.0
PERFORM ANALYSIS
CHANGE
LOAD 241 0.6DL+WL11
REPEAT LOAD
1 0.6 18 1.0
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1 0.6 18 1.0
PERFORM ANALYSIS
CHANGE
LOAD 242 0.6DL+WL12
REPEAT LOAD
1 0.6 19 1.0
PERFORM ANALYSIS
CHANGE
LOAD 243 0.6DL+0.7SL1
REPEAT LOAD
1 0.6 20 0.7
PERFORM ANALYSIS
CHANGE
LOAD 244 0.6DL+0.7SL2
REPEAT LOAD
1 0.6 21 0.7
PERFORM ANALYSIS
CHANGE
LOAD 245 DL+0.75WL1+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 246 DL+0.75WL2+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 247 DL+0.75WL3+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 248 DL+0.75WL4+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 249 DL+0.75WL5+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 250 DL+0.75WL67+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 251 DL+0.75WL7+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 252 DL+0.75WL8+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 253 DL+0.75WL9+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 254 DL+0.75WL10+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 255 DL+0.75WL11+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 256 DL+0.75WL12+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 257 DL+0.75WL1+0.75L+0.75LR+CR2
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LOAD 257 DL+0.75WL1+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 258 DL+0.75WL2+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 259 DL+0.75WL3+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 260 DL+0.75WL4+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 261 DL+0.75WL5+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 262 DL+0.75WL67+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 263 DL+0.75WL7+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 264 DL+0.75WL8+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 265 DL+0.75WL9+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 266 DL+0.75WL10+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 267 DL+0.75WL11+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 268 DL+0.75WL12+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 269 DL+0.75WL1+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 270 DL+0.75WL2+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 271 DL+0.75WL3+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 272 DL+0.75WL4+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75 6 1.0
PERFORM ANALYSIS
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PERFORM ANALYSIS
CHANGE
LOAD 273 DL+0.75WL5+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 274 DL+0.75WL67+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 275 DL+0.75WL7+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 276 DL+0.75WL8+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 277 DL+0.75WL9+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 278 DL+0.75WL10+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 279 DL+0.75WL11+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 280 DL+0.75WL12+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 281 DL+0.75WL1+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 282 DL+0.75WL2+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 283 DL+0.75WL3+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 284 DL+0.75WL4+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 285 DL+0.75WL5+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 286 DL+0.75WL67+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 287 DL+0.75WL7+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 288 DL+0.75WL8+0.75L+0.75LR+CR4
REPEAT LOAD
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REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 289 DL+0.75WL9+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 290 DL+0.75WL10+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 291 DL+0.75WL11+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 292 DL+0.75WL12+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
*************************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD 101 1.4DL
REPEAT LOAD
1 1.4
PERFORM ANALYSIS
CHANGE
LOAD 102 1.4DL+1.4TL1+1.7L+0.5LR
REPEAT LOAD
1 1.4 22 1.4 3 1.7 2 0.5
PERFORM ANALYSIS
CHANGE
LOAD 103 1.4DL+1.4TL2+1.7L+0.5LR
REPEAT LOAD
1 1.4 23 1.4 3 1.7 2 0.5
PERFORM ANALYSIS
CHANGE
LOAD 104 1.2DL+1.6LR+1.0L
REPEAT LOAD
1 1.2 2 1.6 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 105 1.2DL+1.6WL1+1.0L+0.5LR
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 106 1.2DL+1.6WL2+1.0L+0.5LR
REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 107 1.2DL+1.6WL3+1.0L+0.5LR
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 108 1.2DL+1.6WL4+1.0L+0.5LR
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 109 1.2DL+1.6WL5+1.0L+0.5LR
REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 110 1.2DL+1.6WL6+1.0L+0.5LR
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 111 1.2DL+1.6WL7+1.0L+0.5LR
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0
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1 1.2 14 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 112 1.2DL+1.6WL8+1.0L+0.5LR
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 113 1.2DL+1.6WL9+1.0L+0.5LR
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 114 1.2DL+1.6WL10+1.0L+0.5LR
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 115 1.2DL+1.6WL11+1.0L+0.5LR
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 116 1.2DL+1.6WL12+1.0L+0.5LR
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 117 1.2DL+1.0SL1+1.0L
REPEAT LOAD
1 1.2 20 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 118 1.2DL+1.0SL2+1.0L
REPEAT LOAD
1 1.2 21 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 119 0.9DL+1.6WL1
REPEAT LOAD
1 0.9 8 1.6
PERFORM ANALYSIS
CHANGE
LOAD 120 0.9DL+1.6WL2
REPEAT LOAD
1 0.9 9 1.6
PERFORM ANALYSIS
CHANGE
LOAD 121 0.9DL+1.6WL3
REPEAT LOAD
1 0.9 10 1.6
PERFORM ANALYSIS
CHANGE
LOAD 122 0.9DL+1.6WL4
REPEAT LOAD
1 0.9 11 1.6
PERFORM ANALYSIS
CHANGE
LOAD 123 0.9DL+1.6WL5
REPEAT LOAD
1 0.9 12 1.6
PERFORM ANALYSIS
CHANGE
LOAD 124 0.9DL+1.6WL6
REPEAT LOAD
1 0.9 13 1.6
PERFORM ANALYSIS
CHANGE
LOAD 125 0.9DL+1.6WL7
REPEAT LOAD
1 0.9 14 1.6
PERFORM ANALYSIS
CHANGE
LOAD 126 0.9DL+1.6WL8
REPEAT LOAD
1 0.9 15 1.6
PERFORM ANALYSIS
CHANGE
LOAD 127 0.9DL+1.6WL9
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LOAD 127 0.9DL+1.6WL9
REPEAT LOAD
1 0.9 16 1.6
PERFORM ANALYSIS
CHANGE
LOAD 128 0.9DL+1.6WL10
REPEAT LOAD
1 0.9 17 1.6
PERFORM ANALYSIS
CHANGE
LOAD 129 0.9DL+1.6WL11
REPEAT LOAD
1 0.9 18 1.6
PERFORM ANALYSIS
CHANGE
LOAD 130 0.9DL+1.6WL12
REPEAT LOAD
1 0.9 19 1.6
PERFORM ANALYSIS
CHANGE
LOAD 131 0.9DL+1.6SL1
REPEAT LOAD
1 0.9 20 1.6
PERFORM ANALYSIS
CHANGE
LOAD 132 0.9DL+1.6SL2
REPEAT LOAD
1 0.9 21 1.6
PERFORM ANALYSIS
CHANGE
LOAD 133 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 134 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 135 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 136 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 137 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 138 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 139 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 140 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 141 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 142 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
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PERFORM ANALYSIS
CHANGE
LOAD 143 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 144 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 145 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 146 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 147 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 148 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 149 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 150 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 151 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 152 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 153 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 154 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 155 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 156 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 157 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 158 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR3
REPEAT LOAD
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REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 159 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 160 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 161 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 162 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 163 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 164 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 165 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 166 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 167 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 168 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 169 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 170 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 171 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 172 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 173 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
```

```
CHANGE
LOAD 174 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 175 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 176 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 177 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 178 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 179 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 180 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
*****************************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 180
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185
CLB 0.05 MEMB 185
CLS 0.05 ALL
CLT 0.05 MEMB 185 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 -
372 393 TO 399 402 TO 409 412 TO 419 422 TO 429 432
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 4 STAAD INPUT – FOUNDATION F1B

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 04/09/2020
JOB NAME FDN. MKD. F1B
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
SET NL 200
SET ITERLIM 15
SET PRINT 1
UNIT METER KN
JOINT COORDINATES
2 0 911.8 3.175; 6 0.75 911.8 3.175; 10 -0.75 911.8 3.175; 34 -0.5 911.8 3.175;
35 -0.25 911.8 3.175; 41 0.25 911.8 3.175; 42 0.5 911.8 3.175;
128 0 913.8 3.175; 302 -0.5 911.8 3.42438; 303 -0.75 911.8 3.42438;
304 -0.25 911.8 3.42438; 305 0 911.8 3.42438; 306 0.25 911.8 3.42438;
307 0.5 911.8 3.42438; 308 0.75 911.8 3.42438; 309 1 911.8 3.175;
313 1 911.8 3.42438; 316 -1 911.8 3.175; 320 -1 911.8 3.42438;
321 -0.75 911.8 3.67438; 322 -0.5 911.8 3.67438; 323 -0.25 911.8 3.67438;
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332 -0.75 911.8 3.92438; 333 -0.5 911.8 3.92438; 334 -0.25 911.8 3.92438;
335 0 911.8 3.92438; 336 0.25 911.8 3.92438; 337 0.5 911.8 3.92438;
338 0.75 911.8 3.92438; 339 1 911.8 3.92438; 342 -1 911.8 3.92438;
343 -0.75 911.8 4.12538; 344 -0.5 911.8 4.12538; 345 -0.25 911.8 4.12538;
346 0 911.8 4.12538; 347 0.25 911.8 4.12538; 348 0.5 911.8 4.12538;
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398 -0.75 911.8 2.67438; 399 -0.5 911.8 2.67438; 400 -0.25 911.8 2.67438;
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412 0 911.8 2.92438; 413 0.25 911.8 2.92438; 414 0.5 911.8 2.92438;
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498 1.25 911.8 2.17438; 499 1.25 911.8 2.67438; 500 1.25 911.8 2.92438;
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544 1.5 911.8 4.12538; 545 1.5 911.8 2.42438; 546 1.5 911.8 2.17438;
547 1.5 911.8 2.67438; 548 1.5 911.8 2.92438; 549 1.75 911.8 3.42438;
550 1.75 911.8 3.175; 551 1.75 911.8 3.67438; 552 1.75 911.8 3.92438;
553 1.75 911.8 4.12538; 554 1.75 911.8 2.42438; 555 1.75 911.8 2.17438;
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587 2 911.8 3.92438; 588 2 911.8 4.12538; 589 2 911.8 2.42438;
590 2 911.8 2.17438; 591 2 911.8 2.67438; 592 2 911.8 2.92438;
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596 2.375 911.8 3.92438; 597 2.375 911.8 4.12538; 598 2.375 911.8 2.42438;
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607 -2 911.8 3.175; 608 -2 911.8 3.67438; 609 -2 911.8 3.92438;
610 -2 911.8 4.12538; 611 -2 911.8 2.42438; 612 -2 911.8 2.17438;
613 -2 911.8 2.67438; 614 -2 911.8 2.92438; 615 -2.375 911.8 3.42438;
616 -2.375 911.8 3.175; 617 -2.375 911.8 3.67438; 618 -2.375 911.8 3.92438;
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622 -2.375 911.8 2.67438; 623 -2.375 911.8 2.92438; 624 -2 911.8 1.92438;
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```



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625 -2.375 911.8 1.92438; 628 -0.5 911.8 1.77538; 629 -0.75 911.8 1.77538;
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655 -1.75 911.8 1.77538; 658 2 911.8 1.77538; 659 2.375 911.8 1.77538;
662 -2 911.8 1.77538; 663 -2.375 911.8 1.77538;
MEMBER INCIDENCES
185 2 128;
ELEMENT INCIDENCES SHELL
326 303 302 34 10; 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2;
333 306 307 42 41; 334 307 308 6 42; 337 308 313 309 6; 342 320 303 10 316;
343 321 322 302 303; 344 322 323 304 302; 345 323 324 305 304;
346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307;
349 327 328 313 308; 352 331 321 303 320; 353 332 333 322 321;
354 333 334 323 322; 355 334 335 324 323; 356 335 336 325 324;
357 336 337 326 325; 358 337 338 327 326; 359 338 339 328 327;
362 342 332 321 331; 363 343 344 333 332; 364 344 345 334 333;
365 345 346 335 334; 366 346 347 336 335; 367 347 348 337 336;
368 348 349 338 337; 369 349 350 339 338; 372 353 343 332 342;
393 387 388 366 365; 394 388 389 369 366; 395 389 390 371 369;
396 390 391 373 371; 397 391 392 375 373; 398 392 393 377 375;
399 393 394 379 377; 402 397 387 365 384; 403 398 399 388 387;
404 399 400 389 388; 405 400 401 390 389; 406 401 402 391 390;
407 402 403 392 391; 408 403 404 393 392; 409 404 405 394 393;
412 408 398 387 397; 413 409 410 399 398; 414 410 411 400 399;
415 411 412 401 400; 416 412 413 402 401; 417 413 414 403 402;
418 414 415 404 403; 419 415 416 405 404; 422 419 409 398 408;
423 10 34 410 409; 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412;
427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415; 432 316 10 409 419;
497 313 492 493 309; 498 328 494 492 313; 499 339 495 494 328;
500 350 496 495 339; 501 394 497 498 379; 502 405 499 497 394;
503 416 500 499 405; 504 309 493 500 416; 513 509 320 316 510;
514 511 331 320 509; 515 512 342 331 511; 516 513 353 342 512;
517 514 397 384 515; 518 516 408 397 514; 519 517 419 408 516;
520 510 316 419 517; 521 365 366 518 519; 522 366 369 520 518;
523 369 371 521 520; 524 371 373 522 521; 525 373 375 523 522;
526 375 377 524 523; 527 377 379 525 524; 528 384 365 519 526;
529 379 498 527 525; 530 515 384 526 528; 541 492 540 541 493;
542 494 542 540 492; 543 495 543 542 494; 544 496 544 543 495;
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548 493 541 548 500; 549 540 549 550 541; 550 542 551 549 540;
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565 567 514 515 568; 566 569 516 514 567; 567 570 517 516 569;
568 563 510 517 570; 569 571 562 563 572; 570 573 564 562 571;
571 574 565 564 573; 572 575 566 565 574; 573 576 567 568 577;
574 578 569 567 576; 575 579 570 569 578; 576 572 563 570 579;
577 568 515 528 580; 578 577 568 580 581; 581 549 584 585 550;
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585 554 589 590 555; 586 556 591 589 554; 587 557 592 591 556;
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591 587 596 595 586; 592 588 597 596 587; 593 589 598 599 590;
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597 555 590 602 559; 598 590 599 603 602; 601 606 571 572 607;
602 608 573 571 606; 603 609 574 573 608; 604 610 575 574 609;
605 611 576 577 612; 606 613 578 576 611; 607 614 579 578 613;
608 607 572 579 614; 609 615 606 607 616; 610 617 608 606 615;
611 618 609 608 617; 612 619 610 609 618; 613 620 611 612 621;
614 622 613 611 620; 615 623 614 613 622; 616 616 607 614 623;
617 612 577 581 624; 618 621 612 624 625; 621 519 518 628 629;
622 518 520 630 628; 623 520 521 631 630; 624 521 522 632 631;
625 522 523 633 632; 626 523 524 634 633; 627 524 525 635 634;
628 526 519 629 636; 629 525 527 637 635; 630 528 526 636 638;
641 527 558 650 637; 642 558 559 651 650; 645 580 528 638 654;
646 581 580 654 655; 649 559 602 658 651; 650 602 603 659 658;
653 624 581 655 662; 654 625 624 662 663;
START GROUP DEFINITION
ELEMENT
_RAFT 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 -
393 TO 399 402 TO 409 412 TO 419 422 TO 429 432 497 TO 504 513 TO 530 541 -
542 TO 558 561 TO 578 581 TO 598 601 TO 618 621 TO 630 641 642 645 646 649 -
650 653 654
_RAFT_MOM 326 327 333 334 337 342 TO 349 352 TO 359 362 TO 369 372 -
393 TO 399 402 TO 409 412 TO 419 422 TO 424 427 TO 429 432 497 TO 504 513 -
514 TO 530 541 TO 558 561 TO 578 581 TO 598 601 TO 618 621 TO 630 641 642 -
645 646 649 650 653 654
```

```
645 646 649 650 653 654
_RAFT_SHEAR 337 342 349 352 359 362 TO 369 372 393 TO 399 402 409 412 419 -
422 429 432 497 TO 504 513 TO 530 541 TO 558 561 TO 578 581 TO 598 -
601 TO 618 621 TO 630 641 642 645 646 649 650 653 654
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 497 TO 504 513 TO 530 541 TO 558 561 -
562 TO 578 581 TO 598 601 TO 618 621 TO 630 641 642 645 646 649 650 653 -
654 THICKNESS 0.6
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.5 ZD 0.85
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 497 TO 504 513 TO 530 541 TO 558 561 -
562 TO 578 581 TO 598 601 TO 618 621 TO 630 641 642 645 646 649 650 653 -
654 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 497 TO 504 513 TO 530 541 TO 558 561 -
562 TO 578 581 TO 598 601 TO 618 621 TO 630 641 642 645 646 649 650 653 -
654 PR GY -22
* LOAD DUE TO SURCHARGE
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 497 TO 504 513 TO 530 541 TO 558 561 -
562 TO 578 581 TO 598 601 TO 618 621 TO 630 641 642 645 646 649 650 653 -
654 PR GY -20
JOINT LOAD
128 FY -22 FZ -3 MX -23
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE ROOF LIVE LOAD ( LR )
JOINT LOAD
128 FY -61 FZ -2 MX -10
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FY -0.001
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE CRANE LOAD  ( CR1 )
JOINT LOAD
128 FY -35 FZ -9 MX -40
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE CRANE LOAD  ( CR2 )
JOINT LOAD
128 FY -26 FZ -7 MX -33
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE CRANE LOAD  ( CR3 )
JOINT LOAD
128 FY -0.0001
*********************************************************************************************
LOAD 7 LOADTYPE None  TITLE CRANE LOAD  ( CR4 )
JOINT LOAD
128 FY -0.0001
*********************************************************************************************
LOAD 8 LOADTYPE None  TITLE WIND LOAD  ( WL1 )
JOINT LOAD
128 FY -117 FZ -24 MX -193
*********************************************************************************************
LOAD 9 LOADTYPE None  TITLE WIND LOAD  ( WL2 )
JOINT LOAD
128 FY 74 FZ -27 MX -207
*********************************************************************************************
```



```
*********************************************************************************************
LOAD 10 LOADTYPE None  TITLE WIND LOAD  ( WL3 )
JOINT LOAD
128 FY 117 FZ -24 MX -193
*********************************************************************************************
LOAD 11 LOADTYPE None  TITLE WIND LOAD  ( WL4 )
JOINT LOAD
128 FY 74 FZ -27 MX -207
*********************************************************************************************
LOAD 12 LOADTYPE None  TITLE WIND LOAD  ( WL5 )
JOINT LOAD
128 FY 105 FZ 38 MX 270
*********************************************************************************************
LOAD 13 LOADTYPE None  TITLE WIND LOAD  ( WL6 )
JOINT LOAD
128 FY 62 FZ 35 MX 256
*********************************************************************************************
LOAD 14 LOADTYPE None  TITLE WIND LOAD  ( WL7 )
JOINT LOAD
128 FY 105 FZ 38 MX 270
*********************************************************************************************
LOAD 15 LOADTYPE None  TITLE WIND LOAD  ( WL8 )
JOINT LOAD
128 FY 62 FZ 35 MX 256
*********************************************************************************************
LOAD 16 LOADTYPE None  TITLE WIND LOAD  ( WL9 )
JOINT LOAD
128 FY 119 FZ 4 MX 22
*********************************************************************************************
LOAD 17 LOADTYPE None  TITLE WIND LOAD  ( WL10 )
JOINT LOAD
128 FY 76 FZ 2 MX 8
*********************************************************************************************
LOAD 18 LOADTYPE None  TITLE WIND LOAD  ( WL11 )
JOINT LOAD
128 FY 119 FZ 11 MX 63
*********************************************************************************************
LOAD 19 LOADTYPE None  TITLE WIND LOAD  ( WL12 )
JOINT LOAD
128 FY 75 FZ 8 MX 48
*********************************************************************************************
LOAD 20 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FZ -1 MX -7
*********************************************************************************************
LOAD 21 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FZ 1 MX 7
*********************************************************************************************
LOAD 22 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL1 )
JOINT LOAD
128 FY 2
*********************************************************************************************
LOAD 23 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL2 )
JOINT LOAD
128 FY -2
PERFORM ANALYSIS
CHANGE
SUPPORTS
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 497 TO 504 513 TO 530 541 TO 558 561 -
562 TO 578 581 TO 598 601 TO 618 621 TO 630 641 642 645 646 649 650 653 -
654 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD 201 DL
REPEAT LOAD
1 1.0
PERFORM ANALYSIS
CHANGE
LOAD 202 DL+L+TL1
REPEAT LOAD
1 1.0 3 1.0 22 1.0
PERFORM ANALYSIS
CHANGE
LOAD 203 DL+L+TL2
REPEAT LOAD
1 1.0 3 1.0 23 1.0
PERFORM ANALYSIS
```

```
PERFORM ANALYSIS
CHANGE
LOAD 204 DL+LR
REPEAT LOAD
1 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 205 DL+0.75L+0.75TL1+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 22 0.75
PERFORM ANALYSIS
CHANGE
LOAD 206 DL+0.75L+0.75TL2+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 23 0.75
PERFORM ANALYSIS
CHANGE
LOAD 207 DL+0.6WL1
REPEAT LOAD
1 1.0 8 0.6
PERFORM ANALYSIS
CHANGE
LOAD 208 DL+WL2
REPEAT LOAD
1 1.0 9 1.0
PERFORM ANALYSIS
CHANGE
LOAD 209 DL+WL3
REPEAT LOAD
1 1.0 10 1.0
PERFORM ANALYSIS
CHANGE
LOAD 210 DL+WL4
REPEAT LOAD
1 1.0 11 1.0
PERFORM ANALYSIS
CHANGE
LOAD 211 DL+WL5
REPEAT LOAD
1 1.0 12 1.0
PERFORM ANALYSIS
CHANGE
LOAD 212 DL+WL6
REPEAT LOAD
1 1.0 13 1.0
PERFORM ANALYSIS
CHANGE
LOAD 213 DL+WL7
REPEAT LOAD
1 1.0 14 1.0
PERFORM ANALYSIS
CHANGE
LOAD 214 DL+WL8
REPEAT LOAD
1 1.0 15 1.0
PERFORM ANALYSIS
CHANGE
LOAD 215 DL+WL9
REPEAT LOAD
1 1.0 16 1.0
PERFORM ANALYSIS
CHANGE
LOAD 216 DL+WL10
REPEAT LOAD
1 1.0 17 1.0
PERFORM ANALYSIS
CHANGE
LOAD 217 DL+WL11
REPEAT LOAD
1 1.0 18 1.0
PERFORM ANALYSIS
CHANGE
LOAD 218 DL+WL12
REPEAT LOAD
1 1.0 19 1.0
PERFORM ANALYSIS
CHANGE
LOAD 219 DL+0.75WL1+0.75L+0.75LR
REPEAT LOAD
```



```
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75
PERFORM ANALYSIS
CHANGE
LOAD 220 DL+0.75WL2+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75
PERFORM ANALYSIS
CHANGE
LOAD 221 DL+0.75WL3+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75
PERFORM ANALYSIS
CHANGE
LOAD 222 DL+0.75WL4+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75
PERFORM ANALYSIS
CHANGE
LOAD 223 DL+0.75WL5+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75
PERFORM ANALYSIS
CHANGE
LOAD 224 DL+0.75WL67+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75
PERFORM ANALYSIS
CHANGE
LOAD 225 DL+0.75WL7+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75
PERFORM ANALYSIS
CHANGE
LOAD 226 DL+0.75WL8+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75
PERFORM ANALYSIS
CHANGE
LOAD 227 DL+0.75WL9+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75
PERFORM ANALYSIS
CHANGE
LOAD 228 DL+0.75WL10+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75
PERFORM ANALYSIS
CHANGE
LOAD 229 DL+0.75WL11+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75
PERFORM ANALYSIS
CHANGE
LOAD 230 DL+0.75WL12+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75
PERFORM ANALYSIS
CHANGE
LOAD 231 0.6DL+WL1
REPEAT LOAD
1 0.6 8 1.0
PERFORM ANALYSIS
CHANGE
LOAD 232 0.6DL+WL2
REPEAT LOAD
1 0.6 9 1.0
PERFORM ANALYSIS
CHANGE
LOAD 233 0.6DL+WL3
REPEAT LOAD
1 0.6 10 1.0
PERFORM ANALYSIS
CHANGE
LOAD 234 0.6DL+WL4
REPEAT LOAD
1 0.6 11 1.0
PERFORM ANALYSIS
CHANGE
```

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CHANGE
LOAD 235 0.6DL+WL5
REPEAT LOAD
1 0.6 12 1.0
PERFORM ANALYSIS
CHANGE
LOAD 236 0.6DL+WL6
REPEAT LOAD
1 0.6 13 1.0
PERFORM ANALYSIS
CHANGE
LOAD 237 0.6DL+WL7
REPEAT LOAD
1 0.6 14 1.0
PERFORM ANALYSIS
CHANGE
LOAD 238 0.6DL+WL8
REPEAT LOAD
1 0.6 15 1.0
PERFORM ANALYSIS
CHANGE
LOAD 239 0.6DL+WL9
REPEAT LOAD
1 0.6 16 1.0
PERFORM ANALYSIS
CHANGE
LOAD 240 0.6DL+WL10
REPEAT LOAD
1 0.6 17 1.0
PERFORM ANALYSIS
CHANGE
LOAD 241 0.6DL+WL11
REPEAT LOAD
1 0.6 18 1.0
PERFORM ANALYSIS
CHANGE
LOAD 242 0.6DL+WL12
REPEAT LOAD
1 0.6 19 1.0
PERFORM ANALYSIS
CHANGE
LOAD 243 0.6DL+0.7SL1
REPEAT LOAD
1 0.6 20 0.7
PERFORM ANALYSIS
CHANGE
LOAD 244 0.6DL+0.7SL2
REPEAT LOAD
1 0.6 21 0.7
PERFORM ANALYSIS
CHANGE
LOAD 245 DL+0.75WL1+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 246 DL+0.75WL2+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 247 DL+0.75WL3+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 248 DL+0.75WL4+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 249 DL+0.75WL5+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 250 DL+0.75WL67+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
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1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 251 DL+0.75WL7+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 252 DL+0.75WL8+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 253 DL+0.75WL9+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 254 DL+0.75WL10+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 255 DL+0.75WL11+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 256 DL+0.75WL12+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 257 DL+0.75WL1+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 258 DL+0.75WL2+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 259 DL+0.75WL3+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 260 DL+0.75WL4+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 261 DL+0.75WL5+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 262 DL+0.75WL67+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 263 DL+0.75WL7+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 264 DL+0.75WL8+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 265 DL+0.75WL9+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 266 DL+0.75WL10+0.75L+0.75LR+CR2
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LOAD 266 DL+0.75WL10+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 267 DL+0.75WL11+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 268 DL+0.75WL12+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 269 DL+0.75WL1+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 270 DL+0.75WL2+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 271 DL+0.75WL3+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 272 DL+0.75WL4+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 273 DL+0.75WL5+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 274 DL+0.75WL67+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 275 DL+0.75WL7+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 276 DL+0.75WL8+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 277 DL+0.75WL9+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 278 DL+0.75WL10+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 279 DL+0.75WL11+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 280 DL+0.75WL12+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 281 DL+0.75WL1+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75 7 1.0
PERFORM ANALYSIS
```



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PERFORM ANALYSIS
CHANGE
LOAD 282 DL+0.75WL2+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 283 DL+0.75WL3+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 284 DL+0.75WL4+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 285 DL+0.75WL5+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 286 DL+0.75WL67+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 287 DL+0.75WL7+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 288 DL+0.75WL8+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 289 DL+0.75WL9+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 290 DL+0.75WL10+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 291 DL+0.75WL11+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 292 DL+0.75WL12+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
*************************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD 101 1.4DL
REPEAT LOAD
1 1.4
PERFORM ANALYSIS
CHANGE
LOAD 102 1.4DL+1.4TL1+1.7L+0.5LR
REPEAT LOAD
1 1.4 22 1.4 3 1.7 2 0.5
PERFORM ANALYSIS
CHANGE
LOAD 103 1.4DL+1.4TL2+1.7L+0.5LR
REPEAT LOAD
1 1.4 23 1.4 3 1.7 2 0.5
PERFORM ANALYSIS
CHANGE
LOAD 104 1.2DL+1.6LR+1.0L
REPEAT LOAD
1 1.2 2 1.6 3 1.0
PERFORM ANALYSIS
CHANGE
```

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CHANGE
LOAD 105 1.2DL+1.6WL1+1.0L+0.5LR
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 106 1.2DL+1.6WL2+1.0L+0.5LR
REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 107 1.2DL+1.6WL3+1.0L+0.5LR
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 108 1.2DL+1.6WL4+1.0L+0.5LR
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 109 1.2DL+1.6WL5+1.0L+0.5LR
REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 110 1.2DL+1.6WL6+1.0L+0.5LR
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 111 1.2DL+1.6WL7+1.0L+0.5LR
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 112 1.2DL+1.6WL8+1.0L+0.5LR
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 113 1.2DL+1.6WL9+1.0L+0.5LR
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 114 1.2DL+1.6WL10+1.0L+0.5LR
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 115 1.2DL+1.6WL11+1.0L+0.5LR
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 116 1.2DL+1.6WL12+1.0L+0.5LR
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 117 1.2DL+1.0SL1+1.0L
REPEAT LOAD
1 1.2 20 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 118 1.2DL+1.0SL2+1.0L
REPEAT LOAD
1 1.2 21 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 119 0.9DL+1.6WL1
REPEAT LOAD
1 0.9 8 1.6
PERFORM ANALYSIS
CHANGE
LOAD 120 0.9DL+1.6WL2
REPEAT LOAD
1 0.9 9 1.6
```



```
1 0.9 9 1.6
PERFORM ANALYSIS
CHANGE
LOAD 121 0.9DL+1.6WL3
REPEAT LOAD
1 0.9 10 1.6
PERFORM ANALYSIS
CHANGE
LOAD 122 0.9DL+1.6WL4
REPEAT LOAD
1 0.9 11 1.6
PERFORM ANALYSIS
CHANGE
LOAD 123 0.9DL+1.6WL5
REPEAT LOAD
1 0.9 12 1.6
PERFORM ANALYSIS
CHANGE
LOAD 124 0.9DL+1.6WL6
REPEAT LOAD
1 0.9 13 1.6
PERFORM ANALYSIS
CHANGE
LOAD 125 0.9DL+1.6WL7
REPEAT LOAD
1 0.9 14 1.6
PERFORM ANALYSIS
CHANGE
LOAD 126 0.9DL+1.6WL8
REPEAT LOAD
1 0.9 15 1.6
PERFORM ANALYSIS
CHANGE
LOAD 127 0.9DL+1.6WL9
REPEAT LOAD
1 0.9 16 1.6
PERFORM ANALYSIS
CHANGE
LOAD 128 0.9DL+1.6WL10
REPEAT LOAD
1 0.9 17 1.6
PERFORM ANALYSIS
CHANGE
LOAD 129 0.9DL+1.6WL11
REPEAT LOAD
1 0.9 18 1.6
PERFORM ANALYSIS
CHANGE
LOAD 130 0.9DL+1.6WL12
REPEAT LOAD
1 0.9 19 1.6
PERFORM ANALYSIS
CHANGE
LOAD 131 0.9DL+1.6SL1
REPEAT LOAD
1 0.9 20 1.6
PERFORM ANALYSIS
CHANGE
LOAD 132 0.9DL+1.6SL2
REPEAT LOAD
1 0.9 21 1.6
PERFORM ANALYSIS
CHANGE
LOAD 133 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 134 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 135 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 136 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR1
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LOAD 136 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 137 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 138 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 139 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 140 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 141 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 142 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 143 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 144 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 145 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 146 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 147 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 148 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 149 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 150 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 151 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
```



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PERFORM ANALYSIS
CHANGE
LOAD 152 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 153 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 154 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 155 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 156 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 157 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 158 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 159 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 160 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 161 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 162 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 163 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 164 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 165 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 166 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 167 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR3
REPEAT LOAD
```

```
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 168 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 169 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 170 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 171 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 172 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 173 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 174 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 175 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 176 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0 7 1.0

PERFORM ANALYSIS
CHANGE
LOAD 177 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 178 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 179 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 180 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
*****************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 180
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185
CLB 0.05 MEMB 185
CLS 0.05 ALL
```



```
CLS 0.05 ALL
CLT 0.05 MEMB 185 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 -
372 393 TO 399 402 TO 409 412 TO 419 422 TO 429 432 497 TO 504 513 TO 530 -
621 TO 630
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 5 STAAD INPUT – FOUNDATION F2

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 04/09/2020
JOB NAME FDN. MKD. F2
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
SET NL 200
SET ITERLIM 15
SET PRINT 1
UNIT METER KN
JOINT COORDINATES
2 0 911.8 3.175; 6 0.75 911.8 3.175; 10 -0.75 911.8 3.175; 34 -0.5 911.8 3.175;
35 -0.25 911.8 3.175; 41 0.25 911.8 3.175; 42 0.5 911.8 3.175;
128 0 913.8 3.175; 302 -0.5 911.8 3.42438; 303 -0.75 911.8 3.42438;
304 -0.25 911.8 3.42438; 305 0 911.8 3.42438; 306 0.25 911.8 3.42438;
307 0.5 911.8 3.42438; 308 0.75 911.8 3.42438; 309 1 911.8 3.175;
313 1 911.8 3.42438; 316 -1 911.8 3.175; 320 -1 911.8 3.42438;
321 -0.75 911.8 3.67438; 322 -0.5 911.8 3.67438; 323 -0.25 911.8 3.67438;
324 0 911.8 3.67438; 325 0.25 911.8 3.67438; 326 0.5 911.8 3.67438;
327 0.75 911.8 3.67438; 328 1 911.8 3.67438; 331 -1 911.8 3.67438;
332 -0.75 911.8 3.92438; 333 -0.5 911.8 3.92438; 334 -0.25 911.8 3.92438;
335 0 911.8 3.92438; 336 0.25 911.8 3.92438; 337 0.5 911.8 3.92438;
338 0.75 911.8 3.92438; 339 1 911.8 3.92438; 342 -1 911.8 3.92438;
343 -0.75 911.8 4.17438; 344 -0.5 911.8 4.17438; 345 -0.25 911.8 4.17438;
346 0 911.8 4.17438; 347 0.25 911.8 4.17438; 348 0.5 911.8 4.17438;
349 0.75 911.8 4.17438; 350 1 911.8 4.17438; 353 -1 911.8 4.17438;
365 -0.75 911.8 2.17438; 366 -0.5 911.8 2.17438; 369 -0.25 911.8 2.17438;
371 0 911.8 2.17438; 373 0.25 911.8 2.17438; 375 0.5 911.8 2.17438;
377 0.75 911.8 2.17438; 379 1 911.8 2.17438; 384 -1 911.8 2.17438;
387 -0.75 911.8 2.42438; 388 -0.5 911.8 2.42438; 389 -0.25 911.8 2.42438;
390 0 911.8 2.42438; 391 0.25 911.8 2.42438; 392 0.5 911.8 2.42438;
393 0.75 911.8 2.42438; 394 1 911.8 2.42438; 397 -1 911.8 2.42438;
398 -0.75 911.8 2.67438; 399 -0.5 911.8 2.67438; 400 -0.25 911.8 2.67438;
401 0 911.8 2.67438; 402 0.25 911.8 2.67438; 403 0.5 911.8 2.67438;
404 0.75 911.8 2.67438; 405 1 911.8 2.67438; 408 -1 911.8 2.67438;
409 -0.75 911.8 2.92438; 410 -0.5 911.8 2.92438; 411 -0.25 911.8 2.92438;
412 0 911.8 2.92438; 413 0.25 911.8 2.92438; 414 0.5 911.8 2.92438;
415 0.75 911.8 2.92438; 416 1 911.8 2.92438; 419 -1 911.8 2.92438;
420 -0.75 911.8 1.92438; 421 -0.5 911.8 1.92438; 422 -0.5 911.8 1.57438;
423 -0.75 911.8 1.57438; 424 -0.25 911.8 1.92438; 425 -0.25 911.8 1.57438;
426 0 911.8 1.92438; 427 0 911.8 1.57438; 428 0.25 911.8 1.92438;
429 0.25 911.8 1.57438; 430 0.5 911.8 1.92438; 431 0.5 911.8 1.57438;
432 0.75 911.8 1.92438; 433 0.75 911.8 1.57438; 434 1 911.8 1.92438;
435 1 911.8 1.57438; 436 -1 911.8 1.92438; 437 -1 911.8 1.57438;
438 -0.75 911.8 4.42438; 439 -0.5 911.8 4.42438; 440 -0.25 911.8 4.42438;
441 0 911.8 4.42438; 442 0.25 911.8 4.42438; 443 0.5 911.8 4.42438;
444 0.75 911.8 4.42438; 445 1 911.8 4.42438; 446 -1 911.8 4.42438;
447 -0.75 911.8 4.77438; 448 -0.5 911.8 4.77438; 449 -0.25 911.8 4.77438;
450 0 911.8 4.77438; 451 0.25 911.8 4.77438; 452 0.5 911.8 4.77438;
453 0.75 911.8 4.77438; 454 1 911.8 4.77438; 455 -1 911.8 4.77438;
492 1.25 911.8 3.42438; 493 1.25 911.8 3.175; 494 1.25 911.8 3.67438;
495 1.25 911.8 3.92438; 496 1.25 911.8 4.17438; 497 1.25 911.8 2.42438;
498 1.25 911.8 2.17438; 499 1.25 911.8 2.67438; 500 1.25 911.8 2.92438;
501 1.25 911.8 1.92438; 502 1.25 911.8 1.57438; 503 1.25 911.8 4.42438;
504 1.25 911.8 4.77438; 509 -1.25 911.8 3.42438; 510 -1.25 911.8 3.175;
511 -1.25 911.8 3.67438; 512 -1.25 911.8 3.92438; 513 -1.25 911.8 4.17438;
514 -1.25 911.8 2.42438; 515 -1.25 911.8 2.17438; 516 -1.25 911.8 2.67438;
517 -1.25 911.8 2.92438; 518 -1.25 911.8 1.92438; 519 -1.25 911.8 1.57438;
520 -1.25 911.8 4.42438; 521 -1.25 911.8 4.77438;
MEMBER INCIDENCES
185 2 128;
ELEMENT INCIDENCES SHELL
326 303 302 34 10; 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2;
333 306 307 42 41; 334 307 308 6 42; 337 308 313 309 6; 342 320 303 10 316;
343 321 322 302 303; 344 322 323 304 302; 345 323 324 305 304;
346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307;
349 327 328 313 308; 352 331 321 303 320; 353 332 333 322 321;
354 333 334 323 322; 355 334 335 324 323; 356 335 336 325 324;
357 336 337 326 325; 358 337 338 327 326; 359 338 339 328 327;
362 342 332 321 331; 363 343 344 333 332; 364 344 345 334 333;
365 345 346 335 334; 366 346 347 336 335; 367 347 348 337 336;
368 348 349 338 337; 369 349 350 339 338; 372 353 343 332 342;
393 387 388 366 365; 394 388 389 369 366; 395 389 390 371 369;
396 390 391 373 371; 397 391 392 375 373; 398 392 393 377 375;
399 393 394 379 377; 402 397 387 365 384; 403 398 399 388 387;
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399 393 394 379 377; 402 397 387 365 384; 403 398 399 388 387;
404 399 400 389 388; 405 400 401 390 389; 406 401 402 391 390;
407 402 403 392 391; 408 403 404 393 392; 409 404 405 394 393;
412 408 398 387 397; 413 409 410 399 398; 414 410 411 400 399;
415 411 412 401 400; 416 412 413 402 401; 417 413 414 403 402;
418 414 415 404 403; 419 415 416 405 404; 422 419 409 398 408;
423 10 34 410 409; 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412;
427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415; 432 316 10 409 419;
433 420 421 422 423; 434 421 424 425 422; 435 424 426 427 425;
436 426 428 429 427; 437 428 430 431 429; 438 430 432 433 431;
439 432 434 435 433; 440 436 420 423 437; 441 365 366 421 420;
442 366 369 424 421; 443 369 371 426 424; 444 371 373 428 426;
445 373 375 430 428; 446 375 377 432 430; 447 377 379 434 432;
448 384 365 420 436; 449 438 439 344 343; 450 439 440 345 344;
451 440 441 346 345; 452 441 442 347 346; 453 442 443 348 347;
454 443 444 349 348; 455 444 445 350 349; 456 446 438 343 353;
457 447 448 439 438; 458 448 449 440 439; 459 449 450 441 440;
460 450 451 442 441; 461 451 452 443 442; 462 452 453 444 443;
463 453 454 445 444; 464 455 447 438 446; 497 313 492 493 309;
498 328 494 492 313; 499 339 495 494 328; 500 350 496 495 339;
501 394 497 498 379; 502 405 499 497 394; 503 416 500 499 405;
504 309 493 500 416; 505 434 501 502 435; 506 379 498 501 434;
507 445 503 496 350; 508 454 504 503 445; 513 509 320 316 510;
514 511 331 320 509; 515 512 342 331 511; 516 513 353 342 512;
517 514 397 384 515; 518 516 408 397 514; 519 517 419 408 516;
520 510 316 419 517; 521 518 436 437 519; 522 515 384 436 518;
523 520 446 353 513; 524 521 455 446 520;
START GROUP DEFINITION
ELEMENT
_RAFT 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 -
393 TO 399 402 TO 409 412 TO 419 422 TO 429 432 TO 464 497 TO 508 -
513 TO 524
_RAFT_MOM 326 327 333 334 337 342 TO 349 352 TO 359 362 TO 369 372 -
393 TO 399 402 TO 409 412 TO 419 422 TO 424 427 TO 429 432 TO 464 -
497 TO 508 513 TO 524
_RAFT_SHEAR 337 342 349 352 359 362 TO 369 372 393 TO 399 402 409 412 419 -
422 429 432 TO 464 497 TO 508 513 TO 524
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 TO 464 497 TO 508 513 TO 523 -
524 THICKNESS 0.5
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.5 ZD 0.5
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 TO 464 497 TO 508 513 TO 523 -
524 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
**********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 TO 464 497 TO 508 513 TO 523 -
524 PR GY -16
* LOAD DUE TO SURCHARGE
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 TO 464 497 TO 508 513 TO 523 -
524 PR GY -20
JOINT LOAD
128 FY -20 FZ -2
* LOAD DUE TO PIPE
128 FY -40
**********************************************************************************************
LOAD 2 LOADTYPE None  TITLE ROOF LIVE LOAD ( LR )
```

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LOAD 2 LOADTYPE None  TITLE ROOF LIVE LOAD ( LR )
JOINT LOAD
128 FY -29 FZ -5
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FY -0.001
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE CRANE LOAD  ( CR1 )
JOINT LOAD
128 FY -31 FZ -8
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE CRANE LOAD  ( CR2 )
JOINT LOAD
128 FY 1
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE CRANE LOAD  ( CR3 )
JOINT LOAD
128 FY -0.0001
*********************************************************************************************
LOAD 7 LOADTYPE None  TITLE CRANE LOAD  ( CR4 )
JOINT LOAD
128 FY -0.0001
*********************************************************************************************
LOAD 8 LOADTYPE None  TITLE WIND LOAD  ( WL1 )
JOINT LOAD
128 FY 83 FZ -37
*********************************************************************************************
LOAD 9 LOADTYPE None  TITLE WIND LOAD  ( WL2 )
JOINT LOAD
128 FY 47 FZ -56
*********************************************************************************************
LOAD 10 LOADTYPE None  TITLE WIND LOAD  ( WL3 )
JOINT LOAD
128 FY 83 FZ -37
*********************************************************************************************
LOAD 11 LOADTYPE None  TITLE WIND LOAD  ( WL4 )
JOINT LOAD
128 FY 47 FZ -56
*********************************************************************************************
LOAD 12 LOADTYPE None  TITLE WIND LOAD  ( WL5 )
JOINT LOAD
128 FY 33 FZ 35
*********************************************************************************************
LOAD 13 LOADTYPE None  TITLE WIND LOAD  ( WL6 )
JOINT LOAD
128 FY -2 FZ 16
*********************************************************************************************
LOAD 14 LOADTYPE None  TITLE WIND LOAD  ( WL7 )
JOINT LOAD
128 FY 33 FZ 35
*********************************************************************************************
LOAD 15 LOADTYPE None  TITLE WIND LOAD  ( WL8 )
JOINT LOAD
128 FY -2 FZ 16
*********************************************************************************************
LOAD 16 LOADTYPE None  TITLE WIND LOAD  ( WL9 )
JOINT LOAD
128 FX 30 FY 138.5 FZ 29
*********************************************************************************************
LOAD 17 LOADTYPE None  TITLE WIND LOAD  ( WL10 )
JOINT LOAD
128 FX 30 FY 102.5 FZ 11
*********************************************************************************************
LOAD 18 LOADTYPE None  TITLE WIND LOAD  ( WL11 )
JOINT LOAD
128 FX 30 FY 157.5 FZ 28
*********************************************************************************************
LOAD 19 LOADTYPE None  TITLE WIND LOAD  ( WL12 )
JOINT LOAD
128 FX 30 FY 121.5 FZ 9
*********************************************************************************************
LOAD 20 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FY 0.0001
*********************************************************************************************
LOAD 21 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FY 0.0001
```



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128 FY 0.0001
*********************************************************************************************
LOAD 22 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL1 )
JOINT LOAD
128 FY 1
*********************************************************************************************
LOAD 23 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL2 )
JOINT LOAD
128 FY -1
PERFORM ANALYSIS
CHANGE
SUPPORTS
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 TO 464 497 TO 508 513 TO 523 -
524 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD 201 DL
REPEAT LOAD
1 1.0
PERFORM ANALYSIS
CHANGE
LOAD 202 DL+L+TL1
REPEAT LOAD
1 1.0 3 1.0 22 1.0
PERFORM ANALYSIS
CHANGE
LOAD 203 DL+L+TL2
REPEAT LOAD
1 1.0 3 1.0 23 1.0
PERFORM ANALYSIS
CHANGE
LOAD 204 DL+LR
REPEAT LOAD
1 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 205 DL+0.75L+0.75TL1+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 22 0.75
PERFORM ANALYSIS
CHANGE
LOAD 206 DL+0.75L+0.75TL2+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 23 0.75
PERFORM ANALYSIS
CHANGE
LOAD 207 DL+0.6WL1
REPEAT LOAD
1 1.0 8 0.6
PERFORM ANALYSIS
CHANGE
LOAD 208 DL+WL2
REPEAT LOAD
1 1.0 9 1.0
PERFORM ANALYSIS
CHANGE
LOAD 209 DL+WL3
REPEAT LOAD
1 1.0 10 1.0
PERFORM ANALYSIS
CHANGE
LOAD 210 DL+WL4
REPEAT LOAD
1 1.0 11 1.0
PERFORM ANALYSIS
CHANGE
LOAD 211 DL+WL5
REPEAT LOAD
1 1.0 12 1.0
PERFORM ANALYSIS
CHANGE
LOAD 212 DL+WL6
REPEAT LOAD
1 1.0 13 1.0
PERFORM ANALYSIS
CHANGE
LOAD 213 DL+WL7
REPEAT LOAD
```

```
REPEAT LOAD
1 1.0 14 1.0
PERFORM ANALYSIS
CHANGE
LOAD 214 DL+WL8
REPEAT LOAD
1 1.0 15 1.0
PERFORM ANALYSIS
CHANGE
LOAD 215 DL+WL9
REPEAT LOAD
1 1.0 16 1.0
PERFORM ANALYSIS
CHANGE
LOAD 216 DL+WL10
REPEAT LOAD
1 1.0 17 1.0
PERFORM ANALYSIS
CHANGE
LOAD 217 DL+WL11
REPEAT LOAD
1 1.0 18 1.0
PERFORM ANALYSIS
CHANGE
LOAD 218 DL+WL12
REPEAT LOAD
1 1.0 19 1.0
PERFORM ANALYSIS
CHANGE
LOAD 219 DL+0.75WL1+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75
PERFORM ANALYSIS
CHANGE
LOAD 220 DL+0.75WL2+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75
PERFORM ANALYSIS
CHANGE
LOAD 221 DL+0.75WL3+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75
PERFORM ANALYSIS
CHANGE
LOAD 222 DL+0.75WL4+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75
PERFORM ANALYSIS
CHANGE
LOAD 223 DL+0.75WL5+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75
PERFORM ANALYSIS
CHANGE
LOAD 224 DL+0.75WL67+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75
PERFORM ANALYSIS
CHANGE
LOAD 225 DL+0.75WL7+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75
PERFORM ANALYSIS
CHANGE
LOAD 226 DL+0.75WL8+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75
PERFORM ANALYSIS
CHANGE
LOAD 227 DL+0.75WL9+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75
PERFORM ANALYSIS
CHANGE
LOAD 228 DL+0.75WL10+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75
PERFORM ANALYSIS
CHANGE
```



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CHANGE
LOAD 229 DL+0.75WL11+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75
PERFORM ANALYSIS
CHANGE
LOAD 230 DL+0.75WL12+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75
PERFORM ANALYSIS
CHANGE
LOAD 231 0.6DL+WL1
REPEAT LOAD
1 0.6 8 1.0
PERFORM ANALYSIS
CHANGE
LOAD 232 0.6DL+WL2
REPEAT LOAD
1 0.6 9 1.0
PERFORM ANALYSIS
CHANGE
LOAD 233 0.6DL+WL3
REPEAT LOAD
1 0.6 10 1.0
PERFORM ANALYSIS
CHANGE
LOAD 234 0.6DL+WL4
REPEAT LOAD
1 0.6 11 1.0
PERFORM ANALYSIS
CHANGE
LOAD 235 0.6DL+WL5
REPEAT LOAD
1 0.6 12 1.0
PERFORM ANALYSIS
CHANGE
LOAD 236 0.6DL+WL6
REPEAT LOAD
1 0.6 13 1.0
PERFORM ANALYSIS
CHANGE
LOAD 237 0.6DL+WL7
REPEAT LOAD
1 0.6 14 1.0
PERFORM ANALYSIS
CHANGE
LOAD 238 0.6DL+WL8
REPEAT LOAD
1 0.6 15 1.0
PERFORM ANALYSIS
CHANGE
LOAD 239 0.6DL+WL9
REPEAT LOAD
1 0.6 16 1.0
PERFORM ANALYSIS
CHANGE
LOAD 240 0.6DL+WL10
REPEAT LOAD
1 0.6 17 1.0
PERFORM ANALYSIS
CHANGE
LOAD 241 0.6DL+WL11
REPEAT LOAD
1 0.6 18 1.0
PERFORM ANALYSIS
CHANGE
LOAD 242 0.6DL+WL12
REPEAT LOAD
1 0.6 19 1.0
PERFORM ANALYSIS
CHANGE
LOAD 243 0.6DL+0.7SL1
REPEAT LOAD
1 0.6 20 0.7
PERFORM ANALYSIS
CHANGE
LOAD 244 0.6DL+0.7SL2
REPEAT LOAD
1 0.6 21 0.7
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1 0.6 21 0.7
PERFORM ANALYSIS
CHANGE
LOAD 245 DL+0.75WL1+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 246 DL+0.75WL2+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 247 DL+0.75WL3+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 248 DL+0.75WL4+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 249 DL+0.75WL5+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 250 DL+0.75WL67+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 251 DL+0.75WL7+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 252 DL+0.75WL8+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 253 DL+0.75WL9+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 254 DL+0.75WL10+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 255 DL+0.75WL11+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 256 DL+0.75WL12+0.75L+0.75LR+CR1
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 257 DL+0.75WL1+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 258 DL+0.75WL2+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 259 DL+0.75WL3+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 260 DL+0.75WL4+0.75L+0.75LR+CR2
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LOAD 260 DL+0.75WL4+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 261 DL+0.75WL5+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 262 DL+0.75WL67+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 263 DL+0.75WL7+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 264 DL+0.75WL8+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 265 DL+0.75WL9+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 266 DL+0.75WL10+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 267 DL+0.75WL11+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 268 DL+0.75WL12+0.75L+0.75LR+CR2
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 269 DL+0.75WL1+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 270 DL+0.75WL2+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 271 DL+0.75WL3+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 272 DL+0.75WL4+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 273 DL+0.75WL5+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 274 DL+0.75WL67+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 275 DL+0.75WL7+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75 6 1.0
PERFORM ANALYSIS
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PERFORM ANALYSIS
CHANGE
LOAD  276 DL+0.75WL8+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 277 DL+0.75WL9+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  278 DL+0.75WL10+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 279 DL+0.75WL11+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 280 DL+0.75WL12+0.75L+0.75LR+CR3
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 281 DL+0.75WL1+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 8 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  282 DL+0.75WL2+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 9 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 283 DL+0.75WL3+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 10 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 284 DL+0.75WL4+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 11 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  285 DL+0.75WL5+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 12 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 286 DL+0.75WL67+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 13 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 287 DL+0.75WL7+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 14 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 288 DL+0.75WL8+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 15 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 289 DL+0.75WL9+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 16 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 290 DL+0.75WL10+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 17 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 291 DL+0.75WL11+0.75L+0.75LR+CR4
REPEAT LOAD
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REPEAT LOAD
1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 292 DL+0.75WL12+0.75L+0.75LR+CR4
REPEAT LOAD
1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
PERFORM ANALYSIS
CHANGE
*********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD 101 1.4DL
REPEAT LOAD
1 1.4
PERFORM ANALYSIS
CHANGE
LOAD 102 1.4DL+1.4TL1+1.7L+0.5LR
REPEAT LOAD
1 1.4 22 1.4 3 1.7 2 0.5
PERFORM ANALYSIS
CHANGE
LOAD  103 1.4DL+1.4TL2+1.7L+0.5LR
REPEAT LOAD
1 1.4 23 1.4 3 1.7 2 0.5
PERFORM ANALYSIS
CHANGE
LOAD  104 1.2DL+1.6LR+1.0L
REPEAT LOAD
1 1.2 2 1.6 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD  105 1.2DL+1.6WL1+1.0L+0.5LR
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 106 1.2DL+1.6WL2+1.0L+0.5LR
REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 107 1.2DL+1.6WL3+1.0L+0.5LR
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 108 1.2DL+1.6WL4+1.0L+0.5LR
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 109 1.2DL+1.6WL5+1.0L+0.5LR
REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 110 1.2DL+1.6WL6+1.0L+0.5LR
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 111 1.2DL+1.6WL7+1.0L+0.5LR
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 112 1.2DL+1.6WL8+1.0L+0.5LR
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD  113 1.2DL+1.6WL9+1.0L+0.5LR
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD  114 1.2DL+1.6WL10+1.0L+0.5LR
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0
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1 1.2 17 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD  115 1.2DL+1.6WL11+1.0L+0.5LR
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD  116 1.2DL+1.6WL12+1.0L+0.5LR
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD  117 1.2DL+1.0SL1+1.0L
REPEAT LOAD
1 1.2 20 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD  118 1.2DL+1.0SL2+1.0L
REPEAT LOAD
1 1.2 21 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 119 0.9DL+1.6WL1
REPEAT LOAD
1 0.9 8 1.6
PERFORM ANALYSIS
CHANGE
LOAD  120 0.9DL+1.6WL2
REPEAT LOAD
1 0.9 9 1.6
PERFORM ANALYSIS
CHANGE
LOAD 121 0.9DL+1.6WL3
REPEAT LOAD
1 0.9 10 1.6
PERFORM ANALYSIS
CHANGE
LOAD  122 0.9DL+1.6WL4
REPEAT LOAD
1 0.9 11 1.6
PERFORM ANALYSIS
CHANGE
LOAD  123 0.9DL+1.6WL5
REPEAT LOAD
1 0.9 12 1.6
PERFORM ANALYSIS
CHANGE
LOAD  124 0.9DL+1.6WL6
REPEAT LOAD
1 0.9 13 1.6
PERFORM ANALYSIS
CHANGE
LOAD  125 0.9DL+1.6WL7
REPEAT LOAD
1 0.9 14 1.6
PERFORM ANALYSIS
CHANGE
LOAD  126 0.9DL+1.6WL8
REPEAT LOAD
1 0.9 15 1.6
PERFORM ANALYSIS
CHANGE
LOAD  127 0.9DL+1.6WL9
REPEAT LOAD
1 0.9 16 1.6
PERFORM ANALYSIS
CHANGE
LOAD  128 0.9DL+1.6WL10
REPEAT LOAD
1 0.9 17 1.6
PERFORM ANALYSIS
CHANGE
LOAD 129 0.9DL+1.6WL11
REPEAT LOAD
1 0.9 18 1.6
PERFORM ANALYSIS
CHANGE
LOAD  130 0.9DL+1.6WL12
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LOAD  130 0.9DL+1.6WL12
REPEAT LOAD
1 0.9 19 1.6
PERFORM ANALYSIS
CHANGE
LOAD  131 0.9DL+1.6SL1
REPEAT LOAD
1 0.9 20 1.6
PERFORM ANALYSIS
CHANGE
LOAD  132 0.9DL+1.6SL2
REPEAT LOAD
1 0.9 21 1.6
PERFORM ANALYSIS
CHANGE
LOAD  133 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  134 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  135 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 136 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  137 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  138 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  139 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  140 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  141 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  142 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  143 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD  144 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR1
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 145 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
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PERFORM ANALYSIS
CHANGE
LOAD  146 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  147 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  148 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  149 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  150 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  151 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 152 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  153 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  154 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  155 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  156 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR2
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD  157 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  158 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  159 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  160 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  161 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR3
REPEAT LOAD
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REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  162 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  163 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  164 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  165 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  166 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  167 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 168 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR3
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD  169 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 8 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  170 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 9 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  171 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 10 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  172 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 11 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  173 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 12 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  174 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 13 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  175 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 14 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 176 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 15 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
```

```
CHANGE
LOAD  177 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 16 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 178 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 17 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  179 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 18 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD  180 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR4
REPEAT LOAD
1 1.2 19 1.6 2 0.5 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
*************************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 180
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185
CLB 0.05 MEMB 185
CLS 0.05 ALL
CLT 0.05 MEMB 185 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 -
372 393 TO 399 402 TO 409 412 TO 419 422 TO 429 432 497 TO 504 513 TO 520
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 6 STAAD INPUT – FOUNDATION F3

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 04/09/2020
JOB NAME FDN. MKD. F3
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 911.8 3.175; 34 -0.5 911.8 3.175; 35 -0.25 911.8 3.175;
41 0.25 911.8 3.175; 42 0.5 911.8 3.175; 128 0 913.8 3.175;
302 -0.5 911.8 3.42438; 304 -0.25 911.8 3.42438; 305 0 911.8 3.42438;
306 0.25 911.8 3.42438; 307 0.5 911.8 3.42438; 322 -0.5 911.8 3.67438;
323 -0.25 911.8 3.67438; 324 0 911.8 3.67438; 325 0.25 911.8 3.67438;
326 0.5 911.8 3.67438; 399 -0.5 911.8 2.67438; 400 -0.25 911.8 2.67438;
401 0 911.8 2.67438; 402 0.25 911.8 2.67438; 403 0.5 911.8 2.67438;
410 -0.5 911.8 2.92438; 411 -0.25 911.8 2.92438; 412 0 911.8 2.92438;
413 0.25 911.8 2.92438; 414 0.5 911.8 2.92438;
MEMBER INCIDENCES
185 2 128;
ELEMENT INCIDENCES SHELL
327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41;
344 322 323 304 302; 345 323 324 305 304; 346 324 325 306 305;
347 325 326 307 306; 414 410 411 400 399; 415 411 412 401 400;
416 412 413 402 401; 417 413 414 403 402; 424 34 35 411 410; 425 35 2 412 411;
426 2 41 413 412; 427 41 42 414 413;
START GROUP DEFINITION
ELEMENT
_RAFT 327 328 332 333 344 TO 347 414 TO 417 424 TO 427
_RAFT_MOM 327 333 344 TO 347 414 TO 417 424 427
_RAFT_SHEAR 327 333 344 TO 347 414 TO 417 424 427
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
327 328 332 333 344 TO 347 414 TO 417 424 TO 427 THICKNESS 0.4
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.4 ZD 0.4
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
327 328 332 333 344 TO 347 414 TO 417 424 TO 426 -
427 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
327 328 332 333 344 TO 347 414 TO 417 424 TO 427 PR GY -22
* LOAD DUE TO SURCHARGE
327 328 332 333 344 TO 347 414 TO 417 424 TO 427 PR GY -20
JOINT LOAD
128 FY -8 FZ -2
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FY -32 FZ -5
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL
1 1.0
LOAD COMB 202 DL+L
1 1.0 2 1.0
*********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD COMB 101 1.4DL
1 1.4
```



```
1 1.4
LOAD COMB 102 1.4DL+1.7L
1 1.4 2 1.7
*********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 102
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185
CLB 0.05 MEMB 185
CLS 0.05 ALL
CLT 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 7 STAAD INPUT – FOUNDATION F4

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 04/09/2020
JOB NAME FDN. MKD. F4
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
SET NL 200
SET ITERLIM 15
SET PRINT 1
UNIT METER KN
JOINT COORDINATES
2 0 911.8 3.175; 34 -0.5 911.8 3.175; 35 -0.25 911.8 3.175;
41 0.25 911.8 3.175; 42 0.5 911.8 3.175; 128 0 913.8 3.175;
302 -0.5 911.8 3.42438; 304 -0.25 911.8 3.42438; 305 0 911.8 3.42438;
306 0.25 911.8 3.42438; 307 0.5 911.8 3.42438; 322 -0.5 911.8 3.67438;
323 -0.25 911.8 3.67438; 324 0 911.8 3.67438; 325 0.25 911.8 3.67438;
326 0.5 911.8 3.67438; 399 -0.5 911.8 2.67438; 400 -0.25 911.8 2.67438;
401 0 911.8 2.67438; 402 0.25 911.8 2.67438; 403 0.5 911.8 2.67438;
410 -0.5 911.8 2.92438; 411 -0.25 911.8 2.92438; 412 0 911.8 2.92438;
413 0.25 911.8 2.92438; 414 0.5 911.8 2.92438; 415 0.75 911.8 3.42438;
416 0.75 911.8 3.175; 417 1.125 911.8 3.42438; 418 1.125 911.8 3.175;
419 0.75 911.8 3.67438; 420 1.125 911.8 3.67438; 421 0.75 911.8 2.92438;
422 0.75 911.8 2.67438; 423 1.125 911.8 2.92438; 424 1.125 911.8 2.67438;
425 -1.125 911.8 3.42438; 426 -0.75 911.8 3.42438; 427 -0.75 911.8 3.175;
428 -1.125 911.8 3.175; 429 -1.125 911.8 3.67438; 430 -0.75 911.8 3.67438;
431 -1.125 911.8 2.92438; 432 -0.75 911.8 2.92438; 433 -0.75 911.8 2.67438;
434 -1.125 911.8 2.67438; 435 -0.5 911.8 3.92438; 436 -0.25 911.8 3.92438;
437 0 911.8 3.92438; 438 0.25 911.8 3.92438; 439 0.5 911.8 3.92438;
440 -0.5 911.8 4.17438; 441 -0.25 911.8 4.17438; 442 0 911.8 4.17438;
443 0.25 911.8 4.17438; 444 0.5 911.8 4.17438; 445 0.75 911.8 3.92438;
446 1.125 911.8 3.92438; 447 0.75 911.8 4.17438; 448 1.125 911.8 4.17438;
449 -1.125 911.8 3.92438; 450 -0.75 911.8 3.92438; 451 -1.125 911.8 4.17438;
452 -0.75 911.8 4.17438; 453 -0.5 911.8 2.42438; 454 -0.25 911.8 2.42438;
455 -0.25 911.8 2.175; 456 -0.5 911.8 2.175; 457 0 911.8 2.42438;
458 0 911.8 2.175; 459 0.25 911.8 2.42438; 460 0.25 911.8 2.175;
461 0.5 911.8 2.42438; 462 0.5 911.8 2.175; 463 0.75 911.8 2.42438;
464 0.75 911.8 2.175; 465 1.125 911.8 2.42438; 466 1.125 911.8 2.175;
467 -1.125 911.8 2.42438; 468 -0.75 911.8 2.42438; 469 -0.75 911.8 2.175;
470 -1.125 911.8 2.175;
MEMBER INCIDENCES
185 2 128;
ELEMENT INCIDENCES SHELL
327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41;
344 322 323 304 302; 345 323 324 305 304; 346 324 325 306 305;
347 325 326 307 306; 414 410 411 400 399; 415 411 412 401 400;
416 412 413 402 401; 417 413 414 403 402; 424 34 35 411 410; 425 35 2 412 411;
426 2 41 413 412; 427 41 42 414 413; 428 307 415 416 42; 429 415 417 418 416;
430 326 419 415 307; 431 419 420 417 415; 432 414 421 422 403;
433 421 423 424 422; 434 42 416 421 414; 435 416 418 423 421;
436 425 426 427 428; 437 426 302 34 427; 438 429 430 426 425;
439 430 322 302 426; 440 431 432 433 434; 441 432 410 399 433;
442 428 427 432 431; 443 427 34 410 432; 444 435 436 323 322;
445 436 437 324 323; 446 437 438 325 324; 447 438 439 326 325;
448 440 441 436 435; 449 441 442 437 436; 450 442 443 438 437;
451 443 444 439 438; 452 439 445 419 326; 453 445 446 420 419;
454 444 447 445 439; 455 447 448 446 445; 456 449 450 430 429;
457 450 435 322 430; 458 451 452 450 449; 459 452 440 435 450;
460 453 454 455 456; 461 454 457 458 455; 462 457 459 460 458;
463 459 461 462 460; 464 399 400 454 453; 465 400 401 457 454;
466 401 402 459 457; 467 402 403 461 459; 468 461 463 464 462;
469 463 465 466 464; 470 403 422 463 461; 471 422 424 465 463;
472 467 468 469 470; 473 468 453 456 469; 474 434 433 468 467;
475 433 399 453 468;
START GROUP DEFINITION
ELEMENT
_RAFT 327 328 332 333 344 TO 347 414 TO 417 424 TO 475
_RAFT_MOM 327 333 344 TO 347 414 TO 417 424 427 TO 475
_RAFT_SHEAR 428 TO 475
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
327 328 332 333 344 TO 347 414 TO 417 424 TO 475 THICKNESS 0.4
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
```



```
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.7 ZD 0.4
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
327 328 332 333 344 TO 347 414 TO 417 424 TO 474 -
475 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
327 328 332 333 344 TO 347 414 TO 417 424 TO 475 PR GY -22
* LOAD DUE TO SURCHARGE
327 328 332 333 344 TO 347 414 TO 417 424 TO 475 PR GY -20
JOINT LOAD
128 FY -5
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE WIND LOAD ( WL )
JOINT LOAD
128 FX 40
PERFORM ANALYSIS
CHANGE
SUPPORTS
327 328 332 333 344 TO 347 414 TO 417 424 TO 474 -
475 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD  201 DL
REPEAT LOAD
1 1.0
PERFORM ANALYSIS
CHANGE
LOAD  202 DL+WL
REPEAT LOAD
1 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD  203 0.6DL+WL
REPEAT LOAD
1 0.6 2 1.0
PERFORM ANALYSIS
CHANGE
*********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD  101 1.4DL
REPEAT LOAD
1 1.4
PERFORM ANALYSIS
CHANGE
LOAD  102 1.2DL+1.6WL
REPEAT LOAD
1 1.4 2 1.6
PERFORM ANALYSIS
CHANGE
LOAD  103 0.9DL+1.6WL
REPEAT LOAD
1 0.9 2 1.6
PERFORM ANALYSIS
CHANGE
*********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 103
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185
CLB 0.05 MEMB 185
CLS 0.05 ALL
CLT 0.05 MEMB 185 327 328 332 333 344 TO 347 414 TO 417 424 TO 427
MAXMAIN 25 ALL
```

```
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 8 STAAD OUTPUT – FOUNDATION F1

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    OCT 18, 2020                 *
*          Time=    17:47:58                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F1.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 04/09/2020
    4. JOB NAME FDN. MKD. F1
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. SET NL 200
   12. SET ITERLIM 15
   13. SET PRINT 1
   14. UNIT METER KN
   15. JOINT COORDINATES
   16. 2 0 911.8 3.175; 6 0.75 911.8 3.175; 10 -0.75 911.8 3.175; 34 -0.5 911.8 3.175
   17. 35 -0.25 911.8 3.175; 41 0.25 911.8 3.175; 42 0.5 911.8 3.175
   18. 128 0 913.8 3.175; 302 -0.5 911.8 3.49938; 303 -0.75 911.8 3.49938
   19. 304 -0.25 911.8 3.49938; 305 0 911.8 3.49938; 306 0.25 911.8 3.49938
   20. 307 0.5 911.8 3.49938; 308 0.75 911.8 3.49938; 309 1 911.8 3.175
   21. 313 1 911.8 3.49938; 316 -1 911.8 3.175; 320 -1 911.8 3.49938
   22. 321 -0.75 911.8 3.79938; 322 -0.5 911.8 3.79938; 323 -0.25 911.8 3.79938
   23. 324 0 911.8 3.79938; 325 0.25 911.8 3.79938; 326 0.5 911.8 3.79938
   24. 327 0.75 911.8 3.79938; 328 1 911.8 3.79938; 331 -1 911.8 3.79938
   25. 332 -0.75 911.8 4.04938; 333 -0.5 911.8 4.04938; 334 -0.25 911.8 4.04938
   26. 335 0 911.8 4.04938; 336 0.25 911.8 4.04938; 337 0.5 911.8 4.04938
   27. 338 0.75 911.8 4.04938; 339 1 911.8 4.04938; 342 -1 911.8 4.04938
   28. 343 -0.75 911.8 4.29938; 344 -0.5 911.8 4.29938; 345 -0.25 911.8 4.29938
   29. 346 0 911.8 4.29938; 347 0.25 911.8 4.29938; 348 0.5 911.8 4.29938
   30. 349 0.75 911.8 4.29938; 350 1 911.8 4.29938; 353 -1 911.8 4.29938
   31. 365 -0.75 911.8 2.04938; 366 -0.5 911.8 2.04938; 369 -0.25 911.8 2.04938
   32. 371 0 911.8 2.04938; 373 0.25 911.8 2.04938; 375 0.5 911.8 2.04938
   33. 377 0.75 911.8 2.04938; 379 1 911.8 2.04938; 384 -1 911.8 2.04938
   34. 387 -0.75 911.8 2.29938; 388 -0.5 911.8 2.29938; 389 -0.25 911.8 2.29938
   35. 390 0 911.8 2.29938; 391 0.25 911.8 2.29938; 392 0.5 911.8 2.29938
   36. 393 0.75 911.8 2.29938; 394 1 911.8 2.29938; 397 -1 911.8 2.29938
   37. 398 -0.75 911.8 2.54938; 399 -0.5 911.8 2.54938; 400 -0.25 911.8 2.54938
   38. 401 0 911.8 2.54938; 402 0.25 911.8 2.54938; 403 0.5 911.8 2.54938
```



```
   39. 404 0.75 911.8 2.54938; 405 1 911.8 2.54938; 408 -1 911.8 2.54938
   40. 409 -0.75 911.8 2.84938; 410 -0.5 911.8 2.84938; 411 -0.25 911.8 2.84938
   41. 412 0 911.8 2.84938; 413 0.25 911.8 2.84938; 414 0.5 911.8 2.84938
   42. 415 0.75 911.8 2.84938; 416 1 911.8 2.84938; 419 -1 911.8 2.84938
   43. 420 -0.75 911.8 1.79938; 421 -0.5 911.8 1.79938; 422 -0.5 911.8 1.54938
   44. 423 -0.75 911.8 1.54938; 424 -0.25 911.8 1.79938; 425 -0.25 911.8 1.54938
   45. 426 0 911.8 1.79938; 427 0 911.8 1.54938; 428 0.25 911.8 1.79938
   46. 429 0.25 911.8 1.54938; 430 0.5 911.8 1.79938; 431 0.5 911.8 1.54938
   47. 432 0.75 911.8 1.79938; 433 0.75 911.8 1.54938; 434 1 911.8 1.79938
   48. 435 1 911.8 1.54938; 436 -1 911.8 1.79938; 437 -1 911.8 1.54938
   49. 438 -0.75 911.8 4.54938; 439 -0.5 911.8 4.54938; 440 -0.25 911.8 4.54938
   50. 441 0 911.8 4.54938; 442 0.25 911.8 4.54938; 443 0.5 911.8 4.54938
   51. 444 0.75 911.8 4.54938; 445 1 911.8 4.54938; 446 -1 911.8 4.54938
   52. 447 -0.75 911.8 4.79938; 448 -0.5 911.8 4.79938; 449 -0.25 911.8 4.79938
   53. 450 0 911.8 4.79938; 451 0.25 911.8 4.79938; 452 0.5 911.8 4.79938
   54. 453 0.75 911.8 4.79938; 454 1 911.8 4.79938; 455 -1 911.8 4.79938
   55. 456 -0.5 911.8 1.29938; 457 -0.75 911.8 1.29938; 458 -0.25 911.8 1.29938
   56. 459 0 911.8 1.29938; 460 0.25 911.8 1.29938; 461 0.5 911.8 1.29938
   57. 462 0.75 911.8 1.29938; 463 1 911.8 1.29938; 464 -1 911.8 1.29938
   58. 465 -0.75 911.8 5.04938; 466 -0.5 911.8 5.04938; 467 -0.25 911.8 5.04938
   59. 468 0 911.8 5.04938; 469 0.25 911.8 5.04938; 470 0.5 911.8 5.04938
   60. 471 0.75 911.8 5.04938; 472 1 911.8 5.04938; 473 -1 911.8 5.04938
   61. 474 -0.5 911.8 1.04938; 475 -0.75 911.8 1.04938; 476 -0.25 911.8 1.04938
   62. 477 0 911.8 1.04938; 478 0.25 911.8 1.04938; 479 0.5 911.8 1.04938
   63. 480 0.75 911.8 1.04938; 481 1 911.8 1.04938; 482 -1 911.8 1.04938
   64. 483 -0.75 911.8 5.29938; 484 -0.5 911.8 5.29938; 485 -0.25 911.8 5.29938
   65. 486 0 911.8 5.29938; 487 0.25 911.8 5.29938; 488 0.5 911.8 5.29938
   66. 489 0.75 911.8 5.29938; 490 1 911.8 5.29938; 491 -1 911.8 5.29938
   67. 492 1.25 911.8 3.49938; 493 1.25 911.8 3.175; 494 1.25 911.8 3.79938
   68. 495 1.25 911.8 4.04938; 496 1.25 911.8 4.29938; 497 1.25 911.8 2.29938
   69. 498 1.25 911.8 2.04938; 499 1.25 911.8 2.54938; 500 1.25 911.8 2.84938
   70. 501 1.25 911.8 1.79938; 502 1.25 911.8 1.54938; 503 1.25 911.8 4.54938
   71. 504 1.25 911.8 4.79938; 505 1.25 911.8 1.29938; 506 1.25 911.8 5.04938
   72. 507 1.25 911.8 1.04938; 508 1.25 911.8 5.29938; 509 -1.25 911.8 3.49938
   73. 510 -1.25 911.8 3.175; 511 -1.25 911.8 3.79938; 512 -1.25 911.8 4.04938
   74. 513 -1.25 911.8 4.29938; 514 -1.25 911.8 2.29938; 515 -1.25 911.8 2.04938
   75. 516 -1.25 911.8 2.54938; 517 -1.25 911.8 2.84938; 518 -1.25 911.8 1.79938
   76. 519 -1.25 911.8 1.54938; 520 -1.25 911.8 4.54938; 521 -1.25 911.8 4.79938
   77. 522 -1.25 911.8 1.29938; 523 -1.25 911.8 5.04938; 524 -1.25 911.8 1.04938
   78. 525 -1.25 911.8 5.29938; 526 -0.5 911.8 0.79938; 527 -0.75 911.8 0.79938
   79. 528 -0.25 911.8 0.79938; 529 0 911.8 0.79938; 530 0.25 911.8 0.79938
   80. 531 0.5 911.8 0.79938; 532 0.75 911.8 0.79938; 533 1 911.8 0.79938
   81. 534 -1 911.8 0.79938; 544 1.25 911.8 0.79938; 546 -1.25 911.8 0.79938
   82. 548 -0.75 911.8 5.54938; 549 -0.5 911.8 5.54938; 550 -0.25 911.8 5.54938
   83. 551 0 911.8 5.54938; 552 0.25 911.8 5.54938; 553 0.5 911.8 5.54938
   84. 554 0.75 911.8 5.54938; 555 1 911.8 5.54938; 556 -1 911.8 5.54938
   85. 566 1.25 911.8 5.54938; 568 -1.25 911.8 5.54938; 569 1.375 911.8 3.49938
   86. 570 1.375 911.8 3.175; 571 1.375 911.8 3.79938; 572 1.375 911.8 4.04938
   87. 573 1.375 911.8 4.29938; 574 1.375 911.8 2.29938; 575 1.375 911.8 2.04938
   88. 576 1.375 911.8 2.54938; 577 1.375 911.8 2.84938; 578 1.375 911.8 1.79938
   89. 579 1.375 911.8 1.54938; 580 1.375 911.8 4.54938; 581 1.375 911.8 4.79938
   90. 582 1.375 911.8 1.29938; 583 1.375 911.8 5.04938; 584 1.375 911.8 1.04938
   91. 585 1.375 911.8 5.29938; 586 1.375 911.8 0.79938; 587 1.375 911.8 5.54938
   92. 588 -1.375 911.8 3.49938; 589 -1.375 911.8 3.175; 590 -1.375 911.8 3.79938
   93. 591 -1.375 911.8 4.04938; 592 -1.375 911.8 4.29938; 593 -1.375 911.8 2.29938
   94. 594 -1.375 911.8 2.04938; 595 -1.375 911.8 2.54938; 596 -1.375 911.8 2.84938
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 95. 597 -1.375 911.8 1.79938; 598 -1.375 911.8 1.54938; 599 -1.375 911.8 4.54938
 96. 600 -1.375 911.8 4.79938; 601 -1.375 911.8 1.29938; 602 -1.375 911.8 5.04938
 97. 603 -1.375 911.8 1.04938; 604 -1.375 911.8 5.29938; 605 -1.375 911.8 0.79938
 98. 606 -1.375 911.8 5.54938
 99. MEMBER INCIDENCES
100. 185 2 128
101. ELEMENT INCIDENCES SHELL
102. 326 303 302 34 10; 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2
103. 333 306 307 42 41; 334 307 308 6 42; 337 308 313 309 6; 342 320 303 10 316
104. 343 321 322 302 303; 344 322 323 304 302; 345 323 324 305 304
105. 346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307
106. 349 327 328 313 308; 352 331 321 303 320; 353 332 333 322 321
107. 354 333 334 323 322; 355 334 335 324 323; 356 335 336 325 324
108. 357 336 337 326 325; 358 337 338 327 326; 359 338 339 328 327
109. 362 342 332 321 331; 363 343 344 333 332; 364 344 345 334 333
110. 365 345 346 335 334; 366 346 347 336 335; 367 347 348 337 336
111. 368 348 349 338 337; 369 349 350 339 338; 372 353 343 332 342
112. 393 387 388 366 365; 394 388 389 369 366; 395 389 390 371 369
113. 396 390 391 373 371; 397 391 392 375 373; 398 392 393 377 375
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115. 404 399 400 389 388; 405 400 401 390 389; 406 401 402 391 390
116. 407 402 403 392 391; 408 403 404 393 392; 409 404 405 394 393
117. 412 408 398 387 397; 413 409 410 399 398; 414 410 411 400 399
118. 415 411 412 401 400; 416 412 413 402 401; 417 413 414 403 402
119. 418 414 415 404 403; 419 415 416 405 404; 422 419 409 398 408
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121. 427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415; 432 316 10 409 419
122. 433 420 421 422 423; 434 421 424 425 422; 435 424 426 427 425
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127. 448 384 365 420 436; 449 438 439 344 343; 450 439 440 345 344
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132. 463 453 454 445 444; 464 455 447 438 446; 465 423 422 456 457
133. 466 422 425 458 456; 467 425 427 459 458; 468 427 429 460 459
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135. 472 437 423 457 464; 473 465 466 448 447; 474 466 467 449 448
136. 475 467 468 450 449; 476 468 469 451 450; 477 469 470 452 451
137. 478 470 471 453 452; 479 471 472 454 453; 480 473 465 447 455
138. 481 457 456 474 475; 482 456 458 476 474; 483 458 459 477 476
139. 484 459 460 478 477; 485 460 461 479 478; 486 461 462 480 479
140. 487 462 463 481 480; 488 464 457 475 482; 489 483 484 466 465
141. 490 484 485 467 466; 491 485 486 468 467; 492 486 487 469 468
142. 493 487 488 470 469; 494 488 489 471 470; 495 489 490 472 471
143. 496 491 483 465 473; 497 313 492 493 309; 498 328 494 492 313
144. 499 339 495 494 328; 500 350 496 495 339; 501 394 497 498 379
145. 502 405 499 497 394; 503 416 500 499 405; 504 309 493 500 416
146. 505 434 501 502 435; 506 379 498 501 434; 507 445 503 496 350
147. 508 454 504 503 445; 509 435 502 505 463; 510 472 506 504 454
148. 511 463 505 507 481; 512 490 508 506 472; 513 509 320 316 510
149. 514 511 331 320 509; 515 512 342 331 511; 516 513 353 342 512
150. 517 514 397 384 515; 518 516 408 397 514; 519 517 419 408 516
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151. 520 510 316 419 517; 521 518 436 437 519; 522 515 384 436 518
152. 523 520 446 353 513; 524 521 455 446 520; 525 519 437 464 522
153. 526 523 473 455 521; 527 522 464 482 524; 528 525 491 473 523
154. 529 475 474 526 527; 530 474 476 528 526; 531 476 477 529 528
155. 532 477 478 530 529; 533 478 479 531 530; 534 479 480 532 531
156. 535 480 481 533 532; 536 482 475 527 534; 545 481 507 544 533
157. 547 524 482 534 546; 549 548 549 484 483; 550 549 550 485 484
158. 551 550 551 486 485; 552 551 552 487 486; 553 552 553 488 487
159. 554 553 554 489 488; 555 554 555 490 489; 556 556 548 483 491
160. 565 555 566 508 490; 567 568 556 491 525; 568 492 569 570 493
161. 569 494 571 569 492; 570 495 572 571 494; 571 496 573 572 495
162. 572 497 574 575 498; 573 499 576 574 497; 574 500 577 576 499
163. 575 493 570 577 500; 576 501 578 579 502; 577 498 575 578 501
164. 578 503 580 573 496; 579 504 581 580 503; 580 502 579 582 505
165. 581 506 583 581 504; 582 505 582 584 507; 583 508 585 583 506
166. 584 507 584 586 544; 585 566 587 585 508; 586 588 509 510 589
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168. 590 593 514 515 594; 591 595 516 514 593; 592 596 517 516 595
169. 593 589 510 517 596; 594 597 518 519 598; 595 594 515 518 597
170. 596 599 520 513 592; 597 600 521 520 599; 598 598 519 522 601
171. 599 602 523 521 600; 600 601 522 524 603; 601 604 525 523 602
172. 602 603 524 546 605; 603 606 568 525 604
173. START GROUP DEFINITION
174. ELEMENT
175. _RAFT 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 -
176. 393 TO 399 402 TO 409 412 TO 419 422 TO 429 432 TO 536 545 547 549 TO 556 -
177. 565 567 TO 603
178. _RAFT_MOM 326 327 333 334 337 342 TO 349 352 TO 359 362 TO 369 372 -
179. 393 TO 399 402 TO 409 412 TO 419 422 TO 424 427 TO 429 432 TO 536 545 547 -
180. 549 TO 556 565 567 TO 603
181. _RAFT_SHEAR 337 342 349 352 359 362 TO 369 372 393 TO 399 402 409 412 419 -
182. 422 429 432 TO 536 545 547 549 TO 556 565 567 TO 603
183. MEMBER
184. END GROUP DEFINITION
185. ELEMENT PROPERTY
186. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
187. 402 TO 409 412 TO 419 422 TO 429 432 TO 536 545 547 549 TO 556 565 -
188. 567 TO 603 THICKNESS 0.6
189. DEFINE MATERIAL START
190. ISOTROPIC CONCRETE
191. E 2.17185E+007
192. POISSON 0.17
193. DENSITY 25
194. ALPHA 1E-005
195. DAMP 0.05
196. TYPE CONCRETE
197. STRENGTH FCU 27579
198. END DEFINE MATERIAL
199. MEMBER PROPERTY AMERICAN
200. 185 PRIS YD 0.5 ZD 0.85
201. CONSTANTS
202. MATERIAL CONCRETE ALL
203. SUPPORTS
204. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
205. 402 TO 409 412 TO 419 422 TO 429 432 TO 536 545 547 549 TO 556 565 -
206. 567 TO 603 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
```

JOINT	INFLUENCE AREA
2	0.081250
6	0.081250
10	0.081250
34	0.081250
35	0.081250
41	0.081250
42	0.081250
302	0.078048
303	0.078048
304	0.078048
305	0.078048
306	0.078048
307	0.078048
308	0.078048
309	0.081250
313	0.078048
316	0.081250
320	0.078048
321	0.068750
322	0.068750
323	0.068750
324	0.068750
325	0.068750
326	0.068750
327	0.068750
328	0.068750
331	0.068750
332	0.062500
333	0.062500
334	0.062500
335	0.062500
336	0.062500
337	0.062500
338	0.062500
339	0.062500
342	0.062500
343	0.062500
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375	0.062500
377	0.062500

379	0.062500
384	0.062500
387	0.062500
388	0.062500
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391	0.062500
392	0.062500
393	0.062500
394	0.062500
397	0.062500
398	0.068750
399	0.068750
400	0.068750
401	0.068750
402	0.068750
403	0.068750
404	0.068750
405	0.068750
408	0.068750
409	0.078202
410	0.078202
411	0.078202
412	0.078202
413	0.078202
414	0.078202
415	0.078202
416	0.078203
419	0.078203
420	0.062500
421	0.062500
422	0.062500
423	0.062500
424	0.062500
425	0.062500
426	0.062500
427	0.062500
428	0.062500
429	0.062500
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431	0.062500
432	0.062500
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437	0.062500
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440	0.062500
441	0.062500
442	0.062500
443	0.062500
444	0.062500
445	0.062500
446	0.062500

STAAD SPACE -- PAGE NO. 7

447	0.062500
448	0.062500
449	0.062500
450	0.062500
451	0.062500
452	0.062500
453	0.062500
454	0.062500
455	0.062500
456	0.062500
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458	0.062500
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464	0.062500
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469	0.062500
470	0.062500
471	0.062500
472	0.062500
473	0.062500
474	0.062500
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476	0.062500
477	0.062500
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479	0.062500
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486	0.062500
487	0.062500
488	0.062500
489	0.062500
490	0.062500
491	0.062500
492	0.058536
493	0.060938
494	0.051562
495	0.046875
496	0.046875
497	0.046875
498	0.046875
499	0.051562
500	0.058652
501	0.046875
502	0.046875

STAAD SPACE -- PAGE NO. 8

503	0.046875
504	0.046875
505	0.046875
506	0.046875
507	0.046875
508	0.046875
509	0.058536
510	0.060938
511	0.051562
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513	0.046875
514	0.046875
515	0.046875
516	0.051562
517	0.058652
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524	0.046875
525	0.046875
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530	0.031250
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532	0.031250
533	0.031250
534	0.031250
544	0.023438
546	0.023438
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556	0.031250
566	0.023437
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569	0.019512
570	0.020312
571	0.017187
572	0.015625
573	0.015625
574	0.015625
575	0.015625
576	0.017187
577	0.019551
578	0.015625
579	0.015625

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       580          0.015625
       581          0.015625
       582          0.015625
       583          0.015625
       584          0.015625
       585          0.015625
       586          0.007812
       587          0.007812
       588          0.019512
       589          0.020312
       590          0.017187
       591          0.015625
       592          0.015625
       593          0.015625
       594          0.015625
       595          0.017187
       596          0.019551
       597          0.015625
       598          0.015625
       599          0.015625
       600          0.015625
       601          0.015625
       602          0.015625
       603          0.015625
       604          0.015625
       605          0.007812
       606          0.007812
    TOTAL          13.062500
207. ********************************************************************************
208. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
209. SELFWEIGHT Y -1
210. ELEMENT LOAD
211. * LOAD DUE TO BACKFILL
212. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
213. 402 TO 409 412 TO 419 422 TO 429 432 TO 536 545 547 549 TO 556 565 -
214. 567 TO 603 PR GY -16
215. * LOAD DUE TO SURCHARGE
216. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
217. 402 TO 409 412 TO 419 422 TO 429 432 TO 536 545 547 549 TO 556 565 -
218. 567 TO 603 PR GY -20
219. JOINT LOAD
220. 128 FY -22 FZ -3 MX -23
221. ********************************************************************************
222. LOAD 2 LOADTYPE NONE  TITLE ROOF LIVE LOAD ( LR )
223. JOINT LOAD
224. 128 FY -61 FZ -2 MX -10
225. ********************************************************************************
226. LOAD 3 LOADTYPE NONE  TITLE LIVE LOAD ( L )
227. JOINT LOAD
228. 128 FY -0.001
229. ********************************************************************************
230. LOAD 4 LOADTYPE NONE  TITLE CRANE LOAD  ( CR1 )
231. JOINT LOAD
232. 128 FY -35 FZ -9 MX -40
233. ********************************************************************************
234. LOAD 5 LOADTYPE NONE  TITLE CRANE LOAD  ( CR2 )
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235. JOINT LOAD
236. 128 FY -26 FZ -7 MX -33
237. ********************************************************************************
238. LOAD 6 LOADTYPE NONE  TITLE CRANE LOAD  ( CR3 )
239. JOINT LOAD
240. 128 FY -0.0001
241. ********************************************************************************
242. LOAD 7 LOADTYPE NONE  TITLE CRANE LOAD  ( CR4 )
243. JOINT LOAD
244. 128 FY -0.0001
245. ********************************************************************************
246. LOAD 8 LOADTYPE NONE  TITLE WIND LOAD  ( WL1 )
247. JOINT LOAD
248. 128 FY 117 FZ -24 MX -193
249. ********************************************************************************
250. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD  ( WL2 )
251. JOINT LOAD
252. 128 FY 74 FZ -27 MX -207
253. ********************************************************************************
254. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD  ( WL3 )
255. JOINT LOAD
256. 128 FY 117 FZ -24 MX -193
257. ********************************************************************************
258. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD  ( WL4 )
259. JOINT LOAD
260. 128 FY 74 FZ -27 MX -207
261. ********************************************************************************
262. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD  ( WL5 )
263. JOINT LOAD
264. 128 FY 105 FZ 38 MX 270
265. ********************************************************************************
266. LOAD 13 LOADTYPE NONE  TITLE WIND LOAD  ( WL6 )
267. JOINT LOAD
268. 128 FY 62 FZ 35 MX 256
269. ********************************************************************************
270. LOAD 14 LOADTYPE NONE  TITLE WIND LOAD  ( WL7 )
271. JOINT LOAD
272. 128 FY 105 FZ 38 MX 270
273. ********************************************************************************
274. LOAD 15 LOADTYPE NONE  TITLE WIND LOAD  ( WL8 )
275. JOINT LOAD
276. 128 FY 62 FZ 35 MX 256
277. ********************************************************************************
278. LOAD 16 LOADTYPE NONE  TITLE WIND LOAD  ( WL9 )
279. JOINT LOAD
280. 128 FY 119 FZ 4 MX 22
281. ********************************************************************************
282. LOAD 17 LOADTYPE NONE  TITLE WIND LOAD  ( WL10 )
283. JOINT LOAD
284. 128 FY 76 FZ 2 MX 8
285. ********************************************************************************
286. LOAD 18 LOADTYPE NONE  TITLE WIND LOAD  ( WL11 )
287. JOINT LOAD
288. 128 FY 119 FZ 11 MX 63
289. ********************************************************************************
290. LOAD 19 LOADTYPE NONE  TITLE WIND LOAD  ( WL12 )
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291. JOINT LOAD
292. 128 FY 75 FZ 8 MX 48
293. ********************************************************************************
294. LOAD 20 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
295. JOINT LOAD
296. 128 FZ -1 MX -7
297. ********************************************************************************
298. LOAD 21 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
299. JOINT LOAD
300. 128 FZ 1 MX 7
301. ********************************************************************************
302. LOAD 22 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL1 )
303. JOINT LOAD
304. 128 FY 2
305. ********************************************************************************
306. LOAD 23 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL2 )
307. JOINT LOAD
308. 128 FY -2
309. PERFORM ANALYSIS
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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 248 NUMBER OF MEMBERS 1
NUMBER OF PLATES 216 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 247

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 162/ 22/ 69 DOF
TOTAL PRIMARY LOAD CASES = 23, TOTAL DEGREES OF FREEDOM = 747
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 52 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.9/ 142653.0 MB

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310. CHANGE
311. SUPPORTS
312. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
313. 402 TO 409 412 TO 419 422 TO 429 432 TO 536 545 547 549 TO 556 565 -
314. 567 TO 603 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
315. ********************************************************************************
316. * UNFACTORED LOAD COMBINATIONS
317. LOAD 201 DL
318. REPEAT LOAD
319. 1 1.0
320. PERFORM ANALYSIS
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```
**NOTE-Tension/Compression converged after  1 iterations, Case=  201

321. CHANGE
322. LOAD 202 DL+L+TL1
323. REPEAT LOAD
324. 1 1.0 3 1.0 22 1.0
325. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  202

326. CHANGE
327. LOAD 203 DL+L+TL2
328. REPEAT LOAD
329. 1 1.0 3 1.0 23 1.0
330. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  203

331. CHANGE
332. LOAD 204 DL+LR
333. REPEAT LOAD
334. 1 1.0 2 1.0
335. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  204

336. CHANGE
337. LOAD 205 DL+0.75L+0.75TL1+0.75LR
338. REPEAT LOAD
339. 1 1.0 2 0.75 3 0.75 22 0.75
340. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

341. CHANGE
342. LOAD 206 DL+0.75L+0.75TL2+0.75LR
343. REPEAT LOAD
344. 1 1.0 2 0.75 3 0.75 23 0.75
345. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

346. CHANGE
347. LOAD 207 DL+0.6WL1
348. REPEAT LOAD
349. 1 1.0 8 0.6
350. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

351. CHANGE
352. LOAD 208 DL+WL2
353. REPEAT LOAD
354. 1 1.0 9 1.0
355. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

356. CHANGE
357. LOAD 209 DL+WL3
358. REPEAT LOAD
359. 1 1.0 10 1.0
360. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

361. CHANGE
362. LOAD 210 DL+WL4
363. REPEAT LOAD
364. 1 1.0 11 1.0
365. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

366. CHANGE
367. LOAD 211 DL+WL5
368. REPEAT LOAD
369. 1 1.0 12 1.0
370. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 211

371. CHANGE
372. LOAD 212 DL+WL6
373. REPEAT LOAD
374. 1 1.0 13 1.0
375. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

376. CHANGE
377. LOAD 213 DL+WL7
378. REPEAT LOAD
379. 1 1.0 14 1.0
380. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

381. CHANGE
382. LOAD 214 DL+WL8
383. REPEAT LOAD
384. 1 1.0 15 1.0

```
 385. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  214


 386. CHANGE
 387. LOAD 215 DL+WL9
 388. REPEAT LOAD
 389. 1 1.0 16 1.0
 390. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  215


 391. CHANGE
 392. LOAD 216 DL+WL10
 393. REPEAT LOAD
 394. 1 1.0 17 1.0
 395. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  216


 396. CHANGE
 397. LOAD 217 DL+WL11
 398. REPEAT LOAD
 399. 1 1.0 18 1.0
 400. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  217


 401. CHANGE
 402. LOAD 218 DL+WL12
 403. REPEAT LOAD
 404. 1 1.0 19 1.0
 405. PERFORM ANALYSIS
```



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**NOTE-Tension/Compression converged after  1 iterations, Case=  218


 406. CHANGE
 407. LOAD 219 DL+0.75WL1+0.75L+0.75LR
 408. REPEAT LOAD
 409. 1 1.0 2 0.75 3 0.75 8 0.75
 410. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  219


 411. CHANGE
 412. LOAD 220 DL+0.75WL2+0.75L+0.75LR
 413. REPEAT LOAD
 414. 1 1.0 2 0.75 3 0.75 9 0.75
 415. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  220


 416. CHANGE
 417. LOAD 221 DL+0.75WL3+0.75L+0.75LR
 418. REPEAT LOAD
 419. 1 1.0 2 0.75 3 0.75 10 0.75
 420. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  221


 421. CHANGE
 422. LOAD 222 DL+0.75WL4+0.75L+0.75LR
 423. REPEAT LOAD
 424. 1 1.0 2 0.75 3 0.75 11 0.75
 425. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  222
```

```
426. CHANGE
427. LOAD 223 DL+0.75WL5+0.75L+0.75LR
428. REPEAT LOAD
429. 1 1.0 2 0.75 3 0.75 12 0.75
430. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  223


431. CHANGE
432. LOAD 224 DL+0.75WL67+0.75L+0.75LR
433. REPEAT LOAD
434. 1 1.0 2 0.75 3 0.75 13 0.75
435. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  224


436. CHANGE
437. LOAD 225 DL+0.75WL7+0.75L+0.75LR
438. REPEAT LOAD
439. 1 1.0 2 0.75 3 0.75 14 0.75
440. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  225


441. CHANGE
442. LOAD 226 DL+0.75WL8+0.75L+0.75LR
443. REPEAT LOAD
444. 1 1.0 2 0.75 3 0.75 15 0.75
445. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  226


446. CHANGE
447. LOAD 227 DL+0.75WL9+0.75L+0.75LR
448. REPEAT LOAD
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449. 1 1.0 2 0.75 3 0.75 16 0.75
450. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  227


451. CHANGE
452. LOAD 228 DL+0.75WL10+0.75L+0.75LR
453. REPEAT LOAD
454. 1 1.0 2 0.75 3 0.75 17 0.75
455. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  228


456. CHANGE
457. LOAD 229 DL+0.75WL11+0.75L+0.75LR
458. REPEAT LOAD
459. 1 1.0 2 0.75 3 0.75 18 0.75
460. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  229


461. CHANGE
462. LOAD 230 DL+0.75WL12+0.75L+0.75LR
463. REPEAT LOAD
464. 1 1.0 2 0.75 3 0.75 19 0.75
465. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  230


466. CHANGE
467. LOAD 231 0.6DL+WL1
468. REPEAT LOAD
469. 1 0.6 8 1.0
470. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 231

471. CHANGE
472. LOAD 232 0.6DL+WL2
473. REPEAT LOAD
474. 1 0.6 9 1.0
475. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 232

476. CHANGE
477. LOAD 233 0.6DL+WL3
478. REPEAT LOAD
479. 1 0.6 10 1.0
480. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 233

481. CHANGE
482. LOAD 234 0.6DL+WL4
483. REPEAT LOAD
484. 1 0.6 11 1.0
485. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 234

486. CHANGE
487. LOAD 235 0.6DL+WL5
488. REPEAT LOAD
489. 1 0.6 12 1.0
490. PERFORM ANALYSIS

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 235

491. CHANGE
492. LOAD 236 0.6DL+WL6
493. REPEAT LOAD
494. 1 0.6 13 1.0
495. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 236

496. CHANGE
497. LOAD 237 0.6DL+WL7
498. REPEAT LOAD
499. 1 0.6 14 1.0
500. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 237

501. CHANGE
502. LOAD 238 0.6DL+WL8
503. REPEAT LOAD
504. 1 0.6 15 1.0
505. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 238

506. CHANGE
507. LOAD 239 0.6DL+WL9
508. REPEAT LOAD
509. 1 0.6 16 1.0
510. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 239

511. CHANGE
512. LOAD 240 0.6DL+WL10
513. REPEAT LOAD
514. 1 0.6 17 1.0
515. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 240

516. CHANGE
517. LOAD 241 0.6DL+WL11
518. REPEAT LOAD
519. 1 0.6 18 1.0
520. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 241

521. CHANGE
522. LOAD 242 0.6DL+WL12
523. REPEAT LOAD
524. 1 0.6 19 1.0
525. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 242

526. CHANGE
527. LOAD 243 0.6DL+0.7SL1
528. REPEAT LOAD
529. 1 0.6 20 0.7
530. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 243

531. CHANGE
532. LOAD 244 0.6DL+0.7SL2
533. REPEAT LOAD
534. 1 0.6 21 0.7
535. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 244

536. CHANGE
537. LOAD 245 DL+0.75WL1+0.75L+0.75LR+CR1
538. REPEAT LOAD
539. 1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
540. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 245

541. CHANGE
542. LOAD 246 DL+0.75WL2+0.75L+0.75LR+CR1
543. REPEAT LOAD
544. 1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
545. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 246

546. CHANGE
547. LOAD 247 DL+0.75WL3+0.75L+0.75LR+CR1
548. REPEAT LOAD
549. 1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
550. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 247

551. CHANGE
552. LOAD 248 DL+0.75WL4+0.75L+0.75LR+CR1
553. REPEAT LOAD
554. 1 1.0 2 0.75 3 0.75 11 0.75 4 1.0

```
 555. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  248


 556. CHANGE
 557. LOAD 249 DL+0.75WL5+0.75L+0.75LR+CR1
 558. REPEAT LOAD
 559. 1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
 560. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  249


 561. CHANGE
 562. LOAD 250 DL+0.75WL67+0.75L+0.75LR+CR1
 563. REPEAT LOAD
 564. 1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
 565. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  250


 566. CHANGE
 567. LOAD 251 DL+0.75WL7+0.75L+0.75LR+CR1
 568. REPEAT LOAD
 569. 1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
 570. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  251


 571. CHANGE
 572. LOAD 252 DL+0.75WL8+0.75L+0.75LR+CR1
 573. REPEAT LOAD
 574. 1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
 575. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  252


 576. CHANGE
 577. LOAD 253 DL+0.75WL9+0.75L+0.75LR+CR1
 578. REPEAT LOAD
 579. 1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
 580. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  253


 581. CHANGE
 582. LOAD 254 DL+0.75WL10+0.75L+0.75LR+CR1
 583. REPEAT LOAD
 584. 1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
 585. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  254


 586. CHANGE
 587. LOAD 255 DL+0.75WL11+0.75L+0.75LR+CR1
 588. REPEAT LOAD
 589. 1 1.0 2 0.75 3 0.75 18 0.75 4 1.0
 590. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  255


 591. CHANGE
 592. LOAD 256 DL+0.75WL12+0.75L+0.75LR+CR1
 593. REPEAT LOAD
 594. 1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
 595. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  256
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596. CHANGE
597. LOAD 257 DL+0.75WL1+0.75L+0.75LR+CR2
598. REPEAT LOAD
599. 1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
600. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  257


601. CHANGE
602. LOAD 258 DL+0.75WL2+0.75L+0.75LR+CR2
603. REPEAT LOAD
604. 1 1.0 2 0.75 3 0.75 9 0.75 5 1.0
605. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  258


606. CHANGE
607. LOAD 259 DL+0.75WL3+0.75L+0.75LR+CR2
608. REPEAT LOAD
609. 1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
610. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  259


611. CHANGE
612. LOAD 260 DL+0.75WL4+0.75L+0.75LR+CR2
613. REPEAT LOAD
614. 1 1.0 2 0.75 3 0.75 11 0.75 5 1.0
615. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  260


616. CHANGE
617. LOAD 261 DL+0.75WL5+0.75L+0.75LR+CR2
618. REPEAT LOAD
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619. 1 1.0 2 0.75 3 0.75 12 0.75 5 1.0
620. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  261


621. CHANGE
622. LOAD 262 DL+0.75WL67+0.75L+0.75LR+CR2
623. REPEAT LOAD
624. 1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
625. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  262


626. CHANGE
627. LOAD 263 DL+0.75WL7+0.75L+0.75LR+CR2
628. REPEAT LOAD
629. 1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
630. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  263


631. CHANGE
632. LOAD 264 DL+0.75WL8+0.75L+0.75LR+CR2
633. REPEAT LOAD
634. 1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
635. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  264


636. CHANGE
637. LOAD 265 DL+0.75WL9+0.75L+0.75LR+CR2
638. REPEAT LOAD
639. 1 1.0 2 0.75 3 0.75 16 0.75 5 1.0
640. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 265

641. CHANGE
642. LOAD 266 DL+0.75WL10+0.75L+0.75LR+CR2
643. REPEAT LOAD
644. 1 1.0 2 0.75 3 0.75 17 0.75 5 1.0
645. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 266

646. CHANGE
647. LOAD 267 DL+0.75WL11+0.75L+0.75LR+CR2
648. REPEAT LOAD
649. 1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
650. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 267

651. CHANGE
652. LOAD 268 DL+0.75WL12+0.75L+0.75LR+CR2
653. REPEAT LOAD
654. 1 1.0 2 0.75 3 0.75 19 0.75 5 1.0
655. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 268

656. CHANGE
657. LOAD 269 DL+0.75WL1+0.75L+0.75LR+CR3
658. REPEAT LOAD
659. 1 1.0 2 0.75 3 0.75 8 0.75 6 1.0
660. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 269

661. CHANGE
662. LOAD 270 DL+0.75WL2+0.75L+0.75LR+CR3
663. REPEAT LOAD
664. 1 1.0 2 0.75 3 0.75 9 0.75 6 1.0
665. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 270

666. CHANGE
667. LOAD 271 DL+0.75WL3+0.75L+0.75LR+CR3
668. REPEAT LOAD
669. 1 1.0 2 0.75 3 0.75 10 0.75 6 1.0
670. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 271

671. CHANGE
672. LOAD 272 DL+0.75WL4+0.75L+0.75LR+CR3
673. REPEAT LOAD
674. 1 1.0 2 0.75 3 0.75 11 0.75 6 1.0
675. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 272

676. CHANGE
677. LOAD 273 DL+0.75WL5+0.75L+0.75LR+CR3
678. REPEAT LOAD
679. 1 1.0 2 0.75 3 0.75 12 0.75 6 1.0
680. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 273

```
 681. CHANGE
 682. LOAD 274 DL+0.75WL67+0.75L+0.75LR+CR3
 683. REPEAT LOAD
 684. 1 1.0 2 0.75 3 0.75 13 0.75 6 1.0
 685. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  274



 686. CHANGE
 687. LOAD 275 DL+0.75WL7+0.75L+0.75LR+CR3
 688. REPEAT LOAD
 689. 1 1.0 2 0.75 3 0.75 14 0.75 6 1.0
 690. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  275



 691. CHANGE
 692. LOAD  276 DL+0.75WL8+0.75L+0.75LR+CR3
 693. REPEAT LOAD
 694. 1 1.0 2 0.75 3 0.75 15 0.75 6 1.0
 695. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  276



 696. CHANGE
 697. LOAD 277 DL+0.75WL9+0.75L+0.75LR+CR3
 698. REPEAT LOAD
 699. 1 1.0 2 0.75 3 0.75 16 0.75 6 1.0
 700. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  277



 701. CHANGE
 702. LOAD  278 DL+0.75WL10+0.75L+0.75LR+CR3
 703. REPEAT LOAD
 704. 1 1.0 2 0.75 3 0.75 17 0.75 6 1.0
```



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 705. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  278



 706. CHANGE
 707. LOAD 279 DL+0.75WL11+0.75L+0.75LR+CR3
 708. REPEAT LOAD
 709. 1 1.0 2 0.75 3 0.75 18 0.75 6 1.0
 710. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  279



 711. CHANGE
 712. LOAD 280 DL+0.75WL12+0.75L+0.75LR+CR3
 713. REPEAT LOAD
 714. 1 1.0 2 0.75 3 0.75 19 0.75 6 1.0
 715. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  280



 716. CHANGE
 717. LOAD 281 DL+0.75WL1+0.75L+0.75LR+CR4
 718. REPEAT LOAD
 719. 1 1.0 2 0.75 3 0.75 8 0.75 7 1.0
 720. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  281



 721. CHANGE
 722. LOAD  282 DL+0.75WL2+0.75L+0.75LR+CR4
 723. REPEAT LOAD
 724. 1 1.0 2 0.75 3 0.75 9 0.75 7 1.0
 725. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 282

726. CHANGE
727. LOAD 283 DL+0.75WL3+0.75L+0.75LR+CR4
728. REPEAT LOAD
729. 1 1.0 2 0.75 3 0.75 10 0.75 7 1.0
730. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 283

731. CHANGE
732. LOAD 284 DL+0.75WL4+0.75L+0.75LR+CR4
733. REPEAT LOAD
734. 1 1.0 2 0.75 3 0.75 11 0.75 7 1.0
735. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 284

736. CHANGE
737. LOAD 285 DL+0.75WL5+0.75L+0.75LR+CR4
738. REPEAT LOAD
739. 1 1.0 2 0.75 3 0.75 12 0.75 7 1.0
740. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 285

741. CHANGE
742. LOAD 286 DL+0.75WL67+0.75L+0.75LR+CR4
743. REPEAT LOAD
744. 1 1.0 2 0.75 3 0.75 13 0.75 7 1.0
745. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 286

746. CHANGE
747. LOAD 287 DL+0.75WL7+0.75L+0.75LR+CR4
748. REPEAT LOAD
749. 1 1.0 2 0.75 3 0.75 14 0.75 7 1.0
750. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 287

751. CHANGE
752. LOAD 288 DL+0.75WL8+0.75L+0.75LR+CR4
753. REPEAT LOAD
754. 1 1.0 2 0.75 3 0.75 15 0.75 7 1.0
755. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 288

756. CHANGE
757. LOAD 289 DL+0.75WL9+0.75L+0.75LR+CR4
758. REPEAT LOAD
759. 1 1.0 2 0.75 3 0.75 16 0.75 7 1.0
760. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 289

761. CHANGE
762. LOAD 290 DL+0.75WL10+0.75L+0.75LR+CR4
763. REPEAT LOAD
764. 1 1.0 2 0.75 3 0.75 17 0.75 7 1.0
765. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 290

766. CHANGE
767. LOAD 291 DL+0.75WL11+0.75L+0.75LR+CR4
768. REPEAT LOAD

```
769. 1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
770. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  291



771. CHANGE
772. LOAD 292 DL+0.75WL12+0.75L+0.75LR+CR4
773. REPEAT LOAD
774. 1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
775. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  292



776. CHANGE
777. ******************************************************************************
778. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
779. LOAD 101 1.4DL
780. REPEAT LOAD
781. 1 1.4
782. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  101



783. CHANGE
784. LOAD 102 1.4DL+1.4TL1+1.7L+0.5LR
785. REPEAT LOAD
786. 1 1.4 22 1.4 3 1.7 2 0.5
787. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  102



788. CHANGE
789. LOAD  103 1.4DL+1.4TL2+1.7L+0.5LR
790. REPEAT LOAD
791. 1 1.4 23 1.4 3 1.7 2 0.5
792. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  103



793. CHANGE
794. LOAD  104 1.2DL+1.6LR+1.0L
795. REPEAT LOAD
796. 1 1.2 2 1.6 3 1.0
797. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  104



798. CHANGE
799. LOAD  105 1.2DL+1.6WL1+1.0L+0.5LR
800. REPEAT LOAD
801. 1 1.2 8 1.6 2 0.5 3 1.0
802. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  105



803. CHANGE
804. LOAD 106 1.2DL+1.6WL2+1.0L+0.5LR
805. REPEAT LOAD
806. 1 1.2 9 1.6 2 0.5 3 1.0
807. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  106



808. CHANGE
809. LOAD 107 1.2DL+1.6WL3+1.0L+0.5LR
810. REPEAT LOAD
811. 1 1.2 10 1.6 2 0.5 3 1.0
812. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 107

813. CHANGE
814. LOAD 108 1.2DL+1.6WL4+1.0L+0.5LR
815. REPEAT LOAD
816. 1 1.2 11 1.6 2 0.5 3 1.0
817. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 108

818. CHANGE
819. LOAD 109 1.2DL+1.6WL5+1.0L+0.5LR
820. REPEAT LOAD
821. 1 1.2 12 1.6 2 0.5 3 1.0
822. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

823. CHANGE
824. LOAD 110 1.2DL+1.6WL6+1.0L+0.5LR
825. REPEAT LOAD
826. 1 1.2 13 1.6 2 0.5 3 1.0
827. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

828. CHANGE
829. LOAD 111 1.2DL+1.6WL7+1.0L+0.5LR
830. REPEAT LOAD
831. 1 1.2 14 1.6 2 0.5 3 1.0
832. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 111

833. CHANGE
834. LOAD 112 1.2DL+1.6WL8+1.0L+0.5LR
835. REPEAT LOAD
836. 1 1.2 15 1.6 2 0.5 3 1.0
837. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 112

838. CHANGE
839. LOAD 113 1.2DL+1.6WL9+1.0L+0.5LR
840. REPEAT LOAD
841. 1 1.2 16 1.6 2 0.5 3 1.0
842. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 113

843. CHANGE
844. LOAD 114 1.2DL+1.6WL10+1.0L+0.5LR
845. REPEAT LOAD
846. 1 1.2 17 1.6 2 0.5 3 1.0
847. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 114

848. CHANGE
849. LOAD 115 1.2DL+1.6WL11+1.0L+0.5LR
850. REPEAT LOAD
851. 1 1.2 18 1.6 2 0.5 3 1.0
852. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 115

853. CHANGE
854. LOAD 116 1.2DL+1.6WL12+1.0L+0.5LR
855. REPEAT LOAD
856. 1 1.2 19 1.6 2 0.5 3 1.0

```
 857. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  116


 858. CHANGE
 859. LOAD  117 1.2DL+1.0SL1+1.0L
 860. REPEAT LOAD
 861. 1 1.2 20 1.0 3 1.0
 862. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  117


 863. CHANGE
 864. LOAD  118 1.2DL+1.0SL2+1.0L
 865. REPEAT LOAD
 866. 1 1.2 21 1.0 3 1.0
 867. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  118


 868. CHANGE
 869. LOAD 119 0.9DL+1.6WL1
 870. REPEAT LOAD
 871. 1 0.9 8 1.6
 872. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  119


 873. CHANGE
 874. LOAD  120 0.9DL+1.6WL2
 875. REPEAT LOAD
 876. 1 0.9 9 1.6
 877. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  120


 878. CHANGE
 879. LOAD 121 0.9DL+1.6WL3
 880. REPEAT LOAD
 881. 1 0.9 10 1.6
 882. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  121


 883. CHANGE
 884. LOAD  122 0.9DL+1.6WL4
 885. REPEAT LOAD
 886. 1 0.9 11 1.6
 887. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  122


 888. CHANGE
 889. LOAD  123 0.9DL+1.6WL5
 890. REPEAT LOAD
 891. 1 0.9 12 1.6
 892. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  123


 893. CHANGE
 894. LOAD  124 0.9DL+1.6WL6
 895. REPEAT LOAD
 896. 1 0.9 13 1.6
 897. PERFORM ANALYSIS
**START ITERATION NO.           2
```

**NOTE-Tension/Compression converged after 2 iterations, Case= 124

898. CHANGE
899. LOAD 125 0.9DL+1.6WL7
900. REPEAT LOAD
901. 1 0.9 14 1.6
902. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 125

903. CHANGE
904. LOAD 126 0.9DL+1.6WL8
905. REPEAT LOAD
906. 1 0.9 15 1.6
907. PERFORM ANALYSIS
**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 126

908. CHANGE
909. LOAD 127 0.9DL+1.6WL9
910. REPEAT LOAD
911. 1 0.9 16 1.6
912. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 127

913. CHANGE
914. LOAD 128 0.9DL+1.6WL10
915. REPEAT LOAD
916. 1 0.9 17 1.6
917. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 128

918. CHANGE
919. LOAD 129 0.9DL+1.6WL11
920. REPEAT LOAD
921. 1 0.9 18 1.6
922. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 129

923. CHANGE
924. LOAD 130 0.9DL+1.6WL12
925. REPEAT LOAD
926. 1 0.9 19 1.6
927. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 130

928. CHANGE
929. LOAD 131 0.9DL+1.6SL1
930. REPEAT LOAD
931. 1 0.9 20 1.6
932. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 131

933. CHANGE
934. LOAD 132 0.9DL+1.6SL2
935. REPEAT LOAD
936. 1 0.9 21 1.6
937. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 132

```
938. CHANGE
939. LOAD  133 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR1
940. REPEAT LOAD
941. 1 1.2 8 1.6 2 0.5 3 1.0 4 1.0
942. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  133



943. CHANGE
944. LOAD  134 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
945. REPEAT LOAD
946. 1 1.2 9 1.6 2 0.5 3 1.0 4 1.0
947. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  134



948. CHANGE
949. LOAD  135 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR1
950. REPEAT LOAD
951. 1 1.2 10 1.6 2 0.5 3 1.0 4 1.0
952. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  135



953. CHANGE
954. LOAD 136 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR1
955. REPEAT LOAD
956. 1 1.2 11 1.6 2 0.5 3 1.0 4 1.0
957. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  136



958. CHANGE
959. LOAD  137 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR1
960. REPEAT LOAD
961. 1 1.2 12 1.6 2 0.5 3 1.0 4 1.0
```



```
962. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  137



963. CHANGE
964. LOAD  138 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR1
965. REPEAT LOAD
966. 1 1.2 13 1.6 2 0.5 3 1.0 4 1.0
967. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  138



968. CHANGE
969. LOAD  139 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR1
970. REPEAT LOAD
971. 1 1.2 14 1.6 2 0.5 3 1.0 4 1.0
972. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  139



973. CHANGE
974. LOAD  140 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR1
975. REPEAT LOAD
976. 1 1.2 15 1.6 2 0.5 3 1.0 4 1.0
977. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  140



978. CHANGE
979. LOAD  141 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR1
980. REPEAT LOAD
981. 1 1.2 16 1.6 2 0.5 3 1.0 4 1.0
982. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 141

983. CHANGE
984. LOAD 142 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR1
985. REPEAT LOAD
986. 1 1.2 17 1.6 2 0.5 3 1.0 4 1.0
987. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 142

988. CHANGE
989. LOAD 143 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR1
990. REPEAT LOAD
991. 1 1.2 18 1.6 2 0.5 3 1.0 4 1.0
992. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 143

993. CHANGE
994. LOAD 144 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR1
995. REPEAT LOAD
996. 1 1.2 19 1.6 2 0.5 3 1.0 4 1.0
997. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 144

998. CHANGE
999. LOAD 145 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR2
1000. REPEAT LOAD
1001. 1 1.2 8 1.6 2 0.5 3 1.0 5 1.0
1002. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 145

1003. CHANGE
1004. LOAD 146 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR2
1005. REPEAT LOAD
1006. 1 1.2 9 1.6 2 0.5 3 1.0 5 1.0
1007. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 146

1008. CHANGE
1009. LOAD 147 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR2
1010. REPEAT LOAD
1011. 1 1.2 10 1.6 2 0.5 3 1.0 5 1.0
1012. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 147

1013. CHANGE
1014. LOAD 148 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR2
1015. REPEAT LOAD
1016. 1 1.2 11 1.6 2 0.5 3 1.0 5 1.0
1017. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 148

1018. CHANGE
1019. LOAD 149 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR2
1020. REPEAT LOAD
1021. 1 1.2 12 1.6 2 0.5 3 1.0 5 1.0
1022. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 149

1023. CHANGE
1024. LOAD 150 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR2
1025. REPEAT LOAD

```
1026. 1 1.2 13 1.6 2 0.5 3 1.0 5 1.0
1027. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  150



1028. CHANGE
1029. LOAD  151 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR2
1030. REPEAT LOAD
1031. 1 1.2 14 1.6 2 0.5 3 1.0 5 1.0
1032. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  151



1033. CHANGE
1034. LOAD 152 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR2
1035. REPEAT LOAD
1036. 1 1.2 15 1.6 2 0.5 3 1.0 5 1.0
1037. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  152



1038. CHANGE
1039. LOAD  153 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR2
1040. REPEAT LOAD
1041. 1 1.2 16 1.6 2 0.5 3 1.0 5 1.0
1042. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  153



1043. CHANGE
1044. LOAD  154 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR2
1045. REPEAT LOAD
1046. 1 1.2 17 1.6 2 0.5 3 1.0 5 1.0
1047. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  154



1048. CHANGE
1049. LOAD  155 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR2
1050. REPEAT LOAD
1051. 1 1.2 18 1.6 2 0.5 3 1.0 5 1.0
1052. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  155



1053. CHANGE
1054. LOAD  156 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR2
1055. REPEAT LOAD
1056. 1 1.2 19 1.6 2 0.5 3 1.0 5 1.0
1057. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  156



1058. CHANGE
1059. LOAD  157 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR3
1060. REPEAT LOAD
1061. 1 1.2 8 1.6 2 0.5 3 1.0 6 1.0
1062. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  157



1063. CHANGE
1064. LOAD  158 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR3
1065. REPEAT LOAD
1066. 1 1.2 9 1.6 2 0.5 3 1.0 6 1.0
1067. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 158

1068. CHANGE
1069. LOAD 159 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR3
1070. REPEAT LOAD
1071. 1 1.2 10 1.6 2 0.5 3 1.0 6 1.0
1072. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 159

1073. CHANGE
1074. LOAD 160 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR3
1075. REPEAT LOAD
1076. 1 1.2 11 1.6 2 0.5 3 1.0 6 1.0
1077. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 160

1078. CHANGE
1079. LOAD 161 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR3
1080. REPEAT LOAD
1081. 1 1.2 12 1.6 2 0.5 3 1.0 6 1.0
1082. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 161

1083. CHANGE
1084. LOAD 162 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR3
1085. REPEAT LOAD
1086. 1 1.2 13 1.6 2 0.5 3 1.0 6 1.0
1087. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 162

1088. CHANGE
1089. LOAD 163 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR3
1090. REPEAT LOAD
1091. 1 1.2 14 1.6 2 0.5 3 1.0 6 1.0
1092. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 163

1093. CHANGE
1094. LOAD 164 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR3
1095. REPEAT LOAD
1096. 1 1.2 15 1.6 2 0.5 3 1.0 6 1.0
1097. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 164

1098. CHANGE
1099. LOAD 165 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR3
1100. REPEAT LOAD
1101. 1 1.2 16 1.6 2 0.5 3 1.0 6 1.0
1102. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 165

1103. CHANGE
1104. LOAD 166 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR3
1105. REPEAT LOAD
1106. 1 1.2 17 1.6 2 0.5 3 1.0 6 1.0
1107. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 166

1108. CHANGE
1109. LOAD 167 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR3
1110. REPEAT LOAD
1111. 1 1.2 18 1.6 2 0.5 3 1.0 6 1.0

```
1112. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  167



1113. CHANGE
1114. LOAD 168 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR3
1115. REPEAT LOAD
1116. 1 1.2 19 1.6 2 0.5 3 1.0 6 1.0
1117. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  168



1118. CHANGE
1119. LOAD  169 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR4
1120. REPEAT LOAD
1121. 1 1.2 8 1.6 2 0.5 3 1.0 7 1.0
1122. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  169



1123. CHANGE
1124. LOAD  170 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR4
1125. REPEAT LOAD
1126. 1 1.2 9 1.6 2 0.5 3 1.0 7 1.0
1127. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  170



1128. CHANGE
1129. LOAD  171 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR4
1130. REPEAT LOAD
1131. 1 1.2 10 1.6 2 0.5 3 1.0 7 1.0
1132. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  171



1133. CHANGE
1134. LOAD  172 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR4
1135. REPEAT LOAD
1136. 1 1.2 11 1.6 2 0.5 3 1.0 7 1.0
1137. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  172



1138. CHANGE
1139. LOAD  173 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR4
1140. REPEAT LOAD
1141. 1 1.2 12 1.6 2 0.5 3 1.0 7 1.0
1142. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  173



1143. CHANGE
1144. LOAD  174 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR4
1145. REPEAT LOAD
1146. 1 1.2 13 1.6 2 0.5 3 1.0 7 1.0
1147. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  174



1148. CHANGE
1149. LOAD  175 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR4
1150. REPEAT LOAD
1151. 1 1.2 14 1.6 2 0.5 3 1.0 7 1.0
1152. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  175
```

```
1153. CHANGE
1154. LOAD 176 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR4
1155. REPEAT LOAD
1156. 1 1.2 15 1.6 2 0.5 3 1.0 7 1.0
1157. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  176



1158. CHANGE
1159. LOAD  177 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR4
1160. REPEAT LOAD
1161. 1 1.2 16 1.6 2 0.5 3 1.0 7 1.0
1162. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  177



1163. CHANGE
1164. LOAD 178 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR4
1165. REPEAT LOAD
1166. 1 1.2 17 1.6 2 0.5 3 1.0 7 1.0
1167. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  178



1168. CHANGE
1169. LOAD  179 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR4
1170. REPEAT LOAD
1171. 1 1.2 18 1.6 2 0.5 3 1.0 7 1.0
1172. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  179



1173. CHANGE
1174. LOAD  180 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR4
1175. REPEAT LOAD
```



```
1176. 1 1.2 19 1.6 2 0.5 3 1.0 7 1.0
1177. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  180



1178. CHANGE
1179. ******************************************************************************
1180. PERFORM ANALYSIS

  ** ALL CASES BEING MADE ACTIVE BEFORE RE-ANALYSIS. **

**NOTE-Tension/Compression converged after  1 iterations, Case=    1



**NOTE-Tension/Compression converged after  1 iterations, Case=    2



**NOTE-Tension/Compression converged after  1 iterations, Case=    3



**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=    4



**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4
**START ITERATION NO.           5
**START ITERATION NO.           6

**NOTE-Tension/Compression converged after  6 iterations, Case=    5



**NOTE-Tension/Compression converged after  1 iterations, Case=    6



**NOTE-Tension/Compression converged after  1 iterations, Case=    7



**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4
**START ITERATION NO.           5
```

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2155931E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.5981744E+04 L-MATRIX DIAG= 9.4223651E-10 EQN NO 745
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2153137E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.6808135E+06 L-MATRIX DIAG= 2.9713847E-06 EQN NO 747
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 6 iterations, Case= 8

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2155931E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.5981744E+04 L-MATRIX DIAG= 9.4223651E-10 EQN NO 745
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2153137E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.6808135E+06 L-MATRIX DIAG= 2.9713847E-06 EQN NO 747
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 6 iterations, Case= 9

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2155931E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.5981744E+04 L-MATRIX DIAG= 9.4223651E-10 EQN NO 745
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2153137E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.6808135E+06 L-MATRIX DIAG= 2.9713847E-06 EQN NO 747
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 6 iterations, Case= 10

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2155931E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.5981744E+04 L-MATRIX DIAG= 9.4223651E-10 EQN NO 745
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2153137E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.6808135E+06 L-MATRIX DIAG= 2.9713847E-06 EQN NO 747
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 6 iterations, Case= 11

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.3609027E-06 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.5981744E+04 L-MATRIX DIAG= 9.4223651E-10 EQN NO 745
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2153137E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.6808135E+06 L-MATRIX DIAG= 2.9713847E-06 EQN NO 747
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 7
**START ITERATION NO. 8
**START ITERATION NO. 9
**START ITERATION NO. 10
**START ITERATION NO. 11
**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 1.8766150E-06 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 13

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 4.7390349E-06 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 14

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 3.4291297E-06 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.6808135E+06 L-MATRIX DIAG= 3.7555583E-06 EQN NO 747
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 15

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.5981744E+04 L-MATRIX DIAG= 9.4223651E-10 EQN NO 745
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2153137E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.6808135E+06 L-MATRIX DIAG= 2.9713847E-06 EQN NO 747
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 12
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.3609027E-06 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.5981744E+04 L-MATRIX DIAG= 9.4223651E-10 EQN NO 745
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

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ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2153137E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.6808135E+06 L-MATRIX DIAG= 2.9713847E-06 EQN NO 747
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 7
**START ITERATION NO. 8
**START ITERATION NO. 9
**START ITERATION NO. 10
**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.3609027E-06 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

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ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.5981744E+04 L-MATRIX DIAG= 9.4223651E-10 EQN NO 745
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2153137E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.6808135E+06 L-MATRIX DIAG= 2.9713847E-06 EQN NO 747
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 13
**START ITERATION NO. 14
**START ITERATION NO. 15

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 13
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.3609027E-06 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.5981744E+04 L-MATRIX DIAG= 9.4223651E-10 EQN NO 745
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2153137E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.6808135E+06 L-MATRIX DIAG= 2.9713847E-06 EQN NO 747
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 7
**START ITERATION NO. 8
**START ITERATION NO. 9
**START ITERATION NO. 10
**START ITERATION NO. 11
**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 1.8766150E-06 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 13

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 4.7390349E-06 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 14

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 3.4291297E-06 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.6808135E+06 L-MATRIX DIAG= 3.7555583E-06 EQN NO 747
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 15

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.5981744E+04 L-MATRIX DIAG= 9.4223651E-10 EQN NO 745
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2153137E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.6808135E+06 L-MATRIX DIAG= 2.9713847E-06 EQN NO 747
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 14
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.3609027E-06 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.5981744E+04 L-MATRIX DIAG= 9.4223651E-10 EQN NO 745
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2153137E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.6808135E+06 L-MATRIX DIAG= 2.9713847E-06 EQN NO 747
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 7
**START ITERATION NO. 8
**START ITERATION NO. 9
**START ITERATION NO. 10
**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.3609027E-06 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

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 **START ITERATION NO.          12

***WARNING -  INSTABILITY AT JOINT   586   DIRECTION = FY
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.5981744E+04 L-MATRIX DIAG=  9.4223651E-10 EQN NO     745
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.


***WARNING -  INSTABILITY AT JOINT   586   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.7111695E+06 L-MATRIX DIAG=  2.2153137E-05 EQN NO     746
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING -  INSTABILITY AT JOINT   586   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.6808135E+06 L-MATRIX DIAG=  2.9713847E-06 EQN NO     747
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
 **START ITERATION NO.          13
 **START ITERATION NO.          14
 **START ITERATION NO.          15

 **WARNING-Tension/Compression did not converge after 15 iterations, Case=   15
 ** If results are not satisfactory then enter a
 ** SET ITERLIM i command (i>10) before loads to get more iterations.



 **START ITERATION NO.           2

***WARNING -  INSTABILITY AT JOINT   586   DIRECTION = FY
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.5981744E+04 L-MATRIX DIAG=  9.4223651E-10 EQN NO     745
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.


***WARNING -  INSTABILITY AT JOINT   586   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.7111695E+06 L-MATRIX DIAG=  2.2153137E-05 EQN NO     746
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
```



```
***WARNING -  INSTABILITY AT JOINT   586   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.6808135E+06 L-MATRIX DIAG=  2.9713847E-06 EQN NO     747
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

 **NOTE-Tension/Compression converged after  2 iterations, Case=   16



 **START ITERATION NO.           2

***WARNING -  INSTABILITY AT JOINT   586   DIRECTION = FY
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.5981744E+04 L-MATRIX DIAG=  9.4223651E-10 EQN NO     745
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.


***WARNING -  INSTABILITY AT JOINT   586   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.7111695E+06 L-MATRIX DIAG=  2.2153137E-05 EQN NO     746
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING -  INSTABILITY AT JOINT   586   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.6808135E+06 L-MATRIX DIAG=  2.9713847E-06 EQN NO     747
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

 **NOTE-Tension/Compression converged after  2 iterations, Case=   17



 **START ITERATION NO.           2
 **START ITERATION NO.           3

***WARNING -  INSTABILITY AT JOINT   586   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.7111695E+06 L-MATRIX DIAG=  2.3609027E-06 EQN NO     746
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.
```

**START ITERATION NO. 4

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.5981744E+04 L-MATRIX DIAG= 9.4223651E-10 EQN NO 745
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2153137E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.6808135E+06 L-MATRIX DIAG= 2.9713847E-06 EQN NO 747
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 4 iterations, Case= 18

**START ITERATION NO. 2
**START ITERATION NO. 3

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.3609027E-06 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 4

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.5981744E+04 L-MATRIX DIAG= 9.4223651E-10 EQN NO 745
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2153137E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.6808135E+06 L-MATRIX DIAG= 2.9713847E-06 EQN NO 747
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 4 iterations, Case= 19

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2155931E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2155466E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2156630E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2151042E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 9 iterations, Case= 20

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.3609027E-06 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 1.0314398E-07 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 6.9476664E-07 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 1.2805685E-06 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 9 iterations, Case= 21

**START ITERATION NO. 2

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.5981744E+04 L-MATRIX DIAG= 9.4223651E-10 EQN NO 745
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.7111695E+06 L-MATRIX DIAG= 2.2153137E-05 EQN NO 746
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 586 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.6808135E+06 L-MATRIX DIAG= 2.9713847E-06 EQN NO 747
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 2 iterations, Case= 22

**NOTE-Tension/Compression converged after 1 iterations, Case= 23

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

**NOTE-Tension/Compression converged after 1 iterations, Case= 211

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

**NOTE-Tension/Compression converged after 1 iterations, Case= 214

**NOTE-Tension/Compression converged after 1 iterations, Case= 215

**NOTE-Tension/Compression converged after 1 iterations, Case= 217

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

**NOTE-Tension/Compression converged after 1 iterations, Case= 219

**NOTE-Tension/Compression converged after 1 iterations, Case= 220

**NOTE-Tension/Compression converged after 1 iterations, Case= 221

**NOTE-Tension/Compression converged after 1 iterations, Case= 222

**NOTE-Tension/Compression converged after 1 iterations, Case= 223

**NOTE-Tension/Compression converged after 1 iterations, Case= 224

**NOTE-Tension/Compression converged after 1 iterations, Case= 225

**NOTE-Tension/Compression converged after 1 iterations, Case= 226

**NOTE-Tension/Compression converged after 1 iterations, Case= 227

**NOTE-Tension/Compression converged after 1 iterations, Case= 228

**NOTE-Tension/Compression converged after 1 iterations, Case= 229

**NOTE-Tension/Compression converged after 1 iterations, Case= 231

**NOTE-Tension/Compression converged after 1 iterations, Case= 232

**NOTE-Tension/Compression converged after 1 iterations, Case= 233

**NOTE-Tension/Compression converged after 1 iterations, Case= 234

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 235

**NOTE-Tension/Compression converged after 1 iterations, Case= 236

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 237

**NOTE-Tension/Compression converged after 1 iterations, Case= 238

**NOTE-Tension/Compression converged after 1 iterations, Case= 239

**NOTE-Tension/Compression converged after 1 iterations, Case= 240

**NOTE-Tension/Compression converged after 1 iterations, Case= 241

**NOTE-Tension/Compression converged after 1 iterations, Case= 242

**NOTE-Tension/Compression converged after 1 iterations, Case= 243

**NOTE-Tension/Compression converged after 1 iterations, Case= 244

**NOTE-Tension/Compression converged after 1 iterations, Case= 245

**NOTE-Tension/Compression converged after 1 iterations, Case= 246

**NOTE-Tension/Compression converged after 1 iterations, Case= 247

**NOTE-Tension/Compression converged after 1 iterations, Case= 248

**NOTE-Tension/Compression converged after 1 iterations, Case= 249

**NOTE-Tension/Compression converged after 1 iterations, Case= 250

**NOTE-Tension/Compression converged after 1 iterations, Case= 251

**NOTE-Tension/Compression converged after 1 iterations, Case= 252

**NOTE-Tension/Compression converged after 1 iterations, Case= 253

**NOTE-Tension/Compression converged after 1 iterations, Case= 254

**NOTE-Tension/Compression converged after 1 iterations, Case= 255

**NOTE-Tension/Compression converged after 1 iterations, Case= 256

**NOTE-Tension/Compression converged after 1 iterations, Case= 258

**NOTE-Tension/Compression converged after 1 iterations, Case= 259

**NOTE-Tension/Compression converged after 1 iterations, Case= 260

**NOTE-Tension/Compression converged after 1 iterations, Case= 261

**NOTE-Tension/Compression converged after 1 iterations, Case= 262

**NOTE-Tension/Compression converged after 1 iterations, Case= 263

**NOTE-Tension/Compression converged after 1 iterations, Case= 264

**NOTE-Tension/Compression converged after 1 iterations, Case= 265

**NOTE-Tension/Compression converged after 1 iterations, Case= 266

**NOTE-Tension/Compression converged after 1 iterations, Case= 267

**NOTE-Tension/Compression converged after 1 iterations, Case= 268

**NOTE-Tension/Compression converged after 1 iterations, Case= 269

**NOTE-Tension/Compression converged after 1 iterations, Case= 270

**NOTE-Tension/Compression converged after 1 iterations, Case= 272

**NOTE-Tension/Compression converged after 1 iterations, Case= 273

**NOTE-Tension/Compression converged after 1 iterations, Case= 274

**NOTE-Tension/Compression converged after 1 iterations, Case= 275

**NOTE-Tension/Compression converged after 1 iterations, Case= 276

**NOTE-Tension/Compression converged after 1 iterations, Case= 277

**NOTE-Tension/Compression converged after 1 iterations, Case= 278

**NOTE-Tension/Compression converged after 1 iterations, Case= 279

**NOTE-Tension/Compression converged after 1 iterations, Case= 280

**NOTE-Tension/Compression converged after 1 iterations, Case= 281

**NOTE-Tension/Compression converged after 1 iterations, Case= 282

**NOTE-Tension/Compression converged after 1 iterations, Case= 283

**NOTE-Tension/Compression converged after 1 iterations, Case= 284

**NOTE-Tension/Compression converged after 1 iterations, Case= 286

**NOTE-Tension/Compression converged after 1 iterations, Case= 287

**NOTE-Tension/Compression converged after 1 iterations, Case= 288

**NOTE-Tension/Compression converged after 1 iterations, Case= 289

**NOTE-Tension/Compression converged after 1 iterations, Case= 290

**NOTE-Tension/Compression converged after 1 iterations, Case= 291

**NOTE-Tension/Compression converged after 1 iterations, Case= 292

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

**NOTE-Tension/Compression converged after 1 iterations, Case= 102

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

**NOTE-Tension/Compression converged after 1 iterations, Case= 104

**NOTE-Tension/Compression converged after 1 iterations, Case= 105

**NOTE-Tension/Compression converged after 1 iterations, Case= 106

**NOTE-Tension/Compression converged after 1 iterations, Case= 108

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

**NOTE-Tension/Compression converged after 1 iterations, Case= 111

**NOTE-Tension/Compression converged after 1 iterations, Case= 112

**NOTE-Tension/Compression converged after 1 iterations, Case= 113

**NOTE-Tension/Compression converged after 1 iterations, Case= 114

**NOTE-Tension/Compression converged after 1 iterations, Case= 115

**NOTE-Tension/Compression converged after 1 iterations, Case= 116

**NOTE-Tension/Compression converged after 1 iterations, Case= 117

**NOTE-Tension/Compression converged after 1 iterations, Case= 118

**NOTE-Tension/Compression converged after 1 iterations, Case= 119

**NOTE-Tension/Compression converged after 1 iterations, Case= 120

**NOTE-Tension/Compression converged after 1 iterations, Case= 122

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 123

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 124

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 125

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 126

**NOTE-Tension/Compression converged after 1 iterations, Case= 127

**NOTE-Tension/Compression converged after 1 iterations, Case= 128

**NOTE-Tension/Compression converged after 1 iterations, Case= 129

**NOTE-Tension/Compression converged after 1 iterations, Case= 130

**NOTE-Tension/Compression converged after 1 iterations, Case= 131

**NOTE-Tension/Compression converged after 1 iterations, Case= 132

**NOTE-Tension/Compression converged after 1 iterations, Case= 133

**NOTE-Tension/Compression converged after 1 iterations, Case= 134

**NOTE-Tension/Compression converged after 1 iterations, Case= 135

**NOTE-Tension/Compression converged after 1 iterations, Case= 136

**NOTE-Tension/Compression converged after 1 iterations, Case= 137

**NOTE-Tension/Compression converged after 1 iterations, Case= 138

**NOTE-Tension/Compression converged after 1 iterations, Case= 139

**NOTE-Tension/Compression converged after 1 iterations, Case= 140

**NOTE-Tension/Compression converged after 1 iterations, Case= 141

**NOTE-Tension/Compression converged after 1 iterations, Case= 142

**NOTE-Tension/Compression converged after 1 iterations, Case= 143

**NOTE-Tension/Compression converged after 1 iterations, Case= 144

**NOTE-Tension/Compression converged after 1 iterations, Case= 145

**NOTE-Tension/Compression converged after 1 iterations, Case= 146

**NOTE-Tension/Compression converged after 1 iterations, Case= 148

**NOTE-Tension/Compression converged after 1 iterations, Case= 149

**NOTE-Tension/Compression converged after 1 iterations, Case= 150

**NOTE-Tension/Compression converged after 1 iterations, Case= 151

**NOTE-Tension/Compression converged after 1 iterations, Case= 152

**NOTE-Tension/Compression converged after 1 iterations, Case= 153

**NOTE-Tension/Compression converged after 1 iterations, Case= 154

**NOTE-Tension/Compression converged after 1 iterations, Case= 155

**NOTE-Tension/Compression converged after 1 iterations, Case= 156

**NOTE-Tension/Compression converged after 1 iterations, Case= 157

**NOTE-Tension/Compression converged after 1 iterations, Case= 158

**NOTE-Tension/Compression converged after 1 iterations, Case= 159

**NOTE-Tension/Compression converged after 1 iterations, Case= 160

**NOTE-Tension/Compression converged after 1 iterations, Case= 162

**NOTE-Tension/Compression converged after 1 iterations, Case= 163

**NOTE-Tension/Compression converged after 1 iterations, Case= 164

**NOTE-Tension/Compression converged after 1 iterations, Case= 165

**NOTE-Tension/Compression converged after 1 iterations, Case= 166

**NOTE-Tension/Compression converged after 1 iterations, Case= 167

**NOTE-Tension/Compression converged after 1 iterations, Case= 168

**NOTE-Tension/Compression converged after 1 iterations, Case= 169

**NOTE-Tension/Compression converged after 1 iterations, Case= 170

**NOTE-Tension/Compression converged after 1 iterations, Case= 171

**NOTE-Tension/Compression converged after 1 iterations, Case= 172

**NOTE-Tension/Compression converged after 1 iterations, Case= 173

**NOTE-Tension/Compression converged after 1 iterations, Case= 174

**NOTE-Tension/Compression converged after 1 iterations, Case= 176

**NOTE-Tension/Compression converged after 1 iterations, Case= 177

**NOTE-Tension/Compression converged after 1 iterations, Case= 178

**NOTE-Tension/Compression converged after 1 iterations, Case= 179

**NOTE-Tension/Compression converged after 1 iterations, Case= 180

1181. LOAD LIST 101 TO 180
1182. START CONCRETE DESIGN

```
1183. CODE ACI
1184. TRACK 2 ALL
1185. FC 35000 MEMB 185
1186. CLB 0.05 MEMB 185
1187. CLS 0.05 ALL
1188. CLT 0.05 MEMB 185 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 -
1189. 372 393 TO 399 402 TO 409 412 TO 419 422 TO 429 432 497 TO 504 513 TO 520 -
1190. 568 TO 575 586 TO 593
1191. MAXMAIN 25 ALL
1192. MINMAIN 25 ALL
1193. MINSEC 10 ALL
1194. DESIGN COLUMN 185
```



```
================================================================

     COLUMN  NO.   185  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 850.0 X 500.0 MMS, TIED
       AREA OF STEEL REQUIRED =  4496.5  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

 12 - 25 MM               1.386      150      STA     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 400.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
14905.30  11924.24   5166.13   1075.52     208.2
   M0       P-tens.   Des.Pn    Des.Mn     e/h
 506.09  -2436.79     14.16      0.00    0.00000
 ------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
14905.30  11924.24   5535.96   1922.86     347.3
   M0       P-tens.   Des.Pn    Des.Mn     e/h
 909.55  -2436.79     14.16    665.85   23.51702
 ------------------------------------------------------
                          Pn       Mn       Pn       Mn    (@ Z )
          |            11006.99   706.83  5503.50  1068.58
       P0 |*           10089.74   816.19  4586.25  1052.42
          | *           9172.49   897.28  3669.00  1002.17
   Pn,max|__*          8255.24   962.68  2751.75   917.43
          |   *         7337.99  1011.18  1834.50   802.66
    Pn    |    *        6420.74  1044.76   917.25   672.49
   NOMINAL|     *         Pn       Mn       Pn       Mn    (@ Y )
    AXIAL |      *     11006.99  1245.42  5503.50  1921.72
COMPRESSION|       *    10089.74  1429.28  4586.25  1866.93
        Pb|-------*Mb   9172.49  1584.00  3669.00  1783.49
          |      *      8255.24  1705.93  2751.75  1640.56
__________|____*_______ 7337.99  1799.89  1834.50  1450.85
          |  * M0   Mn, 6420.74  1869.83   917.25  1214.05
          | *   BENDING
    P-tens|*     MOMENT
          |


 ********************END OF COLUMN DESIGN RESULTS********************


1195. END CONCRETE DESIGN
```

1196. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 18,2020 TIME= 17:59:29 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 9 STAAD OUTPUT – FOUNDATION F1A

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    OCT 18, 2020                 *
*          Time=    17: 3:43                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F1A.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 04/09/2020
    4. JOB NAME FDN. MKD. F1A
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. SET NL 200
   12. SET ITERLIM 15
   13. SET PRINT 1
   14. UNIT METER KN
   15. JOINT COORDINATES
   16. 2 0 911.8 3.175; 6 0.75 911.8 3.175; 10 -0.9 911.8 3.175; 34 -0.6 911.8 3.175
   17. 35 -0.3 911.8 3.175; 41 0.25 911.8 3.175; 42 0.5 911.8 3.175
   18. 128 0 913.8 3.175; 302 -0.6 911.8 3.49938; 303 -0.9 911.8 3.49938
   19. 304 -0.3 911.8 3.49938; 305 0 911.8 3.49938; 306 0.25 911.8 3.49938
   20. 307 0.5 911.8 3.49938; 308 0.75 911.8 3.49938; 309 0.95 911.8 3.175
   21. 313 0.95 911.8 3.49938; 316 -1.15 911.8 3.175; 320 -1.15 911.8 3.49938
   22. 321 -0.9 911.8 3.79938; 322 -0.6 911.8 3.79938; 323 -0.3 911.8 3.79938
   23. 324 0 911.8 3.79938; 325 0.25 911.8 3.79938; 326 0.5 911.8 3.79938
   24. 327 0.75 911.8 3.79938; 328 0.95 911.8 3.79938; 331 -1.15 911.8 3.79938
   25. 332 -0.9 911.8 4.04938; 333 -0.6 911.8 4.04938; 334 -0.3 911.8 4.04938
   26. 335 0 911.8 4.04938; 336 0.25 911.8 4.04938; 337 0.5 911.8 4.04938
   27. 338 0.75 911.8 4.04938; 339 0.95 911.8 4.04938; 342 -1.15 911.8 4.04938
   28. 343 -0.9 911.8 4.29938; 344 -0.6 911.8 4.29938; 345 -0.3 911.8 4.29938
   29. 346 0 911.8 4.29938; 347 0.25 911.8 4.29938; 348 0.5 911.8 4.29938
   30. 349 0.75 911.8 4.29938; 350 0.95 911.8 4.29938; 353 -1.15 911.8 4.29938
   31. 365 -0.9 911.8 2.04938; 366 -0.6 911.8 2.04938; 369 -0.3 911.8 2.04938
   32. 371 0 911.8 2.04938; 373 0.25 911.8 2.04938; 375 0.5 911.8 2.04938
   33. 377 0.75 911.8 2.04938; 379 0.95 911.8 2.04938; 384 -1.15 911.8 2.04938
   34. 387 -0.9 911.8 2.29938; 388 -0.6 911.8 2.29938; 389 -0.3 911.8 2.29938
   35. 390 0 911.8 2.29938; 391 0.25 911.8 2.29938; 392 0.5 911.8 2.29938
   36. 393 0.75 911.8 2.29938; 394 0.95 911.8 2.29938; 397 -1.15 911.8 2.29938
   37. 398 -0.9 911.8 2.54938; 399 -0.6 911.8 2.54938; 400 -0.3 911.8 2.54938
   38. 401 0 911.8 2.54938; 402 0.25 911.8 2.54938; 403 0.5 911.8 2.54938
```



```
   39. 404 0.75 911.8 2.54938; 405 0.95 911.8 2.54938; 408 -1.15 911.8 2.54938
   40. 409 -0.9 911.8 2.84938; 410 -0.6 911.8 2.84938; 411 -0.3 911.8 2.84938
   41. 412 0 911.8 2.84938; 413 0.25 911.8 2.84938; 414 0.5 911.8 2.84938
   42. 415 0.75 911.8 2.84938; 416 0.95 911.8 2.84938; 419 -1.15 911.8 2.84938
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   46. 429 0.25 911.8 1.54938; 430 0.5 911.8 1.79938; 431 0.5 911.8 1.54938
   47. 432 0.75 911.8 1.79938; 433 0.75 911.8 1.54938; 434 0.95 911.8 1.79938
   48. 435 0.95 911.8 1.54938; 436 -1.15 911.8 1.79938; 437 -1.15 911.8 1.54938
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   51. 444 0.75 911.8 4.54938; 445 0.95 911.8 4.54938; 446 -1.15 911.8 4.54938
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   54. 453 0.75 911.8 4.79938; 454 0.95 911.8 4.79938; 455 -1.15 911.8 4.79938
   55. 456 -0.6 911.8 1.29938; 457 -0.9 911.8 1.29938; 458 -0.3 911.8 1.29938
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   57. 462 0.75 911.8 1.29938; 463 0.95 911.8 1.29938; 464 -1.15 911.8 1.29938
   58. 465 -0.9 911.8 5.04938; 466 -0.6 911.8 5.04938; 467 -0.3 911.8 5.04938
   59. 468 0 911.8 5.04938; 469 0.25 911.8 5.04938; 470 0.5 911.8 5.04938
   60. 471 0.75 911.8 5.04938; 472 0.95 911.8 5.04938; 473 -1.15 911.8 5.04938
   61. 474 -0.6 911.8 1.04938; 475 -0.9 911.8 1.04938; 476 -0.3 911.8 1.04938
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   63. 480 0.75 911.8 1.04938; 481 0.95 911.8 1.04938; 482 -1.15 911.8 1.04938
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   66. 489 0.75 911.8 5.29938; 490 0.95 911.8 5.29938; 491 -1.15 911.8 5.29938
   67. 526 -0.6 911.8 0.67438; 527 -0.9 911.8 0.67438; 528 -0.3 911.8 0.67438
   68. 529 0 911.8 0.67438; 530 0.25 911.8 0.67438; 531 0.5 911.8 0.67438
   69. 532 0.75 911.8 0.67438; 533 0.95 911.8 0.67438; 534 -1.15 911.8 0.67438
   70. 548 -0.9 911.8 5.67438; 549 -0.6 911.8 5.67438; 550 -0.3 911.8 5.67438
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   72. 554 0.75 911.8 5.67438; 555 0.95 911.8 5.67438; 556 -1.15 911.8 5.67438
   73. MEMBER INCIDENCES
   74. 185 2 128
   75. ELEMENT INCIDENCES SHELL
   76. 326 303 302 34 10; 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2
   77. 333 306 307 42 41; 334 307 308 6 42; 337 308 313 309 6; 342 320 303 10 316
   78. 343 321 322 302 303; 344 322 323 304 302; 345 323 324 305 304
   79. 346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307
   80. 349 327 328 313 308; 352 331 321 303 320; 353 332 333 322 321
   81. 354 333 334 323 322; 355 334 335 324 323; 356 335 336 325 324
   82. 357 336 337 326 325; 358 337 338 327 326; 359 338 339 328 327
   83. 362 342 332 321 331; 363 343 344 333 332; 364 344 345 334 333
   84. 365 345 346 335 334; 366 346 347 336 335; 367 347 348 337 336
   85. 368 348 349 338 337; 369 349 350 339 338; 372 353 343 332 342
   86. 393 387 388 366 365; 394 388 389 369 366; 395 389 390 371 369
   87. 396 390 391 373 371; 397 391 392 375 373; 398 392 393 377 375
   88. 399 393 394 379 377; 402 397 387 365 384; 403 398 399 388 387
   89. 404 399 400 389 388; 405 400 401 390 389; 406 401 402 391 390
   90. 407 402 403 392 391; 408 403 404 393 392; 409 404 405 394 393
   91. 412 408 398 387 397; 413 409 410 399 398; 414 410 411 400 399
   92. 415 411 412 401 400; 416 412 413 402 401; 417 413 414 403 402
   93. 418 414 415 404 403; 419 415 416 405 404; 422 419 409 398 408
   94. 423 10 34 410 409; 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412
```

```
 95. 427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415; 432 316 10 409 419
 96. 433 420 421 422 423; 434 421 424 425 422; 435 424 426 427 425
 97. 436 426 428 429 427; 437 428 430 431 429; 438 430 432 433 431
 98. 439 432 434 435 433; 440 436 420 423 437; 441 365 366 421 420
 99. 442 366 369 424 421; 443 369 371 426 424; 444 371 373 428 426
100. 445 373 375 430 428; 446 375 377 432 430; 447 377 379 434 432
101. 448 384 365 420 436; 449 438 439 344 343; 450 439 440 345 344
102. 451 440 441 346 345; 452 441 442 347 346; 453 442 443 348 347
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104. 457 447 448 439 438; 458 448 449 440 439; 459 449 450 441 440
105. 460 450 451 442 441; 461 451 452 443 442; 462 452 453 444 443
106. 463 453 454 445 444; 464 455 447 438 446; 465 423 422 456 457
107. 466 422 425 458 456; 467 425 427 459 458; 468 427 429 460 459
108. 469 429 431 461 460; 470 431 433 462 461; 471 433 435 463 462
109. 472 437 423 457 464; 473 465 466 448 447; 474 466 467 449 448
110. 475 467 468 450 449; 476 468 469 451 450; 477 469 470 452 451
111. 478 470 471 453 452; 479 471 472 454 453; 480 473 465 447 455
112. 481 457 456 474 475; 482 456 458 476 474; 483 458 459 477 476
113. 484 459 460 478 477; 485 460 461 479 478; 486 461 462 480 479
114. 487 462 463 481 480; 488 464 457 475 482; 489 483 484 466 465
115. 490 484 485 467 466; 491 485 486 468 467; 492 486 487 469 468
116. 493 487 488 470 469; 494 488 489 471 470; 495 489 490 472 471
117. 496 491 483 465 473; 529 475 474 526 527; 530 474 476 528 526
118. 531 476 477 529 528; 532 477 478 530 529; 533 478 479 531 530
119. 534 479 480 532 531; 535 480 481 533 532; 536 482 475 527 534
120. 549 548 549 484 483; 550 549 550 485 484; 551 550 551 486 485
121. 552 551 552 487 486; 553 552 553 488 487; 554 553 554 489 488
122. 555 554 555 490 489; 556 556 548 483 491
123. START GROUP DEFINITION
124. ELEMENT
125. _RAFT 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 -
126. 393 TO 399 402 TO 409 412 TO 419 422 TO 429 432 TO 496 529 TO 536 -
127. 549 TO 556
128. _RAFT_MOM 326 327 333 334 337 342 TO 349 352 TO 359 362 TO 369 372 -
129. 393 TO 399 402 TO 409 412 TO 419 422 TO 424 427 TO 429 432 TO 496 -
130. 529 TO 536 549 TO 556
131. _RAFT_SHEAR 326 337 342 343 349 352 353 359 362 TO 369 372 393 TO 399 402 -
132. 403 409 412 413 419 422 423 429 432 TO 496 529 TO 536 549 TO 556
133. MEMBER
134. END GROUP DEFINITION
135. ELEMENT PROPERTY
136. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
137. 402 TO 409 412 TO 419 422 TO 429 432 TO 496 529 TO 536 549 TO 555 -
138. 556 THICKNESS 0.6
139. DEFINE MATERIAL START
140. ISOTROPIC CONCRETE
141. E 2.17185E+007
142. POISSON 0.17
143. DENSITY 25
144. ALPHA 1E-005
145. DAMP 0.05
146. TYPE CONCRETE
147. STRENGTH FCU 27579
148. END DEFINE MATERIAL
149. MEMBER PROPERTY AMERICAN
150. 185 PRIS YD 0.5 ZD 0.85
```



```
151. CONSTANTS
152. MATERIAL CONCRETE ALL
153. SUPPORTS
154. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
155. 402 TO 409 412 TO 419 422 TO 429 432 TO 496 529 TO 536 549 TO 555 -
156. 556 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
```

JOINT	INFLUENCE AREA
2	0.089375
6	0.073125
10	0.089375
34	0.097500
35	0.097500
41	0.081250
42	0.081250
302	0.093657
303	0.085852
304	0.093657
305	0.085852
306	0.078048
307	0.078048
308	0.070243
309	0.032500
313	0.031219
316	0.040625
320	0.039024
321	0.075625
322	0.082500
323	0.082500
324	0.075625
325	0.068750
326	0.068750
327	0.061875
328	0.027500
331	0.034375
332	0.068750
333	0.075000
334	0.075000
335	0.068750
336	0.062500
337	0.062500
338	0.056250
339	0.025000
342	0.031250
343	0.068750
344	0.075000
345	0.075000
346	0.068750
347	0.062500
348	0.062500
349	0.056250
350	0.025000
353	0.031250
365	0.068750

STAAD SPACE -- PAGE NO. 5

366	0.075000
369	0.075000
371	0.068750
373	0.062500
375	0.062500
377	0.056250
379	0.025000
384	0.031250
387	0.068750
388	0.075000
389	0.075000
390	0.068750
391	0.062500
392	0.062500
393	0.056250
394	0.025000
397	0.031250
398	0.075625
399	0.082500
400	0.082500
401	0.075625
402	0.068750
403	0.068750
404	0.061875
405	0.027500
408	0.034375
409	0.086023
410	0.093843
411	0.093843
412	0.086023
413	0.078202
414	0.078202
415	0.070382
416	0.031281
419	0.039101
420	0.068750
421	0.075000
422	0.075000
423	0.068750
424	0.075000
425	0.075000
426	0.068750
427	0.068750
428	0.062500
429	0.062500
430	0.062500
431	0.062500
432	0.056250
433	0.056250
434	0.025000
435	0.025000
436	0.031250
437	0.031250
438	0.068750
439	0.075000
440	0.075000

STAAD SPACE -- PAGE NO. 6

441	0.068750
442	0.062500
443	0.062500
444	0.056250
445	0.025000
446	0.031250
447	0.068750
448	0.075000
449	0.075000
450	0.068750
451	0.062500
452	0.062500
453	0.056250
454	0.025000
455	0.031250
456	0.075000
457	0.068750
458	0.075000
459	0.068750
460	0.062500
461	0.062500
462	0.056250
463	0.025000
464	0.031250
465	0.068750
466	0.075000
467	0.075000
468	0.068750
469	0.062500
470	0.062500
471	0.056250
472	0.025000
473	0.031250
474	0.093750
475	0.085937
476	0.093750
477	0.085937
478	0.078125
479	0.078125
480	0.070312
481	0.031250
482	0.039062
483	0.085937
484	0.093750
485	0.093750
486	0.085937
487	0.078125
488	0.078125
489	0.070312
490	0.031250
491	0.039062
526	0.056250
527	0.051562
528	0.056250
529	0.051563
530	0.046875

```
       531          0.046875
       532          0.042187
       533          0.018750
       534          0.023437
       548          0.051562
       549          0.056250
       550          0.056250
       551          0.051562
       552          0.046875
       553          0.046875
       554          0.042187
       555          0.018750
       556          0.023437
    TOTAL          10.499997
157. ********************************************************************************
158. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
159. SELFWEIGHT Y -1
160. ELEMENT LOAD
161. * LOAD DUE TO BACKFILL
162. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
163. 402 TO 409 412 TO 419 422 TO 429 432 TO 496 529 TO 536 549 TO 555 -
164. 556 PR GY -22
165. * LOAD DUE TO SURCHARGE
166. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
167. 402 TO 409 412 TO 419 422 TO 429 432 TO 496 529 TO 536 549 TO 555 -
168. 556 PR GY -20
169. JOINT LOAD
170. 128 FY -22 FZ -3 MX -23
171. ********************************************************************************
172. LOAD 2 LOADTYPE NONE  TITLE ROOF LIVE LOAD ( LR )
173. JOINT LOAD
174. 128 FY -61 FZ -2 MX -10
175. ********************************************************************************
176. LOAD 3 LOADTYPE NONE  TITLE LIVE LOAD ( L )
177. JOINT LOAD
178. 128 FY -0.001
179. ********************************************************************************
180. LOAD 4 LOADTYPE NONE  TITLE CRANE LOAD  ( CR1 )
181. JOINT LOAD
182. 128 FY -35 FZ -9 MX -40
183. ********************************************************************************
184. LOAD 5 LOADTYPE NONE  TITLE CRANE LOAD  ( CR2 )
185. JOINT LOAD
186. 128 FY -26 FZ -7 MX -33
187. ********************************************************************************
188. LOAD 6 LOADTYPE NONE  TITLE CRANE LOAD  ( CR3 )
189. JOINT LOAD
190. 128 FY -0.0001
191. ********************************************************************************
192. LOAD 7 LOADTYPE NONE  TITLE CRANE LOAD  ( CR4 )
193. JOINT LOAD
194. 128 FY -0.0001
195. ********************************************************************************
196. LOAD 8 LOADTYPE NONE  TITLE WIND LOAD  ( WL1 )
197. JOINT LOAD
198. 128 FY 117 FZ -24 MX -193
```



```
199. ********************************************************************************
200. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD  ( WL2 )
201. JOINT LOAD
202. 128 FY 74 FZ -27 MX -207
203. ********************************************************************************
204. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD  ( WL3 )
205. JOINT LOAD
206. 128 FY 117 FZ -24 MX -193
207. ********************************************************************************
208. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD  ( WL4 )
209. JOINT LOAD
210. 128 FY 74 FZ -27 MX -207
211. ********************************************************************************
212. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD  ( WL5 )
213. JOINT LOAD
214. 128 FY 105 FZ 38 MX 270
215. ********************************************************************************
216. LOAD 13 LOADTYPE NONE  TITLE WIND LOAD  ( WL6 )
217. JOINT LOAD
218. 128 FY 62 FZ 35 MX 256
219. ********************************************************************************
220. LOAD 14 LOADTYPE NONE  TITLE WIND LOAD  ( WL7 )
221. JOINT LOAD
222. 128 FY 105 FZ 38 MX 270
223. ********************************************************************************
224. LOAD 15 LOADTYPE NONE  TITLE WIND LOAD  ( WL8 )
225. JOINT LOAD
226. 128 FY 62 FZ 35 MX 256
227. ********************************************************************************
228. LOAD 16 LOADTYPE NONE  TITLE WIND LOAD  ( WL9 )
229. JOINT LOAD
230. 128 FY 119 FZ 4 MX 22
231. ********************************************************************************
232. LOAD 17 LOADTYPE NONE  TITLE WIND LOAD  ( WL10 )
233. JOINT LOAD
234. 128 FY 76 FZ 2 MX 8
235. ********************************************************************************
236. LOAD 18 LOADTYPE NONE  TITLE WIND LOAD  ( WL11 )
237. JOINT LOAD
238. 128 FY 119 FZ 11 MX 63
239. ********************************************************************************
240. LOAD 19 LOADTYPE NONE  TITLE WIND LOAD  ( WL12 )
241. JOINT LOAD
242. 128 FY 75 FZ 8 MX 48
243. ********************************************************************************
244. LOAD 20 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
245. JOINT LOAD
246. 128 FZ -1 MX -7
247. ********************************************************************************
248. LOAD 21 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
249. JOINT LOAD
250. 128 FZ 1 MX 7
251. ********************************************************************************
252. LOAD 22 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL1 )
253. JOINT LOAD
254. 128 FY 2
```

```
255. ********************************************************************************
256. LOAD 23 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL2 )
257. JOINT LOAD
258. 128 FY -2
259. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 172 NUMBER OF MEMBERS 1
NUMBER OF PLATES 144 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 171

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 79/ 16/ 51 DOF
TOTAL PRIMARY LOAD CASES = 23, TOTAL DEGREES OF FREEDOM = 519
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 27 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.2/ 142665.0 MB

```
260. CHANGE
261. SUPPORTS
262. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
263. 402 TO 409 412 TO 419 422 TO 429 432 TO 496 529 TO 536 549 TO 555 -
264. 556 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
265. ********************************************************************************
266. * UNFACTORED LOAD COMBINATIONS
267. LOAD 201 DL
268. REPEAT LOAD
269. 1 1.0
270. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

271. CHANGE
272. LOAD 202 DL+L+TL1
273. REPEAT LOAD
274. 1 1.0 3 1.0 22 1.0
275. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after 1 iterations, Case= 202

276. CHANGE
277. LOAD 203 DL+L+TL2
278. REPEAT LOAD
279. 1 1.0 3 1.0 23 1.0
280. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

281. CHANGE
282. LOAD 204 DL+LR
283. REPEAT LOAD
284. 1 1.0 2 1.0
285. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

286. CHANGE
287. LOAD 205 DL+0.75L+0.75TL1+0.75LR
288. REPEAT LOAD
289. 1 1.0 2 0.75 3 0.75 22 0.75
290. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

291. CHANGE
292. LOAD 206 DL+0.75L+0.75TL2+0.75LR
293. REPEAT LOAD
294. 1 1.0 2 0.75 3 0.75 23 0.75
295. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 206
```

```
 296. CHANGE
 297. LOAD 207 DL+0.6WL1
 298. REPEAT LOAD
 299. 1 1.0 8 0.6
 300. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  207



 301. CHANGE
 302. LOAD 208 DL+WL2
 303. REPEAT LOAD
 304. 1 1.0 9 1.0
 305. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  208



 306. CHANGE
 307. LOAD 209 DL+WL3
 308. REPEAT LOAD
 309. 1 1.0 10 1.0
 310. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  209



 311. CHANGE
 312. LOAD 210 DL+WL4
 313. REPEAT LOAD
 314. 1 1.0 11 1.0
 315. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  210



 316. CHANGE
 317. LOAD 211 DL+WL5
 318. REPEAT LOAD
```



```
 319. 1 1.0 12 1.0
 320. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  211



 321. CHANGE
 322. LOAD 212 DL+WL6
 323. REPEAT LOAD
 324. 1 1.0 13 1.0
 325. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  212



 326. CHANGE
 327. LOAD 213 DL+WL7
 328. REPEAT LOAD
 329. 1 1.0 14 1.0
 330. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  213



 331. CHANGE
 332. LOAD 214 DL+WL8
 333. REPEAT LOAD
 334. 1 1.0 15 1.0
 335. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  214



 336. CHANGE
 337. LOAD 215 DL+WL9
 338. REPEAT LOAD
 339. 1 1.0 16 1.0
 340. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 215

341. CHANGE
342. LOAD 216 DL+WL10
343. REPEAT LOAD
344. 1 1.0 17 1.0
345. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 216

346. CHANGE
347. LOAD 217 DL+WL11
348. REPEAT LOAD
349. 1 1.0 18 1.0
350. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 217

351. CHANGE
352. LOAD 218 DL+WL12
353. REPEAT LOAD
354. 1 1.0 19 1.0
355. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

356. CHANGE
357. LOAD 219 DL+0.75WL1+0.75L+0.75LR
358. REPEAT LOAD
359. 1 1.0 2 0.75 3 0.75 8 0.75
360. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 219

361. CHANGE
362. LOAD 220 DL+0.75WL2+0.75L+0.75LR
363. REPEAT LOAD
364. 1 1.0 2 0.75 3 0.75 9 0.75
365. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 220

366. CHANGE
367. LOAD 221 DL+0.75WL3+0.75L+0.75LR
368. REPEAT LOAD
369. 1 1.0 2 0.75 3 0.75 10 0.75
370. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 221

371. CHANGE
372. LOAD 222 DL+0.75WL4+0.75L+0.75LR
373. REPEAT LOAD
374. 1 1.0 2 0.75 3 0.75 11 0.75
375. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 222

376. CHANGE
377. LOAD 223 DL+0.75WL5+0.75L+0.75LR
378. REPEAT LOAD
379. 1 1.0 2 0.75 3 0.75 12 0.75
380. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 223

```
 381. CHANGE
 382. LOAD 224 DL+0.75WL67+0.75L+0.75LR
 383. REPEAT LOAD
 384. 1 1.0 2 0.75 3 0.75 13 0.75
 385. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  224



 386. CHANGE
 387. LOAD 225 DL+0.75WL7+0.75L+0.75LR
 388. REPEAT LOAD
 389. 1 1.0 2 0.75 3 0.75 14 0.75
 390. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  225



 391. CHANGE
 392. LOAD 226 DL+0.75WL8+0.75L+0.75LR
 393. REPEAT LOAD
 394. 1 1.0 2 0.75 3 0.75 15 0.75
 395. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  226



 396. CHANGE
 397. LOAD 227 DL+0.75WL9+0.75L+0.75LR
 398. REPEAT LOAD
 399. 1 1.0 2 0.75 3 0.75 16 0.75
 400. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  227



 401. CHANGE
 402. LOAD 228 DL+0.75WL10+0.75L+0.75LR
 403. REPEAT LOAD
 404. 1 1.0 2 0.75 3 0.75 17 0.75
```



```
 405. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  228



 406. CHANGE
 407. LOAD 229 DL+0.75WL11+0.75L+0.75LR
 408. REPEAT LOAD
 409. 1 1.0 2 0.75 3 0.75 18 0.75
 410. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  229



 411. CHANGE
 412. LOAD 230 DL+0.75WL12+0.75L+0.75LR
 413. REPEAT LOAD
 414. 1 1.0 2 0.75 3 0.75 19 0.75
 415. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  230



 416. CHANGE
 417. LOAD 231 0.6DL+WL1
 418. REPEAT LOAD
 419. 1 0.6 8 1.0
 420. PERFORM ANALYSIS
**START ITERATION NO.           2

**NOTE-Tension/Compression converged after  2 iterations, Case=  231



 421. CHANGE
 422. LOAD 232 0.6DL+WL2
 423. REPEAT LOAD
 424. 1 0.6 9 1.0
 425. PERFORM ANALYSIS
```

```
**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=  232



 426. CHANGE
 427. LOAD 233 0.6DL+WL3
 428. REPEAT LOAD
 429. 1 0.6 10 1.0
 430. PERFORM ANALYSIS
**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=  233



 431. CHANGE
 432. LOAD 234 0.6DL+WL4
 433. REPEAT LOAD
 434. 1 0.6 11 1.0
 435. PERFORM ANALYSIS
**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=  234



 436. CHANGE
 437. LOAD 235 0.6DL+WL5
 438. REPEAT LOAD
 439. 1 0.6 12 1.0
 440. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  235



 441. CHANGE
 442. LOAD 236 0.6DL+WL6
 443. REPEAT LOAD
```



```
 444. 1 0.6 13 1.0
 445. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  236



 446. CHANGE
 447. LOAD 237 0.6DL+WL7
 448. REPEAT LOAD
 449. 1 0.6 14 1.0
 450. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  237



 451. CHANGE
 452. LOAD 238 0.6DL+WL8
 453. REPEAT LOAD
 454. 1 0.6 15 1.0
 455. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  238



 456. CHANGE
 457. LOAD 239 0.6DL+WL9
 458. REPEAT LOAD
 459. 1 0.6 16 1.0
 460. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  239
```

```
 461. CHANGE
 462. LOAD 240 0.6DL+WL10
 463. REPEAT LOAD
 464. 1 0.6 17 1.0
 465. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  240



 466. CHANGE
 467. LOAD 241 0.6DL+WL11
 468. REPEAT LOAD
 469. 1 0.6 18 1.0
 470. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  241



 471. CHANGE
 472. LOAD 242 0.6DL+WL12
 473. REPEAT LOAD
 474. 1 0.6 19 1.0
 475. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  242



 476. CHANGE
 477. LOAD 243 0.6DL+0.7SL1
 478. REPEAT LOAD
 479. 1 0.6 20 0.7
 480. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  243



 481. CHANGE
 482. LOAD 244 0.6DL+0.7SL2
 483. REPEAT LOAD
 484. 1 0.6 21 0.7
```



```
 485. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  244



 486. CHANGE
 487. LOAD 245 DL+0.75WL1+0.75L+0.75LR+CR1
 488. REPEAT LOAD
 489. 1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
 490. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  245



 491. CHANGE
 492. LOAD 246 DL+0.75WL2+0.75L+0.75LR+CR1
 493. REPEAT LOAD
 494. 1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
 495. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  246



 496. CHANGE
 497. LOAD 247 DL+0.75WL3+0.75L+0.75LR+CR1
 498. REPEAT LOAD
 499. 1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
 500. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  247



 501. CHANGE
 502. LOAD 248 DL+0.75WL4+0.75L+0.75LR+CR1
 503. REPEAT LOAD
 504. 1 1.0 2 0.75 3 0.75 11 0.75 4 1.0
 505. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 248

506. CHANGE
507. LOAD 249 DL+0.75WL5+0.75L+0.75LR+CR1
508. REPEAT LOAD
509. 1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
510. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 249

511. CHANGE
512. LOAD 250 DL+0.75WL67+0.75L+0.75LR+CR1
513. REPEAT LOAD
514. 1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
515. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 250

516. CHANGE
517. LOAD 251 DL+0.75WL7+0.75L+0.75LR+CR1
518. REPEAT LOAD
519. 1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
520. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 251

521. CHANGE
522. LOAD 252 DL+0.75WL8+0.75L+0.75LR+CR1
523. REPEAT LOAD
524. 1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
525. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 252

526. CHANGE
527. LOAD 253 DL+0.75WL9+0.75L+0.75LR+CR1
528. REPEAT LOAD
529. 1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
530. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 253

531. CHANGE
532. LOAD 254 DL+0.75WL10+0.75L+0.75LR+CR1
533. REPEAT LOAD
534. 1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
535. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 254

536. CHANGE
537. LOAD 255 DL+0.75WL11+0.75L+0.75LR+CR1
538. REPEAT LOAD
539. 1 1.0 2 0.75 3 0.75 18 0.75 4 1.0
540. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 255

541. CHANGE
542. LOAD 256 DL+0.75WL12+0.75L+0.75LR+CR1
543. REPEAT LOAD
544. 1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
545. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 256

546. CHANGE
547. LOAD 257 DL+0.75WL1+0.75L+0.75LR+CR2
548. REPEAT LOAD

```
549. 1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
550. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  257


551. CHANGE
552. LOAD 258 DL+0.75WL2+0.75L+0.75LR+CR2
553. REPEAT LOAD
554. 1 1.0 2 0.75 3 0.75 9 0.75 5 1.0
555. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  258


556. CHANGE
557. LOAD 259 DL+0.75WL3+0.75L+0.75LR+CR2
558. REPEAT LOAD
559. 1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
560. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  259


561. CHANGE
562. LOAD 260 DL+0.75WL4+0.75L+0.75LR+CR2
563. REPEAT LOAD
564. 1 1.0 2 0.75 3 0.75 11 0.75 5 1.0
565. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  260


566. CHANGE
567. LOAD 261 DL+0.75WL5+0.75L+0.75LR+CR2
568. REPEAT LOAD
569. 1 1.0 2 0.75 3 0.75 12 0.75 5 1.0
570. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  261


571. CHANGE
572. LOAD 262 DL+0.75WL67+0.75L+0.75LR+CR2
573. REPEAT LOAD
574. 1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
575. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  262


576. CHANGE
577. LOAD 263 DL+0.75WL7+0.75L+0.75LR+CR2
578. REPEAT LOAD
579. 1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
580. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  263


581. CHANGE
582. LOAD 264 DL+0.75WL8+0.75L+0.75LR+CR2
583. REPEAT LOAD
584. 1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
585. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  264


586. CHANGE
587. LOAD 265 DL+0.75WL9+0.75L+0.75LR+CR2
588. REPEAT LOAD
589. 1 1.0 2 0.75 3 0.75 16 0.75 5 1.0
590. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 265

591. CHANGE
592. LOAD 266 DL+0.75WL10+0.75L+0.75LR+CR2
593. REPEAT LOAD
594. 1 1.0 2 0.75 3 0.75 17 0.75 5 1.0
595. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 266

596. CHANGE
597. LOAD 267 DL+0.75WL11+0.75L+0.75LR+CR2
598. REPEAT LOAD
599. 1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
600. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 267

601. CHANGE
602. LOAD 268 DL+0.75WL12+0.75L+0.75LR+CR2
603. REPEAT LOAD
604. 1 1.0 2 0.75 3 0.75 19 0.75 5 1.0
605. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 268

606. CHANGE
607. LOAD 269 DL+0.75WL1+0.75L+0.75LR+CR3
608. REPEAT LOAD
609. 1 1.0 2 0.75 3 0.75 8 0.75 6 1.0
610. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 269

611. CHANGE
612. LOAD 270 DL+0.75WL2+0.75L+0.75LR+CR3
613. REPEAT LOAD
614. 1 1.0 2 0.75 3 0.75 9 0.75 6 1.0
615. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 270

616. CHANGE
617. LOAD 271 DL+0.75WL3+0.75L+0.75LR+CR3
618. REPEAT LOAD
619. 1 1.0 2 0.75 3 0.75 10 0.75 6 1.0
620. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 271

621. CHANGE
622. LOAD 272 DL+0.75WL4+0.75L+0.75LR+CR3
623. REPEAT LOAD
624. 1 1.0 2 0.75 3 0.75 11 0.75 6 1.0
625. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 272

626. CHANGE
627. LOAD 273 DL+0.75WL5+0.75L+0.75LR+CR3
628. REPEAT LOAD
629. 1 1.0 2 0.75 3 0.75 12 0.75 6 1.0
630. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 273

631. CHANGE
632. LOAD 274 DL+0.75WL67+0.75L+0.75LR+CR3
633. REPEAT LOAD
634. 1 1.0 2 0.75 3 0.75 13 0.75 6 1.0

```
 635. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  274


 636. CHANGE
 637. LOAD 275 DL+0.75WL7+0.75L+0.75LR+CR3
 638. REPEAT LOAD
 639. 1 1.0 2 0.75 3 0.75 14 0.75 6 1.0
 640. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  275


 641. CHANGE
 642. LOAD 276 DL+0.75WL8+0.75L+0.75LR+CR3
 643. REPEAT LOAD
 644. 1 1.0 2 0.75 3 0.75 15 0.75 6 1.0
 645. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  276


 646. CHANGE
 647. LOAD 277 DL+0.75WL9+0.75L+0.75LR+CR3
 648. REPEAT LOAD
 649. 1 1.0 2 0.75 3 0.75 16 0.75 6 1.0
 650. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  277


 651. CHANGE
 652. LOAD 278 DL+0.75WL10+0.75L+0.75LR+CR3
 653. REPEAT LOAD
 654. 1 1.0 2 0.75 3 0.75 17 0.75 6 1.0
 655. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  278


 656. CHANGE
 657. LOAD 279 DL+0.75WL11+0.75L+0.75LR+CR3
 658. REPEAT LOAD
 659. 1 1.0 2 0.75 3 0.75 18 0.75 6 1.0
 660. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  279


 661. CHANGE
 662. LOAD 280 DL+0.75WL12+0.75L+0.75LR+CR3
 663. REPEAT LOAD
 664. 1 1.0 2 0.75 3 0.75 19 0.75 6 1.0
 665. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  280


 666. CHANGE
 667. LOAD 281 DL+0.75WL1+0.75L+0.75LR+CR4
 668. REPEAT LOAD
 669. 1 1.0 2 0.75 3 0.75 8 0.75 7 1.0
 670. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  281


 671. CHANGE
 672. LOAD 282 DL+0.75WL2+0.75L+0.75LR+CR4
 673. REPEAT LOAD
 674. 1 1.0 2 0.75 3 0.75 9 0.75 7 1.0
 675. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  282
```

```
676. CHANGE
677. LOAD 283 DL+0.75WL3+0.75L+0.75LR+CR4
678. REPEAT LOAD
679. 1 1.0 2 0.75 3 0.75 10 0.75 7 1.0
680. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  283



681. CHANGE
682. LOAD 284 DL+0.75WL4+0.75L+0.75LR+CR4
683. REPEAT LOAD
684. 1 1.0 2 0.75 3 0.75 11 0.75 7 1.0
685. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  284



686. CHANGE
687. LOAD 285 DL+0.75WL5+0.75L+0.75LR+CR4
688. REPEAT LOAD
689. 1 1.0 2 0.75 3 0.75 12 0.75 7 1.0
690. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  285



691. CHANGE
692. LOAD 286 DL+0.75WL67+0.75L+0.75LR+CR4
693. REPEAT LOAD
694. 1 1.0 2 0.75 3 0.75 13 0.75 7 1.0
695. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  286



696. CHANGE
697. LOAD 287 DL+0.75WL7+0.75L+0.75LR+CR4
698. REPEAT LOAD
```



```
699. 1 1.0 2 0.75 3 0.75 14 0.75 7 1.0
700. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  287



701. CHANGE
702. LOAD 288 DL+0.75WL8+0.75L+0.75LR+CR4
703. REPEAT LOAD
704. 1 1.0 2 0.75 3 0.75 15 0.75 7 1.0
705. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  288



706. CHANGE
707. LOAD 289 DL+0.75WL9+0.75L+0.75LR+CR4
708. REPEAT LOAD
709. 1 1.0 2 0.75 3 0.75 16 0.75 7 1.0
710. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  289



711. CHANGE
712. LOAD 290 DL+0.75WL10+0.75L+0.75LR+CR4
713. REPEAT LOAD
714. 1 1.0 2 0.75 3 0.75 17 0.75 7 1.0
715. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  290



716. CHANGE
717. LOAD 291 DL+0.75WL11+0.75L+0.75LR+CR4
718. REPEAT LOAD
719. 1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
720. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 291

```
721. CHANGE
722. LOAD 292 DL+0.75WL12+0.75L+0.75LR+CR4
723. REPEAT LOAD
724. 1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
725. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 292

```
726. CHANGE
727. ******************************************************************************
728. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
729. LOAD 101 1.4DL
730. REPEAT LOAD
731. 1 1.4
732. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

```
733. CHANGE
734. LOAD 102 1.4DL+1.4TL1+1.7L+0.5LR
735. REPEAT LOAD
736. 1 1.4 22 1.4 3 1.7 2 0.5
737. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 102

```
738. CHANGE
739. LOAD 103 1.4DL+1.4TL2+1.7L+0.5LR
740. REPEAT LOAD
741. 1 1.4 23 1.4 3 1.7 2 0.5
742. PERFORM ANALYSIS
```


**NOTE-Tension/Compression converged after 1 iterations, Case= 103

```
743. CHANGE
744. LOAD 104 1.2DL+1.6LR+1.0L
745. REPEAT LOAD
746. 1 1.2 2 1.6 3 1.0
747. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 104

```
748. CHANGE
749. LOAD 105 1.2DL+1.6WL1+1.0L+0.5LR
750. REPEAT LOAD
751. 1 1.2 8 1.6 2 0.5 3 1.0
752. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 105

```
753. CHANGE
754. LOAD 106 1.2DL+1.6WL2+1.0L+0.5LR
755. REPEAT LOAD
756. 1 1.2 9 1.6 2 0.5 3 1.0
757. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 106

```
758. CHANGE
759. LOAD 107 1.2DL+1.6WL3+1.0L+0.5LR
760. REPEAT LOAD
761. 1 1.2 10 1.6 2 0.5 3 1.0
762. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 107

763. CHANGE
764. LOAD 108 1.2DL+1.6WL4+1.0L+0.5LR
765. REPEAT LOAD
766. 1 1.2 11 1.6 2 0.5 3 1.0
767. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 108

768. CHANGE
769. LOAD 109 1.2DL+1.6WL5+1.0L+0.5LR
770. REPEAT LOAD
771. 1 1.2 12 1.6 2 0.5 3 1.0
772. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

773. CHANGE
774. LOAD 110 1.2DL+1.6WL6+1.0L+0.5LR
775. REPEAT LOAD
776. 1 1.2 13 1.6 2 0.5 3 1.0
777. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

778. CHANGE
779. LOAD 111 1.2DL+1.6WL7+1.0L+0.5LR
780. REPEAT LOAD
781. 1 1.2 14 1.6 2 0.5 3 1.0
782. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 111

783. CHANGE
784. LOAD 112 1.2DL+1.6WL8+1.0L+0.5LR
785. REPEAT LOAD
786. 1 1.2 15 1.6 2 0.5 3 1.0
787. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 112

788. CHANGE
789. LOAD 113 1.2DL+1.6WL9+1.0L+0.5LR
790. REPEAT LOAD
791. 1 1.2 16 1.6 2 0.5 3 1.0
792. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 113

793. CHANGE
794. LOAD 114 1.2DL+1.6WL10+1.0L+0.5LR
795. REPEAT LOAD
796. 1 1.2 17 1.6 2 0.5 3 1.0
797. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 114

798. CHANGE
799. LOAD 115 1.2DL+1.6WL11+1.0L+0.5LR
800. REPEAT LOAD
801. 1 1.2 18 1.6 2 0.5 3 1.0
802. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 115

803. CHANGE
804. LOAD 116 1.2DL+1.6WL12+1.0L+0.5LR
805. REPEAT LOAD
806. 1 1.2 19 1.6 2 0.5 3 1.0

```
 807. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  116



 808. CHANGE
 809. LOAD 117 1.2DL+1.0SL1+1.0L
 810. REPEAT LOAD
 811. 1 1.2 20 1.0 3 1.0
 812. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  117



 813. CHANGE
 814. LOAD 118 1.2DL+1.0SL2+1.0L
 815. REPEAT LOAD
 816. 1 1.2 21 1.0 3 1.0
 817. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  118



 818. CHANGE
 819. LOAD 119 0.9DL+1.6WL1
 820. REPEAT LOAD
 821. 1 0.9 8 1.6
 822. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  119



 823. CHANGE
 824. LOAD 120 0.9DL+1.6WL2
 825. REPEAT LOAD
 826. 1 0.9 9 1.6
 827. PERFORM ANALYSIS
```



```
**START ITERATION NO.           2

**NOTE-Tension/Compression converged after  2 iterations, Case=  120



 828. CHANGE
 829. LOAD 121 0.9DL+1.6WL3
 830. REPEAT LOAD
 831. 1 0.9 10 1.6
 832. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  121



 833. CHANGE
 834. LOAD 122 0.9DL+1.6WL4
 835. REPEAT LOAD
 836. 1 0.9 11 1.6
 837. PERFORM ANALYSIS
**START ITERATION NO.           2

**NOTE-Tension/Compression converged after  2 iterations, Case=  122



 838. CHANGE
 839. LOAD 123 0.9DL+1.6WL5
 840. REPEAT LOAD
 841. 1 0.9 12 1.6
 842. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=  123



 843. CHANGE
 844. LOAD 124 0.9DL+1.6WL6
```

```
 845. REPEAT LOAD
 846. 1 0.9 13 1.6
 847. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3

 **NOTE-Tension/Compression converged after  3 iterations, Case=  124



 848. CHANGE
 849. LOAD 125 0.9DL+1.6WL7
 850. REPEAT LOAD
 851. 1 0.9 14 1.6
 852. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

 **NOTE-Tension/Compression converged after  4 iterations, Case=  125



 853. CHANGE
 854. LOAD 126 0.9DL+1.6WL8
 855. REPEAT LOAD
 856. 1 0.9 15 1.6
 857. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3

 **NOTE-Tension/Compression converged after  3 iterations, Case=  126



 858. CHANGE
 859. LOAD 127 0.9DL+1.6WL9
 860. REPEAT LOAD
 861. 1 0.9 16 1.6
 862. PERFORM ANALYSIS

 **NOTE-Tension/Compression converged after  1 iterations, Case=  127
```



```
 863. CHANGE
 864. LOAD 128 0.9DL+1.6WL10
 865. REPEAT LOAD
 866. 1 0.9 17 1.6
 867. PERFORM ANALYSIS

 **NOTE-Tension/Compression converged after  1 iterations, Case=  128



 868. CHANGE
 869. LOAD 129 0.9DL+1.6WL11
 870. REPEAT LOAD
 871. 1 0.9 18 1.6
 872. PERFORM ANALYSIS

 **NOTE-Tension/Compression converged after  1 iterations, Case=  129



 873. CHANGE
 874. LOAD 130 0.9DL+1.6WL12
 875. REPEAT LOAD
 876. 1 0.9 19 1.6
 877. PERFORM ANALYSIS

 **NOTE-Tension/Compression converged after  1 iterations, Case=  130



 878. CHANGE
 879. LOAD 131 0.9DL+1.6SL1
 880. REPEAT LOAD
 881. 1 0.9 20 1.6
 882. PERFORM ANALYSIS

 **NOTE-Tension/Compression converged after  1 iterations, Case=  131



 883. CHANGE
 884. LOAD 132 0.9DL+1.6SL2
 885. REPEAT LOAD
```

```
886. 1 0.9 21 1.6
887. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  132



888. CHANGE
889. LOAD 133 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR1
890. REPEAT LOAD
891. 1 1.2 8 1.6 2 0.5 3 1.0 4 1.0
892. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  133



893. CHANGE
894. LOAD 134 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
895. REPEAT LOAD
896. 1 1.2 9 1.6 2 0.5 3 1.0 4 1.0
897. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  134



898. CHANGE
899. LOAD 135 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR1
900. REPEAT LOAD
901. 1 1.2 10 1.6 2 0.5 3 1.0 4 1.0
902. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  135



903. CHANGE
904. LOAD 136 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR1
905. REPEAT LOAD
906. 1 1.2 11 1.6 2 0.5 3 1.0 4 1.0
907. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  136



908. CHANGE
909. LOAD 137 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR1
910. REPEAT LOAD
911. 1 1.2 12 1.6 2 0.5 3 1.0 4 1.0
912. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  137



913. CHANGE
914. LOAD 138 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR1
915. REPEAT LOAD
916. 1 1.2 13 1.6 2 0.5 3 1.0 4 1.0
917. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  138



918. CHANGE
919. LOAD 139 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR1
920. REPEAT LOAD
921. 1 1.2 14 1.6 2 0.5 3 1.0 4 1.0
922. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  139



923. CHANGE
924. LOAD 140 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR1
925. REPEAT LOAD
926. 1 1.2 15 1.6 2 0.5 3 1.0 4 1.0
927. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 140

928. CHANGE
929. LOAD 141 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR1
930. REPEAT LOAD
931. 1 1.2 16 1.6 2 0.5 3 1.0 4 1.0
932. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 141

933. CHANGE
934. LOAD 142 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR1
935. REPEAT LOAD
936. 1 1.2 17 1.6 2 0.5 3 1.0 4 1.0
937. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 142

938. CHANGE
939. LOAD 143 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR1
940. REPEAT LOAD
941. 1 1.2 18 1.6 2 0.5 3 1.0 4 1.0
942. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 143

943. CHANGE
944. LOAD 144 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR1
945. REPEAT LOAD
946. 1 1.2 19 1.6 2 0.5 3 1.0 4 1.0
947. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 144

948. CHANGE
949. LOAD 145 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR2
950. REPEAT LOAD
951. 1 1.2 8 1.6 2 0.5 3 1.0 5 1.0
952. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 145

953. CHANGE
954. LOAD 146 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR2
955. REPEAT LOAD
956. 1 1.2 9 1.6 2 0.5 3 1.0 5 1.0
957. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 146

958. CHANGE
959. LOAD 147 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR2
960. REPEAT LOAD
961. 1 1.2 10 1.6 2 0.5 3 1.0 5 1.0
962. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 147

963. CHANGE
964. LOAD 148 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR2
965. REPEAT LOAD
966. 1 1.2 11 1.6 2 0.5 3 1.0 5 1.0
967. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 148

968. CHANGE
969. LOAD 149 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR2
970. REPEAT LOAD
971. 1 1.2 12 1.6 2 0.5 3 1.0 5 1.0

```
 972. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  149



 973. CHANGE
 974. LOAD 150 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR2
 975. REPEAT LOAD
 976. 1 1.2 13 1.6 2 0.5 3 1.0 5 1.0
 977. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  150



 978. CHANGE
 979. LOAD 151 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR2
 980. REPEAT LOAD
 981. 1 1.2 14 1.6 2 0.5 3 1.0 5 1.0
 982. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  151



 983. CHANGE
 984. LOAD 152 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR2
 985. REPEAT LOAD
 986. 1 1.2 15 1.6 2 0.5 3 1.0 5 1.0
 987. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  152



 988. CHANGE
 989. LOAD 153 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR2
 990. REPEAT LOAD
 991. 1 1.2 16 1.6 2 0.5 3 1.0 5 1.0
 992. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  153



 993. CHANGE
 994. LOAD 154 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR2
 995. REPEAT LOAD
 996. 1 1.2 17 1.6 2 0.5 3 1.0 5 1.0
 997. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  154



 998. CHANGE
 999. LOAD 155 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR2
1000. REPEAT LOAD
1001. 1 1.2 18 1.6 2 0.5 3 1.0 5 1.0
1002. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  155



1003. CHANGE
1004. LOAD 156 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR2
1005. REPEAT LOAD
1006. 1 1.2 19 1.6 2 0.5 3 1.0 5 1.0
1007. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  156



1008. CHANGE
1009. LOAD 157 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR3
1010. REPEAT LOAD
1011. 1 1.2 8 1.6 2 0.5 3 1.0 6 1.0
1012. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  157
```

```
1013. CHANGE
1014. LOAD 158 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR3
1015. REPEAT LOAD
1016. 1 1.2 9 1.6 2 0.5 3 1.0 6 1.0
1017. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  158



1018. CHANGE
1019. LOAD 159 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR3
1020. REPEAT LOAD
1021. 1 1.2 10 1.6 2 0.5 3 1.0 6 1.0
1022. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  159



1023. CHANGE
1024. LOAD 160 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR3
1025. REPEAT LOAD
1026. 1 1.2 11 1.6 2 0.5 3 1.0 6 1.0
1027. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  160



1028. CHANGE
1029. LOAD 161 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR3
1030. REPEAT LOAD
1031. 1 1.2 12 1.6 2 0.5 3 1.0 6 1.0
1032. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  161



1033. CHANGE
1034. LOAD 162 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR3
1035. REPEAT LOAD
```



```
1036. 1 1.2 13 1.6 2 0.5 3 1.0 6 1.0
1037. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  162



1038. CHANGE
1039. LOAD 163 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR3
1040. REPEAT LOAD
1041. 1 1.2 14 1.6 2 0.5 3 1.0 6 1.0
1042. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  163



1043. CHANGE
1044. LOAD 164 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR3
1045. REPEAT LOAD
1046. 1 1.2 15 1.6 2 0.5 3 1.0 6 1.0
1047. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  164



1048. CHANGE
1049. LOAD 165 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR3
1050. REPEAT LOAD
1051. 1 1.2 16 1.6 2 0.5 3 1.0 6 1.0
1052. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  165



1053. CHANGE
1054. LOAD 166 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR3
1055. REPEAT LOAD
1056. 1 1.2 17 1.6 2 0.5 3 1.0 6 1.0
1057. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 166

1058. CHANGE
1059. LOAD 167 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR3
1060. REPEAT LOAD
1061. 1 1.2 18 1.6 2 0.5 3 1.0 6 1.0
1062. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 167

1063. CHANGE
1064. LOAD 168 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR3
1065. REPEAT LOAD
1066. 1 1.2 19 1.6 2 0.5 3 1.0 6 1.0
1067. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 168

1068. CHANGE
1069. LOAD 169 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR4
1070. REPEAT LOAD
1071. 1 1.2 8 1.6 2 0.5 3 1.0 7 1.0
1072. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 169

1073. CHANGE
1074. LOAD 170 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR4
1075. REPEAT LOAD
1076. 1 1.2 9 1.6 2 0.5 3 1.0 7 1.0
1077. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 170

1078. CHANGE
1079. LOAD 171 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR4
1080. REPEAT LOAD
1081. 1 1.2 10 1.6 2 0.5 3 1.0 7 1.0
1082. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 171

1083. CHANGE
1084. LOAD 172 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR4
1085. REPEAT LOAD
1086. 1 1.2 11 1.6 2 0.5 3 1.0 7 1.0
1087. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 172

1088. CHANGE
1089. LOAD 173 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR4
1090. REPEAT LOAD
1091. 1 1.2 12 1.6 2 0.5 3 1.0 7 1.0
1092. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 173

1093. CHANGE
1094. LOAD 174 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR4
1095. REPEAT LOAD
1096. 1 1.2 13 1.6 2 0.5 3 1.0 7 1.0
1097. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 174

```
1098. CHANGE
1099. LOAD 175 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR4
1100. REPEAT LOAD
1101. 1 1.2 14 1.6 2 0.5 3 1.0 7 1.0
1102. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  175



1103. CHANGE
1104. LOAD 176 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR4
1105. REPEAT LOAD
1106. 1 1.2 15 1.6 2 0.5 3 1.0 7 1.0
1107. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  176



1108. CHANGE
1109. LOAD 177 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR4
1110. REPEAT LOAD
1111. 1 1.2 16 1.6 2 0.5 3 1.0 7 1.0
1112. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  177



1113. CHANGE
1114. LOAD 178 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR4
1115. REPEAT LOAD
1116. 1 1.2 17 1.6 2 0.5 3 1.0 7 1.0
1117. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  178



1118. CHANGE
1119. LOAD 179 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR4
1120. REPEAT LOAD
1121. 1 1.2 18 1.6 2 0.5 3 1.0 7 1.0
```



```
1122. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  179



1123. CHANGE
1124. LOAD 180 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR4
1125. REPEAT LOAD
1126. 1 1.2 19 1.6 2 0.5 3 1.0 7 1.0
1127. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  180



1128. CHANGE
1129. ******************************************************************************
1130. PERFORM ANALYSIS

  ** ALL CASES BEING MADE ACTIVE BEFORE RE-ANALYSIS. **

**NOTE-Tension/Compression converged after  1 iterations, Case=    1



**NOTE-Tension/Compression converged after  1 iterations, Case=    2



**NOTE-Tension/Compression converged after  1 iterations, Case=    3



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6

**NOTE-Tension/Compression converged after  6 iterations, Case=    4



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6
```

**NOTE-Tension/Compression converged after 6 iterations, Case= 5

**NOTE-Tension/Compression converged after 1 iterations, Case= 6

**NOTE-Tension/Compression converged after 1 iterations, Case= 7

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= 1.0288553E-05 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= 8.0626924E-06 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= 9.2913397E-06 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 8.9034438E-07 EQN NO 519
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.4314221E+04 L-MATRIX DIAG= 1.9645086E-10 EQN NO 517
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4633787E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 3.6507845E-07 EQN NO 519
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 8 iterations, Case= 8

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= 8.9223031E-06 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.4314221E+04 L-MATRIX DIAG= 1.9645086E-10 EQN NO 517
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4633787E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 3.6507845E-07 EQN NO 519
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 7
**START ITERATION NO. 8
**START ITERATION NO. 9
**START ITERATION NO. 10
**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= 8.9223031E-06 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.4314221E+04 L-MATRIX DIAG= 1.9645086E-10 EQN NO 517
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4633787E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 3.6507845E-07 EQN NO 519
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 13
**START ITERATION NO. 14
**START ITERATION NO. 15

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 9
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= 1.0288553E-05 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= 8.0626924E-06 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= 9.2913397E-06 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 8.9034438E-07 EQN NO 519
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.4314221E+04 L-MATRIX DIAG= 1.9645086E-10 EQN NO 517
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4633787E-07 EQN NO 518

***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 3.6507845E-07 EQN NO 519
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 8 iterations, Case= 10

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= 8.9223031E-06 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.4314221E+04 L-MATRIX DIAG= 1.9645086E-10 EQN NO 517
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4633787E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 3.6507845E-07 EQN NO 519
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 7

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 **START ITERATION NO.           8
 **START ITERATION NO.           9
 **START ITERATION NO.          10
 **START ITERATION NO.          11

***WARNING -  INSTABILITY AT JOINT   555   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.9062673E+06 L-MATRIX DIAG=  8.9223031E-06 EQN NO     518
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.          12

***WARNING -  INSTABILITY AT JOINT   555   DIRECTION = FY
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.4314221E+04 L-MATRIX DIAG=  1.9645086E-10 EQN NO     517
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.


***WARNING -  INSTABILITY AT JOINT   555   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.9062673E+06 L-MATRIX DIAG= -6.4633787E-07 EQN NO     518
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING -  INSTABILITY AT JOINT   555   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.7308082E+06 L-MATRIX DIAG=  3.6507845E-07 EQN NO     519
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
 **START ITERATION NO.          13
 **START ITERATION NO.          14
 **START ITERATION NO.          15

 **WARNING-Tension/Compression did not converge after 15 iterations, Case=   11
 ** If results are not satisfactory then enter a
 ** SET ITERLIM i command (i>10) before loads to get more iterations.



 **START ITERATION NO.           2
 **START ITERATION NO.           3
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 **START ITERATION NO.           4
 **START ITERATION NO.           5

***WARNING -  INSTABILITY AT JOINT   555   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.9062673E+06 L-MATRIX DIAG= -6.4540654E-07 EQN NO     518
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.           6

***WARNING -  INSTABILITY AT JOINT   555   DIRECTION = FY
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.4314221E+04 L-MATRIX DIAG=  1.9645086E-10 EQN NO     517
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.


***WARNING -  INSTABILITY AT JOINT   555   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.9062673E+06 L-MATRIX DIAG= -6.4633787E-07 EQN NO     518
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING -  INSTABILITY AT JOINT   555   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.7308082E+06 L-MATRIX DIAG=  3.6507845E-07 EQN NO     519
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

 **NOTE-Tension/Compression converged after  6 iterations, Case=   12



 **START ITERATION NO.           2
 **START ITERATION NO.           3
 **START ITERATION NO.           4
 **START ITERATION NO.           5

***WARNING -  INSTABILITY AT JOINT   555   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.9062673E+06 L-MATRIX DIAG= -6.4540654E-07 EQN NO     518
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
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NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.4314221E+04 L-MATRIX DIAG= 1.9645086E-10 EQN NO 517
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4633787E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 3.6507845E-07 EQN NO 519
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 6 iterations, Case= 13

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4540654E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.4314221E+04 L-MATRIX DIAG= 1.9645086E-10 EQN NO 517
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4633787E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 3.6507845E-07 EQN NO 519
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 6 iterations, Case= 14

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4540654E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.4314221E+04 L-MATRIX DIAG= 1.9645086E-10 EQN NO 517
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4633787E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 3.6507845E-07 EQN NO 519
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 6 iterations, Case= 15

**START ITERATION NO. 2

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.4314221E+04 L-MATRIX DIAG= 1.9645086E-10 EQN NO 517
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4633787E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 3.6507845E-07 EQN NO 519
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 2 iterations, Case= 16

**START ITERATION NO. 2

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.4314221E+04 L-MATRIX DIAG= 1.9645086E-10 EQN NO 517
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4633787E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 3.6507845E-07 EQN NO 519
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 2 iterations, Case= 17

**START ITERATION NO. 2
**START ITERATION NO. 3

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4563937E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 4

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4470805E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4587221E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 8.6845830E-07 EQN NO 519
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.4314221E+04 L-MATRIX DIAG= 1.9645086E-10 EQN NO 517
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4633787E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 3.6507845E-07 EQN NO 519
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 6 iterations, Case= 18

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4563937E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4470805E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4587221E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 8.6845830E-07 EQN NO 519
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.4314221E+04 L-MATRIX DIAG= 1.9645086E-10 EQN NO 517
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4633787E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 3.6507845E-07 EQN NO 519

***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 7 iterations, Case= 19

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= 8.9223031E-06 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= 9.2685223E-06 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 6 iterations, Case= 20

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4540654E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4726919E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4657070E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4540654E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 2.6915222E-07 EQN NO 519
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4540654E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4494088E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4633787E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 2.5145710E-07 EQN NO 519
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4633787E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 13

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4633787E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 2.9895455E-07 EQN NO 519
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 14

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4633787E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 14 iterations, Case= 21

**START ITERATION NO. 2

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.4314221E+04 L-MATRIX DIAG= 1.9645086E-10 EQN NO 517
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.9062673E+06 L-MATRIX DIAG= -6.4633787E-07 EQN NO 518
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 555 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.7308082E+06 L-MATRIX DIAG= 3.6507845E-07 EQN NO 519
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 2 iterations, Case= 22

**NOTE-Tension/Compression converged after 1 iterations, Case= 23

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

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**NOTE-Tension/Compression converged after 1 iterations, Case= 211

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**NOTE-Tension/Compression converged after 1 iterations, Case= 213

**NOTE-Tension/Compression converged after 1 iterations, Case= 214

**NOTE-Tension/Compression converged after 1 iterations, Case= 216

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**NOTE-Tension/Compression converged after 1 iterations, Case= 220

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**NOTE-Tension/Compression converged after 1 iterations, Case= 226

**NOTE-Tension/Compression converged after 1 iterations, Case= 227

**NOTE-Tension/Compression converged after 1 iterations, Case= 228

**NOTE-Tension/Compression converged after 1 iterations, Case= 230

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 231

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 232

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 233

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 234

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 235

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 236

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 237

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 238

**NOTE-Tension/Compression converged after 1 iterations, Case= 239

**NOTE-Tension/Compression converged after 1 iterations, Case= 240

**NOTE-Tension/Compression converged after 1 iterations, Case= 241

**NOTE-Tension/Compression converged after 1 iterations, Case= 242

**NOTE-Tension/Compression converged after 1 iterations, Case= 243

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**NOTE-Tension/Compression converged after 1 iterations, Case= 115

**NOTE-Tension/Compression converged after 1 iterations, Case= 116

**NOTE-Tension/Compression converged after 1 iterations, Case= 118

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 119

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 120

**START ITERATION NO. 2
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**NOTE-Tension/Compression converged after 3 iterations, Case= 121

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**NOTE-Tension/Compression converged after 2 iterations, Case= 122

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 123

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 124

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 125

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 126

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**NOTE-Tension/Compression converged after 1 iterations, Case= 179

**NOTE-Tension/Compression converged after 1 iterations, Case= 180

```
1131. LOAD LIST 101 TO 180
1132. START CONCRETE DESIGN
```



```
1133. CODE ACI
1134. TRACK 2 ALL
1135. FC 35000 MEMB 185
1136. CLB 0.05 MEMB 185
1137. CLS 0.05 ALL
1138. CLT 0.05 MEMB 185 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 -
1139. 372 393 TO 399 402 TO 409 412 TO 419 422 TO 429 432
1140. MAXMAIN 25 ALL
1141. MINMAIN 25 ALL
1142. MINSEC 10 ALL
1143. DESIGN COLUMN 185
```

==========

COLUMN NO. 185 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, RECT SIZE - 850.0 X 500.0 MMS, TIED
AREA OF STEEL REQUIRED = 4496.5 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 25 MM	1.386	150	STA	0.650

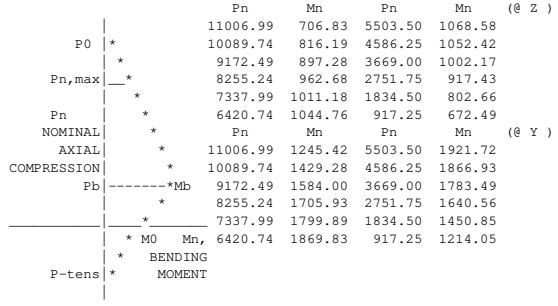
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
14905.30	11924.24	5166.13	1075.52	208.2
M0	P-tens.	Des.Pn	Des.Mn	e/h
506.09	-2436.79	14.16	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
14905.30	11924.24	5535.96	1922.86	347.3
M0	P-tens.	Des.Pn	Des.Mn	e/h
909.55	-2436.79	14.16	665.85	23.51702



**********END OF COLUMN DESIGN RESULTS**********

1144. END CONCRETE DESIGN

1145. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 18,2020 TIME= 17:15:44 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 10 STAAD OUTPUT – FOUNDATION F1B

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    OCT 18, 2020                 *
*          Time=    17:20: 8                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F1B.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 04/09/2020
    4. JOB NAME FDN. MKD. F1B
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. SET NL 200
   12. SET ITERLIM 15
   13. SET PRINT 1
   14. UNIT METER KN
   15. JOINT COORDINATES
   16. 2 0 911.8 3.175; 6 0.75 911.8 3.175; 10 -0.75 911.8 3.175; 34 -0.5 911.8 3.175
   17. 35 -0.25 911.8 3.175; 41 0.25 911.8 3.175; 42 0.5 911.8 3.175
   18. 128 0 913.8 3.175; 302 -0.5 911.8 3.42438; 303 -0.75 911.8 3.42438
   19. 304 -0.25 911.8 3.42438; 305 0 911.8 3.42438; 306 0.25 911.8 3.42438
   20. 307 0.5 911.8 3.42438; 308 0.75 911.8 3.42438; 309 1 911.8 3.175
   21. 313 1 911.8 3.42438; 316 -1 911.8 3.175; 320 -1 911.8 3.42438
   22. 321 -0.75 911.8 3.67438; 322 -0.5 911.8 3.67438; 323 -0.25 911.8 3.67438
   23. 324 0 911.8 3.67438; 325 0.25 911.8 3.67438; 326 0.5 911.8 3.67438
   24. 327 0.75 911.8 3.67438; 328 1 911.8 3.67438; 331 -1 911.8 3.67438
   25. 332 -0.75 911.8 3.92438; 333 -0.5 911.8 3.92438; 334 -0.25 911.8 3.92438
   26. 335 0 911.8 3.92438; 336 0.25 911.8 3.92438; 337 0.5 911.8 3.92438
   27. 338 0.75 911.8 3.92438; 339 1 911.8 3.92438; 342 -1 911.8 3.92438
   28. 343 -0.75 911.8 4.12538; 344 -0.5 911.8 4.12538; 345 -0.25 911.8 4.12538
   29. 346 0 911.8 4.12538; 347 0.25 911.8 4.12538; 348 0.5 911.8 4.12538
   30. 349 0.75 911.8 4.12538; 350 1 911.8 4.12538; 353 -1 911.8 4.12538
   31. 365 -0.75 911.8 2.17438; 366 -0.5 911.8 2.17438; 369 -0.25 911.8 2.17438
   32. 371 0 911.8 2.17438; 373 0.25 911.8 2.17438; 375 0.5 911.8 2.17438
   33. 377 0.75 911.8 2.17438; 379 1 911.8 2.17438; 384 -1 911.8 2.17438
   34. 387 -0.75 911.8 2.42438; 388 -0.5 911.8 2.42438; 389 -0.25 911.8 2.42438
   35. 390 0 911.8 2.42438; 391 0.25 911.8 2.42438; 392 0.5 911.8 2.42438
   36. 393 0.75 911.8 2.42438; 394 1 911.8 2.42438; 397 -1 911.8 2.42438
   37. 398 -0.75 911.8 2.67438; 399 -0.5 911.8 2.67438; 400 -0.25 911.8 2.67438
   38. 401 0 911.8 2.67438; 402 0.25 911.8 2.67438; 403 0.5 911.8 2.67438
```



```
   39. 404 0.75 911.8 2.67438; 405 1 911.8 2.67438; 408 -1 911.8 2.67438
   40. 409 -0.75 911.8 2.92438; 410 -0.5 911.8 2.92438; 411 -0.25 911.8 2.92438
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   42. 415 0.75 911.8 2.92438; 416 1 911.8 2.92438; 419 -1 911.8 2.92438
   43. 492 1.25 911.8 3.42438; 493 1.25 911.8 3.175; 494 1.25 911.8 3.67438
   44. 495 1.25 911.8 3.92438; 496 1.25 911.8 4.12538; 497 1.25 911.8 2.42438
   45. 498 1.25 911.8 2.17438; 499 1.25 911.8 2.67438; 500 1.25 911.8 2.92438
   46. 509 -1.25 911.8 3.42438; 510 -1.25 911.8 3.175; 511 -1.25 911.8 3.67438
   47. 512 -1.25 911.8 3.92438; 513 -1.25 911.8 4.12538; 514 -1.25 911.8 2.42438
   48. 515 -1.25 911.8 2.17438; 516 -1.25 911.8 2.67438; 517 -1.25 911.8 2.92438
   49. 518 -0.5 911.8 1.92438; 519 -0.75 911.8 1.92438; 520 -0.25 911.8 1.92438
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   51. 524 0.75 911.8 1.92438; 525 1 911.8 1.92438; 526 -1 911.8 1.92438
   52. 527 1.25 911.8 1.92438; 528 -1.25 911.8 1.92438; 540 1.5 911.8 3.42438
   53. 541 1.5 911.8 3.175; 542 1.5 911.8 3.67438; 543 1.5 911.8 3.92438
   54. 544 1.5 911.8 4.12538; 545 1.5 911.8 2.42438; 546 1.5 911.8 2.17438
   55. 547 1.5 911.8 2.67438; 548 1.5 911.8 2.92438; 549 1.75 911.8 3.42438
   56. 550 1.75 911.8 3.175; 551 1.75 911.8 3.67438; 552 1.75 911.8 3.92438
   57. 553 1.75 911.8 4.12538; 554 1.75 911.8 2.42438; 555 1.75 911.8 2.17438
   58. 556 1.75 911.8 2.67438; 557 1.75 911.8 2.92438; 558 1.5 911.8 1.92438
   59. 559 1.75 911.8 1.92438; 562 -1.5 911.8 3.42438; 563 -1.5 911.8 3.175
   60. 564 -1.5 911.8 3.67438; 565 -1.5 911.8 3.92438; 566 -1.5 911.8 4.12538
   61. 567 -1.5 911.8 2.42438; 568 -1.5 911.8 2.17438; 569 -1.5 911.8 2.67438
   62. 570 -1.5 911.8 2.92438; 571 -1.75 911.8 3.42438; 572 -1.75 911.8 3.175
   63. 573 -1.75 911.8 3.67438; 574 -1.75 911.8 3.92438; 575 -1.75 911.8 4.12538
   64. 576 -1.75 911.8 2.42438; 577 -1.75 911.8 2.17438; 578 -1.75 911.8 2.67438
   65. 579 -1.75 911.8 2.92438; 580 -1.5 911.8 1.92438; 581 -1.75 911.8 1.92438
   66. 584 2 911.8 3.42438; 585 2 911.8 3.175; 586 2 911.8 3.67438
   67. 587 2 911.8 3.92438; 588 2 911.8 4.12538; 589 2 911.8 2.42438
   68. 590 2 911.8 2.17438; 591 2 911.8 2.67438; 592 2 911.8 2.92438
   69. 593 2.375 911.8 3.42438; 594 2.375 911.8 3.175; 595 2.375 911.8 3.67438
   70. 596 2.375 911.8 3.92438; 597 2.375 911.8 4.12538; 598 2.375 911.8 2.42438
   71. 599 2.375 911.8 2.17438; 600 2.375 911.8 2.67438; 601 2.375 911.8 2.92438
   72. 602 2 911.8 1.92438; 603 2.375 911.8 1.92438; 606 -2 911.8 3.42438
   73. 607 -2 911.8 3.175; 608 -2 911.8 3.67438; 609 -2 911.8 3.92438
   74. 610 -2 911.8 4.12538; 611 -2 911.8 2.42438; 612 -2 911.8 2.17438
   75. 613 -2 911.8 2.67438; 614 -2 911.8 2.92438; 615 -2.375 911.8 3.42438
   76. 616 -2.375 911.8 3.175; 617 -2.375 911.8 3.67438; 618 -2.375 911.8 3.92438
   77. 619 -2.375 911.8 4.12538; 620 -2.375 911.8 2.42438; 621 -2.375 911.8 2.17438
   78. 622 -2.375 911.8 2.67438; 623 -2.375 911.8 2.92438; 624 -2 911.8 1.92438
   79. 625 -2.375 911.8 1.92438; 628 -0.5 911.8 1.77538; 629 -0.75 911.8 1.77538
   80. 630 -0.25 911.8 1.77538; 631 0 911.8 1.77538; 632 0.25 911.8 1.77538
   81. 633 0.5 911.8 1.77538; 634 0.75 911.8 1.77538; 635 1 911.8 1.77538
   82. 636 -1 911.8 1.77538; 637 1.25 911.8 1.77538; 638 -1.25 911.8 1.77538
   83. 650 1.5 911.8 1.77538; 651 1.75 911.8 1.77538; 654 -1.5 911.8 1.77538
   84. 655 -1.75 911.8 1.77538; 658 2 911.8 1.77538; 659 2.375 911.8 1.77538
   85. 662 -2 911.8 1.77538; 663 -2.375 911.8 1.77538
   86. MEMBER INCIDENCES
   87. 185 2 128
   88. ELEMENT INCIDENCES SHELL
   89. 326 303 302 34 10; 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2
   90. 333 306 307 42 41; 334 307 308 6 42; 337 308 313 309 6; 342 320 303 10 316
   91. 343 321 322 302 303; 344 322 323 304 302; 345 323 324 305 304
   92. 346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307
   93. 349 327 328 313 308; 352 331 321 303 320; 353 332 333 322 321
   94. 354 333 334 323 322; 355 334 335 324 323; 356 335 336 325 324
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 95. 357 336 337 326 325; 358 337 338 327 326; 359 338 339 328 327
 96. 362 342 332 321 331; 363 343 344 333 332; 364 344 345 334 333
 97. 365 345 346 335 334; 366 346 347 336 335; 367 347 348 337 336
 98. 368 348 349 338 337; 369 349 350 339 338; 372 353 343 332 342
 99. 393 387 388 366 365; 394 388 389 369 366; 395 389 390 371 369
100. 396 390 391 373 371; 397 391 392 375 373; 398 392 393 377 375
101. 399 393 394 379 377; 402 397 387 365 384; 403 398 399 388 387
102. 404 399 400 389 388; 405 400 401 390 389; 406 401 402 391 390
103. 407 402 403 392 391; 408 403 404 393 392; 409 404 405 394 393
104. 412 408 398 387 397; 413 409 410 399 398; 414 410 411 400 399
105. 415 411 412 401 400; 416 412 413 402 401; 417 413 414 403 402
106. 418 414 415 404 403; 419 415 416 405 404; 422 419 409 398 408
107. 423 10 34 410 409; 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412
108. 427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415; 432 316 10 409 419
109. 497 313 492 493 309; 498 328 494 492 313; 499 339 495 494 328
110. 500 350 496 495 339; 501 394 497 498 379; 502 405 499 497 394
111. 503 416 500 499 405; 504 309 493 500 416; 513 509 320 316 510
112. 514 511 331 320 509; 515 512 342 331 511; 516 513 353 342 512
113. 517 514 397 384 515; 518 516 408 397 514; 519 517 419 408 516
114. 520 510 316 419 517; 521 365 366 518 519; 522 366 369 520 518
115. 523 369 371 521 520; 524 371 373 522 521; 525 373 375 523 522
116. 526 375 377 524 523; 527 377 379 525 524; 528 384 365 519 526
117. 529 379 498 527 525; 530 515 384 526 528; 541 492 540 541 493
118. 542 494 542 540 492; 543 495 543 542 494; 544 496 544 543 495
119. 545 497 545 546 498; 546 499 547 545 497; 547 500 548 547 499
120. 548 493 541 548 500; 549 540 549 550 541; 550 542 551 549 540
121. 551 543 552 551 542; 552 544 553 552 543; 553 545 554 555 546
122. 554 547 556 554 545; 555 548 557 556 547; 556 541 550 557 548
123. 557 498 546 558 527; 558 546 555 559 558; 561 562 509 510 563
124. 562 564 511 509 562; 563 565 512 511 564; 564 566 513 512 565
125. 565 567 514 515 568; 566 569 516 514 567; 567 570 517 516 569
126. 568 563 510 517 570; 569 571 562 563 572; 570 573 564 562 571
127. 571 574 565 564 573; 572 575 566 565 574; 573 576 567 568 577
128. 574 578 569 567 576; 575 579 570 569 578; 576 572 563 570 579
129. 577 568 515 528 580; 578 577 568 580 581; 581 549 584 585 550
130. 582 551 586 584 549; 583 552 587 586 551; 584 553 588 587 552
131. 585 554 589 590 555; 586 556 591 589 554; 587 557 592 591 556
132. 588 550 585 592 557; 589 584 593 594 585; 590 586 595 593 584
133. 591 587 596 595 586; 592 588 597 596 587; 593 589 598 599 590
134. 594 591 600 598 589; 595 592 601 600 591; 596 585 594 601 592
135. 597 555 590 602 559; 598 590 599 603 602; 601 606 571 572 607
136. 602 608 573 571 606; 603 609 574 573 608; 604 610 575 574 609
137. 605 611 576 577 612; 606 613 578 576 611; 607 614 579 578 613
138. 608 607 572 579 614; 609 615 606 607 616; 610 617 608 606 615
139. 611 618 609 608 617; 612 619 610 609 618; 613 620 611 612 621
140. 614 622 613 611 620; 615 623 614 613 622; 616 616 607 614 623
141. 617 612 577 581 624; 618 621 612 624 625; 621 519 518 628 629
142. 622 518 520 630 628; 623 520 521 631 630; 624 521 522 632 631
143. 625 522 523 633 632; 626 523 524 634 633; 627 524 525 635 634
144. 628 526 519 629 636; 629 525 527 637 635; 630 528 526 636 638
145. 641 527 558 650 637; 642 558 559 651 650; 645 580 528 638 654
146. 646 581 580 654 655; 649 559 602 658 651; 650 602 603 659 658
147. 653 624 581 655 662; 654 625 624 662 663
148. START GROUP DEFINITION
149. ELEMENT
150. _RAFT 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 -
```



```
151. 393 TO 399 402 TO 409 412 TO 419 422 TO 429 432 497 TO 504 513 TO 530 541 -
152. 542 TO 558 561 TO 578 581 TO 598 601 TO 618 621 TO 630 641 642 645 646 649 -
153. 650 653 654
154. _RAFT_MOM 326 327 333 334 337 342 TO 349 352 TO 359 362 TO 369 372 -
155. 393 TO 399 402 TO 409 412 TO 419 422 TO 424 427 TO 429 432 497 TO 504 513 -
156. 514 TO 530 541 TO 558 561 TO 578 581 TO 598 601 TO 618 621 TO 630 641 642 -
157. 645 646 649 650 653 654
158. _RAFT_SHEAR 337 342 349 352 359 362 TO 369 372 393 TO 399 402 409 412 419 -
159. 422 429 432 497 TO 504 513 TO 530 541 TO 558 561 TO 578 581 TO 598 -
160. 601 TO 618 621 TO 630 641 642 645 646 649 650 653 654
161. MEMBER
162. END GROUP DEFINITION
163. ELEMENT PROPERTY
164. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
165. 402 TO 409 412 TO 419 422 TO 429 432 497 TO 504 513 TO 530 541 TO 558 561 -
166. 562 TO 578 581 TO 598 601 TO 618 621 TO 630 641 642 645 646 649 650 653 -
167. 654 THICKNESS 0.6
168. DEFINE MATERIAL START
169. ISOTROPIC CONCRETE
170. E 2.17185E+007
171. POISSON 0.17
172. DENSITY 25
173. ALPHA 1E-005
174. DAMP 0.05
175. TYPE CONCRETE
176. STRENGTH FCU 27579
177. END DEFINE MATERIAL
178. MEMBER PROPERTY AMERICAN
179. 185 PRIS YD 0.5 ZD 0.85
180. CONSTANTS
181. MATERIAL CONCRETE ALL
182. SUPPORTS
183. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
184. 402 TO 409 412 TO 419 422 TO 429 432 497 TO 504 513 TO 530 541 TO 558 561 -
185. 562 TO 578 581 TO 598 601 TO 618 621 TO 630 641 642 645 646 649 650 653 -
186. 654 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
```

JOINT	INFLUENCE AREA
2	0.062500
6	0.062500
10	0.062500
34	0.062500
35	0.062500
41	0.062500
42	0.062500
302	0.062422
303	0.062422
304	0.062422
305	0.062422
306	0.062422
307	0.062422
308	0.062422
309	0.062500
313	0.062423

STAAD SPACE -- PAGE NO. 5

316	0.062500
320	0.062423
321	0.062500
322	0.062500
323	0.062500
324	0.062500
325	0.062500
326	0.062500
327	0.062500
328	0.062500
331	0.062500
332	0.056375
333	0.056375
334	0.056375
335	0.056375
336	0.056375
337	0.056375
338	0.056375
339	0.056375
342	0.056375
343	0.025125
344	0.025125
345	0.025125
346	0.025125
347	0.025125
348	0.025125
349	0.025125
350	0.025125
353	0.025125
365	0.062500
366	0.062500
369	0.062500
371	0.062500
373	0.062500
375	0.062500
377	0.062500
379	0.062500
384	0.062500
387	0.062500
388	0.062500
389	0.062500
390	0.062500
391	0.062500
392	0.062500
393	0.062500
394	0.062500
397	0.062500
398	0.062500
399	0.062500
400	0.062500
401	0.062500
402	0.062500
403	0.062500
404	0.062500
405	0.062500
408	0.062500

STAAD SPACE -- PAGE NO. 6

409	0.062578
410	0.062578
411	0.062578
412	0.062578
413	0.062578
414	0.062578
415	0.062578
416	0.062578
419	0.062578
492	0.062422
493	0.062500
494	0.062500
495	0.056375
496	0.025125
497	0.062500
498	0.062500
499	0.062500
500	0.062578
509	0.062422
510	0.062500
511	0.062500
512	0.056375
513	0.025125
514	0.062500
515	0.062500
516	0.062500
517	0.062578
518	0.049875
519	0.049875
520	0.049875
521	0.049875
522	0.049875
523	0.049875
524	0.049875
525	0.049875
526	0.049875
527	0.049875
528	0.049875
540	0.062422
541	0.062500
542	0.062500
543	0.056375
544	0.025125
545	0.062500
546	0.062500
547	0.062500
548	0.062577
549	0.062422
550	0.062500
551	0.062500
552	0.056375
553	0.025125
554	0.062500
555	0.062500
556	0.062500
557	0.062578

558	0.049875
559	0.049875
562	0.062422
563	0.062500
564	0.062500
565	0.056375
566	0.025125
567	0.062500
568	0.062500
569	0.062500
570	0.062577
571	0.062422
572	0.062500
573	0.062500
574	0.056375
575	0.025125
576	0.062500
577	0.062500
578	0.062500
579	0.062578
580	0.049875
581	0.049875
584	0.078028
585	0.078125
586	0.078125
587	0.070469
588	0.031406
589	0.078125
590	0.078125
591	0.078125
592	0.078222
593	0.046817
594	0.046875
595	0.046875
596	0.042281
597	0.018844
598	0.046875
599	0.046875
600	0.046875
601	0.046933
602	0.062344
603	0.037406
606	0.078028
607	0.078125
608	0.078125
609	0.070469
610	0.031406
611	0.078125
612	0.078125
613	0.078125
614	0.078222
615	0.046817
616	0.046875
617	0.046875
618	0.042281
619	0.018844

620	0.046875
621	0.046875
622	0.046875
623	0.046933
624	0.062344
625	0.037406
628	0.018625
629	0.018625
630	0.018625
631	0.018625
632	0.018625
633	0.018625
634	0.018625
635	0.018625
636	0.018625
637	0.018625
638	0.018625
650	0.018625
651	0.018625
654	0.018625
655	0.018625
658	0.023281
659	0.013969
662	0.023281
663	0.013969
TOTAL	11.162502

```
187. ********************************************************************************
188. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
189. SELFWEIGHT Y -1
190. ELEMENT LOAD
191. * LOAD DUE TO BACKFILL
192. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
193. 402 TO 409 412 TO 419 422 TO 429 432 497 TO 504 513 TO 530 541 TO 558 561 -
194. 562 TO 578 581 TO 598 601 TO 618 621 TO 630 641 642 645 646 649 650 653 -
195. 654 PR GY -22
196. * LOAD DUE TO SURCHARGE
197. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
198. 402 TO 409 412 TO 419 422 TO 429 432 497 TO 504 513 TO 530 541 TO 558 561 -
199. 562 TO 578 581 TO 598 601 TO 618 621 TO 630 641 642 645 646 649 650 653 -
200. 654 PR GY -20
201. JOINT LOAD
202. 128 FY -22 FZ -3 MX -23
203. ********************************************************************************
204. LOAD 2 LOADTYPE NONE  TITLE ROOF LIVE LOAD ( LR )
205. JOINT LOAD
206. 128 FY -61 FZ -2 MX -10
207. ********************************************************************************
208. LOAD 3 LOADTYPE NONE  TITLE LIVE LOAD ( L )
209. JOINT LOAD
210. 128 FY -0.001
211. ********************************************************************************
212. LOAD 4 LOADTYPE NONE  TITLE CRANE LOAD  ( CR1 )
213. JOINT LOAD
214. 128 FY -35 FZ -9 MX -40
215. ********************************************************************************
216. LOAD 5 LOADTYPE NONE  TITLE CRANE LOAD  ( CR2 )
```

```
217. JOINT LOAD
218. 128 FY -26 FZ -7 MX -33
219. ********************************************************************************
220. LOAD 6 LOADTYPE NONE  TITLE CRANE LOAD  ( CR3 )
221. JOINT LOAD
222. 128 FY -0.0001
223. ********************************************************************************
224. LOAD 7 LOADTYPE NONE  TITLE CRANE LOAD  ( CR4 )
225. JOINT LOAD
226. 128 FY -0.0001
227. ********************************************************************************
228. LOAD 8 LOADTYPE NONE  TITLE WIND LOAD  ( WL1 )
229. JOINT LOAD
230. 128 FY -117 FZ -24 MX -193
231. ********************************************************************************
232. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD  ( WL2 )
233. JOINT LOAD
234. 128 FY 74 FZ -27 MX -207
235. ********************************************************************************
236. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD  ( WL3 )
237. JOINT LOAD
238. 128 FY 117 FZ -24 MX -193
239. ********************************************************************************
240. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD  ( WL4 )
241. JOINT LOAD
242. 128 FY 74 FZ -27 MX -207
243. ********************************************************************************
244. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD  ( WL5 )
245. JOINT LOAD
246. 128 FY 105 FZ 38 MX 270
247. ********************************************************************************
248. LOAD 13 LOADTYPE NONE  TITLE WIND LOAD  ( WL6 )
249. JOINT LOAD
250. 128 FY 62 FZ 35 MX 256
251. ********************************************************************************
252. LOAD 14 LOADTYPE NONE  TITLE WIND LOAD  ( WL7 )
253. JOINT LOAD
254. 128 FY 105 FZ 38 MX 270
255. ********************************************************************************
256. LOAD 15 LOADTYPE NONE  TITLE WIND LOAD  ( WL8 )
257. JOINT LOAD
258. 128 FY 62 FZ 35 MX 256
259. ********************************************************************************
260. LOAD 16 LOADTYPE NONE  TITLE WIND LOAD  ( WL9 )
261. JOINT LOAD
262. 128 FY 119 FZ 4 MX 22
263. ********************************************************************************
264. LOAD 17 LOADTYPE NONE  TITLE WIND LOAD  ( WL10 )
265. JOINT LOAD
266. 128 FY 76 FZ 2 MX 8
267. ********************************************************************************
268. LOAD 18 LOADTYPE NONE  TITLE WIND LOAD  ( WL11 )
269. JOINT LOAD
270. 128 FY 119 FZ 11 MX 63
271. ********************************************************************************
272. LOAD 19 LOADTYPE NONE  TITLE WIND LOAD  ( WL12 )
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273. JOINT LOAD
274. 128 FY 75 FZ 8 MX 48
275. ********************************************************************************
276. LOAD 20 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
277. JOINT LOAD
278. 128 FZ -1 MX -7
279. ********************************************************************************
280. LOAD 21 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
281. JOINT LOAD
282. 128 FZ 1 MX 7
283. ********************************************************************************
284. LOAD 22 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL1 )
285. JOINT LOAD
286. 128 FY 2
287. ********************************************************************************
288. LOAD 23 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL2 )
289. JOINT LOAD
290. 128 FY -2
291. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 210 NUMBER OF MEMBERS 1
NUMBER OF PLATES 180 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 209

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 98/ 19/ 60 DOF
TOTAL PRIMARY LOAD CASES = 23, TOTAL DEGREES OF FREEDOM = 633
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 38 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.5/ 142671.0 MB

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292. CHANGE
293. SUPPORTS
294. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
295. 402 TO 409 412 TO 419 422 TO 429 432 497 TO 504 513 TO 530 541 TO 558 561 -
296. 562 TO 578 581 TO 598 601 TO 618 621 TO 630 641 642 645 646 649 650 653 -
297. 654 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
298. ********************************************************************************
299. * UNFACTORED LOAD COMBINATIONS
300. LOAD 201 DL
301. REPEAT LOAD
302. 1 1.0
303. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

304. CHANGE
305. LOAD 202 DL+L+TL1
306. REPEAT LOAD
307. 1 1.0 3 1.0 22 1.0
308. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

309. CHANGE
310. LOAD 203 DL+L+TL2
311. REPEAT LOAD
312. 1 1.0 3 1.0 23 1.0
313. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

314. CHANGE
315. LOAD 204 DL+LR
316. REPEAT LOAD
317. 1 1.0 2 1.0
318. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

319. CHANGE
320. LOAD 205 DL+0.75L+0.75TL1+0.75LR
321. REPEAT LOAD
322. 1 1.0 2 0.75 3 0.75 22 0.75
323. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

324. CHANGE
325. LOAD 206 DL+0.75L+0.75TL2+0.75LR
326. REPEAT LOAD
327. 1 1.0 2 0.75 3 0.75 23 0.75
328. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

329. CHANGE
330. LOAD 207 DL+0.6WL1
331. REPEAT LOAD
332. 1 1.0 8 0.6
333. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

334. CHANGE
335. LOAD 208 DL+WL2
336. REPEAT LOAD
337. 1 1.0 9 1.0
338. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 208

339. CHANGE
340. LOAD 209 DL+WL3
341. REPEAT LOAD
342. 1 1.0 10 1.0
343. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 209

344. CHANGE
345. LOAD 210 DL+WL4
346. REPEAT LOAD
347. 1 1.0 11 1.0
348. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 210

349. CHANGE
350. LOAD 211 DL+WL5
351. REPEAT LOAD
352. 1 1.0 12 1.0
353. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 211

354. CHANGE
355. LOAD 212 DL+WL6
356. REPEAT LOAD
357. 1 1.0 13 1.0
358. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 212

359. CHANGE
360. LOAD 213 DL+WL7
361. REPEAT LOAD
362. 1 1.0 14 1.0
363. PERFORM ANALYSIS

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 213

364. CHANGE
365. LOAD 214 DL+WL8
366. REPEAT LOAD
367. 1 1.0 15 1.0
368. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 214

369. CHANGE
370. LOAD 215 DL+WL9
371. REPEAT LOAD
372. 1 1.0 16 1.0
373. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 215

374. CHANGE
375. LOAD 216 DL+WL10
376. REPEAT LOAD
377. 1 1.0 17 1.0
378. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 216

379. CHANGE
380. LOAD 217 DL+WL11
381. REPEAT LOAD
382. 1 1.0 18 1.0
383. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 217

384. CHANGE
385. LOAD 218 DL+WL12
386. REPEAT LOAD
387. 1 1.0 19 1.0
388. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

389. CHANGE
390. LOAD 219 DL+0.75WL1+0.75L+0.75LR
391. REPEAT LOAD
392. 1 1.0 2 0.75 3 0.75 8 0.75
393. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 219

394. CHANGE
395. LOAD 220 DL+0.75WL2+0.75L+0.75LR
396. REPEAT LOAD
397. 1 1.0 2 0.75 3 0.75 9 0.75
398. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 220

399. CHANGE
400. LOAD 221 DL+0.75WL3+0.75L+0.75LR
401. REPEAT LOAD
402. 1 1.0 2 0.75 3 0.75 10 0.75
403. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 221

404. CHANGE
405. LOAD 222 DL+0.75WL4+0.75L+0.75LR
406. REPEAT LOAD
407. 1 1.0 2 0.75 3 0.75 11 0.75
408. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 222

409. CHANGE
410. LOAD 223 DL+0.75WL5+0.75L+0.75LR
411. REPEAT LOAD
412. 1 1.0 2 0.75 3 0.75 12 0.75
413. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 223

414. CHANGE
415. LOAD 224 DL+0.75WL67+0.75L+0.75LR
416. REPEAT LOAD
417. 1 1.0 2 0.75 3 0.75 13 0.75
418. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 224

419. CHANGE
420. LOAD 225 DL+0.75WL7+0.75L+0.75LR
421. REPEAT LOAD
422. 1 1.0 2 0.75 3 0.75 14 0.75
423. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 225

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 424. CHANGE
 425. LOAD 226 DL+0.75WL8+0.75L+0.75LR
 426. REPEAT LOAD
 427. 1 1.0 2 0.75 3 0.75 15 0.75
 428. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  226



 429. CHANGE
 430. LOAD 227 DL+0.75WL9+0.75L+0.75LR
 431. REPEAT LOAD
 432. 1 1.0 2 0.75 3 0.75 16 0.75
 433. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  227



 434. CHANGE
 435. LOAD 228 DL+0.75WL10+0.75L+0.75LR
 436. REPEAT LOAD
 437. 1 1.0 2 0.75 3 0.75 17 0.75
 438. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  228



 439. CHANGE
 440. LOAD 229 DL+0.75WL11+0.75L+0.75LR
 441. REPEAT LOAD
 442. 1 1.0 2 0.75 3 0.75 18 0.75
 443. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  229



 444. CHANGE
 445. LOAD 230 DL+0.75WL12+0.75L+0.75LR
 446. REPEAT LOAD
 447. 1 1.0 2 0.75 3 0.75 19 0.75
```



```
 448. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  230



 449. CHANGE
 450. LOAD 231 0.6DL+WL1
 451. REPEAT LOAD
 452. 1 0.6 8 1.0
 453. PERFORM ANALYSIS
**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=  231



 454. CHANGE
 455. LOAD 232 0.6DL+WL2
 456. REPEAT LOAD
 457. 1 0.6 9 1.0
 458. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  232



 459. CHANGE
 460. LOAD 233 0.6DL+WL3
 461. REPEAT LOAD
 462. 1 0.6 10 1.0
 463. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  233
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```
 464. CHANGE
 465. LOAD 234 0.6DL+WL4
 466. REPEAT LOAD
 467. 1 0.6 11 1.0
 468. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  234



 469. CHANGE
 470. LOAD 235 0.6DL+WL5
 471. REPEAT LOAD
 472. 1 0.6 12 1.0
 473. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  235



 474. CHANGE
 475. LOAD 236 0.6DL+WL6
 476. REPEAT LOAD
 477. 1 0.6 13 1.0
 478. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  236



 479. CHANGE
 480. LOAD 237 0.6DL+WL7
 481. REPEAT LOAD
 482. 1 0.6 14 1.0
 483. PERFORM ANALYSIS
```



```
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  237



 484. CHANGE
 485. LOAD 238 0.6DL+WL8
 486. REPEAT LOAD
 487. 1 0.6 15 1.0
 488. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  238



 489. CHANGE
 490. LOAD 239 0.6DL+WL9
 491. REPEAT LOAD
 492. 1 0.6 16 1.0
 493. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  239



 494. CHANGE
 495. LOAD 240 0.6DL+WL10
 496. REPEAT LOAD
 497. 1 0.6 17 1.0
 498. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  240



 499. CHANGE
 500. LOAD 241 0.6DL+WL11
```

```
 501. REPEAT LOAD
 502. 1 0.6 18 1.0
 503. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  241


 504. CHANGE
 505. LOAD 242 0.6DL+WL12
 506. REPEAT LOAD
 507. 1 0.6 19 1.0
 508. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  242


 509. CHANGE
 510. LOAD 243 0.6DL+0.7SL1
 511. REPEAT LOAD
 512. 1 0.6 20 0.7
 513. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  243


 514. CHANGE
 515. LOAD 244 0.6DL+0.7SL2
 516. REPEAT LOAD
 517. 1 0.6 21 0.7
 518. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  244


 519. CHANGE
 520. LOAD 245 DL+0.75WL1+0.75L+0.75LR+CR1
 521. REPEAT LOAD
 522. 1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
 523. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  245


 524. CHANGE
 525. LOAD 246 DL+0.75WL2+0.75L+0.75LR+CR1
 526. REPEAT LOAD
 527. 1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
 528. PERFORM ANALYSIS
**START ITERATION NO.           2

**NOTE-Tension/Compression converged after  2 iterations, Case=  246


 529. CHANGE
 530. LOAD 247 DL+0.75WL3+0.75L+0.75LR+CR1
 531. REPEAT LOAD
 532. 1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
 533. PERFORM ANALYSIS
**START ITERATION NO.           2

**NOTE-Tension/Compression converged after  2 iterations, Case=  247


 534. CHANGE
 535. LOAD 248 DL+0.75WL4+0.75L+0.75LR+CR1
 536. REPEAT LOAD
 537. 1 1.0 2 0.75 3 0.75 11 0.75 4 1.0
 538. PERFORM ANALYSIS
**START ITERATION NO.           2

**NOTE-Tension/Compression converged after  2 iterations, Case=  248


 539. CHANGE
 540. LOAD 249 DL+0.75WL5+0.75L+0.75LR+CR1
 541. REPEAT LOAD
 542. 1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
 543. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 249

544. CHANGE
545. LOAD 250 DL+0.75WL67+0.75L+0.75LR+CR1
546. REPEAT LOAD
547. 1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
548. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 250

549. CHANGE
550. LOAD 251 DL+0.75WL7+0.75L+0.75LR+CR1
551. REPEAT LOAD
552. 1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
553. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 251

554. CHANGE
555. LOAD 252 DL+0.75WL8+0.75L+0.75LR+CR1
556. REPEAT LOAD
557. 1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
558. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 252

559. CHANGE
560. LOAD 253 DL+0.75WL9+0.75L+0.75LR+CR1
561. REPEAT LOAD
562. 1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
563. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 253

564. CHANGE
565. LOAD 254 DL+0.75WL10+0.75L+0.75LR+CR1
566. REPEAT LOAD
567. 1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
568. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 254

569. CHANGE
570. LOAD 255 DL+0.75WL11+0.75L+0.75LR+CR1
571. REPEAT LOAD
572. 1 1.0 2 0.75 3 0.75 18 0.75 4 1.0
573. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 255

574. CHANGE
575. LOAD 256 DL+0.75WL12+0.75L+0.75LR+CR1
576. REPEAT LOAD
577. 1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
578. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 256

579. CHANGE
580. LOAD 257 DL+0.75WL1+0.75L+0.75LR+CR2
581. REPEAT LOAD
582. 1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
583. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 257

```
 584. CHANGE
 585. LOAD 258 DL+0.75WL2+0.75L+0.75LR+CR2
 586. REPEAT LOAD
 587. 1 1.0 2 0.75 3 0.75 9 0.75 5 1.0
 588. PERFORM ANALYSIS
**START ITERATION NO.           2

 **NOTE-Tension/Compression converged after  2 iterations, Case=  258



 589. CHANGE
 590. LOAD 259 DL+0.75WL3+0.75L+0.75LR+CR2
 591. REPEAT LOAD
 592. 1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
 593. PERFORM ANALYSIS
**START ITERATION NO.           2

 **NOTE-Tension/Compression converged after  2 iterations, Case=  259



 594. CHANGE
 595. LOAD 260 DL+0.75WL4+0.75L+0.75LR+CR2
 596. REPEAT LOAD
 597. 1 1.0 2 0.75 3 0.75 11 0.75 5 1.0
 598. PERFORM ANALYSIS
**START ITERATION NO.           2

 **NOTE-Tension/Compression converged after  2 iterations, Case=  260



 599. CHANGE
 600. LOAD 261 DL+0.75WL5+0.75L+0.75LR+CR2
 601. REPEAT LOAD
 602. 1 1.0 2 0.75 3 0.75 12 0.75 5 1.0
 603. PERFORM ANALYSIS

 **NOTE-Tension/Compression converged after  1 iterations, Case=  261



 604. CHANGE
```



```
 605. LOAD 262 DL+0.75WL67+0.75L+0.75LR+CR2
 606. REPEAT LOAD
 607. 1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
 608. PERFORM ANALYSIS

 **NOTE-Tension/Compression converged after  1 iterations, Case=  262



 609. CHANGE
 610. LOAD 263 DL+0.75WL7+0.75L+0.75LR+CR2
 611. REPEAT LOAD
 612. 1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
 613. PERFORM ANALYSIS

 **NOTE-Tension/Compression converged after  1 iterations, Case=  263



 614. CHANGE
 615. LOAD 264 DL+0.75WL8+0.75L+0.75LR+CR2
 616. REPEAT LOAD
 617. 1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
 618. PERFORM ANALYSIS

 **NOTE-Tension/Compression converged after  1 iterations, Case=  264



 619. CHANGE
 620. LOAD 265 DL+0.75WL9+0.75L+0.75LR+CR2
 621. REPEAT LOAD
 622. 1 1.0 2 0.75 3 0.75 16 0.75 5 1.0
 623. PERFORM ANALYSIS

 **NOTE-Tension/Compression converged after  1 iterations, Case=  265



 624. CHANGE
 625. LOAD 266 DL+0.75WL10+0.75L+0.75LR+CR2
 626. REPEAT LOAD
 627. 1 1.0 2 0.75 3 0.75 17 0.75 5 1.0
 628. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 266

629. CHANGE
630. LOAD 267 DL+0.75WL11+0.75L+0.75LR+CR2
631. REPEAT LOAD
632. 1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
633. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 267

634. CHANGE
635. LOAD 268 DL+0.75WL12+0.75L+0.75LR+CR2
636. REPEAT LOAD
637. 1 1.0 2 0.75 3 0.75 19 0.75 5 1.0
638. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 268

639. CHANGE
640. LOAD 269 DL+0.75WL1+0.75L+0.75LR+CR3
641. REPEAT LOAD
642. 1 1.0 2 0.75 3 0.75 8 0.75 6 1.0
643. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 269

644. CHANGE
645. LOAD 270 DL+0.75WL2+0.75L+0.75LR+CR3
646. REPEAT LOAD
647. 1 1.0 2 0.75 3 0.75 9 0.75 6 1.0
648. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 270

649. CHANGE
650. LOAD 271 DL+0.75WL3+0.75L+0.75LR+CR3
651. REPEAT LOAD
652. 1 1.0 2 0.75 3 0.75 10 0.75 6 1.0
653. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 271

654. CHANGE
655. LOAD 272 DL+0.75WL4+0.75L+0.75LR+CR3
656. REPEAT LOAD
657. 1 1.0 2 0.75 3 0.75 11 0.75 6 1.0
658. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 272

659. CHANGE
660. LOAD 273 DL+0.75WL5+0.75L+0.75LR+CR3
661. REPEAT LOAD
662. 1 1.0 2 0.75 3 0.75 12 0.75 6 1.0
663. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 273

664. CHANGE
665. LOAD 274 DL+0.75WL67+0.75L+0.75LR+CR3
666. REPEAT LOAD
667. 1 1.0 2 0.75 3 0.75 13 0.75 6 1.0
668. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 274

```
669. CHANGE
670. LOAD 275 DL+0.75WL7+0.75L+0.75LR+CR3
671. REPEAT LOAD
672. 1 1.0 2 0.75 3 0.75 14 0.75 6 1.0
673. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  275



674. CHANGE
675. LOAD 276 DL+0.75WL8+0.75L+0.75LR+CR3
676. REPEAT LOAD
677. 1 1.0 2 0.75 3 0.75 15 0.75 6 1.0
678. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  276



679. CHANGE
680. LOAD 277 DL+0.75WL9+0.75L+0.75LR+CR3
681. REPEAT LOAD
682. 1 1.0 2 0.75 3 0.75 16 0.75 6 1.0
683. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  277



684. CHANGE
685. LOAD 278 DL+0.75WL10+0.75L+0.75LR+CR3
686. REPEAT LOAD
687. 1 1.0 2 0.75 3 0.75 17 0.75 6 1.0
688. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  278



689. CHANGE
690. LOAD 279 DL+0.75WL11+0.75L+0.75LR+CR3
691. REPEAT LOAD
692. 1 1.0 2 0.75 3 0.75 18 0.75 6 1.0
```



```
693. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  279



694. CHANGE
695. LOAD 280 DL+0.75WL12+0.75L+0.75LR+CR3
696. REPEAT LOAD
697. 1 1.0 2 0.75 3 0.75 19 0.75 6 1.0
698. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  280



699. CHANGE
700. LOAD 281 DL+0.75WL1+0.75L+0.75LR+CR4
701. REPEAT LOAD
702. 1 1.0 2 0.75 3 0.75 8 0.75 7 1.0
703. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  281



704. CHANGE
705. LOAD 282 DL+0.75WL2+0.75L+0.75LR+CR4
706. REPEAT LOAD
707. 1 1.0 2 0.75 3 0.75 9 0.75 7 1.0
708. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  282



709. CHANGE
710. LOAD 283 DL+0.75WL3+0.75L+0.75LR+CR4
711. REPEAT LOAD
712. 1 1.0 2 0.75 3 0.75 10 0.75 7 1.0
713. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 283

714. CHANGE
715. LOAD 284 DL+0.75WL4+0.75L+0.75LR+CR4
716. REPEAT LOAD
717. 1 1.0 2 0.75 3 0.75 11 0.75 7 1.0
718. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 284

719. CHANGE
720. LOAD 285 DL+0.75WL5+0.75L+0.75LR+CR4
721. REPEAT LOAD
722. 1 1.0 2 0.75 3 0.75 12 0.75 7 1.0
723. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 285

724. CHANGE
725. LOAD 286 DL+0.75WL67+0.75L+0.75LR+CR4
726. REPEAT LOAD
727. 1 1.0 2 0.75 3 0.75 13 0.75 7 1.0
728. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 286

729. CHANGE
730. LOAD 287 DL+0.75WL7+0.75L+0.75LR+CR4
731. REPEAT LOAD
732. 1 1.0 2 0.75 3 0.75 14 0.75 7 1.0
733. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 287

734. CHANGE
735. LOAD 288 DL+0.75WL8+0.75L+0.75LR+CR4
736. REPEAT LOAD
737. 1 1.0 2 0.75 3 0.75 15 0.75 7 1.0
738. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 288

739. CHANGE
740. LOAD 289 DL+0.75WL9+0.75L+0.75LR+CR4
741. REPEAT LOAD
742. 1 1.0 2 0.75 3 0.75 16 0.75 7 1.0
743. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 289

744. CHANGE
745. LOAD 290 DL+0.75WL10+0.75L+0.75LR+CR4
746. REPEAT LOAD
747. 1 1.0 2 0.75 3 0.75 17 0.75 7 1.0
748. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 290

749. CHANGE
750. LOAD 291 DL+0.75WL11+0.75L+0.75LR+CR4
751. REPEAT LOAD
752. 1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
753. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 291

754. CHANGE
755. LOAD 292 DL+0.75WL12+0.75L+0.75LR+CR4
756. REPEAT LOAD

```
 757. 1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
 758. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  292



 759. CHANGE
 760. ******************************************************************************
 761. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
 762. LOAD 101 1.4DL
 763. REPEAT LOAD
 764. 1 1.4
 765. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  101



 766. CHANGE
 767. LOAD 102 1.4DL+1.4TL1+1.7L+0.5LR
 768. REPEAT LOAD
 769. 1 1.4 22 1.4 3 1.7 2 0.5
 770. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  102



 771. CHANGE
 772. LOAD 103 1.4DL+1.4TL2+1.7L+0.5LR
 773. REPEAT LOAD
 774. 1 1.4 23 1.4 3 1.7 2 0.5
 775. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  103



 776. CHANGE
 777. LOAD 104 1.2DL+1.6LR+1.0L
 778. REPEAT LOAD
 779. 1 1.2 2 1.6 3 1.0
 780. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  104



 781. CHANGE
 782. LOAD 105 1.2DL+1.6WL1+1.0L+0.5LR
 783. REPEAT LOAD
 784. 1 1.2 8 1.6 2 0.5 3 1.0
 785. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  105



 786. CHANGE
 787. LOAD 106 1.2DL+1.6WL2+1.0L+0.5LR
 788. REPEAT LOAD
 789. 1 1.2 9 1.6 2 0.5 3 1.0
 790. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4
**START ITERATION NO.           5

**NOTE-Tension/Compression converged after  5 iterations, Case=  106



 791. CHANGE
 792. LOAD 107 1.2DL+1.6WL3+1.0L+0.5LR
 793. REPEAT LOAD
 794. 1 1.2 10 1.6 2 0.5 3 1.0
 795. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  107



 796. CHANGE
 797. LOAD 108 1.2DL+1.6WL4+1.0L+0.5LR
 798. REPEAT LOAD
```

```
 799. 1 1.2 11 1.6 2 0.5 3 1.0
 800. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  108



 801. CHANGE
 802. LOAD 109 1.2DL+1.6WL5+1.0L+0.5LR
 803. REPEAT LOAD
 804. 1 1.2 12 1.6 2 0.5 3 1.0
 805. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  109



 806. CHANGE
 807. LOAD 110 1.2DL+1.6WL6+1.0L+0.5LR
 808. REPEAT LOAD
 809. 1 1.2 13 1.6 2 0.5 3 1.0
 810. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  110



 811. CHANGE
 812. LOAD 111 1.2DL+1.6WL7+1.0L+0.5LR
 813. REPEAT LOAD
 814. 1 1.2 14 1.6 2 0.5 3 1.0
 815. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
```



```
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  111



 816. CHANGE
 817. LOAD 112 1.2DL+1.6WL8+1.0L+0.5LR
 818. REPEAT LOAD
 819. 1 1.2 15 1.6 2 0.5 3 1.0
 820. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  112



 821. CHANGE
 822. LOAD 113 1.2DL+1.6WL9+1.0L+0.5LR
 823. REPEAT LOAD
 824. 1 1.2 16 1.6 2 0.5 3 1.0
 825. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  113



 826. CHANGE
 827. LOAD 114 1.2DL+1.6WL10+1.0L+0.5LR
 828. REPEAT LOAD
 829. 1 1.2 17 1.6 2 0.5 3 1.0
 830. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  114



 831. CHANGE
 832. LOAD 115 1.2DL+1.6WL11+1.0L+0.5LR
 833. REPEAT LOAD
 834. 1 1.2 18 1.6 2 0.5 3 1.0
 835. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 115

836. CHANGE
837. LOAD 116 1.2DL+1.6WL12+1.0L+0.5LR
838. REPEAT LOAD
839. 1 1.2 19 1.6 2 0.5 3 1.0
840. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 116

841. CHANGE
842. LOAD 117 1.2DL+1.0SL1+1.0L
843. REPEAT LOAD
844. 1 1.2 20 1.0 3 1.0
845. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 117

846. CHANGE
847. LOAD 118 1.2DL+1.0SL2+1.0L
848. REPEAT LOAD
849. 1 1.2 21 1.0 3 1.0
850. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 118

851. CHANGE
852. LOAD 119 0.9DL+1.6WL1
853. REPEAT LOAD
854. 1 0.9 8 1.6
855. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 119

856. CHANGE
857. LOAD 120 0.9DL+1.6WL2
858. REPEAT LOAD
859. 1 0.9 9 1.6
860. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 120

861. CHANGE
862. LOAD 121 0.9DL+1.6WL3
863. REPEAT LOAD
864. 1 0.9 10 1.6
865. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 121

866. CHANGE
867. LOAD 122 0.9DL+1.6WL4
868. REPEAT LOAD
869. 1 0.9 11 1.6
870. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 122

871. CHANGE

```
 872. LOAD 123 0.9DL+1.6WL5
 873. REPEAT LOAD
 874. 1 0.9 12 1.6
 875. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6

**NOTE-Tension/Compression converged after  6 iterations, Case=  123



 876. CHANGE
 877. LOAD 124 0.9DL+1.6WL6
 878. REPEAT LOAD
 879. 1 0.9 13 1.6
 880. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  124



 881. CHANGE
 882. LOAD 125 0.9DL+1.6WL7
 883. REPEAT LOAD
 884. 1 0.9 14 1.6
 885. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6

**NOTE-Tension/Compression converged after  6 iterations, Case=  125



 886. CHANGE
 887. LOAD 126 0.9DL+1.6WL8
 888. REPEAT LOAD
 889. 1 0.9 15 1.6
 890. PERFORM ANALYSIS
```



```
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  126



 891. CHANGE
 892. LOAD 127 0.9DL+1.6WL9
 893. REPEAT LOAD
 894. 1 0.9 16 1.6
 895. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  127



 896. CHANGE
 897. LOAD 128 0.9DL+1.6WL10
 898. REPEAT LOAD
 899. 1 0.9 17 1.6
 900. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  128



 901. CHANGE
 902. LOAD 129 0.9DL+1.6WL11
 903. REPEAT LOAD
 904. 1 0.9 18 1.6
 905. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  129



 906. CHANGE
 907. LOAD 130 0.9DL+1.6WL12
 908. REPEAT LOAD
 909. 1 0.9 19 1.6
 910. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 130

911. CHANGE
912. LOAD 131 0.9DL+1.6SL1
913. REPEAT LOAD
914. 1 0.9 20 1.6
915. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 131

916. CHANGE
917. LOAD 132 0.9DL+1.6SL2
918. REPEAT LOAD
919. 1 0.9 21 1.6
920. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 132

921. CHANGE
922. LOAD 133 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR1
923. REPEAT LOAD
924. 1 1.2 8 1.6 2 0.5 3 1.0 4 1.0
925. PERFORM ANALYSIS
**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 133

926. CHANGE
927. LOAD 134 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
928. REPEAT LOAD
929. 1 1.2 9 1.6 2 0.5 3 1.0 4 1.0
930. PERFORM ANALYSIS

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 134

931. CHANGE
932. LOAD 135 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR1
933. REPEAT LOAD
934. 1 1.2 10 1.6 2 0.5 3 1.0 4 1.0
935. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 135

936. CHANGE
937. LOAD 136 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR1
938. REPEAT LOAD
939. 1 1.2 11 1.6 2 0.5 3 1.0 4 1.0
940. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 136

941. CHANGE
942. LOAD 137 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR1
943. REPEAT LOAD
944. 1 1.2 12 1.6 2 0.5 3 1.0 4 1.0
945. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 137

946. CHANGE

```
 947. LOAD 138 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR1
 948. REPEAT LOAD
 949. 1 1.2 13 1.6 2 0.5 3 1.0 4 1.0
 950. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=  138



 951. CHANGE
 952. LOAD 139 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR1
 953. REPEAT LOAD
 954. 1 1.2 14 1.6 2 0.5 3 1.0 4 1.0
 955. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=  139



 956. CHANGE
 957. LOAD 140 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR1
 958. REPEAT LOAD
 959. 1 1.2 15 1.6 2 0.5 3 1.0 4 1.0
 960. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=  140



 961. CHANGE
 962. LOAD 141 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR1
 963. REPEAT LOAD
 964. 1 1.2 16 1.6 2 0.5 3 1.0 4 1.0
 965. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  141



 966. CHANGE
 967. LOAD 142 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR1
 968. REPEAT LOAD
 969. 1 1.2 17 1.6 2 0.5 3 1.0 4 1.0
 970. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  142



 971. CHANGE
 972. LOAD 143 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR1
 973. REPEAT LOAD
 974. 1 1.2 18 1.6 2 0.5 3 1.0 4 1.0
 975. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  143



 976. CHANGE
 977. LOAD 144 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR1
 978. REPEAT LOAD
 979. 1 1.2 19 1.6 2 0.5 3 1.0 4 1.0
 980. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  144



 981. CHANGE
 982. LOAD 145 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR2
 983. REPEAT LOAD
 984. 1 1.2 8 1.6 2 0.5 3 1.0 5 1.0
 985. PERFORM ANALYSIS
**START ITERATION NO.           2

**NOTE-Tension/Compression converged after  2 iterations, Case=  145
```

```
 986. CHANGE
 987. LOAD 146 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR2
 988. REPEAT LOAD
 989. 1 1.2 9 1.6 2 0.5 3 1.0 5 1.0
 990. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  146



 991. CHANGE
 992. LOAD 147 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR2
 993. REPEAT LOAD
 994. 1 1.2 10 1.6 2 0.5 3 1.0 5 1.0
 995. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=  147



 996. CHANGE
 997. LOAD 148 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR2
 998. REPEAT LOAD
 999. 1 1.2 11 1.6 2 0.5 3 1.0 5 1.0
1000. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  148



1001. CHANGE
1002. LOAD 149 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR2
1003. REPEAT LOAD
1004. 1 1.2 12 1.6 2 0.5 3 1.0 5 1.0
1005. PERFORM ANALYSIS
```



```
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=  149



1006. CHANGE
1007. LOAD 150 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR2
1008. REPEAT LOAD
1009. 1 1.2 13 1.6 2 0.5 3 1.0 5 1.0
1010. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  150



1011. CHANGE
1012. LOAD 151 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR2
1013. REPEAT LOAD
1014. 1 1.2 14 1.6 2 0.5 3 1.0 5 1.0
1015. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=  151



1016. CHANGE
1017. LOAD 152 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR2
1018. REPEAT LOAD
1019. 1 1.2 15 1.6 2 0.5 3 1.0 5 1.0
1020. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  152



1021. CHANGE
1022. LOAD 153 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR2
1023. REPEAT LOAD
```

```
1024. 1 1.2 16 1.6 2 0.5 3 1.0 5 1.0
1025. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  153



1026. CHANGE
1027. LOAD 154 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR2
1028. REPEAT LOAD
1029. 1 1.2 17 1.6 2 0.5 3 1.0 5 1.0
1030. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  154



1031. CHANGE
1032. LOAD 155 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR2
1033. REPEAT LOAD
1034. 1 1.2 18 1.6 2 0.5 3 1.0 5 1.0
1035. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  155



1036. CHANGE
1037. LOAD 156 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR2
1038. REPEAT LOAD
1039. 1 1.2 19 1.6 2 0.5 3 1.0 5 1.0
1040. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  156



1041. CHANGE
1042. LOAD 157 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR3
1043. REPEAT LOAD
1044. 1 1.2 8 1.6 2 0.5 3 1.0 6 1.0
1045. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  157



1046. CHANGE
1047. LOAD 158 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR3
1048. REPEAT LOAD
1049. 1 1.2 9 1.6 2 0.5 3 1.0 6 1.0
1050. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4
**START ITERATION NO.           5

**NOTE-Tension/Compression converged after  5 iterations, Case=  158



1051. CHANGE
1052. LOAD 159 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR3
1053. REPEAT LOAD
1054. 1 1.2 10 1.6 2 0.5 3 1.0 6 1.0
1055. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  159



1056. CHANGE
1057. LOAD 160 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR3
1058. REPEAT LOAD
1059. 1 1.2 11 1.6 2 0.5 3 1.0 6 1.0
1060. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4
**START ITERATION NO.           5

**NOTE-Tension/Compression converged after  5 iterations, Case=  160
```

```
1061. CHANGE
1062. LOAD 161 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR3
1063. REPEAT LOAD
1064. 1 1.2 12 1.6 2 0.5 3 1.0 6 1.0
1065. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=  161



1066. CHANGE
1067. LOAD 162 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR3
1068. REPEAT LOAD
1069. 1 1.2 13 1.6 2 0.5 3 1.0 6 1.0
1070. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  162



1071. CHANGE
1072. LOAD 163 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR3
1073. REPEAT LOAD
1074. 1 1.2 14 1.6 2 0.5 3 1.0 6 1.0
1075. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=  163



1076. CHANGE
1077. LOAD 164 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR3
1078. REPEAT LOAD
1079. 1 1.2 15 1.6 2 0.5 3 1.0 6 1.0
1080. PERFORM ANALYSIS
```



```
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  164



1081. CHANGE
1082. LOAD 165 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR3
1083. REPEAT LOAD
1084. 1 1.2 16 1.6 2 0.5 3 1.0 6 1.0
1085. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  165



1086. CHANGE
1087. LOAD 166 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR3
1088. REPEAT LOAD
1089. 1 1.2 17 1.6 2 0.5 3 1.0 6 1.0
1090. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  166



1091. CHANGE
1092. LOAD 167 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR3
1093. REPEAT LOAD
1094. 1 1.2 18 1.6 2 0.5 3 1.0 6 1.0
1095. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  167



1096. CHANGE
1097. LOAD 168 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR3
1098. REPEAT LOAD
1099. 1 1.2 19 1.6 2 0.5 3 1.0 6 1.0
1100. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 168

1101. CHANGE
1102. LOAD 169 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR4
1103. REPEAT LOAD
1104. 1 1.2 8 1.6 2 0.5 3 1.0 7 1.0
1105. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 169

1106. CHANGE
1107. LOAD 170 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR4
1108. REPEAT LOAD
1109. 1 1.2 9 1.6 2 0.5 3 1.0 7 1.0
1110. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 170

1111. CHANGE
1112. LOAD 171 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR4
1113. REPEAT LOAD
1114. 1 1.2 10 1.6 2 0.5 3 1.0 7 1.0
1115. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 171

1116. CHANGE
1117. LOAD 172 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR4
1118. REPEAT LOAD
1119. 1 1.2 11 1.6 2 0.5 3 1.0 7 1.0
1120. PERFORM ANALYSIS

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 172

1121. CHANGE
1122. LOAD 173 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR4
1123. REPEAT LOAD
1124. 1 1.2 12 1.6 2 0.5 3 1.0 7 1.0
1125. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 173

1126. CHANGE
1127. LOAD 174 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR4
1128. REPEAT LOAD
1129. 1 1.2 13 1.6 2 0.5 3 1.0 7 1.0
1130. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 174

1131. CHANGE
1132. LOAD 175 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR4
1133. REPEAT LOAD
1134. 1 1.2 14 1.6 2 0.5 3 1.0 7 1.0
1135. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 175

1136. CHANGE
1137. LOAD 176 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR4
1138. REPEAT LOAD
1139. 1 1.2 15 1.6 2 0.5 3 1.0 7 1.0
1140. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 176

1141. CHANGE
1142. LOAD 177 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR4
1143. REPEAT LOAD
1144. 1 1.2 16 1.6 2 0.5 3 1.0 7 1.0
1145. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 177

1146. CHANGE
1147. LOAD 178 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR4
1148. REPEAT LOAD
1149. 1 1.2 17 1.6 2 0.5 3 1.0 7 1.0
1150. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 178

1151. CHANGE
1152. LOAD 179 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR4
1153. REPEAT LOAD
1154. 1 1.2 18 1.6 2 0.5 3 1.0 7 1.0
1155. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 179

1156. CHANGE
1157. LOAD 180 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR4
1158. REPEAT LOAD
1159. 1 1.2 19 1.6 2 0.5 3 1.0 7 1.0
1160. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 180

1161. CHANGE
1162. **
1163. PERFORM ANALYSIS

** ALL CASES BEING MADE ACTIVE BEFORE RE-ANALYSIS. **

**NOTE-Tension/Compression converged after 1 iterations, Case= 1

**NOTE-Tension/Compression converged after 1 iterations, Case= 2

**NOTE-Tension/Compression converged after 1 iterations, Case= 3

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.2740645E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 6 iterations, Case= 4

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.2740645E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 6 iterations, Case= 5

**NOTE-Tension/Compression converged after 1 iterations, Case= 6

**NOTE-Tension/Compression converged after 1 iterations, Case= 7

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.2740645E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 6 iterations, Case= 8

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.2740645E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.0012994E+04 L-MATRIX DIAG= 6.0754246E-10 EQN NO 631
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.0701049E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2310213E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 6 iterations, Case= 9

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.2740645E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.0012994E+04 L-MATRIX DIAG= 6.0754246E-10 EQN NO 631
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.0701049E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2310213E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 6 iterations, Case= 10

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.2740645E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.0012994E+04 L-MATRIX DIAG= 6.0754246E-10 EQN NO 631
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.0701049E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2310213E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 6 iterations, Case= 11

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= -8.3353370E-08 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.0012994E+04 L-MATRIX DIAG= 6.0754246E-10 EQN NO 631
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.0701049E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2310213E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 7
**START ITERATION NO. 8
**START ITERATION NO. 9
**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2328839E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2326511E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= -3.9627776E-07 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2319526E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 13

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.0012994E+04 L-MATRIX DIAG= 6.0754246E-10 EQN NO 631
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.0701049E-06 EQN NO 632

***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2310213E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 14
**START ITERATION NO. 15

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 12
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= -8.3353370E-08 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.0012994E+04 L-MATRIX DIAG= 6.0754246E-10 EQN NO 631
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.0701049E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2310213E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 7
**START ITERATION NO. 8
**START ITERATION NO. 9
**START ITERATION NO. 10
**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= -4.8847869E-07 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= -2.8265640E-07 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 13

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= -3.9627776E-07 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2319526E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 14

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.0012994E+04 L-MATRIX DIAG= 6.0754246E-10 EQN NO 631
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.0701049E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2310213E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 15

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 13
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= -8.3353370E-08 EQN NO 632

***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.0012994E+04 L-MATRIX DIAG= 6.0754246E-10 EQN NO 631
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.0701049E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2310213E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 7
**START ITERATION NO. 8
**START ITERATION NO. 9
**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2328839E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

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 **START ITERATION NO.          11

***WARNING -  INSTABILITY AT JOINT   663   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.0281236E+06 L-MATRIX DIAG=  7.2326511E-06 EQN NO     633
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.          12

***WARNING -  INSTABILITY AT JOINT   663   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  3.3502758E+06 L-MATRIX DIAG= -3.9627776E-07 EQN NO     632
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.


***WARNING -  INSTABILITY AT JOINT   663   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.0281236E+06 L-MATRIX DIAG=  7.2319526E-06 EQN NO     633
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
 **START ITERATION NO.          13

***WARNING -  INSTABILITY AT JOINT   663   DIRECTION = FY
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  3.0012994E+04 L-MATRIX DIAG=  6.0754246E-10 EQN NO     631
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.
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***WARNING -  INSTABILITY AT JOINT   663   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  3.3502758E+06 L-MATRIX DIAG=  3.0701049E-06 EQN NO     632
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING -  INSTABILITY AT JOINT   663   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.0281236E+06 L-MATRIX DIAG=  7.2310213E-06 EQN NO     633
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
 **START ITERATION NO.          14
 **START ITERATION NO.          15

 **WARNING-Tension/Compression did not converge after 15 iterations, Case=   14
 ** If results are not satisfactory then enter a
 ** SET ITERLIM i command (i>10) before loads to get more iterations.



 **START ITERATION NO.           2
 **START ITERATION NO.           3
 **START ITERATION NO.           4
 **START ITERATION NO.           5

***WARNING -  INSTABILITY AT JOINT   663   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  3.3502758E+06 L-MATRIX DIAG= -8.3353370E-08 EQN NO     632
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.           6

***WARNING -  INSTABILITY AT JOINT   663   DIRECTION = FY
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  3.0012994E+04 L-MATRIX DIAG=  6.0754246E-10 EQN NO     631
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.
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***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.0701049E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2310213E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 7
**START ITERATION NO. 8
**START ITERATION NO. 9
**START ITERATION NO. 10
**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= -4.8847869E-07 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= -2.8265640E-07 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 13

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= -3.9627776E-07 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2319526E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 14

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.0012994E+04 L-MATRIX DIAG= 6.0754246E-10 EQN NO 631
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.0701049E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2310213E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 15

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 15
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**START ITERATION NO. 2

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.0012994E+04 L-MATRIX DIAG= 6.0754246E-10 EQN NO 631
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.0701049E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2310213E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 2 iterations, Case= 16

**START ITERATION NO. 2

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.0012994E+04 L-MATRIX DIAG= 6.0754246E-10 EQN NO 631
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.0701049E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2310213E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 2 iterations, Case= 17

**START ITERATION NO. 2
**START ITERATION NO. 3

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= -8.3353370E-08 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 4

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.0012994E+04 L-MATRIX DIAG= 6.0754246E-10 EQN NO 631
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.0701049E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2310213E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 4 iterations, Case= 18

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= -8.3353370E-08 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.0012994E+04 L-MATRIX DIAG= 6.0754246E-10 EQN NO 631
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.0701049E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2310213E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 5 iterations, Case= 19

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.2740645E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.2591633E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.1301752E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 2.4866313E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.5450794E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 9 iterations, Case= 20

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= -8.3353370E-08 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 5.6112185E-07 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= -7.8883022E-07 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 5.8207661E-08 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.2782555E-07 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 9 iterations, Case= 21

**START ITERATION NO. 2

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.0012994E+04 L-MATRIX DIAG= 6.0754246E-10 EQN NO 631
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.3502758E+06 L-MATRIX DIAG= 3.0701049E-06 EQN NO 632
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 663 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.0281236E+06 L-MATRIX DIAG= 7.2310213E-06 EQN NO 633
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 2 iterations, Case= 22

**NOTE-Tension/Compression converged after 1 iterations, Case= 23

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 208

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 209

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 210

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 211

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 212

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 213

```
**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  214



**NOTE-Tension/Compression converged after  1 iterations, Case=  215



**NOTE-Tension/Compression converged after  1 iterations, Case=  216



**NOTE-Tension/Compression converged after  1 iterations, Case=  217



**NOTE-Tension/Compression converged after  1 iterations, Case=  218



**NOTE-Tension/Compression converged after  1 iterations, Case=  219



**NOTE-Tension/Compression converged after  1 iterations, Case=  220



**NOTE-Tension/Compression converged after  1 iterations, Case=  221



**NOTE-Tension/Compression converged after  1 iterations, Case=  222



**NOTE-Tension/Compression converged after  1 iterations, Case=  223



**NOTE-Tension/Compression converged after  1 iterations, Case=  224



**NOTE-Tension/Compression converged after  1 iterations, Case=  225



**NOTE-Tension/Compression converged after  1 iterations, Case=  226
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  227



**NOTE-Tension/Compression converged after  1 iterations, Case=  228



**NOTE-Tension/Compression converged after  1 iterations, Case=  229



**NOTE-Tension/Compression converged after  1 iterations, Case=  230


**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=  231


**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  232


**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  233


**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  234


**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  235


**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
```

**NOTE-Tension/Compression converged after 4 iterations, Case= 236

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 237

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 238

**NOTE-Tension/Compression converged after 1 iterations, Case= 239

**NOTE-Tension/Compression converged after 1 iterations, Case= 240

**NOTE-Tension/Compression converged after 1 iterations, Case= 241

**NOTE-Tension/Compression converged after 1 iterations, Case= 242

**NOTE-Tension/Compression converged after 1 iterations, Case= 243

**NOTE-Tension/Compression converged after 1 iterations, Case= 244

**NOTE-Tension/Compression converged after 1 iterations, Case= 245

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 246

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 247

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 248

**NOTE-Tension/Compression converged after 1 iterations, Case= 249

**NOTE-Tension/Compression converged after 1 iterations, Case= 250

**NOTE-Tension/Compression converged after 1 iterations, Case= 251

**NOTE-Tension/Compression converged after 1 iterations, Case= 252

**NOTE-Tension/Compression converged after 1 iterations, Case= 253

**NOTE-Tension/Compression converged after 1 iterations, Case= 254

**NOTE-Tension/Compression converged after 1 iterations, Case= 255

**NOTE-Tension/Compression converged after 1 iterations, Case= 256

**NOTE-Tension/Compression converged after 1 iterations, Case= 257

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 258

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 259

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 1 iterations, Case= 261

**NOTE-Tension/Compression converged after 1 iterations, Case= 262

**NOTE-Tension/Compression converged after 1 iterations, Case= 263

**NOTE-Tension/Compression converged after 1 iterations, Case= 264

**NOTE-Tension/Compression converged after 1 iterations, Case= 265

**NOTE-Tension/Compression converged after 1 iterations, Case= 266

**NOTE-Tension/Compression converged after 1 iterations, Case= 267

**NOTE-Tension/Compression converged after 1 iterations, Case= 268

**NOTE-Tension/Compression converged after 1 iterations, Case= 269

**NOTE-Tension/Compression converged after 1 iterations, Case= 270

**NOTE-Tension/Compression converged after 1 iterations, Case= 271

**NOTE-Tension/Compression converged after 1 iterations, Case= 272

**NOTE-Tension/Compression converged after 1 iterations, Case= 273

**NOTE-Tension/Compression converged after 1 iterations, Case= 275

**NOTE-Tension/Compression converged after 1 iterations, Case= 276

**NOTE-Tension/Compression converged after 1 iterations, Case= 277

**NOTE-Tension/Compression converged after 1 iterations, Case= 278

**NOTE-Tension/Compression converged after 1 iterations, Case= 279

**NOTE-Tension/Compression converged after 1 iterations, Case= 280

**NOTE-Tension/Compression converged after 1 iterations, Case= 281

**NOTE-Tension/Compression converged after 1 iterations, Case= 282

**NOTE-Tension/Compression converged after 1 iterations, Case= 283

**NOTE-Tension/Compression converged after 1 iterations, Case= 284

**NOTE-Tension/Compression converged after 1 iterations, Case= 285

**NOTE-Tension/Compression converged after 1 iterations, Case= 286

**NOTE-Tension/Compression converged after 1 iterations, Case= 287

**NOTE-Tension/Compression converged after 1 iterations, Case= 289

**NOTE-Tension/Compression converged after 1 iterations, Case= 290

**NOTE-Tension/Compression converged after 1 iterations, Case= 291

**NOTE-Tension/Compression converged after 1 iterations, Case= 292

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

**NOTE-Tension/Compression converged after 1 iterations, Case= 102

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

**NOTE-Tension/Compression converged after 1 iterations, Case= 104

**NOTE-Tension/Compression converged after 1 iterations, Case= 105

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 106

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 107

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 5 iterations, Case= 108

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 109

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 110

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 111

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 112

**NOTE-Tension/Compression converged after 1 iterations, Case= 113

**NOTE-Tension/Compression converged after 1 iterations, Case= 114

**NOTE-Tension/Compression converged after 1 iterations, Case= 115

**NOTE-Tension/Compression converged after 1 iterations, Case= 116

**NOTE-Tension/Compression converged after 1 iterations, Case= 117

**NOTE-Tension/Compression converged after 1 iterations, Case= 118

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 3 iterations, Case= 119

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 120

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 121

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 122

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 123

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 124

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 125

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 126

**NOTE-Tension/Compression converged after 1 iterations, Case= 127

**NOTE-Tension/Compression converged after 1 iterations, Case= 128

**NOTE-Tension/Compression converged after 1 iterations, Case= 129

**NOTE-Tension/Compression converged after 1 iterations, Case= 130

**NOTE-Tension/Compression converged after 1 iterations, Case= 131

**NOTE-Tension/Compression converged after 1 iterations, Case= 132

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 133

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 134

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 135

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 136

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 4 iterations, Case= 137

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 138

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 139

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 140

**NOTE-Tension/Compression converged after 1 iterations, Case= 141

**NOTE-Tension/Compression converged after 1 iterations, Case= 142

**NOTE-Tension/Compression converged after 1 iterations, Case= 143

**NOTE-Tension/Compression converged after 1 iterations, Case= 144

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 145

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 146

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 147

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 148

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 149

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 150

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 151

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 152

**NOTE-Tension/Compression converged after 1 iterations, Case= 153

**NOTE-Tension/Compression converged after 1 iterations, Case= 154

**NOTE-Tension/Compression converged after 1 iterations, Case= 155

**NOTE-Tension/Compression converged after 1 iterations, Case= 156

**NOTE-Tension/Compression converged after 1 iterations, Case= 157

```
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  158



**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  159



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  160



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  161



**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  162



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  163



**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  164



**NOTE-Tension/Compression converged after  1 iterations, Case=  165



**NOTE-Tension/Compression converged after  1 iterations, Case=  166
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```
**NOTE-Tension/Compression converged after  1 iterations, Case=  167



**NOTE-Tension/Compression converged after  1 iterations, Case=  168



**NOTE-Tension/Compression converged after  1 iterations, Case=  169



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  170



**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  171



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  172



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  173



**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  174



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  175



**START ITERATION NO.          2
**START ITERATION NO.          3
```

**NOTE-Tension/Compression converged after 3 iterations, Case= 176

**NOTE-Tension/Compression converged after 1 iterations, Case= 177

**NOTE-Tension/Compression converged after 1 iterations, Case= 178

**NOTE-Tension/Compression converged after 1 iterations, Case= 179

**NOTE-Tension/Compression converged after 1 iterations, Case= 180

1164. LOAD LIST 101 TO 180
1165. START CONCRETE DESIGN

1166. CODE ACI
1167. TRACK 2 ALL
1168. FC 35000 MEMB 185
1169. CLB 0.05 MEMB 185
1170. CLS 0.05 ALL
1171. CLT 0.05 MEMB 185 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 -
1172. 372 393 TO 399 402 TO 409 412 TO 419 422 TO 429 432 497 TO 504 513 TO 530 -
1173. 621 TO 630
1174. MAXMAIN 25 ALL
1175. MINMAIN 25 ALL
1176. MINSEC 10 ALL
1177. DESIGN COLUMN 185

==========

COLUMN NO. 185 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, RECT SIZE - 850.0 X 500.0 MMS, TIED
AREA OF STEEL REQUIRED = 4496.5 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 25 MM	1.386	150	STA	0.650

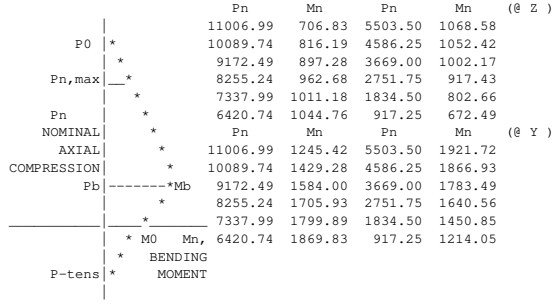
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
14905.30	11924.24	5166.13	1075.52	208.2
M0	P-tens.	Des.Pn	Des.Mn	e/h
506.09	-2436.79	14.16	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
14905.30	11924.24	5535.96	1922.86	347.3
M0	P-tens.	Des.Pn	Des.Mn	e/h
909.55	-2436.79	14.16	665.85	23.51702



**********END OF COLUMN DESIGN RESULTS**********

1178. END CONCRETE DESIGN

1179. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 18,2020 TIME= 17:35:32 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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* http://www.bentley.com *
****

ATTACHMENT 11 STAAD OUTPUT – FOUNDATION F2

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    OCT  1, 2020                 *
*          Time=    16:39:38                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F2.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 04/09/2020
    4. JOB NAME FDN. MKD. F2
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. SET NL 200
   12. SET ITERLIM 15
   13. SET PRINT 1
   14. UNIT METER KN
   15. JOINT COORDINATES
   16. 2 0 911.8 3.175; 6 0.75 911.8 3.175; 10 -0.75 911.8 3.175; 34 -0.5 911.8 3.175
   17. 35 -0.25 911.8 3.175; 41 0.25 911.8 3.175; 42 0.5 911.8 3.175
   18. 128 0 913.8 3.175; 302 -0.5 911.8 3.42438; 303 -0.75 911.8 3.42438
   19. 304 -0.25 911.8 3.42438; 305 0 911.8 3.42438; 306 0.25 911.8 3.42438
   20. 307 0.5 911.8 3.42438; 308 0.75 911.8 3.42438; 309 1 911.8 3.175
   21. 313 1 911.8 3.42438; 316 -1 911.8 3.175; 320 -1 911.8 3.42438
   22. 321 -0.75 911.8 3.67438; 322 -0.5 911.8 3.67438; 323 -0.25 911.8 3.67438
   23. 324 0 911.8 3.67438; 325 0.25 911.8 3.67438; 326 0.5 911.8 3.67438
   24. 327 0.75 911.8 3.67438; 328 1 911.8 3.67438; 331 -1 911.8 3.67438
   25. 332 -0.75 911.8 3.92438; 333 -0.5 911.8 3.92438; 334 -0.25 911.8 3.92438
   26. 335 0 911.8 3.92438; 336 0.25 911.8 3.92438; 337 0.5 911.8 3.92438
   27. 338 0.75 911.8 3.92438; 339 1 911.8 3.92438; 342 -1 911.8 3.92438
   28. 343 -0.75 911.8 4.17438; 344 -0.5 911.8 4.17438; 345 -0.25 911.8 4.17438
   29. 346 0 911.8 4.17438; 347 0.25 911.8 4.17438; 348 0.5 911.8 4.17438
   30. 349 0.75 911.8 4.17438; 350 1 911.8 4.17438; 353 -1 911.8 4.17438
   31. 365 -0.75 911.8 2.17438; 366 -0.5 911.8 2.17438; 369 -0.25 911.8 2.17438
   32. 371 0 911.8 2.17438; 373 0.25 911.8 2.17438; 375 0.5 911.8 2.17438
   33. 377 0.75 911.8 2.17438; 379 1 911.8 2.17438; 384 -1 911.8 2.17438
   34. 387 -0.75 911.8 2.42438; 388 -0.5 911.8 2.42438; 389 -0.25 911.8 2.42438
   35. 390 0 911.8 2.42438; 391 0.25 911.8 2.42438; 392 0.5 911.8 2.42438
   36. 393 0.75 911.8 2.42438; 394 1 911.8 2.42438; 397 -1 911.8 2.42438
   37. 398 -0.75 911.8 2.67438; 399 -0.5 911.8 2.67438; 400 -0.25 911.8 2.67438
   38. 401 0 911.8 2.67438; 402 0.25 911.8 2.67438; 403 0.5 911.8 2.67438
```



```
   39. 404 0.75 911.8 2.67438; 405 1 911.8 2.67438; 408 -1 911.8 2.67438
   40. 409 -0.75 911.8 2.92438; 410 -0.5 911.8 2.92438; 411 -0.25 911.8 2.92438
   41. 412 0 911.8 2.92438; 413 0.25 911.8 2.92438; 414 0.5 911.8 2.92438
   42. 415 0.75 911.8 2.92438; 416 1 911.8 2.92438; 419 -1 911.8 2.92438
   43. 420 -0.75 911.8 1.92438; 421 -0.5 911.8 1.92438; 422 -0.5 911.8 1.57438
   44. 423 -0.75 911.8 1.57438; 424 -0.25 911.8 1.92438; 425 -0.25 911.8 1.57438
   45. 426 0 911.8 1.92438; 427 0 911.8 1.57438; 428 0.25 911.8 1.92438
   46. 429 0.25 911.8 1.57438; 430 0.5 911.8 1.92438; 431 0.5 911.8 1.57438
   47. 432 0.75 911.8 1.92438; 433 0.75 911.8 1.57438; 434 1 911.8 1.92438
   48. 435 1 911.8 1.57438; 436 -1 911.8 1.92438; 437 -1 911.8 1.57438
   49. 438 -0.75 911.8 4.42438; 439 -0.5 911.8 4.42438; 440 -0.25 911.8 4.42438
   50. 441 0 911.8 4.42438; 442 0.25 911.8 4.42438; 443 0.5 911.8 4.42438
   51. 444 0.75 911.8 4.42438; 445 1 911.8 4.42438; 446 -1 911.8 4.42438
   52. 447 -0.75 911.8 4.77438; 448 -0.5 911.8 4.77438; 449 -0.25 911.8 4.77438
   53. 450 0 911.8 4.77438; 451 0.25 911.8 4.77438; 452 0.5 911.8 4.77438
   54. 453 0.75 911.8 4.77438; 454 1 911.8 4.77438; 455 -1 911.8 4.77438
   55. 492 1.25 911.8 3.42438; 493 1.25 911.8 3.175; 494 1.25 911.8 3.67438
   56. 495 1.25 911.8 3.92438; 496 1.25 911.8 4.17438; 497 1.25 911.8 2.42438
   57. 498 1.25 911.8 2.17438; 499 1.25 911.8 2.67438; 500 1.25 911.8 2.92438
   58. 501 1.25 911.8 1.92438; 502 1.25 911.8 1.57438; 503 1.25 911.8 4.42438
   59. 504 1.25 911.8 4.77438; 509 -1.25 911.8 3.42438; 510 -1.25 911.8 3.175
   60. 511 -1.25 911.8 3.67438; 512 -1.25 911.8 3.92438; 513 -1.25 911.8 4.17438
   61. 514 -1.25 911.8 2.42438; 515 -1.25 911.8 2.17438; 516 -1.25 911.8 2.67438
   62. 517 -1.25 911.8 2.92438; 518 -1.25 911.8 1.92438; 519 -1.25 911.8 1.57438
   63. 520 -1.25 911.8 4.42438; 521 -1.25 911.8 4.77438
   64. MEMBER INCIDENCES
   65. 185 2 128
   66. ELEMENT INCIDENCES SHELL
   67. 326 303 302 34 10; 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2
   68. 333 306 307 42 41; 334 307 308 6 42; 337 308 313 309 6; 342 320 303 10 316
   69. 343 321 322 302 303; 344 322 323 304 302; 345 323 324 305 304
   70. 346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307
   71. 349 327 328 313 308; 352 331 321 303 320; 353 332 333 322 321
   72. 354 333 334 323 322; 355 334 335 324 323; 356 335 336 325 324
   73. 357 336 337 326 325; 358 337 338 327 326; 359 338 339 328 327
   74. 362 342 332 321 331; 363 343 344 333 332; 364 344 345 334 333
   75. 365 345 346 335 334; 366 346 347 336 335; 367 347 348 337 336
   76. 368 348 349 338 337; 369 349 350 339 338; 372 353 343 332 342
   77. 393 387 388 366 365; 394 388 389 369 366; 395 389 390 371 369
   78. 396 390 391 373 371; 397 391 392 375 373; 398 392 393 377 375
   79. 399 393 394 379 377; 402 397 387 365 384; 403 398 399 388 387
   80. 404 399 400 389 388; 405 400 401 390 389; 406 401 402 391 390
   81. 407 402 403 392 391; 408 403 404 393 392; 409 404 405 394 393
   82. 412 408 398 387 397; 413 409 410 399 398; 414 410 411 400 399
   83. 415 411 412 401 400; 416 412 413 402 401; 417 413 414 403 402
   84. 418 414 415 404 403; 419 415 416 405 404; 422 419 409 398 408
   85. 423 10 34 410 409; 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412
   86. 427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415; 432 316 10 409 419
   87. 433 420 421 422 423; 434 421 424 425 422; 435 424 426 427 425
   88. 436 426 428 429 427; 437 428 430 431 429; 438 430 432 433 431
   89. 439 432 434 435 433; 440 436 420 423 437; 441 365 366 421 420
   90. 442 366 369 424 421; 443 369 371 426 424; 444 371 373 428 426
   91. 445 373 375 430 428; 446 375 377 432 430; 447 377 379 434 432
   92. 448 384 365 420 436; 449 438 439 344 343; 450 439 440 345 344
   93. 451 440 441 346 345; 452 441 442 347 346; 453 442 443 348 347
   94. 454 443 444 349 348; 455 444 445 350 349; 456 446 438 343 353
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 95. 457 447 448 439 438; 458 448 449 440 439; 459 449 450 441 440
 96. 460 450 451 442 441; 461 451 452 443 442; 462 452 453 444 443
 97. 463 453 454 445 444; 464 455 447 438 446; 497 313 492 493 309
 98. 498 328 494 492 313; 499 339 495 494 328; 500 350 496 495 339
 99. 501 394 497 498 379; 502 405 499 497 394; 503 416 500 499 405
100. 504 309 493 500 416; 505 434 501 502 435; 506 379 498 501 434
101. 507 445 503 496 350; 508 454 504 503 445; 513 509 320 316 510
102. 514 511 331 320 509; 515 512 342 331 511; 516 513 353 342 512
103. 517 514 397 384 515; 518 516 408 397 514; 519 517 419 408 516
104. 520 510 316 419 517; 521 518 436 437 519; 522 515 384 436 518
105. 523 520 446 353 513; 524 521 455 446 520
106. START GROUP DEFINITION
107. ELEMENT
108. _RAFT 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 -
109. 393 TO 399 402 TO 409 412 TO 419 422 TO 429 432 TO 464 497 TO 508 -
110. 513 TO 524
111. _RAFT_MOM 326 327 333 334 337 342 TO 349 352 TO 359 362 TO 369 372 -
112. 393 TO 399 402 TO 409 412 TO 419 422 TO 424 427 TO 429 432 TO 464 -
113. 497 TO 508 513 TO 524
114. _RAFT_SHEAR 337 342 349 352 359 362 TO 369 372 393 TO 399 402 409 412 419 -
115. 422 429 432 TO 464 497 TO 508 513 TO 524
116. MEMBER
117. END GROUP DEFINITION
118. ELEMENT PROPERTY
119. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
120. 402 TO 409 412 TO 419 422 TO 429 432 TO 464 497 TO 508 513 TO 523 -
121. 524 THICKNESS 0.5
122. DEFINE MATERIAL START
123. ISOTROPIC CONCRETE
124. E 2.17185E+007
125. POISSON 0.17
126. DENSITY 25
127. ALPHA 1E-005
128. DAMP 0.05
129. TYPE CONCRETE
130. STRENGTH FCU 27579
131. END DEFINE MATERIAL
132. MEMBER PROPERTY AMERICAN
133. 185 PRIS YD 0.5 ZD 0.5
134. CONSTANTS
135. MATERIAL CONCRETE ALL
136. SUPPORTS
137. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
138. 402 TO 409 412 TO 419 422 TO 429 432 TO 464 497 TO 508 513 TO 523 -
139. 524 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
```

JOINT	INFLUENCE AREA
2	0.062500
6	0.062500
10	0.062500
34	0.062500
35	0.062500
41	0.062500
42	0.062500

302	0.062422
303	0.062422
304	0.062422
305	0.062422
306	0.062422
307	0.062422
308	0.062422
309	0.062500
313	0.062423
316	0.062500
320	0.062423
321	0.062500
322	0.062500
323	0.062500
324	0.062500
325	0.062500
326	0.062500
327	0.062500
328	0.062500
331	0.062500
332	0.062500
333	0.062500
334	0.062500
335	0.062500
336	0.062500
337	0.062500
338	0.062500
339	0.062500
342	0.062500
343	0.062500
344	0.062500
345	0.062500
346	0.062500
347	0.062500
348	0.062500
349	0.062500
350	0.062500
353	0.062500
365	0.062500
366	0.062500
369	0.062500
371	0.062500
373	0.062500
375	0.062500
377	0.062500
379	0.062500
384	0.062500
387	0.062500
388	0.062500
389	0.062500
390	0.062500
391	0.062500
392	0.062500
393	0.062500
394	0.062500
397	0.062500

398	0.062500
399	0.062500
400	0.062500
401	0.062500
402	0.062500
403	0.062500
404	0.062500
405	0.062500
408	0.062500
409	0.062578
410	0.062578
411	0.062578
412	0.062578
413	0.062578
414	0.062578
415	0.062578
416	0.062578
419	0.062578
420	0.075000
421	0.075000
422	0.043750
423	0.043750
424	0.075000
425	0.043750
426	0.075000
427	0.043750
428	0.075000
429	0.043750
430	0.075000
431	0.043750
432	0.075000
433	0.043750
434	0.075000
435	0.043750
436	0.075000
437	0.043750
438	0.075000
439	0.075000
440	0.075000
441	0.075000
442	0.075000
443	0.075000
444	0.075000
445	0.075000
446	0.075000
447	0.043750
448	0.043750
449	0.043750
450	0.043750
451	0.043750
452	0.043750
453	0.043750
454	0.043750
455	0.043750
492	0.031211
493	0.031250

494	0.031250
495	0.031250
496	0.031250
497	0.031250
498	0.031250
499	0.031250
500	0.031289
501	0.037500
502	0.021875
503	0.037500
504	0.021875
509	0.031211
510	0.031250
511	0.031250
512	0.031250
513	0.031250
514	0.031250
515	0.031250
516	0.031250
517	0.031289
518	0.037500
519	0.021875
520	0.037500
521	0.021875
TOTAL	7.999995

```
140. ********************************************************************************
141. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
142. SELFWEIGHT Y -1
143. ELEMENT LOAD
144. * LOAD DUE TO BACKFILL
145. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
146. 402 TO 409 412 TO 419 422 TO 429 432 TO 464 497 TO 508 513 TO 523 -
147. 524 PR GY -16
148. * LOAD DUE TO SURCHARGE
149. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
150. 402 TO 409 412 TO 419 422 TO 429 432 TO 464 497 TO 508 513 TO 523 -
151. 524 PR GY -20
152. JOINT LOAD
153. 128 FY -20 FZ -2
154. * LOAD DUE TO PIPE
155. 128 FY -40
156. ********************************************************************************
157. LOAD 2 LOADTYPE NONE  TITLE ROOF LIVE LOAD ( LR )
158. JOINT LOAD
159. 128 FY -29 FZ -5
160. ********************************************************************************
161. LOAD 3 LOADTYPE NONE  TITLE LIVE LOAD ( L )
162. JOINT LOAD
163. 128 FY -0.001
164. ********************************************************************************
165. LOAD 4 LOADTYPE NONE  TITLE CRANE LOAD  ( CR1 )
166. JOINT LOAD
167. 128 FY -31 FZ -8
168. ********************************************************************************
169. LOAD 5 LOADTYPE NONE  TITLE CRANE LOAD  ( CR2 )
170. JOINT LOAD
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171. 128 FY 1
172. ********************************************************************************
173. LOAD 6 LOADTYPE NONE  TITLE CRANE LOAD  ( CR3 )
174. JOINT LOAD
175. 128 FY -0.0001
176. ********************************************************************************
177. LOAD 7 LOADTYPE NONE  TITLE CRANE LOAD  ( CR4 )
178. JOINT LOAD
179. 128 FY -0.0001
180. ********************************************************************************
181. LOAD 8 LOADTYPE NONE  TITLE WIND LOAD  ( WL1 )
182. JOINT LOAD
183. 128 FY 83 FZ -37
184. ********************************************************************************
185. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD  ( WL2 )
186. JOINT LOAD
187. 128 FY 47 FZ -56
188. ********************************************************************************
189. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD  ( WL3 )
190. JOINT LOAD
191. 128 FY 83 FZ -37
192. ********************************************************************************
193. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD  ( WL4 )
194. JOINT LOAD
195. 128 FY 47 FZ -56
196. ********************************************************************************
197. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD  ( WL5 )
198. JOINT LOAD
199. 128 FY 33 FZ 35
200. ********************************************************************************
201. LOAD 13 LOADTYPE NONE  TITLE WIND LOAD  ( WL6 )
202. JOINT LOAD
203. 128 FY -2 FZ 16
204. ********************************************************************************
205. LOAD 14 LOADTYPE NONE  TITLE WIND LOAD  ( WL7 )
206. JOINT LOAD
207. 128 FY 33 FZ 35
208. ********************************************************************************
209. LOAD 15 LOADTYPE NONE  TITLE WIND LOAD  ( WL8 )
210. JOINT LOAD
211. 128 FY -2 FZ 16
212. ********************************************************************************
213. LOAD 16 LOADTYPE NONE  TITLE WIND LOAD  ( WL9 )
214. JOINT LOAD
215. 128 FX 30 FY 138.5 FZ 29
216. ********************************************************************************
217. LOAD 17 LOADTYPE NONE  TITLE WIND LOAD  ( WL10 )
218. JOINT LOAD
219. 128 FX 30 FY 102.5 FZ 11
220. ********************************************************************************
221. LOAD 18 LOADTYPE NONE  TITLE WIND LOAD  ( WL11 )
222. JOINT LOAD
223. 128 FX 30 FY 157.5 FZ 28
224. ********************************************************************************
225. LOAD 19 LOADTYPE NONE  TITLE WIND LOAD  ( WL12 )
226. JOINT LOAD
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227. 128 FX 30 FY 121.5 FZ 9
228. ********************************************************************************
229. LOAD 20 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
230. JOINT LOAD
231. 128 FY 0.0001
232. ********************************************************************************
233. LOAD 21 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
234. JOINT LOAD
235. 128 FY 0.0001
236. ********************************************************************************
237. LOAD 22 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL1 )
238. JOINT LOAD
239. 128 FY 1
240. ********************************************************************************
241. LOAD 23 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL2 )
242. JOINT LOAD
243. 128 FY -1.
244. PERFORM ANALYSIS


        P R O B L E M   S T A T I S T I C S
        -----------------------------------

  NUMBER OF JOINTS        144  NUMBER OF MEMBERS       1
  NUMBER OF PLATES        120  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS    143

       SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

  ORIGINAL/FINAL BAND-WIDTH=   122/    19/     63 DOF
TOTAL      PRIMARY LOAD CASES =    23, TOTAL DEGREES OF FREEDOM =     435
TOTAL LOAD COMBINATION  CASES =     0  SO FAR.
  SIZE OF STIFFNESS MATRIX =       28 DOUBLE  KILO-WORDS
  REQRD/AVAIL. DISK SPACE  =     13.1/ 143489.4 MB



245. CHANGE
246. SUPPORTS
247. 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 372 393 TO 399 -
248. 402 TO 409 412 TO 419 422 TO 429 432 TO 464 497 TO 508 513 TO 523 -
249. 524 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
250. ********************************************************************************
251. * UNFACTORED LOAD COMBINATIONS
252. LOAD 201 DL
253. REPEAT LOAD
254. 1 1.0
255. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

256. CHANGE
257. LOAD 202 DL+L+TL1
258. REPEAT LOAD
259. 1 1.0 3 1.0 22 1.0
260. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

261. CHANGE
262. LOAD 203 DL+L+TL2
263. REPEAT LOAD
264. 1 1.0 3 1.0 23 1.0
265. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

266. CHANGE
267. LOAD 204 DL+LR
268. REPEAT LOAD
269. 1 1.0 2 1.0
270. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

271. CHANGE
272. LOAD 205 DL+0.75L+0.75TL1+0.75LR
273. REPEAT LOAD
274. 1 1.0 2 0.75 3 0.75 22 0.75
275. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

276. CHANGE
277. LOAD 206 DL+0.75L+0.75TL2+0.75LR
278. REPEAT LOAD
279. 1 1.0 2 0.75 3 0.75 23 0.75
280. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

281. CHANGE
282. LOAD 207 DL+0.6WL1
283. REPEAT LOAD
284. 1 1.0 8 0.6
285. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

286. CHANGE
287. LOAD 208 DL+WL2
288. REPEAT LOAD
289. 1 1.0 9 1.0
290. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

291. CHANGE
292. LOAD 209 DL+WL3
293. REPEAT LOAD
294. 1 1.0 10 1.0
295. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

296. CHANGE
297. LOAD 210 DL+WL4
298. REPEAT LOAD

```
299. 1 1.0 11 1.0
300. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  210



301. CHANGE
302. LOAD 211 DL+WL5
303. REPEAT LOAD
304. 1 1.0 12 1.0
305. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  211



306. CHANGE
307. LOAD 212 DL+WL6
308. REPEAT LOAD
309. 1 1.0 13 1.0
310. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  212



311. CHANGE
312. LOAD 213 DL+WL7
313. REPEAT LOAD
314. 1 1.0 14 1.0
315. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  213



316. CHANGE
317. LOAD 214 DL+WL8
318. REPEAT LOAD
319. 1 1.0 15 1.0
320. PERFORM ANALYSIS
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```
**NOTE-Tension/Compression converged after  1 iterations, Case=  214



321. CHANGE
322. LOAD 215 DL+WL9
323. REPEAT LOAD
324. 1 1.0 16 1.0
325. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  215



326. CHANGE
327. LOAD 216 DL+WL10
328. REPEAT LOAD
329. 1 1.0 17 1.0
330. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  216



331. CHANGE
332. LOAD 217 DL+WL11
333. REPEAT LOAD
334. 1 1.0 18 1.0
335. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  217



336. CHANGE
337. LOAD 218 DL+WL12
338. REPEAT LOAD
339. 1 1.0 19 1.0
340. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

341. CHANGE
342. LOAD 219 DL+0.75WL1+0.75L+0.75LR
343. REPEAT LOAD
344. 1 1.0 2 0.75 3 0.75 8 0.75
345. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 219

346. CHANGE
347. LOAD 220 DL+0.75WL2+0.75L+0.75LR
348. REPEAT LOAD
349. 1 1.0 2 0.75 3 0.75 9 0.75
350. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 220

351. CHANGE
352. LOAD 221 DL+0.75WL3+0.75L+0.75LR
353. REPEAT LOAD
354. 1 1.0 2 0.75 3 0.75 10 0.75
355. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 221

356. CHANGE
357. LOAD 222 DL+0.75WL4+0.75L+0.75LR
358. REPEAT LOAD
359. 1 1.0 2 0.75 3 0.75 11 0.75
360. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 222

361. CHANGE
362. LOAD 223 DL+0.75WL5+0.75L+0.75LR
363. REPEAT LOAD
364. 1 1.0 2 0.75 3 0.75 12 0.75
365. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 223

366. CHANGE
367. LOAD 224 DL+0.75WL67+0.75L+0.75LR
368. REPEAT LOAD
369. 1 1.0 2 0.75 3 0.75 13 0.75
370. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 224

371. CHANGE
372. LOAD 225 DL+0.75WL7+0.75L+0.75LR
373. REPEAT LOAD
374. 1 1.0 2 0.75 3 0.75 14 0.75
375. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 225

376. CHANGE
377. LOAD 226 DL+0.75WL8+0.75L+0.75LR
378. REPEAT LOAD
379. 1 1.0 2 0.75 3 0.75 15 0.75
380. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 226

381. CHANGE
382. LOAD 227 DL+0.75WL9+0.75L+0.75LR
383. REPEAT LOAD
384. 1 1.0 2 0.75 3 0.75 16 0.75

```
 385. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  227



 386. CHANGE
 387. LOAD 228 DL+0.75WL10+0.75L+0.75LR
 388. REPEAT LOAD
 389. 1 1.0 2 0.75 3 0.75 17 0.75
 390. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  228



 391. CHANGE
 392. LOAD 229 DL+0.75WL11+0.75L+0.75LR
 393. REPEAT LOAD
 394. 1 1.0 2 0.75 3 0.75 18 0.75
 395. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  229



 396. CHANGE
 397. LOAD 230 DL+0.75WL12+0.75L+0.75LR
 398. REPEAT LOAD
 399. 1 1.0 2 0.75 3 0.75 19 0.75
 400. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  230



 401. CHANGE
 402. LOAD 231 0.6DL+WL1
 403. REPEAT LOAD
 404. 1 0.6 8 1.0
 405. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  231



 406. CHANGE
 407. LOAD 232 0.6DL+WL2
 408. REPEAT LOAD
 409. 1 0.6 9 1.0
 410. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  232



 411. CHANGE
 412. LOAD 233 0.6DL+WL3
 413. REPEAT LOAD
 414. 1 0.6 10 1.0
 415. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  233



 416. CHANGE
 417. LOAD 234 0.6DL+WL4
 418. REPEAT LOAD
 419. 1 0.6 11 1.0
 420. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  234



 421. CHANGE
 422. LOAD 235 0.6DL+WL5
 423. REPEAT LOAD
 424. 1 0.6 12 1.0
 425. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  235
```

```
 426. CHANGE
 427. LOAD 236 0.6DL+WL6
 428. REPEAT LOAD
 429. 1 0.6 13 1.0
 430. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  236


 431. CHANGE
 432. LOAD 237 0.6DL+WL7
 433. REPEAT LOAD
 434. 1 0.6 14 1.0
 435. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  237


 436. CHANGE
 437. LOAD 238 0.6DL+WL8
 438. REPEAT LOAD
 439. 1 0.6 15 1.0
 440. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  238


 441. CHANGE
 442. LOAD 239 0.6DL+WL9
 443. REPEAT LOAD
 444. 1 0.6 16 1.0
 445. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  239
```



```
 446. CHANGE
 447. LOAD 240 0.6DL+WL10
 448. REPEAT LOAD
 449. 1 0.6 17 1.0
 450. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  240


 451. CHANGE
 452. LOAD 241 0.6DL+WL11
 453. REPEAT LOAD
 454. 1 0.6 18 1.0
 455. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  241


 456. CHANGE
 457. LOAD 242 0.6DL+WL12
 458. REPEAT LOAD
 459. 1 0.6 19 1.0
 460. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  242


 461. CHANGE
 462. LOAD 243 0.6DL+0.7SL1
 463. REPEAT LOAD
 464. 1 0.6 20 0.7
 465. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  243


 466. CHANGE
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```
467. LOAD 244 0.6DL+0.7SL2
468. REPEAT LOAD
469. 1 0.6 21 0.7
470. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  244



471. CHANGE
472. LOAD 245 DL+0.75WL1+0.75L+0.75LR+CR1
473. REPEAT LOAD
474. 1 1.0 2 0.75 3 0.75 8 0.75 4 1.0
475. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  245



476. CHANGE
477. LOAD 246 DL+0.75WL2+0.75L+0.75LR+CR1
478. REPEAT LOAD
479. 1 1.0 2 0.75 3 0.75 9 0.75 4 1.0
480. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  246



481. CHANGE
482. LOAD 247 DL+0.75WL3+0.75L+0.75LR+CR1
483. REPEAT LOAD
484. 1 1.0 2 0.75 3 0.75 10 0.75 4 1.0
485. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  247



486. CHANGE
487. LOAD 248 DL+0.75WL4+0.75L+0.75LR+CR1
488. REPEAT LOAD
489. 1 1.0 2 0.75 3 0.75 11 0.75 4 1.0
490. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  248



491. CHANGE
492. LOAD 249 DL+0.75WL5+0.75L+0.75LR+CR1
493. REPEAT LOAD
494. 1 1.0 2 0.75 3 0.75 12 0.75 4 1.0
495. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  249



496. CHANGE
497. LOAD 250 DL+0.75WL67+0.75L+0.75LR+CR1
498. REPEAT LOAD
499. 1 1.0 2 0.75 3 0.75 13 0.75 4 1.0
500. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  250



501. CHANGE
502. LOAD 251 DL+0.75WL7+0.75L+0.75LR+CR1
503. REPEAT LOAD
504. 1 1.0 2 0.75 3 0.75 14 0.75 4 1.0
505. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  251



506. CHANGE
507. LOAD 252 DL+0.75WL8+0.75L+0.75LR+CR1
508. REPEAT LOAD
509. 1 1.0 2 0.75 3 0.75 15 0.75 4 1.0
510. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 252

511. CHANGE
512. LOAD 253 DL+0.75WL9+0.75L+0.75LR+CR1
513. REPEAT LOAD
514. 1 1.0 2 0.75 3 0.75 16 0.75 4 1.0
515. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 253

516. CHANGE
517. LOAD 254 DL+0.75WL10+0.75L+0.75LR+CR1
518. REPEAT LOAD
519. 1 1.0 2 0.75 3 0.75 17 0.75 4 1.0
520. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 254

521. CHANGE
522. LOAD 255 DL+0.75WL11+0.75L+0.75LR+CR1
523. REPEAT LOAD
524. 1 1.0 2 0.75 3 0.75 18 0.75 4 1.0
525. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 255

526. CHANGE
527. LOAD 256 DL+0.75WL12+0.75L+0.75LR+CR1
528. REPEAT LOAD
529. 1 1.0 2 0.75 3 0.75 19 0.75 4 1.0
530. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 256

531. CHANGE
532. LOAD 257 DL+0.75WL1+0.75L+0.75LR+CR2
533. REPEAT LOAD
534. 1 1.0 2 0.75 3 0.75 8 0.75 5 1.0
535. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 257

536. CHANGE
537. LOAD 258 DL+0.75WL2+0.75L+0.75LR+CR2
538. REPEAT LOAD
539. 1 1.0 2 0.75 3 0.75 9 0.75 5 1.0
540. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 258

541. CHANGE
542. LOAD 259 DL+0.75WL3+0.75L+0.75LR+CR2
543. REPEAT LOAD
544. 1 1.0 2 0.75 3 0.75 10 0.75 5 1.0
545. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 259

546. CHANGE
547. LOAD 260 DL+0.75WL4+0.75L+0.75LR+CR2
548. REPEAT LOAD
549. 1 1.0 2 0.75 3 0.75 11 0.75 5 1.0
550. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 260

551. CHANGE
552. LOAD 261 DL+0.75WL5+0.75L+0.75LR+CR2
553. REPEAT LOAD
554. 1 1.0 2 0.75 3 0.75 12 0.75 5 1.0

```
 555. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  261


 556. CHANGE
 557. LOAD 262 DL+0.75WL67+0.75L+0.75LR+CR2
 558. REPEAT LOAD
 559. 1 1.0 2 0.75 3 0.75 13 0.75 5 1.0
 560. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  262


 561. CHANGE
 562. LOAD 263 DL+0.75WL7+0.75L+0.75LR+CR2
 563. REPEAT LOAD
 564. 1 1.0 2 0.75 3 0.75 14 0.75 5 1.0
 565. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  263


 566. CHANGE
 567. LOAD 264 DL+0.75WL8+0.75L+0.75LR+CR2
 568. REPEAT LOAD
 569. 1 1.0 2 0.75 3 0.75 15 0.75 5 1.0
 570. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  264


 571. CHANGE
 572. LOAD 265 DL+0.75WL9+0.75L+0.75LR+CR2
 573. REPEAT LOAD
 574. 1 1.0 2 0.75 3 0.75 16 0.75 5 1.0
 575. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  265


 576. CHANGE
 577. LOAD 266 DL+0.75WL10+0.75L+0.75LR+CR2
 578. REPEAT LOAD
 579. 1 1.0 2 0.75 3 0.75 17 0.75 5 1.0
 580. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  266


 581. CHANGE
 582. LOAD 267 DL+0.75WL11+0.75L+0.75LR+CR2
 583. REPEAT LOAD
 584. 1 1.0 2 0.75 3 0.75 18 0.75 5 1.0
 585. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  267


 586. CHANGE
 587. LOAD 268 DL+0.75WL12+0.75L+0.75LR+CR2
 588. REPEAT LOAD
 589. 1 1.0 2 0.75 3 0.75 19 0.75 5 1.0
 590. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  268


 591. CHANGE
 592. LOAD 269 DL+0.75WL1+0.75L+0.75LR+CR3
 593. REPEAT LOAD
 594. 1 1.0 2 0.75 3 0.75 8 0.75 6 1.0
 595. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  269
```

```
596. CHANGE
597. LOAD 270 DL+0.75WL2+0.75L+0.75LR+CR3
598. REPEAT LOAD
599. 1 1.0 2 0.75 3 0.75 9 0.75 6 1.0
600. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  270



601. CHANGE
602. LOAD 271 DL+0.75WL3+0.75L+0.75LR+CR3
603. REPEAT LOAD
604. 1 1.0 2 0.75 3 0.75 10 0.75 6 1.0
605. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  271



606. CHANGE
607. LOAD 272 DL+0.75WL4+0.75L+0.75LR+CR3
608. REPEAT LOAD
609. 1 1.0 2 0.75 3 0.75 11 0.75 6 1.0
610. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  272



611. CHANGE
612. LOAD 273 DL+0.75WL5+0.75L+0.75LR+CR3
613. REPEAT LOAD
614. 1 1.0 2 0.75 3 0.75 12 0.75 6 1.0
615. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  273



616. CHANGE
617. LOAD 274 DL+0.75WL67+0.75L+0.75LR+CR3
618. REPEAT LOAD
```



```
619. 1 1.0 2 0.75 3 0.75 13 0.75 6 1.0
620. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  274



621. CHANGE
622. LOAD 275 DL+0.75WL7+0.75L+0.75LR+CR3
623. REPEAT LOAD
624. 1 1.0 2 0.75 3 0.75 14 0.75 6 1.0
625. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  275



626. CHANGE
627. LOAD  276 DL+0.75WL8+0.75L+0.75LR+CR3
628. REPEAT LOAD
629. 1 1.0 2 0.75 3 0.75 15 0.75 6 1.0
630. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  276



631. CHANGE
632. LOAD 277 DL+0.75WL9+0.75L+0.75LR+CR3
633. REPEAT LOAD
634. 1 1.0 2 0.75 3 0.75 16 0.75 6 1.0
635. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  277



636. CHANGE
637. LOAD  278 DL+0.75WL10+0.75L+0.75LR+CR3
638. REPEAT LOAD
639. 1 1.0 2 0.75 3 0.75 17 0.75 6 1.0
640. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 278

641. CHANGE
642. LOAD 279 DL+0.75WL11+0.75L+0.75LR+CR3
643. REPEAT LOAD
644. 1 1.0 2 0.75 3 0.75 18 0.75 6 1.0
645. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 279

646. CHANGE
647. LOAD 280 DL+0.75WL12+0.75L+0.75LR+CR3
648. REPEAT LOAD
649. 1 1.0 2 0.75 3 0.75 19 0.75 6 1.0
650. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 280

651. CHANGE
652. LOAD 281 DL+0.75WL1+0.75L+0.75LR+CR4
653. REPEAT LOAD
654. 1 1.0 2 0.75 3 0.75 8 0.75 7 1.0
655. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 281

656. CHANGE
657. LOAD 282 DL+0.75WL2+0.75L+0.75LR+CR4
658. REPEAT LOAD
659. 1 1.0 2 0.75 3 0.75 9 0.75 7 1.0
660. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 282

661. CHANGE
662. LOAD 283 DL+0.75WL3+0.75L+0.75LR+CR4
663. REPEAT LOAD
664. 1 1.0 2 0.75 3 0.75 10 0.75 7 1.0
665. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 283

666. CHANGE
667. LOAD 284 DL+0.75WL4+0.75L+0.75LR+CR4
668. REPEAT LOAD
669. 1 1.0 2 0.75 3 0.75 11 0.75 7 1.0
670. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 284

671. CHANGE
672. LOAD 285 DL+0.75WL5+0.75L+0.75LR+CR4
673. REPEAT LOAD
674. 1 1.0 2 0.75 3 0.75 12 0.75 7 1.0
675. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 285

676. CHANGE
677. LOAD 286 DL+0.75WL67+0.75L+0.75LR+CR4
678. REPEAT LOAD
679. 1 1.0 2 0.75 3 0.75 13 0.75 7 1.0
680. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 286

```
 681. CHANGE
 682. LOAD 287 DL+0.75WL7+0.75L+0.75LR+CR4
 683. REPEAT LOAD
 684. 1 1.0 2 0.75 3 0.75 14 0.75 7 1.0
 685. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  287



 686. CHANGE
 687. LOAD 288 DL+0.75WL8+0.75L+0.75LR+CR4
 688. REPEAT LOAD
 689. 1 1.0 2 0.75 3 0.75 15 0.75 7 1.0
 690. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  288



 691. CHANGE
 692. LOAD 289 DL+0.75WL9+0.75L+0.75LR+CR4
 693. REPEAT LOAD
 694. 1 1.0 2 0.75 3 0.75 16 0.75 7 1.0
 695. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  289



 696. CHANGE
 697. LOAD 290 DL+0.75WL10+0.75L+0.75LR+CR4
 698. REPEAT LOAD
 699. 1 1.0 2 0.75 3 0.75 17 0.75 7 1.0
 700. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  290



 701. CHANGE
 702. LOAD 291 DL+0.75WL11+0.75L+0.75LR+CR4
 703. REPEAT LOAD
 704. 1 1.0 2 0.75 3 0.75 18 0.75 7 1.0
```



```
 705. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  291



 706. CHANGE
 707. LOAD 292 DL+0.75WL12+0.75L+0.75LR+CR4
 708. REPEAT LOAD
 709. 1 1.0 2 0.75 3 0.75 19 0.75 7 1.0
 710. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  292



 711. CHANGE
 712. ******************************************************************************
 713. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
 714. LOAD 101 1.4DL
 715. REPEAT LOAD
 716. 1 1.4
 717. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  101



 718. CHANGE
 719. LOAD 102 1.4DL+1.4TL1+1.7L+0.5LR
 720. REPEAT LOAD
 721. 1 1.4 22 1.4 3 1.7 2 0.5
 722. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  102



 723. CHANGE
 724. LOAD  103 1.4DL+1.4TL2+1.7L+0.5LR
 725. REPEAT LOAD
 726. 1 1.4 23 1.4 3 1.7 2 0.5
 727. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

728. CHANGE
729. LOAD 104 1.2DL+1.6LR+1.0L
730. REPEAT LOAD
731. 1 1.2 2 1.6 3 1.0
732. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 104

733. CHANGE
734. LOAD 105 1.2DL+1.6WL1+1.0L+0.5LR
735. REPEAT LOAD
736. 1 1.2 8 1.6 2 0.5 3 1.0
737. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 105

738. CHANGE
739. LOAD 106 1.2DL+1.6WL2+1.0L+0.5LR
740. REPEAT LOAD
741. 1 1.2 9 1.6 2 0.5 3 1.0
742. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 106

743. CHANGE
744. LOAD 107 1.2DL+1.6WL3+1.0L+0.5LR
745. REPEAT LOAD
746. 1 1.2 10 1.6 2 0.5 3 1.0
747. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 107

748. CHANGE
749. LOAD 108 1.2DL+1.6WL4+1.0L+0.5LR
750. REPEAT LOAD
751. 1 1.2 11 1.6 2 0.5 3 1.0
752. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 108

753. CHANGE
754. LOAD 109 1.2DL+1.6WL5+1.0L+0.5LR
755. REPEAT LOAD
756. 1 1.2 12 1.6 2 0.5 3 1.0
757. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

758. CHANGE
759. LOAD 110 1.2DL+1.6WL6+1.0L+0.5LR
760. REPEAT LOAD
761. 1 1.2 13 1.6 2 0.5 3 1.0
762. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

763. CHANGE
764. LOAD 111 1.2DL+1.6WL7+1.0L+0.5LR
765. REPEAT LOAD
766. 1 1.2 14 1.6 2 0.5 3 1.0
767. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 111

```
 768. CHANGE
 769. LOAD 112 1.2DL+1.6WL8+1.0L+0.5LR
 770. REPEAT LOAD
 771. 1 1.2 15 1.6 2 0.5 3 1.0
 772. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  112



 773. CHANGE
 774. LOAD  113 1.2DL+1.6WL9+1.0L+0.5LR
 775. REPEAT LOAD
 776. 1 1.2 16 1.6 2 0.5 3 1.0
 777. PERFORM ANALYSIS
**START ITERATION NO.           2

**NOTE-Tension/Compression converged after  2 iterations, Case=  113



 778. CHANGE
 779. LOAD  114 1.2DL+1.6WL10+1.0L+0.5LR
 780. REPEAT LOAD
 781. 1 1.2 17 1.6 2 0.5 3 1.0
 782. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  114



 783. CHANGE
 784. LOAD  115 1.2DL+1.6WL11+1.0L+0.5LR
 785. REPEAT LOAD
 786. 1 1.2 18 1.6 2 0.5 3 1.0
 787. PERFORM ANALYSIS
**START ITERATION NO.           2

**NOTE-Tension/Compression converged after  2 iterations, Case=  115



 788. CHANGE
 789. LOAD  116 1.2DL+1.6WL12+1.0L+0.5LR
```



```
 790. REPEAT LOAD
 791. 1 1.2 19 1.6 2 0.5 3 1.0
 792. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  116



 793. CHANGE
 794. LOAD  117 1.2DL+1.0SL1+1.0L
 795. REPEAT LOAD
 796. 1 1.2 20 1.0 3 1.0
 797. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  117



 798. CHANGE
 799. LOAD  118 1.2DL+1.0SL2+1.0L
 800. REPEAT LOAD
 801. 1 1.2 21 1.0 3 1.0
 802. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  118



 803. CHANGE
 804. LOAD 119 0.9DL+1.6WL1
 805. REPEAT LOAD
 806. 1 0.9 8 1.6
 807. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  119



 808. CHANGE
 809. LOAD  120 0.9DL+1.6WL2
 810. REPEAT LOAD
 811. 1 0.9 9 1.6
 812. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 120

813. CHANGE
814. LOAD 121 0.9DL+1.6WL3
815. REPEAT LOAD
816. 1 0.9 10 1.6
817. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 121

818. CHANGE
819. LOAD 122 0.9DL+1.6WL4
820. REPEAT LOAD
821. 1 0.9 11 1.6
822. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 122

823. CHANGE
824. LOAD 123 0.9DL+1.6WL5
825. REPEAT LOAD
826. 1 0.9 12 1.6
827. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 123

828. CHANGE
829. LOAD 124 0.9DL+1.6WL6
830. REPEAT LOAD
831. 1 0.9 13 1.6
832. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 124

833. CHANGE
834. LOAD 125 0.9DL+1.6WL7
835. REPEAT LOAD
836. 1 0.9 14 1.6
837. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 125

838. CHANGE
839. LOAD 126 0.9DL+1.6WL8
840. REPEAT LOAD
841. 1 0.9 15 1.6
842. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 126

843. CHANGE
844. LOAD 127 0.9DL+1.6WL9
845. REPEAT LOAD
846. 1 0.9 16 1.6
847. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 127

848. CHANGE
849. LOAD 128 0.9DL+1.6WL10
850. REPEAT LOAD
851. 1 0.9 17 1.6
852. PERFORM ANALYSIS
**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 128

853. CHANGE
854. LOAD 129 0.9DL+1.6WL11
855. REPEAT LOAD
856. 1 0.9 18 1.6
857. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 129

858. CHANGE
859. LOAD 130 0.9DL+1.6WL12
860. REPEAT LOAD
861. 1 0.9 19 1.6
862. PERFORM ANALYSIS
**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 130

863. CHANGE
864. LOAD 131 0.9DL+1.6SL1
865. REPEAT LOAD
866. 1 0.9 20 1.6
867. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 131

868. CHANGE
869. LOAD 132 0.9DL+1.6SL2
870. REPEAT LOAD
871. 1 0.9 21 1.6
872. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 132

873. CHANGE
874. LOAD 133 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR1
875. REPEAT LOAD
876. 1 1.2 8 1.6 2 0.5 3 1.0 4 1.0
877. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 133

878. CHANGE
879. LOAD 134 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR1
880. REPEAT LOAD
881. 1 1.2 9 1.6 2 0.5 3 1.0 4 1.0
882. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 134

883. CHANGE
884. LOAD 135 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR1
885. REPEAT LOAD
886. 1 1.2 10 1.6 2 0.5 3 1.0 4 1.0
887. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 135

888. CHANGE
889. LOAD 136 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR1
890. REPEAT LOAD
891. 1 1.2 11 1.6 2 0.5 3 1.0 4 1.0
892. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 136

```
 893. CHANGE
 894. LOAD  137 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR1
 895. REPEAT LOAD
 896. 1 1.2 12 1.6 2 0.5 3 1.0 4 1.0
 897. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  137



 898. CHANGE
 899. LOAD  138 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR1
 900. REPEAT LOAD
 901. 1 1.2 13 1.6 2 0.5 3 1.0 4 1.0
 902. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  138



 903. CHANGE
 904. LOAD  139 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR1
 905. REPEAT LOAD
 906. 1 1.2 14 1.6 2 0.5 3 1.0 4 1.0
 907. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  139



 908. CHANGE
 909. LOAD  140 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR1
 910. REPEAT LOAD
 911. 1 1.2 15 1.6 2 0.5 3 1.0 4 1.0
 912. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  140



 913. CHANGE
 914. LOAD  141 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR1
 915. REPEAT LOAD
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 916. 1 1.2 16 1.6 2 0.5 3 1.0 4 1.0
 917. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  141



 918. CHANGE
 919. LOAD  142 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR1
 920. REPEAT LOAD
 921. 1 1.2 17 1.6 2 0.5 3 1.0 4 1.0
 922. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  142



 923. CHANGE
 924. LOAD  143 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR1
 925. REPEAT LOAD
 926. 1 1.2 18 1.6 2 0.5 3 1.0 4 1.0
 927. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  143



 928. CHANGE
 929. LOAD  144 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR1
 930. REPEAT LOAD
 931. 1 1.2 19 1.6 2 0.5 3 1.0 4 1.0
 932. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  144



 933. CHANGE
 934. LOAD 145 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR2
 935. REPEAT LOAD
 936. 1 1.2 8 1.6 2 0.5 3 1.0 5 1.0
 937. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 145

938. CHANGE
939. LOAD 146 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR2
940. REPEAT LOAD
941. 1 1.2 9 1.6 2 0.5 3 1.0 5 1.0
942. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 146

943. CHANGE
944. LOAD 147 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR2
945. REPEAT LOAD
946. 1 1.2 10 1.6 2 0.5 3 1.0 5 1.0
947. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 147

948. CHANGE
949. LOAD 148 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR2
950. REPEAT LOAD
951. 1 1.2 11 1.6 2 0.5 3 1.0 5 1.0
952. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 148

953. CHANGE
954. LOAD 149 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR2
955. REPEAT LOAD
956. 1 1.2 12 1.6 2 0.5 3 1.0 5 1.0
957. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 149

958. CHANGE
959. LOAD 150 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR2
960. REPEAT LOAD
961. 1 1.2 13 1.6 2 0.5 3 1.0 5 1.0
962. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 150

963. CHANGE
964. LOAD 151 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR2
965. REPEAT LOAD
966. 1 1.2 14 1.6 2 0.5 3 1.0 5 1.0
967. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 151

968. CHANGE
969. LOAD 152 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR2
970. REPEAT LOAD
971. 1 1.2 15 1.6 2 0.5 3 1.0 5 1.0
972. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 152

973. CHANGE
974. LOAD 153 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR2
975. REPEAT LOAD
976. 1 1.2 16 1.6 2 0.5 3 1.0 5 1.0
977. PERFORM ANALYSIS
**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 153

```
 978. CHANGE
 979. LOAD  154 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR2
 980. REPEAT LOAD
 981. 1 1.2 17 1.6 2 0.5 3 1.0 5 1.0
 982. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  154



 983. CHANGE
 984. LOAD  155 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR2
 985. REPEAT LOAD
 986. 1 1.2 18 1.6 2 0.5 3 1.0 5 1.0
 987. PERFORM ANALYSIS
**START ITERATION NO.           2

**NOTE-Tension/Compression converged after  2 iterations, Case=  155



 988. CHANGE
 989. LOAD  156 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR2
 990. REPEAT LOAD
 991. 1 1.2 19 1.6 2 0.5 3 1.0 5 1.0
 992. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  156



 993. CHANGE
 994. LOAD  157 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR3
 995. REPEAT LOAD
 996. 1 1.2 8 1.6 2 0.5 3 1.0 6 1.0
 997. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  157



 998. CHANGE
 999. LOAD  158 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR3
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1000. REPEAT LOAD
1001. 1 1.2 9 1.6 2 0.5 3 1.0 6 1.0
1002. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  158



1003. CHANGE
1004. LOAD  159 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR3
1005. REPEAT LOAD
1006. 1 1.2 10 1.6 2 0.5 3 1.0 6 1.0
1007. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  159



1008. CHANGE
1009. LOAD  160 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR3
1010. REPEAT LOAD
1011. 1 1.2 11 1.6 2 0.5 3 1.0 6 1.0
1012. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  160



1013. CHANGE
1014. LOAD  161 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR3
1015. REPEAT LOAD
1016. 1 1.2 12 1.6 2 0.5 3 1.0 6 1.0
1017. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  161



1018. CHANGE
1019. LOAD  162 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR3
1020. REPEAT LOAD
1021. 1 1.2 13 1.6 2 0.5 3 1.0 6 1.0
1022. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 162

1023. CHANGE
1024. LOAD 163 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR3
1025. REPEAT LOAD
1026. 1 1.2 14 1.6 2 0.5 3 1.0 6 1.0
1027. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 163

1028. CHANGE
1029. LOAD 164 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR3
1030. REPEAT LOAD
1031. 1 1.2 15 1.6 2 0.5 3 1.0 6 1.0
1032. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 164

1033. CHANGE
1034. LOAD 165 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR3
1035. REPEAT LOAD
1036. 1 1.2 16 1.6 2 0.5 3 1.0 6 1.0
1037. PERFORM ANALYSIS
**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 165

1038. CHANGE
1039. LOAD 166 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR3
1040. REPEAT LOAD
1041. 1 1.2 17 1.6 2 0.5 3 1.0 6 1.0
1042. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 166

1043. CHANGE
1044. LOAD 167 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR3
1045. REPEAT LOAD
1046. 1 1.2 18 1.6 2 0.5 3 1.0 6 1.0
1047. PERFORM ANALYSIS
**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 167

1048. CHANGE
1049. LOAD 168 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR3
1050. REPEAT LOAD
1051. 1 1.2 19 1.6 2 0.5 3 1.0 6 1.0
1052. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 168

1053. CHANGE
1054. LOAD 169 1.2DL+1.6WL1+1.0L+0.5LR+1.0CR4
1055. REPEAT LOAD
1056. 1 1.2 8 1.6 2 0.5 3 1.0 7 1.0
1057. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 169

1058. CHANGE
1059. LOAD 170 1.2DL+1.6WL2+1.0L+0.5LR+1.0CR4
1060. REPEAT LOAD
1061. 1 1.2 9 1.6 2 0.5 3 1.0 7 1.0
1062. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 170

1063. CHANGE
1064. LOAD 171 1.2DL+1.6WL3+1.0L+0.5LR+1.0CR4
1065. REPEAT LOAD
1066. 1 1.2 10 1.6 2 0.5 3 1.0 7 1.0
1067. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 171

1068. CHANGE
1069. LOAD 172 1.2DL+1.6WL4+1.0L+0.5LR+1.0CR4
1070. REPEAT LOAD
1071. 1 1.2 11 1.6 2 0.5 3 1.0 7 1.0
1072. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 172

1073. CHANGE
1074. LOAD 173 1.2DL+1.6WL5+1.0L+0.5LR+1.0CR4
1075. REPEAT LOAD
1076. 1 1.2 12 1.6 2 0.5 3 1.0 7 1.0
1077. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 173

1078. CHANGE
1079. LOAD 174 1.2DL+1.6WL6+1.0L+0.5LR+1.0CR4
1080. REPEAT LOAD
1081. 1 1.2 13 1.6 2 0.5 3 1.0 7 1.0
1082. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 174

1083. CHANGE
1084. LOAD 175 1.2DL+1.6WL7+1.0L+0.5LR+1.0CR4
1085. REPEAT LOAD
1086. 1 1.2 14 1.6 2 0.5 3 1.0 7 1.0
1087. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 175

1088. CHANGE
1089. LOAD 176 1.2DL+1.6WL8+1.0L+0.5LR+1.0CR4
1090. REPEAT LOAD
1091. 1 1.2 15 1.6 2 0.5 3 1.0 7 1.0
1092. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 176

1093. CHANGE
1094. LOAD 177 1.2DL+1.6WL9+1.0L+0.5LR+1.0CR4
1095. REPEAT LOAD
1096. 1 1.2 16 1.6 2 0.5 3 1.0 7 1.0
1097. PERFORM ANALYSIS
**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 177

1098. CHANGE
1099. LOAD 178 1.2DL+1.6WL10+1.0L+0.5LR+1.0CR4
1100. REPEAT LOAD
1101. 1 1.2 17 1.6 2 0.5 3 1.0 7 1.0
1102. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 178

1103. CHANGE
1104. LOAD 179 1.2DL+1.6WL11+1.0L+0.5LR+1.0CR4
1105. REPEAT LOAD

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1106. 1 1.2 18 1.6 2 0.5 3 1.0 7 1.0
1107. PERFORM ANALYSIS
**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=  179



1108. CHANGE
1109. LOAD  180 1.2DL+1.6WL12+1.0L+0.5LR+1.0CR4
1110. REPEAT LOAD
1111. 1 1.2 19 1.6 2 0.5 3 1.0 7 1.0
1112. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  180



1113. CHANGE
1114. ******************************************************************************
1115. PERFORM ANALYSIS

  ** ALL CASES BEING MADE ACTIVE BEFORE RE-ANALYSIS. **

**NOTE-Tension/Compression converged after  1 iterations, Case=    1



**NOTE-Tension/Compression converged after  1 iterations, Case=    2



**NOTE-Tension/Compression converged after  1 iterations, Case=    3



**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=    4



**START ITERATION NO.          2

***WARNING -  INSTABILITY AT JOINT   504   DIRECTION = FY
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.6981339E+04 L-MATRIX DIAG=  2.9103830E-10 EQN NO     433
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
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  **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
           DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
           BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
           STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
           IS   1.000E-09

           THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
           ELEMENTS AS WELL AS TRUE SINGULARITIES.


***WARNING -  INSTABILITY AT JOINT   504   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.1895042E+06 L-MATRIX DIAG=  1.2433156E-06 EQN NO     434
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING -  INSTABILITY AT JOINT   504   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.4256656E+06 L-MATRIX DIAG=  1.0386575E-06 EQN NO     435
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

 **NOTE-Tension/Compression converged after  2 iterations, Case=    5



 **NOTE-Tension/Compression converged after  1 iterations, Case=    6



 **NOTE-Tension/Compression converged after  1 iterations, Case=    7



 **START ITERATION NO.          2
 **START ITERATION NO.          3

***WARNING -  INSTABILITY AT JOINT   504   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.1895042E+06 L-MATRIX DIAG=  4.4470653E-07 EQN NO     434
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

  **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
           DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
           BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
           STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
           IS   1.000E-09

           THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
           ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.          4

***WARNING -  INSTABILITY AT JOINT   504   DIRECTION = FY
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.6981339E+04 L-MATRIX DIAG=  2.9103830E-10 EQN NO     433
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
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NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.2433156E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 1.0386575E-06 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 4 iterations, Case= 8

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 4.4470653E-07 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6981339E+04 L-MATRIX DIAG= 2.9103830E-10 EQN NO 433
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.2433156E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 1.0386575E-06 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 6
**START ITERATION NO. 7
**START ITERATION NO. 8
**START ITERATION NO. 9
**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 7.0431270E-07 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.6218983E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

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 **START ITERATION NO.          12

***WARNING -  INSTABILITY AT JOINT   504   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.1895042E+06 L-MATRIX DIAG=  7.8533776E-07 EQN NO     434
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.


***WARNING -  INSTABILITY AT JOINT   504   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.4256656E+06 L-MATRIX DIAG=  5.3946860E-07 EQN NO     435
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
 **START ITERATION NO.          13

***WARNING -  INSTABILITY AT JOINT   504   DIRECTION = FY
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.6981339E+04 L-MATRIX DIAG=  2.9103830E-10 EQN NO     433
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.


***WARNING -  INSTABILITY AT JOINT   504   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.1895042E+06 L-MATRIX DIAG=  1.2433156E-06 EQN NO     434
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING -  INSTABILITY AT JOINT   504   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.4256656E+06 L-MATRIX DIAG=  1.0386575E-06 EQN NO     435
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
 **START ITERATION NO.          14
 **START ITERATION NO.          15

 **WARNING-Tension/Compression did not converge after 15 iterations, Case=    9
 ** If results are not satisfactory then enter a
 ** SET ITERLIM i command (i>10) before loads to get more iterations.


 **START ITERATION NO.           2
```



```
 **START ITERATION NO.           3

***WARNING -  INSTABILITY AT JOINT   504   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.1895042E+06 L-MATRIX DIAG=  4.4470653E-07 EQN NO     434
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.           4

***WARNING -  INSTABILITY AT JOINT   504   DIRECTION = FY
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.6981339E+04 L-MATRIX DIAG=  2.9103830E-10 EQN NO     433
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.


***WARNING -  INSTABILITY AT JOINT   504   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.1895042E+06 L-MATRIX DIAG=  1.2433156E-06 EQN NO     434
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING -  INSTABILITY AT JOINT   504   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.4256656E+06 L-MATRIX DIAG=  1.0386575E-06 EQN NO     435
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

 **NOTE-Tension/Compression converged after  4 iterations, Case=   10



 **START ITERATION NO.           2
 **START ITERATION NO.           3
 **START ITERATION NO.           4

***WARNING -  INSTABILITY AT JOINT   504   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.1895042E+06 L-MATRIX DIAG=  4.4470653E-07 EQN NO     434
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
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NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6981339E+04 L-MATRIX DIAG= 2.9103830E-10 EQN NO 433
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.2433156E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 1.0386575E-06 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 6
**START ITERATION NO. 7
**START ITERATION NO. 8
**START ITERATION NO. 9
**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 7.0431270E-07 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.6218983E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 7.8533776E-07 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 5.3946860E-07 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 13

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6981339E+04 L-MATRIX DIAG= 2.9103830E-10 EQN NO 433
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.2433156E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 1.0386575E-06 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 14
**START ITERATION NO. 15

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 11
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.2428500E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6981339E+04 L-MATRIX DIAG= 2.9103830E-10 EQN NO 433
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.2433156E-06 EQN NO 434

***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 1.0386575E-06 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 5 iterations, Case= 12

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.2428500E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 5 iterations, Case= 13

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.2428500E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6981339E+04 L-MATRIX DIAG= 2.9103830E-10 EQN NO 433

***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.2433156E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 1.0386575E-06 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 5 iterations, Case= 14

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.2428500E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 5 iterations, Case= 15

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 1.0244548E-06 EQN NO 435

***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.2433156E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 1.0386575E-06 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6981339E+04 L-MATRIX DIAG= 2.9103830E-10 EQN NO 433
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.2433156E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 1.0386575E-06 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 7 iterations, Case= 16

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 7.0873648E-07 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 1.0244548E-06 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.2433156E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 1.0386575E-06 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6981339E+04 L-MATRIX DIAG= 2.9103830E-10 EQN NO 433
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.2433156E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 1.0386575E-06 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 7 iterations, Case= 17

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.2433156E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 1.0386575E-06 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6981339E+04 L-MATRIX DIAG= 2.9103830E-10 EQN NO 433
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.2433156E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 1.0386575E-06 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 6 iterations, Case= 18

**START ITERATION NO. 2
**START ITERATION NO. 3

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 9.1642141E-07 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 4

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 1.0244548E-06 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.2433156E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 1.0386575E-06 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6981339E+04 L-MATRIX DIAG= 2.9103830E-10 EQN NO 433
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.2433156E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 1.0386575E-06 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 6 iterations, Case= 19

**NOTE-Tension/Compression converged after 1 iterations, Case= 20

**NOTE-Tension/Compression converged after 1 iterations, Case= 21

**START ITERATION NO. 2

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6981339E+04 L-MATRIX DIAG= 2.9103830E-10 EQN NO 433
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1895042E+06 L-MATRIX DIAG= 1.2433156E-06 EQN NO 434
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 504 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.4256656E+06 L-MATRIX DIAG= 1.0386575E-06 EQN NO 435
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 2 iterations, Case= 22

**NOTE-Tension/Compression converged after 1 iterations, Case= 23

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

**NOTE-Tension/Compression converged after 1 iterations, Case= 214

**NOTE-Tension/Compression converged after 1 iterations, Case= 215

**NOTE-Tension/Compression converged after 1 iterations, Case= 216

**NOTE-Tension/Compression converged after 1 iterations, Case= 217

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

**NOTE-Tension/Compression converged after 1 iterations, Case= 219

**NOTE-Tension/Compression converged after 1 iterations, Case= 220

**NOTE-Tension/Compression converged after 1 iterations, Case= 221

**NOTE-Tension/Compression converged after 1 iterations, Case= 222

**NOTE-Tension/Compression converged after 1 iterations, Case= 223

**NOTE-Tension/Compression converged after 1 iterations, Case= 224

**NOTE-Tension/Compression converged after 1 iterations, Case= 226

**NOTE-Tension/Compression converged after 1 iterations, Case= 227

**NOTE-Tension/Compression converged after 1 iterations, Case= 228

**NOTE-Tension/Compression converged after 1 iterations, Case= 229

**NOTE-Tension/Compression converged after 1 iterations, Case= 230

**NOTE-Tension/Compression converged after 1 iterations, Case= 231

**NOTE-Tension/Compression converged after 1 iterations, Case= 232

**NOTE-Tension/Compression converged after 1 iterations, Case= 233

**NOTE-Tension/Compression converged after 1 iterations, Case= 234

**NOTE-Tension/Compression converged after 1 iterations, Case= 235

**NOTE-Tension/Compression converged after 1 iterations, Case= 236

**NOTE-Tension/Compression converged after 1 iterations, Case= 237

**NOTE-Tension/Compression converged after 1 iterations, Case= 238

**NOTE-Tension/Compression converged after 4 iterations, Case= 239

**NOTE-Tension/Compression converged after 1 iterations, Case= 240

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 241

**NOTE-Tension/Compression converged after 1 iterations, Case= 242

**NOTE-Tension/Compression converged after 1 iterations, Case= 243

**NOTE-Tension/Compression converged after 1 iterations, Case= 244

**NOTE-Tension/Compression converged after 1 iterations, Case= 245

**NOTE-Tension/Compression converged after 1 iterations, Case= 246

**NOTE-Tension/Compression converged after 1 iterations, Case= 247

**NOTE-Tension/Compression converged after 1 iterations, Case= 248

**NOTE-Tension/Compression converged after 1 iterations, Case= 249

**NOTE-Tension/Compression converged after 1 iterations, Case= 250

**NOTE-Tension/Compression converged after 1 iterations, Case= 251

**NOTE-Tension/Compression converged after 1 iterations, Case= 252

**NOTE-Tension/Compression converged after 1 iterations, Case= 253

**NOTE-Tension/Compression converged after 1 iterations, Case= 254

**NOTE-Tension/Compression converged after 1 iterations, Case= 255

**NOTE-Tension/Compression converged after 1 iterations, Case= 256

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**NOTE-Tension/Compression converged after 1 iterations, Case= 258

**NOTE-Tension/Compression converged after 1 iterations, Case= 259

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**NOTE-Tension/Compression converged after 1 iterations, Case= 286

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**NOTE-Tension/Compression converged after 1 iterations, Case= 291

**NOTE-Tension/Compression converged after 1 iterations, Case= 292

**NOTE-Tension/Compression converged after 1 iterations, Case= 102

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

**NOTE-Tension/Compression converged after 1 iterations, Case= 104

**NOTE-Tension/Compression converged after 1 iterations, Case= 105

**NOTE-Tension/Compression converged after 1 iterations, Case= 106

**NOTE-Tension/Compression converged after 1 iterations, Case= 107

**NOTE-Tension/Compression converged after 1 iterations, Case= 108

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

**NOTE-Tension/Compression converged after 1 iterations, Case= 111

**NOTE-Tension/Compression converged after 1 iterations, Case= 112

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 113

**NOTE-Tension/Compression converged after 1 iterations, Case= 114

**NOTE-Tension/Compression converged after 2 iterations, Case= 115

**NOTE-Tension/Compression converged after 1 iterations, Case= 116

**NOTE-Tension/Compression converged after 1 iterations, Case= 117

**NOTE-Tension/Compression converged after 1 iterations, Case= 118

**NOTE-Tension/Compression converged after 1 iterations, Case= 119

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**NOTE-Tension/Compression converged after 1 iterations, Case= 122

**NOTE-Tension/Compression converged after 1 iterations, Case= 123

**NOTE-Tension/Compression converged after 1 iterations, Case= 124

**NOTE-Tension/Compression converged after 1 iterations, Case= 125

**NOTE-Tension/Compression converged after 1 iterations, Case= 126

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 127

**NOTE-Tension/Compression converged after 2 iterations, Case= 128

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 129

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 130

**NOTE-Tension/Compression converged after 1 iterations, Case= 131

**NOTE-Tension/Compression converged after 1 iterations, Case= 132

**NOTE-Tension/Compression converged after 1 iterations, Case= 133

**NOTE-Tension/Compression converged after 1 iterations, Case= 134

**NOTE-Tension/Compression converged after 1 iterations, Case= 135

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**NOTE-Tension/Compression converged after 1 iterations, Case= 137

**NOTE-Tension/Compression converged after 1 iterations, Case= 138

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**NOTE-Tension/Compression converged after 1 iterations, Case= 149

**NOTE-Tension/Compression converged after 1 iterations, Case= 150

**NOTE-Tension/Compression converged after 1 iterations, Case= 151

**NOTE-Tension/Compression converged after 1 iterations, Case= 152

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 153

**NOTE-Tension/Compression converged after 1 iterations, Case= 154

**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 155

**NOTE-Tension/Compression converged after 1 iterations, Case= 156

**NOTE-Tension/Compression converged after 1 iterations, Case= 157

**NOTE-Tension/Compression converged after 1 iterations, Case= 158

**NOTE-Tension/Compression converged after 1 iterations, Case= 159

**NOTE-Tension/Compression converged after 1 iterations, Case= 160

**NOTE-Tension/Compression converged after 1 iterations, Case= 161

**NOTE-Tension/Compression converged after 1 iterations, Case= 162

**NOTE-Tension/Compression converged after 1 iterations, Case= 163

**NOTE-Tension/Compression converged after 1 iterations, Case= 164

**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 165

**NOTE-Tension/Compression converged after 1 iterations, Case= 166

**NOTE-Tension/Compression converged after 2 iterations, Case= 167

**NOTE-Tension/Compression converged after 1 iterations, Case= 168

**NOTE-Tension/Compression converged after 1 iterations, Case= 169

**NOTE-Tension/Compression converged after 1 iterations, Case= 170

**NOTE-Tension/Compression converged after 1 iterations, Case= 171

**NOTE-Tension/Compression converged after 1 iterations, Case= 172

**NOTE-Tension/Compression converged after 1 iterations, Case= 173

**NOTE-Tension/Compression converged after 1 iterations, Case= 174

**NOTE-Tension/Compression converged after 1 iterations, Case= 175

**NOTE-Tension/Compression converged after 1 iterations, Case= 176

**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 177

**NOTE-Tension/Compression converged after 1 iterations, Case= 178

**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 179

**NOTE-Tension/Compression converged after 1 iterations, Case= 180

```
1116. LOAD LIST 101 TO 180
1117. START CONCRETE DESIGN
```



```
1118. CODE ACI
1119. TRACK 2 ALL
1120. FC 35000 MEMB 185
1121. CLB 0.05 MEMB 185
1122. CLS 0.05 ALL
1123. CLT 0.05 MEMB 185 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 TO 369 -
1124. 372 393 TO 399 402 TO 409 412 TO 419 422 TO 429 432 497 TO 504 513 TO 520
1125. MAXMAIN 25 ALL
1126. MINMAIN 25 ALL
1127. MINSEC 10 ALL
1128. DESIGN COLUMN 185
```

COLUMN NO. 185 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
AREA OF STEEL REQUIRED = 3620.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.571	134	STA	0.650

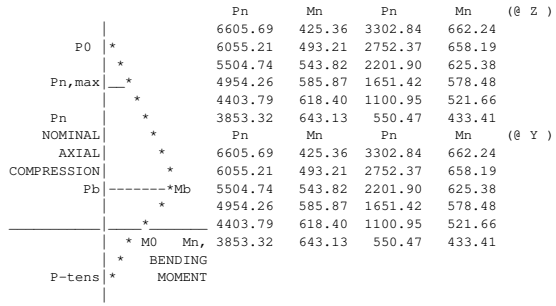
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8945.20	7156.16	3013.84	670.78	222.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
333.62	-1624.53	88.16	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8945.20	7156.16	3013.84	670.78	222.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
333.62	-1624.53	88.16	315.39	1.78879



**********END OF COLUMN DESIGN RESULTS**********

1129. END CONCRETE DESIGN

1130. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 1,2020 TIME= 16:51:21 ****

For technical assistance on STAAD.Pro, please visit
http://selectservices.bentley.com/en-US/

Details about additional assistance from
Bentley and Partners can be found at program menu
Help->Technical Support

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ATTACHMENT 12 STAAD OUTPUT – FOUNDATION F3

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    OCT 18, 2020                   *
*           Time=    17:59:45                       *
*                                                   *
*      USER ID: LNT ECC DIV                         *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F3.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 04/09/2020
     4. JOB NAME FDN. MKD. F3
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045
     7. JOB REV 0
     8. ENGINEER NAME SOR
     9. END JOB INFORMATION
    10. INPUT WIDTH 79
    11. UNIT METER KN
    12. JOINT COORDINATES
    13. 2 0 911.8 3.175; 34 -0.5 911.8 3.175; 35 -0.25 911.8 3.175
    14. 41 0.25 911.8 3.175; 42 0.5 911.8 3.175; 128 0 913.8 3.175
    15. 302 -0.5 911.8 3.42438; 304 -0.25 911.8 3.42438; 305 0 911.8 3.42438
    16. 306 0.25 911.8 3.42438; 307 0.5 911.8 3.42438; 322 -0.5 911.8 3.67438
    17. 323 -0.25 911.8 3.67438; 324 0 911.8 3.67438; 325 0.25 911.8 3.67438
    18. 326 0.5 911.8 3.67438; 399 -0.5 911.8 2.67438; 400 -0.25 911.8 2.67438
    19. 401 0 911.8 2.67438; 402 0.25 911.8 2.67438; 403 0.5 911.8 2.67438
    20. 410 -0.5 911.8 2.92438; 411 -0.25 911.8 2.92438; 412 0 911.8 2.92438
    21. 413 0.25 911.8 2.92438; 414 0.5 911.8 2.92438
    22. MEMBER INCIDENCES
    23. 185 2 128
    24. ELEMENT INCIDENCES SHELL
    25. 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41
    26. 344 322 323 304 302; 345 323 324 305 304; 346 324 325 306 305
    27. 347 325 326 307 306; 414 410 411 400 399; 415 411 412 401 400
    28. 416 412 413 402 401; 417 413 414 403 402; 424 34 35 411 410; 425 35 2 412 411
    29. 426 2 41 413 412; 427 41 42 414 413
    30. START GROUP DEFINITION
    31. ELEMENT
    32. _RAFT 327 328 332 333 344 TO 347 414 TO 417 424 TO 427
    33. _RAFT_MOM 327 333 344 TO 347 414 TO 417 424 427
    34. _RAFT_SHEAR 327 333 344 TO 347 414 TO 417 424 427
    35. MEMBER
    36. END GROUP DEFINITION
    37. ELEMENT PROPERTY
    38. 327 328 332 333 344 TO 347 414 TO 417 424 TO 427 THICKNESS 0.4
```

```
39. DEFINE MATERIAL START
40. ISOTROPIC CONCRETE
41. E 2.17185E+007
42. POISSON 0.17
43. DENSITY 25
44. ALPHA 1E-005
45. DAMP 0.05
46. TYPE CONCRETE
47. STRENGTH FCU 27579
48. END DEFINE MATERIAL
49. MEMBER PROPERTY AMERICAN
50. 185 PRIS YD 0.4 ZD 0.4
51. CONSTANTS
52. MATERIAL CONCRETE ALL
53. SUPPORTS
54. 327 328 332 333 344 TO 347 414 TO 417 424 TO 426 -
55. 427 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
```

JOINT	INFLUENCE AREA
2	0.062500
34	0.031250
35	0.062500
41	0.062500
42	0.031250
302	0.031211
304	0.062422
305	0.062422
306	0.062422
307	0.031211
322	0.015625
323	0.031250
324	0.031250
325	0.031250
326	0.015625
399	0.015625
400	0.031250
401	0.031250
402	0.031250
403	0.015625
410	0.031289
411	0.062578
412	0.062578
413	0.062578
414	0.031289
TOTAL	1.000000

```
56. ********************************************************************************
57. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
58. SELFWEIGHT Y -1
59. ELEMENT LOAD
60. * LOAD DUE TO BACKFILL
61. 327 328 332 333 344 TO 347 414 TO 417 424 TO 427 PR GY -22
62. * LOAD DUE TO SURCHARGE
63. 327 328 332 333 344 TO 347 414 TO 417 424 TO 427 PR GY -20
64. JOINT LOAD
```

```
 65. 128 FY -8 FZ -2
 66. ********************************************************************************
 67. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( L )
 68. JOINT LOAD
 69. 128 FY -32 FZ -5
 70. ********************************************************************************
 71. * UNFACTORED LOAD COMBINATIONS
 72. LOAD COMB 201 DL
 73. 1 1.0
 74. LOAD COMB 202 DL+L
 75. 1 1.0 2 1.0
 76. ********************************************************************************
 77. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
 78. LOAD COMB 101 1.4DL
 79. 1 1.4
 80. LOAD COMB 102 1.4DL+1.7L
 81. 1 1.4 2 1.7
 82. ********************************************************************************
 83. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**26**	**NUMBER OF MEMBERS**	**1**
NUMBER OF PLATES	**16**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**25**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 24/ 10/ 36 DOF
TOTAL PRIMARY LOAD CASES = 2, TOTAL DEGREES OF FREEDOM = 81
TOTAL LOAD COMBINATION CASES = 4 SO FAR.
SIZE OF STIFFNESS MATRIX = 3 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.1/ 142631.8 MB

```
 84. LOAD LIST 101 TO 102
 85. START CONCRETE DESIGN
```

```
86. CODE ACI
87. TRACK 2 ALL
88. FC 35000 MEMB 185
89. CLB 0.05 MEMB 185
90. CLS 0.05 ALL
91. CLT 0.05 ALL
92. MAXMAIN 25 ALL
93. MINMAIN 25 ALL
94. MINSEC 10 ALL
95. DESIGN COLUMN 185
```

==========

COLUMN NO. 185 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
4 - 25 MM	1.227	101	END	0.650

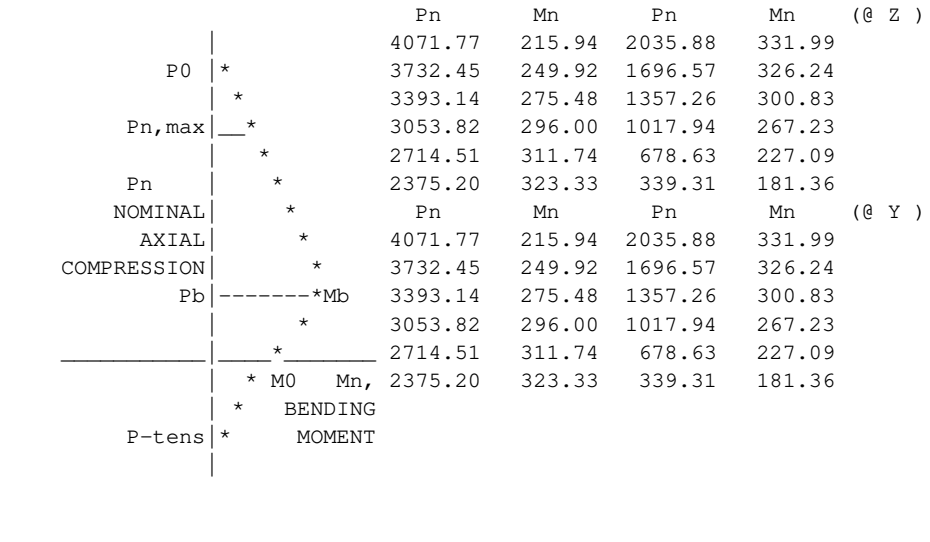
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5513.85	4411.08	1861.79	335.48	180.2
M0	P-tens.	Des.Pn	Des.Mn	e/h
132.10	-812.26	17.23	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5513.85	4411.08	1861.79	335.48	180.2
M0	P-tens.	Des.Pn	Des.Mn	e/h
132.10	-812.26	17.23	0.00	0.00000



**********END OF COLUMN DESIGN RESULTS**********

96. END CONCRETE DESIGN
97. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 18,2020 TIME= 17:59:50 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 13 STAAD OUTPUT – FOUNDATION F4

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    OCT  1, 2020                 *
*          Time=    17:10:18                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F4.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 04/09/2020
    4. JOB NAME FDN. MKD. F4
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. SET NL 200
   12. SET ITERLIM 15
   13. SET PRINT 1
   14. UNIT METER KN
   15. JOINT COORDINATES
   16. 2 0 911.8 3.175; 34 -0.5 911.8 3.175; 35 -0.25 911.8 3.175
   17. 41 0.25 911.8 3.175; 42 0.5 911.8 3.175; 128 0 913.8 3.175
   18. 302 -0.5 911.8 3.42438; 304 -0.25 911.8 3.42438; 305 0 911.8 3.42438
   19. 306 0.25 911.8 3.42438; 307 0.5 911.8 3.42438; 322 -0.5 911.8 3.67438
   20. 323 -0.25 911.8 3.67438; 324 0 911.8 3.67438; 325 0.25 911.8 3.67438
   21. 326 0.5 911.8 3.67438; 399 -0.5 911.8 2.67438; 400 -0.25 911.8 2.67438
   22. 401 0 911.8 2.67438; 402 0.25 911.8 2.67438; 403 0.5 911.8 2.67438
   23. 410 -0.5 911.8 2.92438; 411 -0.25 911.8 2.92438; 412 0 911.8 2.92438
   24. 413 0.25 911.8 2.92438; 414 0.5 911.8 2.92438; 415 0.75 911.8 3.42438
   25. 416 0.75 911.8 3.175; 417 1.125 911.8 3.42438; 418 1.125 911.8 3.175
   26. 419 0.75 911.8 3.67438; 420 1.125 911.8 3.67438; 421 0.75 911.8 2.92438
   27. 422 0.75 911.8 2.67438; 423 1.125 911.8 2.92438; 424 1.125 911.8 2.67438
   28. 425 -1.125 911.8 3.42438; 426 -0.75 911.8 3.42438; 427 -0.75 911.8 3.175
   29. 428 -1.125 911.8 3.175; 429 -1.125 911.8 3.67438; 430 -0.75 911.8 3.67438
   30. 431 -1.125 911.8 2.92438; 432 -0.75 911.8 2.92438; 433 -0.75 911.8 2.67438
   31. 434 -1.125 911.8 2.67438; 435 -0.5 911.8 3.92438; 436 -0.25 911.8 3.92438
   32. 437 0 911.8 3.92438; 438 0.25 911.8 3.92438; 439 0.5 911.8 3.92438
   33. 440 -0.5 911.8 4.17438; 441 -0.25 911.8 4.17438; 442 0 911.8 4.17438
   34. 443 0.25 911.8 4.17438; 444 0.5 911.8 4.17438; 445 0.75 911.8 3.92438
   35. 446 1.125 911.8 3.92438; 447 0.75 911.8 4.17438; 448 1.125 911.8 4.17438
   36. 449 -1.125 911.8 3.92438; 450 -0.75 911.8 3.92438; 451 -1.125 911.8 4.17438
   37. 452 -0.75 911.8 4.17438; 453 -0.5 911.8 2.42438; 454 -0.25 911.8 2.42438
   38. 455 -0.25 911.8 2.175; 456 -0.5 911.8 2.175; 457 0 911.8 2.42438
```



```
   39. 458 0 911.8 2.175; 459 0.25 911.8 2.42438; 460 0.25 911.8 2.175
   40. 461 0.5 911.8 2.42438; 462 0.5 911.8 2.175; 463 0.75 911.8 2.42438
   41. 464 0.75 911.8 2.175; 465 1.125 911.8 2.42438; 466 1.125 911.8 2.175
   42. 467 -1.125 911.8 2.42438; 468 -0.75 911.8 2.42438; 469 -0.75 911.8 2.175
   43. 470 -1.125 911.8 2.175
   44. MEMBER INCIDENCES
   45. 185 2 128
   46. ELEMENT INCIDENCES SHELL
   47. 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41
   48. 344 322 323 304 302; 345 323 324 305 304; 346 324 325 306 305
   49. 347 325 326 307 306; 414 410 411 400 399; 415 411 412 401 400
   50. 416 412 413 402 401; 417 413 414 403 402; 424 34 35 411 410; 425 35 2 412 411
   51. 426 2 41 413 412; 427 41 42 414 413; 428 307 415 416 42; 429 415 417 418 416
   52. 430 326 419 415 307; 431 419 420 417 415; 432 414 421 422 403
   53. 433 421 423 424 422; 434 42 416 421 414; 435 416 418 423 421
   54. 436 425 426 427 428; 437 426 302 34 427; 438 429 430 426 425
   55. 439 430 322 302 426; 440 431 432 433 434; 441 432 410 399 433
   56. 442 428 427 432 431; 443 427 34 410 432; 444 435 436 323 322
   57. 445 436 437 324 323; 446 437 438 325 324; 447 438 439 326 325
   58. 448 440 441 436 435; 449 441 442 437 436; 450 442 443 438 437
   59. 451 443 444 439 438; 452 439 445 419 326; 453 445 446 420 419
   60. 454 444 447 445 439; 455 447 448 446 445; 456 449 450 430 429
   61. 457 450 435 322 430; 458 451 452 450 449; 459 452 440 435 450
   62. 460 453 454 455 456; 461 454 457 458 455; 462 457 459 460 458
   63. 463 459 461 462 460; 464 399 400 454 453; 465 400 401 457 454
   64. 466 401 402 459 457; 467 402 403 461 459; 468 461 463 464 462
   65. 469 463 465 466 464; 470 403 422 463 461; 471 422 424 465 463
   66. 472 467 468 469 470; 473 468 453 456 469; 474 434 433 468 467
   67. 475 433 399 453 468
   68. START GROUP DEFINITION
   69. ELEMENT
   70. _RAFT 327 328 332 333 344 TO 347 414 TO 417 424 TO 475
   71. _RAFT_MOM 327 333 344 TO 347 414 TO 417 424 427 TO 475
   72. _RAFT_SHEAR 428 TO 475
   73. MEMBER
   74. END GROUP DEFINITION
   75. ELEMENT PROPERTY
   76. 327 328 332 333 344 TO 347 414 TO 417 424 TO 475 THICKNESS 0.4
   77. DEFINE MATERIAL START
   78. ISOTROPIC CONCRETE
   79. E 2.17185E+007
   80. POISSON 0.17
   81. DENSITY 25
   82. ALPHA 1E-005
   83. DAMP 0.05
   84. TYPE CONCRETE
   85. STRENGTH FCU 27579
   86. END DEFINE MATERIAL
   87. MEMBER PROPERTY AMERICAN
   88. 185 PRIS YD 0.7 ZD 0.4
   89. CONSTANTS
   90. MATERIAL CONCRETE ALL
   91. SUPPORTS
   92. 327 328 332 333 344 TO 347 414 TO 417 424 TO 474 -
   93. 475 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
```

JOINT	INFLUENCE AREA
2	0.062500
34	0.062500
35	0.062500
41	0.062500
42	0.062500
302	0.062422
304	0.062422
305	0.062422
306	0.062422
307	0.062422
322	0.062500
323	0.062500
324	0.062500
325	0.062500
326	0.062500
399	0.062500
400	0.062500
401	0.062500
402	0.062500
403	0.062500
410	0.062578
411	0.062578
412	0.062578
413	0.062578
414	0.062578
415	0.078028
416	0.078125
417	0.046817
418	0.046875
419	0.078125
420	0.046875
421	0.078222
422	0.078125
423	0.046933
424	0.046875
425	0.046817
426	0.078028
427	0.078125
428	0.046875
429	0.046875
430	0.078125
431	0.046933
432	0.078222
433	0.078125
434	0.046875
435	0.062500
436	0.062500
437	0.062500
438	0.062500
439	0.062500
440	0.031250
441	0.031250

442	0.031250
443	0.031250
444	0.031250
445	0.078125
446	0.046875
447	0.039063
448	0.023438
449	0.046875
450	0.078125
451	0.023438
452	0.039063
453	0.062422
454	0.062422
455	0.031172
456	0.031172
457	0.062422
458	0.031172
459	0.062422
460	0.031172
461	0.062422
462	0.031172
463	0.078028
464	0.038966
465	0.046817
466	0.023379
467	0.046817
468	0.078028
469	0.038966
470	0.023379
TOTAL	4.498604

```
 94. ********************************************************************************
 95. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
 96. SELFWEIGHT Y -1
 97. ELEMENT LOAD
 98. * LOAD DUE TO BACKFILL
 99. 327 328 332 333 344 TO 347 414 TO 417 424 TO 475 PR GY -22
100. * LOAD DUE TO SURCHARGE
101. 327 328 332 333 344 TO 347 414 TO 417 424 TO 475 PR GY -20
102. JOINT LOAD
103. 128 FY -5
104. ********************************************************************************
105. LOAD 2 LOADTYPE NONE  TITLE WIND LOAD ( WL )
106. JOINT LOAD
107. 128 FX 40
108. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 82 NUMBER OF MEMBERS 1
NUMBER OF PLATES 64 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 81

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 63/ 16/ 54 DOF
TOTAL PRIMARY LOAD CASES = 2, TOTAL DEGREES OF FREEDOM = 249
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 14 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.5/ 143474.9 MB

```
109. CHANGE
110. SUPPORTS
111. 327 328 332 333 344 TO 347 414 TO 417 424 TO 474 -
112. 475 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
113. ******************************************************************************
114. * UNFACTORED LOAD COMBINATIONS
115. LOAD  201 DL
116. REPEAT LOAD
117. 1 1.0
118. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  201

119. CHANGE
120. LOAD  202 DL+WL
121. REPEAT LOAD
122. 1 1.0 2 1.0
123. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  202

124. CHANGE
125. LOAD  203 0.6DL+WL
126. REPEAT LOAD
127. 1 0.6 2 1.0
```



```
128. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  203

129. CHANGE
130. ******************************************************************************
131. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
132. LOAD  101 1.4DL
133. REPEAT LOAD
134. 1 1.4
135. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  101

136. CHANGE
137. LOAD  102 1.2DL+1.6WL
138. REPEAT LOAD
139. 1 1.4 2 1.6
140. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  102

141. CHANGE
142. LOAD  103 0.9DL+1.6WL
143. REPEAT LOAD
144. 1 0.9 2 1.6
145. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  103

146. CHANGE
147. ******************************************************************************
```

148. PERFORM ANALYSIS

** ALL CASES BEING MADE ACTIVE BEFORE RE-ANALYSIS. **

**NOTE-Tension/Compression converged after 1 iterations, Case= 1

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

***WARNING - INSTABILITY AT JOINT 470 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 7.0799798E+05 L-MATRIX DIAG= -7.0547685E-08 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 470 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 7.0799798E+05 L-MATRIX DIAG= 1.9907020E-08 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 470 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 7.0799798E+05 L-MATRIX DIAG= 3.9581209E-09 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 470 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 8.4939199E+05 L-MATRIX DIAG= 4.0745363E-08 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 470 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 7.0799798E+05 L-MATRIX DIAG= -1.4156103E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 470 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 7.0799798E+05 L-MATRIX DIAG= -1.9499566E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 470 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 7.0799798E+05 L-MATRIX DIAG= -3.1071249E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 470 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 8.4939199E+05 L-MATRIX DIAG= 5.8207661E-10 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 470 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 7.0799798E+05 L-MATRIX DIAG= -2.8323848E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 470 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 7.0799798E+05 L-MATRIX DIAG= -2.5937334E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 470 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 8.4939199E+05 L-MATRIX DIAG= -1.5133992E-08 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 470 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 7.0799798E+05 L-MATRIX DIAG= -2.6705675E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 13

***WARNING - INSTABILITY AT JOINT 470 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.3209176E+04 L-MATRIX DIAG= 1.0913936E-11 EQN NO 247
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 470 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 8.4939199E+05 L-MATRIX DIAG= -1.5133992E-08 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 470 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 7.0799798E+05 L-MATRIX DIAG= 2.8545037E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 14
**START ITERATION NO. 15

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 2
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

```
**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  203



**NOTE-Tension/Compression converged after  1 iterations, Case=  101



**NOTE-Tension/Compression converged after  1 iterations, Case=  102


**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  103


 149. LOAD LIST 101 103
 150. START CONCRETE DESIGN
```



```
 151. CODE ACI
 152. TRACK 2 ALL
 153. FC 35000 MEMB 185
 154. CLB 0.05 MEMB 185
 155. CLS 0.05 ALL
 156. CLT 0.05 MEMB 185 327 328 332 333 344 TO 347 414 TO 417 424 TO 427
 157. MAXMAIN 25 ALL
 158. MINMAIN 25 ALL
 159. MINSEC 10 ALL
 160. DESIGN COLUMN 185
```

==========

COLUMN NO. 185 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, RECT SIZE - 400.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2800.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.402	101	END	0.650

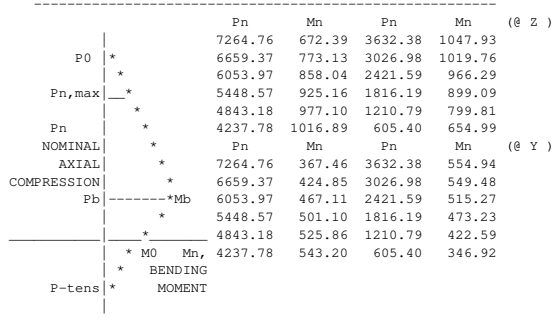
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
9837.70	7870.16	3544.31	1051.93	296.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
490.07	-1624.53	10.77	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
9837.70	7870.16	3265.06	560.22	171.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
261.67	-1624.53	10.77	0.00	0.00000



**********END OF COLUMN DESIGN RESULTS**********


```
161. END CONCRETE DESIGN
162. FINISH


          ********** END OF THE STAAD.Pro RUN **********

            **** DATE= OCT  1,2020   TIME= 17:10:57 ****

       ************************************************************
       *   For technical assistance on STAAD.Pro, please visit    *
       *   http://selectservices.bentley.com/en-US/               *
       *                                                          *
       *   Details about additional assistance from               *
       *   Bentley and Partners can be found at program menu      *
       *   Help->Technical Support                                *
       *                                                          *
       *      Copyright (c) 1997-2014 Bentley Systems, Inc.       *
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       ************************************************************
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